



DesignWare® Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC

Databook

dwc_32g_phy_tsmc12ffc_x2ns – Product Code: H976-0

Copyright Notice and Proprietary Information

© 2023 Synopsys, Inc. All rights reserved. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>

All other product or company names may be trademarks of their respective owners.

Free and Open-Source Software Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

Synopsys, Inc.

www.synopsys.com

Contents

Revision History	9
Preface	11
Databook Organization	11
Web Resources	12
Reference Documentation	12
STARs on the Web (SotW)	13
Synopsys Statement on Inclusivity and Diversity	13
Customer Support	14
Chapter 1	
Product Overview	15
1.1 Quick Reference	16
1.2 General Product Description	17
1.3 Features	18
1.4 Applications	19
1.5 Standards Compliance	20
1.5.1 Configuration Settings	25
1.5.2 PCIe 5.0	25
1.5.3 CCIX	27
1.5.4 Ethernet – 1000BASE-KX	28
1.5.5 Ethernet – 2.5GBASE-KX and 2.5GSEI	29
1.5.6 Ethernet – 5GBASE-KR and 5GSEI	30
1.5.7 Ethernet – 10GBASE-KX4 and XAUI	31
1.5.8 10G Ethernet – 10GBASE-KR	32
1.5.9 10G Ethernet – 40GBASE-KR4	33
1.5.10 10G Ethernet – 40GBASE-CR4 and 100GBASE-CR10	34
1.5.11 10G Ethernet-CAUI-10 C2C	35
1.5.12 10G Ethernet-XLAUI C2C and C2M	36
1.5.13 10G Ethernet-XLPPI and CPPI	37
1.5.14 25G Ethernet-25GBASE-KR and 25GBASE-KR-S	37
1.5.15 25G Ethernet-25GBASE-CR and 25GBASE-CR-S	39
1.5.16 25G Ethernet-CAUI-4 C2C and C2M	40
1.5.17 25G Ethernet-25GAUI, 50GAUI-2, 100GAUI-4, 200GAUI-8, 400GAUI-16 C2C and C2M	40
1.5.18 25G Ethernet-LAUI-2 C2C and C2M	42
1.5.19 100G Ethernet – 100GBASE-KR4	43
1.5.20 100G Ethernet – 100GBASE-CR4	44

1.5.21	SGMII	45
1.5.22	QSGMII	46
1.5.23	2.5G-SXGMII	47
1.5.24	5G-SXGMII and 5G-DXGMII	48
1.5.25	10G-SXGMII, 10G-DXGMII and 10G-QXGMII	49
1.5.26	JESD204C	50
1.5.27	CPRI	51
1.5.28	RXAUI/Double XAUI	52
1.5.29	SATA	53
1.6	Process Information	54
1.7	Devices List	54
Chapter 2		
Deliverables		57
2.1	Introduction	58
2.2	Front-End Views	58
2.3	Back-End Views	59
2.4	Signal-Integrity Views	59
2.5	Test Views	59
Chapter 3		
Functional Description		61
3.1	PHY	62
3.2	PMA	62
3.2.1	Receiver	64
3.2.2	RX VCO	68
3.2.3	Transmitter	69
3.2.4	Support Block	74
3.3	Raw PCS	76
Chapter 4		
PHY Usage and Configuration		79
4.1	PHY Power-Up	80
4.2	PHY Initialization Sequence	80
4.2.1	Resistor Tuning	82
4.3	Lane Configuration Update	83
4.4	MPLL Configuration Update	86
4.5	TX Datapath	87
4.5.1	TX Clock Alignment	87
4.5.2	Relationship Between tx_clk, tx_asic_div_clk, and tx_width	88
4.6	RX Datapath	90
4.7	RX Signal Detector	92
4.8	RX CDR PPM Detector	94
4.8.1	Configuring the RX CDR PPM Detector	94
4.9	Fast Context Switching	95
4.9.1	Usage Overview	95
4.9.2	Initial Calibration	96
4.9.3	Subsequent Rate Changes	98
4.9.4	Force Recalibration	103
4.10	TX and RX Power States	105

4.10.1 TX and RX Power State Description	105
4.10.2 Power State Transition Times for TX and RX	106
4.11 TX Equalization	108
4.11.1 TX Equalization Coefficient Mapping	108
4.11.2 TX EQ Initialization and Presets	110
4.12 TX and RX Adaptation	114
4.12.1 TX Adaptation	114
4.12.2 RX Adaptation Sequence	117
4.12.3 PCIe—TX Equalization Link Adaptation Using Figure of Merit	118
4.12.4 Long Reach Ethernet—TX Equalization Adaptation Using Direction Change	119
4.13 Receiver Detection	122
4.14 Control Register Interface	123
4.14.1 CR Parallel Interface	124
4.14.2 JTAG Interface	126
4.15 ROM Interface	128
4.16 External SRAM Support	129
4.16.1 Bypassing SRAM Bootloading	129
4.17 SRAM/ROM Support and Bootloading Options	133
4.18 Firmware	134
4.18.1 Checking the Firmware Version	134
4.18.2 Firmware Clock Interface	135
4.19 PERST	136
4.20 Aggregation and Bifurcation Support	138
4.21 Power Gating	141
4.22 Disabling a PHY	145
4.23 Configuring the PHY	146
4.24 Context Restore	148
4.24.1 Functional Description	148
4.24.2 Interfaces	148
4.24.3 Customer Debug Feature	148
4.24.4 Generating the SRAM and ROM Images	150

Chapter 5

Test and Diagnostics	155
5.1 Test Modes	156
5.1.1 Boundary Scan (ACJTAG)	156
5.1.2 Burn-In	159
5.1.3 IDDQ	159
5.1.4 Flyover	160
5.1.5 Scan (At-Speed, Stuck-At)	162
5.1.6 Scan Dump	163
5.1.7 On-Chip Clocking Support	163
5.1.8 Analog DC Test Capabilities	164
5.2 Diagnostics	176
5.2.1 Loopback Functions	177
5.2.2 BERT	179
5.2.3 RX Margining	182
5.2.4 RX Internal Eye Monitor	185
5.2.5 Suggested Steps In Debugging Link-Up Issues	185

5.2.6 SRAM Adaptation Recording	190
5.2.7 Temperature Sensor	190
Chapter 6	
Verilog Model Description	191
6.1 Overview	192
6.2 Verilog Function	192
6.3 Serial Line Modeling	193
6.4 Verilog Model Files and Simulators	194
6.5 Fast Simulation Mode	195
6.5.1 Interrupt Disable Mode	201
6.5.2 Skip Speed Change Calibration	201
6.6 PCIe Simulation Environment	201
Chapter 7	
Physical-Level Implementation	203
7.1 PMA Assembly	204
7.1.1 Flip-Chip Pinout	204
7.1.2 Bump Pad RDL and UBM Sizing	205
7.1.3 PMA Size	205
7.2 PMA Placement	206
7.2.1 PMA Rotation and Mirroring	206
7.2.2 PMA Abutment	207
7.3 PMA Fill	209
7.4 Operating Conditions	209
7.4.1 Power Supply Limits	209
7.4.2 Temperature	209
7.4.3 Electromigration	210
7.5 Power Supply Connectivity	211
7.5.1 Power and Ground Isolation	211
7.5.2 On-Chip Supply Decoupling	211
7.6 ESD	212
7.6.1 ESD Requirements	212
7.6.2 ESD Protection Strategy	212
7.6.3 ESD Back-to-Back Diode Connection to Core VSS	213
7.7 Reference Clock	215
7.8 Reference Resistor	215
7.8.1 Reference Resistor Contention Bus	215
7.9 Test Bus Connectivity	219
7.10 Raw PCS Synthesis	220
7.10.1 Synthesis Database	220
7.10.2 PHY (RAW PCS and PMA) Constraints	221
Chapter 8	
Board- and Package-Level Implementation	243
8.1 Package Type	244
8.2 Signal Integrity – RX and TX Differential Signal Guidelines	244
8.2.1 Package Guidelines – RX and TX Maximum Allowed Package Loss	244
8.2.2 Signal Integrity Analysis	244
8.3 Power Integrity – Supplies Requirements	245

8.3.1 Power Supply Requirements	245
8.3.2 Package and Board Guidelines – Power Supply Sharing	249
8.3.3 Recommended Maximum Off-Silicon Die Power Supply Inductance	250
8.3.4 Package Guidelines – Avoidance of Substrate Routing Above Inductors	251
8.4 External Reference Resistor	253
8.4.1 Package and Board Requirements – Reference Resistor	253
8.4.2 Reference Resistor Implementation for a Single PMA	254
8.4.3 Sharing a Reference Resistor Between Multiple PMAs	254
8.5 Reference Clock	256
8.6 Board Requirements – External Components	256
Chapter 9	
Reference Clock Implementation	259
9.1 Reference Clock Requirements	260
9.1.1 Reference Clock Requirements PCIe (100 MHz)	260
9.1.2 Reference Clock Amplitude Requirements	260
9.1.3 Generic Reference Clock Requirement for Protocols	261
9.1.4 Repeat Clock Specification	262
9.2 Reference Clock Implementation	264
9.2.1 External Off-Chip Clock Sources	266
9.2.2 On-Chip Clock Sources	267
9.2.3 Reference Clock Repeater	269
9.2.4 Reference Clock Distribution	270
Chapter 10	
Electrical Specifications	273
10.1 Power Dissipation Overview	274
10.2 Power Dissipation for PCIe	275
10.3 Power Dissipation for CCIX	280
10.4 Power Dissipation for Ethernet	281
Appendix A	
Support and Lane Configuration Settings	285
Appendix B	
Macro ifdef Settings	287
Appendix C	
Installation Quick Reference	291

Revision History

The following table lists the revision history of the PHY for each release. For a detailed description of PHY component updates, refer to the PHY release notes.

**Note**

- In some instances, documentation-only updates occur. The DesignWare IP product (https://www.synopsys.com/dw/ipdir.php?ds=dwc_multi_protocol_32g_phy) contains the latest information.
- Documentation-only updates are designated using the following numbering structure: <phy_version>_d<x> (for example, 1.00a_d1).

PHY Version	Date	Description
3.01a	April 4, 2023	Initial release

Preface

This databook describes the Synopsys DesignWare® Cores Multi-Protocol 32G PHY, which is a high-performance, low-power transceiver for high-bandwidth low latency applications.

Databook Organization

This databook is organized as follows:

- [Chapter 1, “Product Overview”](#), provides an introduction to the DesignWare® Cores Multi-Protocol 32G PHY and its features.
- [Chapter 2, “Deliverables”](#), describes the deliverables included in the PHY full-package release.
- [Chapter 3, “Functional Description”](#), provides a functional overview of the PHY and describes its test mode.
- [Chapter 4, “PHY Usage and Configuration”](#), explains the usage of PHY features and lists the PHY configuration settings for the protocols and data rates that are supported.
- [Chapter 5, “Test and Diagnostics”](#), provides the test modes supported in the PHY and the input configuration settings for each mode; diagnostic features to enhance the testbench characterization and ATE testing.
- [Chapter 6, “Verilog Model Description”](#), provides an overview of simulating the PHY Verilog model.
- [Chapter 7, “Physical-Level Implementation”](#), describes the physical implementation of the PHY.
- [Chapter 8, “Board- and Package-Level Implementation”](#), describes requirements and guidelines for implementing the PHY at the board and package level.
- [Chapter 9, “Reference Clock Implementation”](#), describes the reference clock that the PHY supports.
- [Chapter 10, “Electrical Specifications”](#), provides PHY electrical specifications for the TSMC12FFCLL technology.
- [Appendix A, “Support and Lane Configuration Settings”](#), provides support and lane configuration values for configuring the PHY for your application.
- [Appendix B, “Macro ifdef Settings”](#), provides the configuration and compile macro ifdef settings for your PHY.
- [Appendix C, “Installation Quick Reference”](#), provides the installation quick reference for your PHY.

Web Resources

- DesignWare IP product information: <https://www.synopsys.com/designware-ip.html>
- Your custom DesignWare IP page: <https://www.synopsys.com/dw/mydesignware.php>
- Documentation through SolvNetPlus: <https://solvnetplus.synopsys.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html>

Reference Documentation

- PCI Express Base Specification, Revision 5.0 Version 1.0, May 2019, PCI-SIG
- PCI Express Card Electromechanical Specification, Revision 3.0, July 2013, PCI-SIG
- PCI Express Mini Card Electromechanical Specification, Revision 2.0, April 2012, PCI-SIG
- CCIX Specification, Revision 1.1, Version 1.0, Sept 6, 2019
- Clause 70 of the IEEE 802.3-2018 specification
- Clause 71 of the IEEE 802.3-2018 specification
- Clauses 47 of the IEEE 802.3-2018 specification
- RXAUI - Reduced Pin XAUI (2 x 6.25Gbps), Dune Networks, V2.1, January 2008
- Clause 70 of the IEEE 802.3-2018 specification
- Clause 84, 72 of the IEEE 802.3-2018 specification
- Annex 83B of the IEEE 802.3-2018 specification
- Annex 83A of the IEEE 802.3-2018 specification
- Annex 86A of the IEEE 802.3-2018 specification
- Clause 72 of the IEEE 802.3-2018 specification
- Annex 83A of the IEEE 802.3-2018 specification
- Clause 110 of the IEEE 802.3-2018 specification
- Clause 111 of the IEEE 802.3-2018 specification
- Clause 92/92A of the IEEE 802.3-2018 specification
- Clause 93/93A,B,C of the IEEE 802.3-2018 specification
- Annex 83D of the IEEE 802.3-2018 specification
- Annex 83E of the IEEE 802.3-2018 specification
- Annex 109A of the IEEE 802.3-2018 specification
- Annex 109B of the IEEE 802.3-2018 specification
- Annex 135D of the IEEE 802.3cd specification
- Annex 135E of the IEEE 802.3cd specification
- Annex 120B of the IEEE 802.3-2018 specification
- Annex 120C of the IEEE 802.3-2018 specification
- Annex 135C of the IEEE 802.3cd specification
- Annex 135B of the IEEE 802.3cd specification

- Clauses 128 of the IEEE 802.3cb specification
- Annex 128A of the IEEE 802.3cb specification
- Annex 130A of the IEEE 802.3cb specification
- Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems
- Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems
- USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017)
USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016)
- Common Public Radio Interface (CPRI) specification V7.0, October 2015, CPRI Cooperation
- Clauses 48 and 71 of the IEEE 802.3-2018 specification
- IEEE Standard 1149.1-2001, IEEE
- IEEE Standard 1149.6-2003, IEEE
- *DFT Compiler User Guide*, Volume 1: Scan (XG Mode), Synopsys, Inc.

STARs on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the DesignWare IP product information:

<https://www.synopsys.com/designware-ip.html>

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Customer Support

To obtain support for your product, contact Support Center using one of the following methods:

- For the fastest response, enter a case through SolvNetPlus:

- a. <https://solvnetplus.synopsys.com>

**Note**

SolvNetPlus does not support Internet Explorer.

- b. Click the Cases menu and then click Create a New Case (below the list of cases).
 - c. Complete the mandatory fields that are marked with an asterisk and click Save.

Make sure to include the following:

- Product L1: DesignWare Cores
 - Product L2: SERDES_32G_PHY

For more information about general usage information, refer to the following article in SolvNetPlus:

<https://solvnetplus.synopsys.com/s/article/SolvNetPlus-Usage-Help-Resources>

- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product L1 and Product L2 names, and Version number in your e-mail so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
 - Attach any debug files you created.
- Or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<https://www.synopsys.com/support/global-support-centers.html>

1

Product Overview

This chapter includes the following sections:

- [“Quick Reference”](#)
- [“General Product Description”](#)
- [“Features”](#)
- [“Applications”](#)
- [“Standards Compliance”](#)
- [“Process Information”](#)
- [“Devices List”](#)

1.1 Quick Reference

This section provides the most relevant details of the DesignWare Cores Multi-Protocol 32G PHY, which are provided along with other details throughout this databook.

Features	
Receiver: AFE containing CTLE and VGAs, and 12-tap DFE (8 fixed and 4 floating taps) Transmitter: 3-tap FFE, low power voltage mode driver Transmitter clocking: Two independent PLLs, based on LC-tank oscillators in the support core and a ring oscillator (RO) based PLL per lane ■ Direct clocking mode for PCIe provides clocks directly from the LC-tank PLLs ■ Cascaded clocking for advanced PCIe bifurcation cases and for other protocols uses the LC-tank PLLs and lane RO PLLs configured in series Lane configurations: x1, x2, x4, x8 ■ Direct TX clocking is supported for x1, x2, x4 hard macro ■ Cascaded clocking is supported for x1, x2, x4, x8 hard macro ■ Support for lane aggregation and bifurcation	
Data Rates	EM
1.25–32 Gbps	88KPOH@110°C
Junction Temperature	Process Technology
-40–125°C	TSMC 12 nm Finfet
Power Supplies	Metal Stacks Supported
VP: 0.9 V -5% to 0.9 V +5% VPH: 1.2/1.5/1.8 V -5% to 1.2/1.5/1.8 V +10% VDDCORE: 0.8/0.9 V -10% to 0.8/0.9 V +5%	11M_2Xa1Xd_h_3Xe_vhv_2Y2R, 9M_2Xa1Xd_h_3Xe_vhv_2Z
ESD	Test Modes Supported
CDM: 6A peak discharge current HBM: 2,000 V	Boundary Scan (ACJTAG), Burn-In, IDDQ, Scan (At-Speed, Stuck-At), Diagnostics (serial and parallel loopback), and Normal Operation
Specifications	
For a list of supported protocols, refer to Table 1-2 (“Supported Level of Simulation and Hardware Validation”).	

1.2 General Product Description

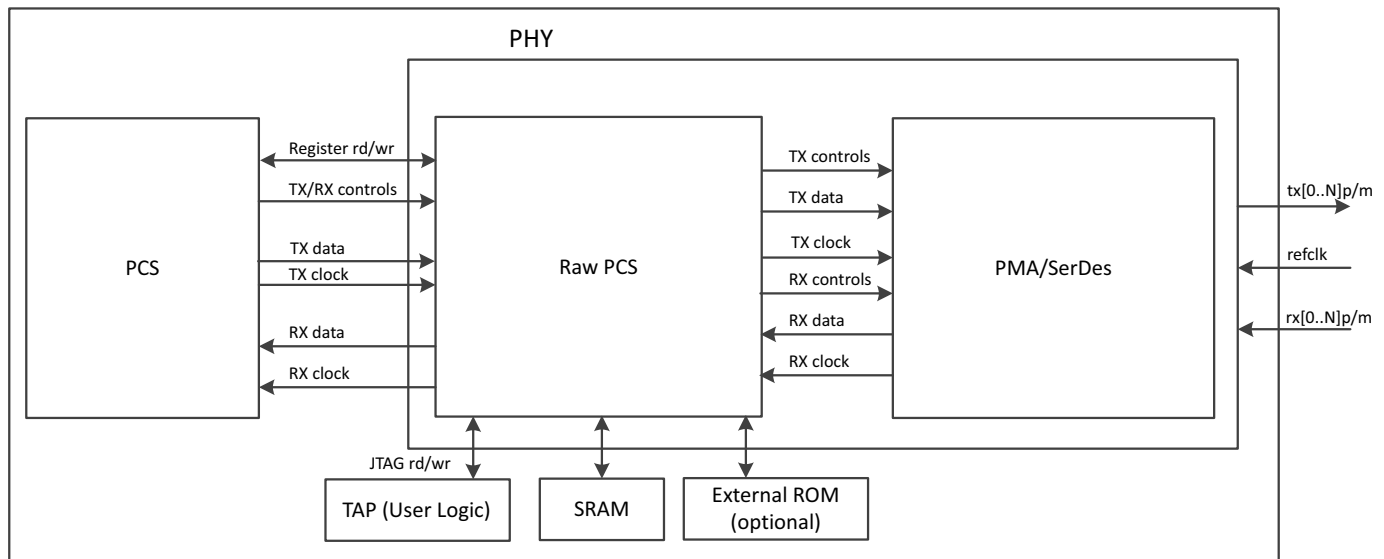
The DesignWare Cores Multi-Protocol 32G PHY provides full-duplex transceivers and consists of the following:

- A hard macro (physical medium attachment (PMA))/SerDes: `dwc_e32mp_phy_pma_x[1,2]_ns`
- A soft RTL module (raw physical coding sublayer (PCS)): `dwc_e32mp_phy_xn_ns_pcs_raw`

Where N = 1, 2, 4, 6, or 8

[Figure 1-1](#) shows the hierarchy of the PHY and its instantiation to create a PHY. The PHY architecture supports aggregation of multiple PMA to create higher-lane configurations.

Figure 1-1 DesignWare Cores Multi-Protocol 32G PHY Usage and Integration Model



Note

- PHYs with number of lanes greater than N (shown in [Figure 1-1](#)) can be generated by aggregating multiple instances of the PHY.
- Mandatory SRAM size is a 32 K single-port SRAM, consisting of 16-bit locations.
- The Synopsys PHY does not use any special layers (esd, deep-nwell, and so on). It uses only the standard process kit.
- The x1, x2, and x4 hard macros support both direct clocking and cascaded TX clocking.
- The x8 hard macro only supports cascaded TX clocking.

The PHY supports a source-synchronous clocking interface on the transmit (TX) and receive (RX) datapaths. The TX parallel data (per lane) is synchronous to the TX clock input of the corresponding lane. Similarly, the RX parallel data is output-synchronized (per lane) to an RX clock that is recovered from the serial data stream that corresponds to that lane.

The PHY supports different bit-width options for the TX and RX datapath interfaces. [Table 1-1](#) lists the general relation between baud rates and corresponding parallel data widths that are supported. For specific bit-widths supported for each protocol, see [Section 4.20, “Aggregation and Bifurcation Support”](#).

Table 1-1 Data Widths Supported

Serial Bit Rate (Gbps)	TX/RX Interface Width
1.25– 8	8-bit, 10-bit, 16-bit, 20-bit, 32-bit, 40-bit, 64-bit
8–16	16-bit, 20-bit, 32-bit, 40-bit, 64-bit
16–32	32-bit, 40-bit, 64-bit

1.3 Features

The PHY provides the following features:

- Highly customizable physical medium attachment (PMA) configuration (controlled by PCS):
 - x1, x2, x4, or x8 hard macro
 - Direct and cascaded TX clocking modes
 - Power consumption minimized using direct clocking
 - Larger baud-rate coverage and concurrent baud-rate support with cascaded clocking
 - Direct clocking mode supports up to x1, x2, and x4 hard macro.
 - Cascaded clocking mode supported by x1, x2, x4, and x8 hard macro.
 - Aggregation and bifurcation up to x16 PHY configuration
 - x16 configuration support is through aggregation of four x4 PHYs
- Single, dual, and quad channel macros configurations
- Extensive PMA debug capability via read/write and read-only registers in PCS
- Support for data-rates within the following ranges:
 - Direct clocking mode: 2.5, 5, 8, 16, 32 Gbps
 - Cascaded PLL mode: 1.25 Gbps to 32 Gbps.
- Configurable TX clocking:
 - Cascaded clock mode using central clocking core LC-PLL and lane RO-PLL
 - Direct clocking mode using LC-PLL only - for power savings for PCIe in x1, x2, and x4 configurations
- Register-based control of all PCS-to-PMA signals
- Adaptive and configurable RX Continuous-Time Linear Equalizer (CTLE) and Decision Feedback Equalizer (DFE)
- Adaptive and programmable TX equalization
- IEEE 1149.1 and 1149.6 (AC JTAG) boundary scan
- Built-In Self-Test (BIST) features for production; at-speed testing on any digital tester
- Advanced, built-in diagnostics including on-chip sampling scope
- Visibility and controllability of hard macro functions through programmable registers in the design:
 - Overrides on all ASIC side functional control inputs for easy debug
 - Access register space through a parallel interface

- Access register space through a JTAG port
- Configurable PMA and RAW PCS with extensive debug options:
 - Independent TX and RX control per lane
 - Configurable TX and RX power modes
 - Pseudo-Random Bit Sequence (PRBS) generation and checker
PRBS31, PRBS23, PRBS16, PRBS15, PRBS11, PRBS9, PRBS7
 - Programmable 10-bit pattern generation with error injection capability
- A wide range of reference clock frequencies with optional fractional frequency correction capability
- A wide range of divided clock frequencies for external-to-PHY usage with optional Spread-Spectrum Clock (SSC) capability
- Built-in, on-chip SSC generation and full configuration from -5000 to +5000 ppm with a maximum range of 5000 ppm; SSC generation across hard macros can not be synchronized. As such, PHY input `mplla/b_ssc_en` should be driven to 0 in case of aggregating multiple PHYs. If this feature is needed, contact Synopsys.
- Temperature sensor

**Note**

With fractional-N or internal SSC generation enabled, increased TX output jitter results. For more information, contact Synopsys.

- A reference clock repeater for other modules
- Supports flip-chip packaging

1.4 Applications

The DesignWare Cores Multi-Protocol 32G PHY IP (PHY) is targeted for connectivity in chip-to-chip or card-to-card communication across a combination of printed circuit board, connectors, backplane wiring, or cable.

Designed for low power, efficient area, low test costs, and process centering for optimal yield and interoperability, the PHY enables designers to introduce competitive products using the latest standards.

1.5 Standards Compliance

The PHY is designed to be compliant with the standards listed within the subsections of this section with requisite exceptions listed. [Table 1-2](#) indicates the level of simulation and hardware validation that Synopsys performs for each listed standard.

Table 1-2 Supported Level of Simulation and Hardware Validation

Protocol	Specification	Non-Continuous Data Rate (Gbps)	Simulation Validation			Silicon Validation		
			Data Rate	Electrical	Synopsys Controller/PHY Interoperation	Data Rate	Electrical	Synopsys Controller/PHY Interoperation
PCIe 5.0	PCI Express Base Specification, Revision 5.0 Version 1.0, May 2019, PCI-SIG	2.5, 5, 8, 16, 32	Yes	Yes	Yes	Yes	Yes	Yes
CCIX	CCIX Specification, Revision 1.1, Version 1.0, Sept 6, 2019	2.5, 5, 8, 16, 20, 25, 32	Yes	Yes	No ^a	Yes	No ^b	No ^a
1000BASE-KX	Clause 70 of the IEEE 802.3-2018 specification	1.25	Yes	Yes	Yes	Yes	No ^b	No
10GBASE-KX4	Clause 71 of the IEEE 802.3-2018 specification	4x3.125	Yes	Yes	Yes	Yes	No ^b	No
XAUI	Clauses 47 of the IEEE 802.3-2018 specification	3.125	Yes	Yes	Yes	Yes	No ^b	No
2.5GBASE-KX	Clauses 128 of the IEEE 802.3cb specification	3.125	Yes	Yes	Yes	Yes	No ^b	No
2.5GSEI	Annex 128A of the IEEE 802.3cb specification	3.125	Yes	Yes	Yes	Yes	No ^b	No
5GBASE-KR	Clauses 130 of the IEEE 802.3cb specification	5.15625	Yes	Yes	Yes	Yes	No ^b	No
5GSEI	Annex 130A of the IEEE 802.3cb specification	5.15625	Yes	Yes	Yes	Yes	No ^b	No

Protocol	Specification		Simulation Validation			Silicon Validation		
		Non-Continuous Data Rate (Gbps)	Data Rate	Electrical	Synopsys Controller/PHY Interoperation	Data Rate	Electrical	Synopsys Controller/PHY Interoperation
RXAUI / Double XAUI	Clauses 48 and 71 of the IEEE 802.3-2018 specification	6.25	Yes	Yes	Yes	Yes	No ^b	No
10GBASE-KR	Clause 72 of the IEEE 802.3-2018 specification	10.3125	Yes	Yes	Yes	Yes	No ^b	No ^a
40GBASE-KR4	Clause 84, 72 of the IEEE 802.3-2018 specification	4 x 10.3125	Yes	Yes	Yes	Yes	No ^b	No ^a
40GBASE-CR4 100GBASE-CR10	Clause 85 of the IEEE 802.3-2018 specification	4 x 10.3125 10 x 10.3125	Yes	Yes	Yes No	Yes	No ^b	No ^a
XLAUI C2M	Annex 83B of the IEEE 802.3-2018 specification	4 x 10.3125	Yes	Yes	Yes	Yes	No ^b	No ^a
XLAUI C2C	Annex 83A of the IEEE 802.3-2018 specification	4 x 10.3125	Yes	Yes	Yes	Yes	No ^b	No ^a
XLPP/CPPI (nPP)	Annex 86A of the IEEE 802.3-2018 specification	4 x 10.3125 10 x 10.3125	Yes	Yes	Yes No	Yes	No ^b	No ^a
CAUI-10 C2C	Annex 83A of the IEEE 802.3-2018 specification	10 x 10.3125	Yes	Yes	No	Yes	No ^b	No ^a
25GBASE-CR 25GBASE-CR-S	Clause 110 of the IEEE 802.3-2018 specification	25.78125	Yes	Yes	Yes	Yes	No ^b	No ^a
25GBASE-KR 25GBASE-KR-S	Clause 111 of the IEEE 802.3-2018 specification	25.78125	Yes	Yes	Yes	Yes	No ^b	No ^a
100GBASE-CR4	Clause 92/92A of the IEEE 802.3-2018 specification	4 x 25.78125	Yes	Yes	Yes	Yes	No ^b	No ^a

Protocol	Specification		Simulation Validation			Silicon Validation		
		Non-Continuous Data Rate (Gbps)	Data Rate	Electrical	Synopsys Controller/PHY Interoperation	Data Rate	Electrical	Synopsys Controller/PHY Interoperation
100GBASE-KR4	Clause 93/93A,B,C of the IEEE 802.3-2018 specification	4 x 25.78125	Yes	Yes	Yes	Yes	Yes	No ^a
CAUI-4 C2C and C2M	Annex 83D of the IEEE 802.3-2018 specification Annex 83E of the IEEE 802.3-2018 specification	25.78125	Yes	Yes	Yes	Yes	No ^b	No ^a
25GAUI C2C and 25GAUI C2M	Annex 109A of the IEEE 802.3-2018 specification Annex 109B of the IEEE 802.3-2018 specification	25.78125	Yes	Yes	Yes	Yes	No ^b	No ^a
50GAUI-2 C2C 100GAUI-4 C2C	Annex 135D of the IEEE 802.3cd specification	2 x 26.5625 4 x 26.5625	Yes	Yes	Yes	Yes	No ^b	No ^a
50GAUI-2 C2M 100GAUI-4 C2M	Annex 135E of the IEEE 802.3cd specification	2 x 26.5625 4 x 26.5625	Yes	Yes	Yes	Yes	No ^b	No ^a
200GAUI-8 C2C 400GAUI-16 C2C	Annex 120B of the IEEE 802.3-2018 specification	8 x 26.5625 16 x 26.5625	Yes	Yes	Yes	Yes	No ^b	No ^a
200GAUI-8 C2M 400GAUI-16 C2M	Annex 120C of the IEEE 802.3-2018 specification	8 x 26.5625 16 x 26.5625	Yes	Yes	Yes	Yes	No ^b	No ^a
LAUI-2 C2C and LAUI-2 C2M	Annex 135B of the IEEE 802.3cd specification Annex 135C of the IEEE 802.3cd specification	2 x 25.78125	Yes	Yes	Yes	Yes	No	No ^a

Protocol	Specification	Simulation Validation				Silicon Validation		
		Non-Continuous Data Rate (Gbps)	Data Rate	Electrical	Synopsys Controller/PHY Interoperation	Data Rate	Electrical	Synopsys Controller/PHY Interoperation
SGMII	Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems	1.25G	Yes	Yes	Yes	Yes	No	No
QSGMII	Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems	5G	Yes	Yes	Yes	Yes	No	No
2.5G-SXGMII	USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017) USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016)	2.578125	Yes	Yes	Yes	Yes	No	No
5G-SXGMII, 5G-DXGMII	USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017) USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016)	5.15625	Yes	Yes	Yes	Yes	No	No
10G-SXGMII, 10G-DXGMII, 10G-QXGMII,	USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017) USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016)	10.3125	Yes	Yes	Yes	Yes	No	No
SATA	Serial ATA Revision 3.3, February 2016	1.5, 3, 6.0G	Yes	Yes	No	Yes	No	No

Protocol	Specification		Simulation Validation			Silicon Validation		
		Non-Continuous Data Rate (Gbps)	Data Rate	Electrical	Synopsys Controller/PHY Interoperation	Data Rate	Electrical	Synopsys Controller/PHY Interoperation
JESD204C	Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converters JESD204C (Revision of JESD204B.01), December 2017	Class B: 1-12.5 Gbps (standard discrete rates: 2.4, 4.8, 9.6, 3.125, 6.25, 12.5) Class C: 6.375-32.44032 Gbps (standard discrete rates: 8.11008, 9.8304, 14.7456, 16.22016, 19.6608, 24.33024, 32.44032)	Yes	Yes	No	Yes	No	No
CPRI	Common Public Radio Interface (CPRI) specification V7.0, October 2015, CPRI Cooperation	1.2288, 2.4576, 3.072, 4.9152, 6.144, 8.11008, 9.8304, 10.1376, 12.16512, 24.33024	Yes	Yes	No	Yes	No	No

a. For Synopsys Controller/PHY interoperation testing details, refer to controller documentation.

b. For further details on electrical characterization tests, contact Synopsys.

1.5.1 Configuration Settings

For each standard, there are multiple possibilities with respect to configuration based on:

- Baudrate
- Reference clock frequency
- Data path width
- Clocking configuration

For more information on per protocol support block and lane configuration settings, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual*

(dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).



Note

The customer environment determines some of the configuration settings, such as the TX equalization, in case of protocols that do not adapt the TX.

1.5.2 PCIe 5.0

The PHY supports the PCIe 5.0 (32 Gbps) protocol and data rate, and is backward compatible with the PCIe 4.0 (16 Gbps), PCIe 3.1 (8 Gbps), PCIe 2.1 (5 Gbps), and PCIe 1.1 (2.5 Gbps) protocol and data rates. The PHY also supports PCIe L1-substate power managements and Separate RefClk Independent SSC Architecture (SRIS).

The PHY complies with the required features specified in the PCI Express Base Specification, Revision 5.0 Version 1.0, May 2019, PCI-SIG.



Note

■ Reference Clock Frequency

- The PHY supports 25 MHz and 100 MHz reference clock (refclk).
- 25 MHz refclk requires use of a higher frequency firmware clock (fw_clk). For more details, contact Synopsys.
- For support of other reference clock frequencies, contact Synopsys.

■ SRIS

- SRIS/SRNS is supported for both 25-MHz and 100-MHz reference clocks (refclk).
- The PHY can internally generate spread spectrum TX output from 25-MHz and 100-MHz unspread reference clock.
- If you are doing a U.2/U.3 implementation, contact Synopsys for support.
- If you are doing a retimer implementation, contact Synopsys for support.

1.5.2.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-3](#) notes the equalization setup for the receiver and transmitter.

Table 1-3 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	Gen1,2: AFE uses a fixed setting Gen3,4,5: AFE is adapted
Receiver DFE equalization	Gen1,2: DFE is bypassed Gen3,4,5: DFE is adapted
Transmitter equalization	Gen1: 3.5 dB fixed de-emphasis Gen2: 6 dB fixed de-emphasis Gen3,4,5: TX link adaptation using FOM based method with fixed presets

The PHY handles adaptation of receiver and transmitter without any manual intervention using FOM based adaptation of TX presets. For more information on FOM based link adaptation, see Section 4.12.3, “PCIe – TX Equalization Link Adaptation Using Figure of Merit”.

**Note**

PHY does not support direction change method for PCIe link adaptation.

1.5.2.2 Exceptions

Table 1-4 notes exceptions to the PCIe 5.0 specification.

Table 1-4 Exceptions to Specification Allowed for PCIe 5.0

Category	Exception Allowed
EI detection	- Only inferred electrical idle detection is supported. - EI Entry – EIOS is used to detect entry into electrical Idle. In the absence of receiving an EIOS, electrical idle may be inferred. For details of “Internal Electrical Idle”, refer to the PCIe Base Specification. The LOS (sigdet) is not used. - EI Exit – LOS (sigdet) can be used for low frequency patterns to detect exit from electrical idle. In PCIe, EIEOS contains a low frequency pattern.
PLL bandwidth	For PCIe Gen3 and Gen4, the PLL bandwidth settings comply with the PCI Express Base Specification, Revision 5.0 Version 1.0, May 2019, PCI-SIG.

**Note**

For use of PCIe with a DC coupled link, contact Synopsys.

1.5.3 CCIX

The Multi-Protocol 32G PHY supports CCIX with data rates of 2.5, 5, 8, 16, 20, 25, and 32 Gbps.

The PHY complies with the required features specified in the CCIX Specification, Revision 1.1, Version 1.0, Sept 6, 2019.



Note

- The PHY supports EDR25-SR channels.
- If you need EDR25-LR channel support, contact Synopsys.

1.5.3.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

Table 1-5 notes the equalization setup for the receiver and transmitter.

Table 1-5 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using FOM based method with fixed presets

The PHY handles adaptation of receiver and transmitter without any manual intervention using FOM based adaptation of TX presets. For more information on FOM based link adaptation, see Section 4.12.3, “PCIe – TX Equalization Link Adaptation Using Figure of Merit”.



Note

- PHY does not support direction change method for CCIX link adaptation.

1.5.3.2 Exceptions

There are no noted exceptions to the CCIX Specification, Revision 1.1, Version 1.0, Sept 6, 2019.



Note

- Data rates above 16.0 Gbps require supply overdrive and constrained temperature operation. (For product overview information including supply voltage and junction temperature, see Section 1.1, “Quick Reference”.)
- For the use of CCIX with reference clock frequencies lower than 100 MHz, contact Synopsys.

1.5.4 Ethernet – 1000BASE-KX

The PHY supports 1000BASE-KX with a data rate of 1.25 Gbps.

The PHY complies with the required features of Clause 70 of the IEEE 802.3-2018 specification.

1.5.4.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-6](#) notes the equalization setup for the receiver and transmitter.

Table 1-6 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting
Receiver DFE equalization	DFE is bypassed
Transmitter equalization	TX uses a fixed setting

The PHY can operate without adaptation of the receiver or transmitter.

1.5.4.2 Exceptions

[Table 1-7](#) notes exceptions to the 100BASE-KX specification.

Table 1-7 Exceptions to Specification Allowed for 1000BASE-KX

Category	Exception Allowed
TX rise/fall	Faster than specification

1.5.5 Ethernet – 2.5GBASE-KX and 2.5GSEI

The PHY supports 2.5GBASE-KX and 2.5GSEI with data rate of 3.125 Gbps.

The PHY complies with the required feature of Clause 128 and Annex 128A of the IEEE 802.3cb-2018 specification.

1.5.5.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-8](#) notes the equalization setup for the receiver and transmitter.

Table 1-8 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting
Receiver DFE equalization	DFE is bypassed
Transmitter equalization	TX uses a fixed setting

The PHY can operate without adaptation of the receiver or transmitter.

1.5.5.2 Exceptions

[Table 1-9](#) notes exceptions to the 2.5GBASE-KX and 2.5GSEI specification.

Table 1-9 Exceptions to Specification Allowed for 2.5GBASE-KX and 2.5GSEI

Category	Exception Allowed
TX rise/fall	Faster than specification

1.5.6 Ethernet – 5GBASE-KR and 5GSEI

The PHY supports 5GBASE-KR and 5GSEI with data rate of 5.15625 Gbps.

The PHY complies with the required feature of Clause 130 and Annex 130A of the IEEE 802.3cb-2018 specification.

1.5.6.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-10](#) notes the equalization setup for the receiver and transmitter.

Table 1-10 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting.
Receiver DFE equalization	DFE is bypassed.
Transmitter equalization	TX uses a fixed setting.

The PHY can operate without adaptation of the receiver or transmitter.

1.5.6.2 Exceptions

[Table 1-11](#) notes exceptions to the 5GBASE-KR and 5GSEI specification.

Table 1-11 Exceptions to Specification Allowed for 5GBASE-KR and 5GSEI

Category	Exception Allowed
TX rise/fall	Faster than specification



Note

Adaption is not currently supported for quarter-rate protocols, which might limit channel support. For more information, contact Synopsys.

1.5.7 Ethernet – 10GBASE-KX4 and XAUI

The PHY supports 10GBASE-KX4 with a data rate of 4 x 3.125 Gbps and XAUI with a data rate of 3.125 Gbps.

The PHY complies with the required features of Clauses 47 of the IEEE 802.3-2018 specification.

1.5.7.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-12](#) notes the equalization setup for the receiver and transmitter.

Table 1-12 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting.
Receiver DFE equalization	DFE is bypassed.
Transmitter equalization	TX uses a fixed setting.

The PHY can operate without adaptation of the receiver or transmitter.

1.5.7.2 Exceptions

[Table 1-13](#) notes exceptions to the 10GBASE-KX4 and XAUI specification.

Table 1-13 Exceptions to Specification Allowed for 10GBASE-KX4 and XAUI

Category	Exception Allowed
TX rise/fall	Faster than specification
Lane-to-lane skew	Exceeds limit

1.5.8 10G Ethernet – 10GBASE-KR

The PHY supports 10GBASE-KR with a data rate of 10.3125 Gbps.

The PHY complies with Clause 72 of the IEEE 802.3-2018 specification. The PHY can support auto-negotiation and link training (AN/LT) when paired with a controller that supports these features.

1.5.8.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-14](#) notes the equalization setup for the receiver and transmitter.

Table 1-14 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#).

1.5.8.2 Exceptions

[Table 1-15](#) notes exceptions to the 10GBASE-KR specification.

Table 1-15 Exceptions to Specification Allowed for 10GBASE-KR

Category	Exception Allowed
TX rise/fall edge rate control	Slower than specification
TX voltage magnitude change from coefficient update	Higher than specification
R _{PRE} for maximum C(0) R _{POST} minimum	Exceeds the preshoot limit for maximum C(0) setting Exceeds the minimum deemphasis limit



Note

For support of Energy Efficient Ethernet (EEE) please contact Synopsys.

1.5.9 10G Ethernet – 40GBASE-KR4

The PHY supports 40GBASE-KR4 with a data rate of 10.3125 Gbps.

The PHY complies with Clause 84, 72 of the IEEE 802.3-2018 specification. The PHY can support auto-negotiation and link training (AN/LT) when paired with a controller that supports these features and forward error correction (FEC).

1.5.9.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-16](#) notes the equalization setup for the receiver and transmitter.

Table 1-16 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#).

1.5.9.2 Exceptions

[Table 1-17](#) notes exceptions to the 40GBASE-KR4 specification.

Table 1-17 Exceptions to Specification Allowed for 40GBASE-KR4

Category	Exception Allowed
TX rise/fall edge rate control	Faster than specification



Note

For support of Energy Efficient Ethernet (EEE) please contact Synopsys.

1.5.10 10G Ethernet – 40GBASE-CR4 and 100GBASE-CR10

The PHY supports 40GBASE-CR4 with a data rate of 4 x 10.3125 Gbps and 100GBASE-CR10 with a data rate of 10 x 10.3125 Gbps.

The PHY complies with Clause 85 of the IEEE 802.3-2018 specification. The PHY can support auto-negotiation and link training (AN/LT) when paired with a controller that supports these features and forward error correction (FEC).

1.5.10.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-18](#) notes the equalization setup for the receiver and transmitter.

Table 1-18 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#).

1.5.10.2 Exceptions

There are no noted exceptions to the specification.



Note

For support of Energy Efficient Ethernet (EEE) please contact Synopsys.

1.5.11 10G Ethernet–CAUI-10 C2C

The PHY supports CAUI chip-to-chip with a data rate of 10 x 10.3125 Gbps.

The PHY complies with Annex 83A of the IEEE 802.3-2018 specification.

1.5.11.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-19](#) notes the equalization setup for the receiver and transmitter.

Table 1-19 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.11.2 Exceptions

There are no noted exceptions to the specification.

1.5.12 10G Ethernet–XLAUI C2C and C2M

The PHY supports XLAUI chip-to-chip and chip-to-module with a data rate of 4 x 10.3125 Gbps.

The PHY complies with Annex 83A of the IEEE 802.3-2018 specification (C2C) and Annex 83B of the IEEE 802.3-2018 specification (C2M).

1.5.12.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-20](#) notes the equalization setup for the receiver and transmitter.

Table 1-20 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.12.2 Exceptions

There are no noted exceptions to the specification.

1.5.13 10G Ethernet–XLPPI and CPPI

The PHY supports XLPPI with a data rate of 4 x 10.3125 Gbps and 10 x 10.3125 Gbps.

The PHY complies with Annex 86A of the IEEE 802.3-2018 specification.

1.5.13.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

Table 1-21 notes the equalization setup for the receiver and transmitter.

Table 1-21 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.13.2 Exceptions

There are no noted exceptions to the specification.

1.5.14 25G Ethernet–25GBASE-KR and 25GBASE-KR-S

The PHY supports 25GBASE-KR and 25GBASE-KR-S with a data rate of 25.78125 Gbps.

The PHY complies with Clause 111 of the IEEE 802.3-2018 specification. The PHY can support auto-negotiation and link training (AN/LT) when paired with a controller that supports these features.



Note

- The PHY supports up to 30 dB channel in no FEC and BASE-R FEC cases.
- The PHY supports up to 35 dB channel in RS-FEC case.
- The channel is defined from package pin to package pin—the PHY allows an additional 3 dB of loss in the RX package traces and 2 dB in the TX package traces.
- Contact Synopsys to confirm support of your specific channel.

1.5.14.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-22](#) notes the equalization setup for the receiver and transmitter.

Table 1-22 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#).

1.5.14.2 Exceptions

[Table 1-23](#) notes exceptions to the 25GBASE-KR and 25GBASE-KR-S specification.

Table 1-23 Exceptions to Specification Allowed for 25GBASE-KR and 25GBASE-KR-S

Category	Exception Allowed
TX voltage magnitude change from coefficient update	Higher than specification
R _{PRE} for maximum C(0) R _{POST} minimum	Exceeds the preshoot limit for maximum C(0) setting Exceeds the minimum deemphasis limit
Linear fit pulse peak	Less than 0.75*vf Note: can pass 0.71 peak/vf ratio of 100GBASE-KR4



Note

For support of Energy Efficient Ethernet (EEE) please contact Synopsys.

1.5.15 25G Ethernet–25GBASE-CR and 25GBASE-CR-S

The PHY supports 25GBASE-CR and 25GBASE-CR-S with a data rate of 25.78125 Gbps.

The PHY complies with Clause 110 of the IEEE 802.3-2018 specification. The PHY can support auto-negotiation and link training (AN/LT) when paired with a controller that supports these features.



Note

- The PHY supports up to 30 dB channel in no FEC and BASE-R FEC cases.
- The channel is defined from package pin to package pin—the PHY allows an additional 3 dB of loss in the RX package traces and 2 dB in the TX package traces.
- Full support of 25GBASE-CR/CR-S is pending silicon verification.
- Contact Synopsys to confirm support of your specific channel.

1.5.15.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

Table 1-24 notes the equalization setup for the receiver and transmitter.

Table 1-24 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see Chapter 4, “PHY Usage and Configuration”.

1.5.15.2 Exceptions

Table 1-25 notes exceptions to the 25GBASE-CR and 25GBASE-CR-S specification.

Table 1-25 Exceptions to Specification Allowed for 25GBASE-CR and 25GBASE-CR-S

Category	Exception Allowed
TX voltage magnitude change from coefficient update	Higher than specification
R _{PRE} for maximum C(0) R _{POST} minimum	Exceeds the preshoot limit for maximum C(0) setting Exceeds the minimum deemphasis limit



Note

For support of Energy Efficient Ethernet (EEE) please contact Synopsys.

1.5.16 25G Ethernet–CAUI-4 C2C and C2M

The PHY supports CAUI-4 chip-to-chip and chip-to-module with a data rate of 25.78125 Gbps.

The PHY complies with Annex 83D of the IEEE 802.3-2018 specification (C2C) and Annex 83E of the IEEE 802.3-2018 specification (C2M).

1.5.16.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

Table 1-26 notes the equalization setup for the receiver and transmitter.

Table 1-26 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.16.2 Exceptions

Table 1-27 notes exceptions to the 25G Ethernet–CAUI-4 C2C and C2M specification.

Table 1-27 Exceptions to Specification Allowed for 25G Ethernet–CAUI-4 C2C and C2M

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER

1.5.17 25G Ethernet–25GAUI, 50GAUI-2, 100GAUI-4, 200GAUI-8, 400GAUI-16 C2C and C2M

The PHY supports 25GAUI chip-to-chip and chip-to-module.

The PHY complies with all of the specifications noted in Table 1-28.

Table 1-28 Supported Data Rates for 25GAUI Ethernet Protocols

Protocol	Data Rate	IEEE Specification
25GAUI C2C	1 x 25.78125 Gbps	Annex 109A of the IEEE 802.3-2018 specification
25GAUI C2M	1 x 25.78125 Gbps	Annex 109B of the IEEE 802.3-2018 specification
50GAUI-2 C2C	2 x 26.5625 Gbps	Annex 135D of the IEEE 802.3cd specification

Protocol	Data Rate	IEEE Specification
50GAUI-2 C2M	2 x 26.5625 Gbps	Annex 135E of the IEEE 802.3cd specification
100GAUI-4 C2C	4 x 26.5625 Gbps	Annex 135D of the IEEE 802.3cd specification
100GAUI-4 C2M	4 x 26.5625 Gbps	Annex 135E of the IEEE 802.3cd specification
200GAUI-8 C2C	8 x 26.5625 Gbps	Annex 120B of the IEEE 802.3-2018 specification
200GAUI-8 C2M	8 x 26.5625 Gbps	Annex 120C of the IEEE 802.3-2018 specification
400GAUI-16 C2C	16 x 26.5625 Gbps	Annex 120B of the IEEE 802.3-2018 specification
400GAUI-16 C2M	16 x 26.5625 Gbps	Annex 120C of the IEEE 802.3-2018 specification

1.5.17.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-29](#) notes the equalization setup for the receiver and transmitter.

Table 1-29 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.17.2 Exceptions

[Table 1-30](#) notes exceptions to the 25G Ethernet–25GAUI, 50GAUI-2, 100GAUI-4, 200GAUI-8, 400GAUI-16 C2C and C2M specification.

Table 1-30 Exceptions to Specification Allowed for 25G Ethernet–25GAUI, 50GAUI-2, 100GAUI-4, 200GAUI-8, 400GAUI-16 C2C and C2M

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER

1.5.18 25G Ethernet–LAUI-2 C2C and C2M

The PHY supports LAUI-2 chip-to-chip and chip-to-module with a data rate of 2 x 25.78125 Gbps.

The PHY complies with Annex 135B of the IEEE 802.3cd specification (C2C) and Annex 135C of the IEEE 802.3cd specification (C2M).

1.5.18.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-31](#) notes the equalization setup for the receiver and transmitter.

Table 1-31 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.18.2 Exceptions

[Table 1-32](#) notes exceptions to the 25G Ethernet–LAUI-2 C2C and C2M specification.

Table 1-32 Exceptions to Specification Allowed for 25G Ethernet–LAUI-2 C2C and C2M

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER

1.5.19 100G Ethernet – 100GBASE-KR4

The PHY supports 100GBASE-KR4 with a data rate of 4 x 25.78125 Gbps.

The PHY complies with Clauses 93/93A, B, and C of the IEEE 802.3-2018 specification. The PHY and controller solution supports back channel adaptation and Energy-Efficient Ethernet (EEE).



Note

- The PHY supports up to 30 dB channel in no FEC case.
- The PHY supports up to 35 dB channel in RS-FEC case.
- The channel is defined from package pin to package pin – the PHY allows an additional 3 dB of loss in the RX package traces and 2 dB in the TX package traces.
- Contact Synopsys to confirm support of your specific channel

1.5.19.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-33](#) notes the equalization setup for the receiver and transmitter.

Table 1-33 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of the receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#)

1.5.19.2 Exceptions

[Table 1-34](#) notes exceptions to the 100GBASE-KR specification.

Table 1-34 Exceptions to Specification Allowed for 100GBASE-KR

Category	Exception Allowed
TX voltage magnitude change from coefficient update	Higher than specification
R _{PRE} for maximum C(0) R _{POST} minimum	Exceeds the preshoot limit for maximum C(0) setting Exceeds the minimum deemphasis limit

1.5.20 100G Ethernet – 100GBASE-CR4

The PHY supports 100GBASE-CR4 with a data rate of 4 x 25.78125 Gbps.

The PHY complies with Clauses 92/92A of the IEEE 802.3-2018 specification. The PHY and controller solution supports back channel adaptation and Energy-Efficient Ethernet (EEE).



Note

- Full support of 100GBASE-CR4 is pending silicon validation.
- Contact Synopsys to confirm support of your specific channel

1.5.20.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-35](#) notes the equalization setup for the receiver and transmitter.

Table 1-35 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	TX link adaptation using INC/DEC method from INIT or PRESET

The PHY handles adaptation of the receiver and transmitter without any manual intervention using INC/DEC based method defined by the specification. For more details on INC/DEC based link adaptation, see [Chapter 4, “PHY Usage and Configuration”](#)

1.5.20.2 Exceptions

There are no noted exceptions to the specification.

1.5.21 SGMII

The PHY supports Serial Gigabit Media Independent Interface (SGMII) at 1.25 Gbps.

The PHY meets the required features as stated in Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems.

1.5.21.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-36](#) notes the equalization setup for the receiver and transmitter.

Table 1-36 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting.
Receiver DFE equalization	DFE is bypassed.
Transmitter equalization	TX uses a fixed setting.

The PHY can operate without adaptation of the receiver or transmitter.

1.5.21.2 Exceptions

[Table 1-37](#) notes exceptions to the SGMII specification.

Table 1-37 Exceptions to Specification Allowed for SGMII

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER
TX Vol, Vos	Lower than specification
TX rise/fall	Faster than specification
TX skew	Exceeds limit
RX Vi	Input voltage range lower than specification
RX hysteresis	Not supported
Clock forwarding with data	PHY only supports the recovered clock mode of link communication; the clock is not forwarded along with data.

1.5.22 QSGMII

The PHY supports Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems that provides four channels of 1.25-Gbps serial data rate datapath between four SGMII interfaces to be sent over a single MAC serial interface operating at 5 Gbps.

1.5.22.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-38](#) notes the equalization setup for the receiver and transmitter.

Table 1-38 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting.
Receiver DFE equalization	DFE is bypassed.
Transmitter equalization	TX uses a fixed setting.

The PHY can operate without adaptation of the receiver or transmitter.

1.5.22.2 Exceptions

[Table 1-39](#) notes exceptions to the QSGMII specification.

Table 1-39 Exceptions to Specification Allowed for QSGMII

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER
TX rise/fall	Faster than specification
TX common mode noise	Exceeds 5% T_V dif.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

1.5.23 2.5G-SXGMII

The PHY supports USXGMII at 2.578125 Gbps.

The PHY is compliant with:

USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017)

USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016).

1.5.23.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-40](#) notes the equalization setup for the receiver and transmitter.

Table 1-40 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting
Receiver DFE equalization	DFE is bypassed
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.23.2 Exceptions

Exceptions to the specification:

Table 1-41 Exceptions to Specification Allowed for 2.5G-SXGMII

Category	Exception Allowed
BER	The PHY designed and tested for 1e-12 BER (specification calls for 1e-15 BER).
ICR	The maximum supported insertion loss-to-crosstalk ratio (ICR) is 23 dB. Note that for a given insertion loss, a lower ICR number gets translated to a higher value of crosstalk.
TX equalization	Manually tune TX equalization for the link or implement your own auto-negotiation.
Internal eye measurement	The PHY has scope slicers and a configurable stats collection block; external software is required to construct the eye diagram.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

1.5.24 5G-SXGMII and 5G-DXGMII

The PHY supports USXGMII at 5.15625 Gbps.

The PHY is compliant with:

USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017)

USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016).

1.5.24.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-42](#) notes the equalization setup for the receiver and transmitter.

Table 1-42 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE uses a fixed setting
Receiver DFE equalization	DFE is bypassed
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.24.2 Exceptions

Exceptions to the specification:

Table 1-43 Exceptions to Specification Allowed for 5G-SXGMII and 5G DXSGMII

Category	Exception Allowed
BER	The PHY designed and tested for 1e-12 BER (specification calls for 1e-15 BER).
ICR	The maximum supported insertion loss-to-crosstalk ratio (ICR) is 23 dB. Note that for a given insertion loss, a lower ICR number gets translated to a higher value of crosstalk.
TX equalization	Manually tune TX equalization for the link or implement your own auto-negotiation.
Internal eye measurement	The PHY has scope slicers and a configurable stats collection block; external software is required to construct the eye diagram.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

1.5.25 10G-SXGMII, 10G-DXGMII and 10G-QXGMII

The PHY supports USXGMII at 10.3125 Gbps.

The PHY is compliant with:

USXGMII Multiport Copper PHY specification v2.11 (EDCS-1517762, dated 5/11/2017)

USXGMII Single-port Copper PHY specification v2.2 (EDCS-1150953, dated 5/23/2016).

1.5.25.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-44](#) notes the equalization setup for the receiver and transmitter.

Table 1-44 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.25.2 Exceptions

Exceptions to the specification:

Table 1-45 Exceptions to Specification Allowed for 10G-SXGMII 10G-DXGMII, and 10G QXSGMII

Category	Exception Allowed
BER	The PHY designed and tested for 1e-12 BER (specification calls for 1e-15 BER).
ICR	The maximum supported insertion loss-to-crosstalk ratio (ICR) is 23 dB. Note that for a given insertion loss, a lower ICR number gets translated to a higher value of crosstalk.
TX equalization	Manually tune TX equalization for the link or implement your own auto-negotiation.
Internal eye measurement	The PHY has scope slicers and a configurable stats collection block; external software is required to construct the eye diagram.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

1.5.26 JESD204C

The PHY supports Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converters JESD204C (Revision of JESD204B.01), December 2017.



Note

- Synopsys currently only evaluated the following rates: 8.11008, 9.8304, 14.7456, 16.22016, 19.6608, 24.33024, 32.44032 Gbps. Supporting other rates is possible, however, additional work is necessary to qualify the settings and the rates. Contact Synopsys to confirm your data rate and refclk support.
- Category B channels are supported.
- Category C channels are supported for classes C-S and C-M.

1.5.26.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-46](#) notes the equalization setup for the receiver and transmitter.

Table 1-46 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	For rates <=6.375 Gbps, AFE uses a fixed setting For rates > 6.375 Gbps, AFE is adapted
Receiver DFE equalization	For rates <=6.375 Gbps, DFE is bypassed For rates > 6.375 Gbps, DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.26.2 Exceptions

Exceptions to the specification:

Table 1-47 Exceptions to Specification Allowed for JESD204C

Category	Exception Allowed
Supported bit error rate	PHY is designed for 1e-12 BER
TX rise/fall	Slower than specification
TX common mode noise	Exceeds 5% T_V dif.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

Category	Exception Allowed
Return loss	RX and TX differential and differential-to-common mode return loss exceed specifications.
Frame clock, multi-frame clock and SYNC handling	Frame clock, multi-frame clock and SYNC handling are outside the scope of the macro.

**Note**

The Synopsys PHY has the ability to measure eye opening using external software.

1.5.27 CPRI

The PHY supports the Common Public Radio Interface (CPRI) specification V7.0, October 2015, CPRI Cooperation

Synopsys currently only evaluated the following rates: 1.2288, 2.4576, 3.072, 4.9152, 6.144, 8.11008, 9.8304, 10.1376, 12.16512, and 24.33024 Gbps. Supporting other rates is possible, however, additional work is necessary to qualify the settings and the rates. Contact Synopsys to confirm your data rate and refclk support.

The CPRI specification references many different electrical specifications (such as, IEEE 802.3, Fibre Channel, OIF, and SFF). The PHY supports CPRI via other supported Ethernet protocols.

The PHY supports the electrical specifications referred to in [Table 1-2](#).

If there are specifications that are not covered but you desire to use them in a CPRI application, contact Synopsys for support.

1.5.27.1 Configuration

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

[Table 1-48](#) notes the equalization setup for the receiver and transmitter.

Table 1-48 Equalization Setup for the Receiver And Transmitter

Parameter	Settings
Receiver AFE equalization	AFE is adapted
Receiver DFE equalization	DFE is adapted
Transmitter equalization	Transmitter equalization optimized by customer

The transmitter equalization is optimized for your given channel. Perform this optimization using either the provided IBIS-AMI model or you can optimize in the hardware. As no adaptation is used for the transmitter, determine a single setting that works for all PVT cases.

1.5.27.2 Exceptions

For CPRI electrical exceptions see the individual specifications in [Table 1-2](#).

The Synopsys PHY does not support CPRI option 1 running at 614.4 Mbps.

1.5.28 RXAUI/Double XAUI

The PHY supports a 2x-scaled mode of XAUI, where all time-based portions of the XAUI electrical specification are scaled by a factor of 0.5. This mode is known as RXAUI (also Double XAUI) at twice the data rate. Serial data is striped over two serial interfaces operating at 6.25 Gbps. The PHY complies with Clauses 48 and 71 of the IEEE 802.3-2018 specification.

1.5.28.1 Reference Clock Frequencies Supported

For the supported reference clocks, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

1.5.28.2 Exceptions

[Table 1-49](#) provides the allowed exceptions to the RXAUI/Double XAUI specification.

Table 1-49 Exceptions to Specification Allowed for RXAUI/Double XAUI

Category	Exception Allowed
TX rise/fall	Faster than specification
Lane-to-lane skew	Exceeds limit
TX common mode noise	Exceeds 5%*T_Vdif.
Coupling	■ AC coupling is supported. ■ For DC coupling support, contact Synopsys.

1.5.29 SATA

The PHY supports the SATA 6G protocol and data rate and is backwards compatible to the SATA 3G and SATA 1.5G protocols and data rates.

The PHY complies with all of the required features of the Serial ATA Revision 3.3, February 2016. The PHY also supports Gen3i (6 Gbps), Gen2i and Gen2m (3 Gbps), Gen 1i and Gen1m (1.5 Gbps).

1.5.29.1 Reference Clock Frequencies Supported

For the supported reference clocks, the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

1.5.29.2 Exceptions

[Table 1-50](#) provides the allowable exceptions to the SATA 6G specification.

Table 1-50 Allowed Exceptions to the SATA 6G Specification

Category	Exception Allowed
Reference clock	When 30-MHz clock is used, OOB sampling window might not be met.
TX rise and fall time	TX rise and fall time is faster than stated in the specification.
No Gen 1x/2x	These have been removed from the SATA specification.

1.6 Process Information

[Table 1-51](#) describes the process information for the PHY. For availability in alternate process nodes or variants, contact your sales representative.

Table 1-51 Process Information

Foundry	Process	Process Core Voltage	Process I/O Voltage	Metal Layers
TSMC	12 nm	0.9 V	1.2/1.5/1.8 V	11M_2Xa1Xd_h_3Xe_vhv_2Y2R, 9M_2Xa1Xd_h_3Xe_vhv_2Z



Note

- A list of GDSII layers used in the IP is included in a text file within the gds sub-directories of the release package. For more information, see [Chapter 2, “Deliverables”](#).
- For PHY supply voltages, see Section [7.4.1, “Power Supply Limits”](#).

1.7 Devices List

[Table 1-52](#) lists the primitive devices used by this PHY.

Table 1-52 Primitive Devices Used

Devices	Description
crtmom_2t	2T RTMOM
moscap_rf_nw	CORE RF MOSCAP Varactor wo DNW
nch_18_mac	I/O 1.8V VT NMOS
nch_18ud12_mac	Thick Oxide UnderDrive 1.2V NMOS
nch_18ud15_mac	Thick Oxide UnderDrive 1.5V NMOS
nch_lvt_mac	CORE Low VT NMOS
nch_svt_mac	CORE Standards VT NMOS
nch_ulvt_mac	Ultra-Low-Vt NFET
ndio_gated18_mac	2T IO 1.8V gated N+/PW DIODE
ndio_hia18_mac	CORE 2T ESD diode N+/PW
ndio_mac	N+/PW Junction DIODE
nmoscap	NMOS in N-Well Varactor
nmoscap_18	1.8V MOSCAP Varactor
pch_18_mac	I/O 1.8V VT PMOS
pch_18ud12_mac	Thick Oxide UnderDrive 1.2V PMOS

Devices	Description
pch_18ud15_mac	Thick Oxide UnderDrive 1.5V PMOS
pch_lvt_mac	CORE Low VT PMOS
pch_svt_mac	CORE Standard VT PMOS
pch_ulvt_mac	Ultra-Low-Vt PFET
pdio_gated18_mac	2T IO 1.8V gated P+/NW DIODE
pdio_hia18_mac	CORE 2T ESD diode P+/NW
pnpi2_mac	Cascade PNP (P+/NW/Psub)
rhim_m	P+ Poly Non-Salicided resistor 3T
rm10w	M10 resistor
rm11w	M11 resistor
rm1w	M1 resistor
rm2w	M2 resistor
rm3w	M3 resistor
rm4w	M4 resistor
rm5w	M5 resistor
rm6w	M6 resistor
rm7w	M7 resistor
rm8w	M8 resistor
rm9w	M9 resistor
rmap	AP metal resistor

Table 1-53 provides devices used in the IP implementation which may incur an additional mask cost.

Table 1-53 Additional Mask Cost

Feature	Foundry Layer Name	Extra (Cost) Layer
High Resistance	RH_TN	117;6

2

Deliverables

This chapter includes the following sections:

- [“Introduction”](#)
- [“Front-End Views”](#)
- [“Back-End Views”](#)
- [“Signal-Integrity Views”](#)
- [“Test Views”](#)

2.1 Introduction

The full-package for each DesignWare Cores Multi-Protocol 32G PHY release contains all of the design views and information required to instantiate the PHY in a top-level chip design. It also allows for the simulation of both the digital functionality of the PHY and the signal integrity of the link, and provides the information to enable production testing of the PHY.

For information about the PHY release directory structure, see the top-level Readme file included in the product deliverables.

The deliverables are grouped into four basic categories based on usage:

- “Front-End Views” – Includes deliverables for system simulation and floor planning
- “Back-End Views” – Includes deliverables for physical design and finishing
- “Signal-Integrity Views” – Includes deliverables for simulation of link performance at the analog level
- “Test Views” – Includes deliverables for testing

2.2 Front-End Views

Front-end views are provided for both the hard macro portions of the PHY and the raw PCS soft-IP top-level wrapper.

A front-end view includes the following:

- Verilog testbench examples



Note

The packaged testbench is a sample testbench for you to use for reference and pipe-cleaning purposes. The testbench does not exhaustively include all validation tests for all protocols.

- Digital simulation environment
- Synthesis timing models in .db and .lib formats
- Behavioral models for analog blocks
- Library Exchange Format (LEF) file with pin sizes and locations
- Protocol-specific testbench examples
- Synthesis constraints for the PCS
- IP-XACT XML file with register details
- TCL and CSV format pin descriptions
- *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Databook*
- *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*
- For protocol support block and lane configuration settings, see the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory)

2.3 Back-End Views

Back-end views are provided for the hard macro portions of the PHY. These deliverables tend to be metal stack dependent, so multiple files for each hard macro are provided. The top-level cell name inside all of the files is the same, but the content is not—you must select the appropriate file that matches your metal stack-up.

A back-end view includes the following:

- GDSII layout and layer map files
- Layout-Versus-Schematic (LVS) netlist in SPICE format, and LVS report
- LVS samples
- Design Rule Check (DRC) report
- DRC samples
- Generic technology (GTECH) netlists and models for simulation
- Gate-level Verilog
- Gate-level netlist and Standard Delay Format (SDF) timing file

2.4 Signal-Integrity Views

A signal-integrity view includes the following:

- S-Parameter models of RX and TX to facilitate package design
- IBIS-AMI models of RX and TX for signal integrity simulations
- Current waveforms, along with on-die power grid equivalent capacitance and resistance (C_{die} and R_{die} , respectively) for power delivery network simulations in the package

2.5 Test Views

A test view includes the following:

- Scan test environment with Automatic Test Pattern Generation (ATPG)
- Scan Verilog netlist

3

Functional Description

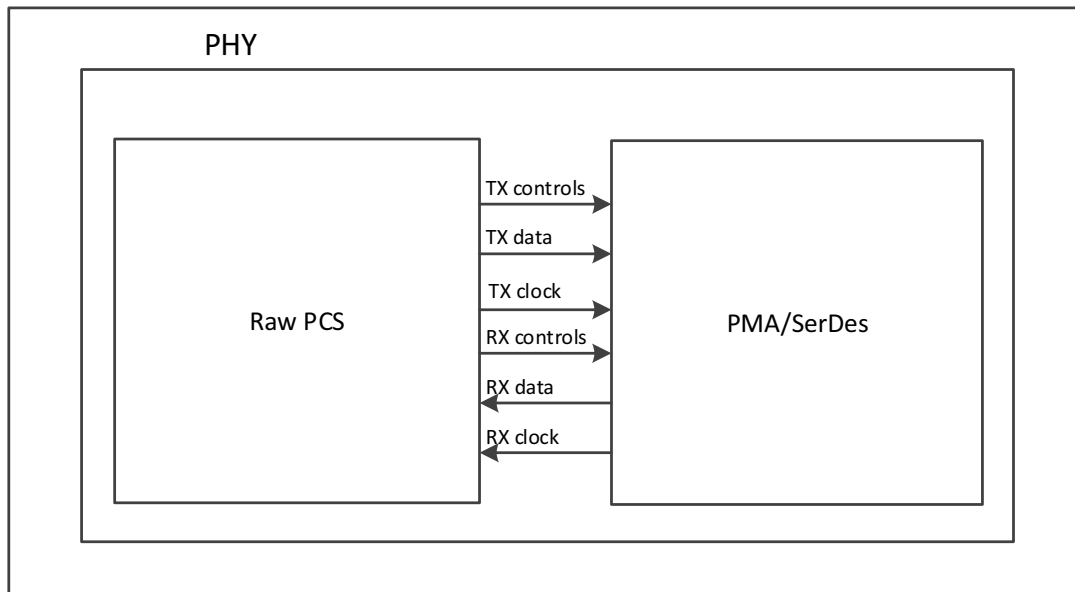
This chapter includes the following sections:

- “PHY”
- “PMA”
- “Raw PCS”

3.1 PHY

The DesignWare Cores Multi-Protocol 32G PHY consists of the hard macro (PMA) and the soft RTL macro (raw PCS). The major blocks are shown in [Figure 3-1](#).

Figure 3-1 DesignWare Cores Multi-Protocol 32G PHY

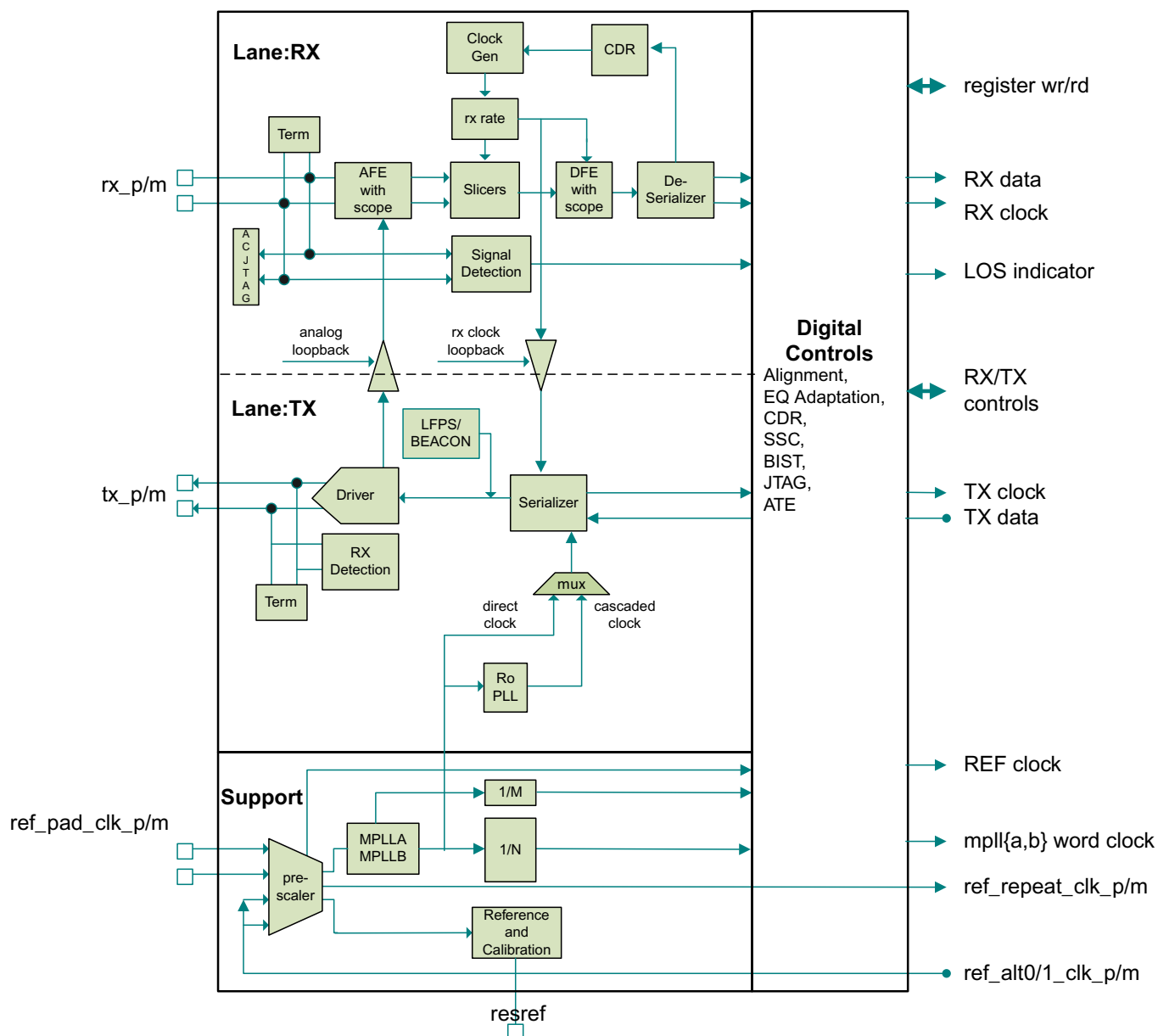


The following sections provide the functional description of the PMA and the Raw PCS.

3.2 PMA

Each PMA comprises a full duplex transceiver for each lane. As shown in [Figure 3-2](#), the PMA comprises the following main blocks:

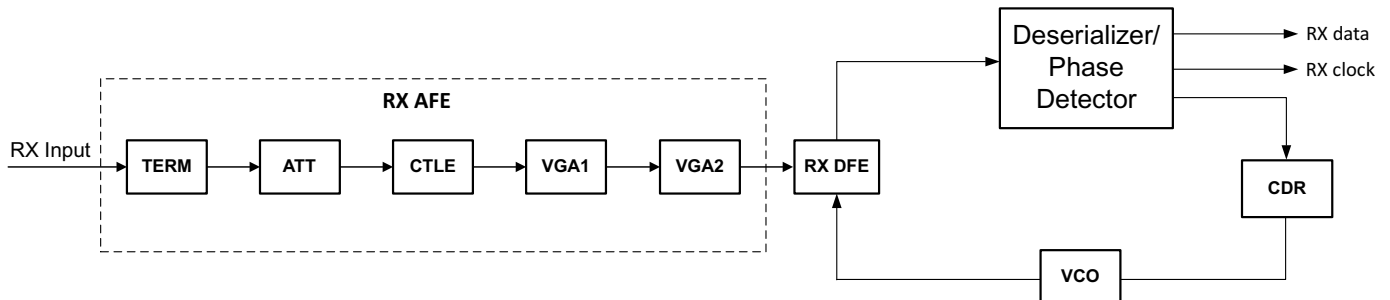
- Lane
 - Receiver (RX)
 - Transmitter (TX)
- Support
 - Reference clock
 - MPLLs
 - Termination Resistance Tuning
 - Bandgap based biasing

Figure 3-2 Single-Lane PMA Architecture

3.2.1 Receiver

Figure 3-3 provides the block diagram of the receiver (RX).

Figure 3-3 RX Block Diagram



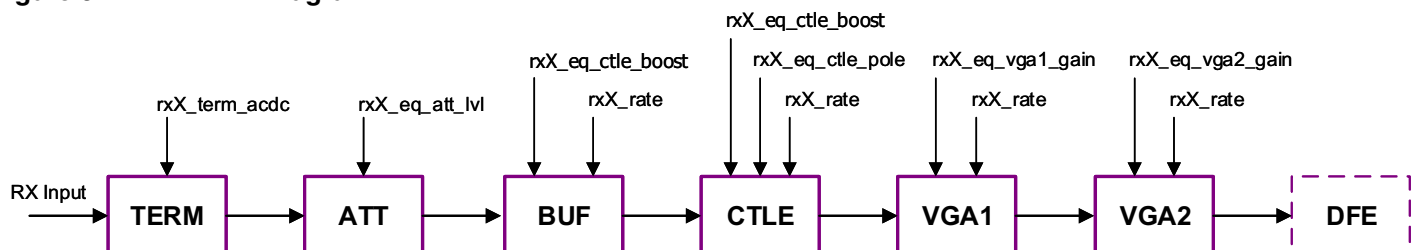
The receiver functions as follows:

1. The PMA receives differential serial data from a pair of external pads. Receiver characteristics, such as equalization and termination, are user-controllable.
2. A Continuous-Time Linear Equalizer (CTLE) is first used to compensate for channel loss.
3. A Decision Feedback Equalizer (DFE) is also implemented after CTLE to optimize the data eye further for high-speed data that exceeds ~6.25 Gbps.
4. The RX output signal is passed to the Clock and Data Recovery (CDR) circuit, which recovers the embedded clock in the data. A dedicated Voltage Controlled Oscillator (VCO) is designed as the source of the recovering clock.
5. The CDR loop adjusts the VCO clock frequency and phase until the frequency matches incoming data and the phase is aligned with incoming data.
6. The recovered clock is used to retiming received data and send it to the deserializer, which produces parallel data and a parallel data clock for the relevant PCS lane.

3.2.1.1 RX Analog Front End

The Analog Front End (AFE) circuitry receives off-chip data and boosts high-frequency components to compensate part of the channel loss with programmable or adaptive CTLE.

Figure 3-4 RX AFE Diagram



As shown in [Figure 3-4](#), the AFE consists of the following blocks:

- Termination (TERM)

The TERM block is tuned by the PHY using an off-chip reference resistor.

To configure the RX termination in grounded or floating mode (see [Figure 3-5](#)), use `rxX_term_acdc`:

- 0=floating mode termination
- 1=mode termination to ground

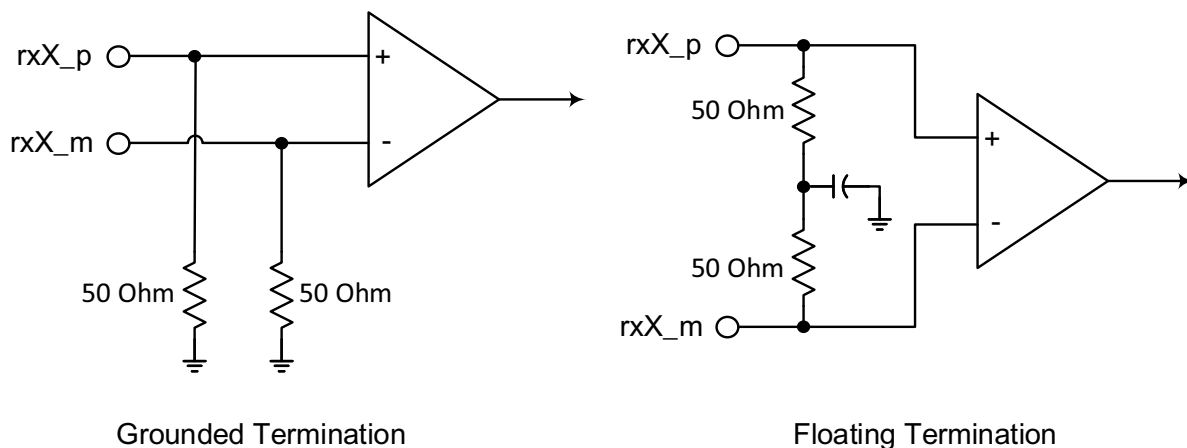
If you use DC floating termination, the common mode is set by the far end transmitter and must be within 0–600 mV range.



Note

- The PHY supports DC-coupled link into RX floating termination.
- If you require use of DC-coupled link into grounded termination, consult Synopsys.

Figure 3-5 RX Termination



- Attenuator (ATT)

The ATT stage optionally attenuates the incoming signal to ensure that the signal remains linear through subsequent stages. The AFE attenuation is controlled using `rxX_eq_att_level`, where a higher setting means higher attenuation. The attenuation range is approximately 9 dB with a step size of 1.25 dB.

- Buffer (BUF) and Continuous-Time Linear Equalizer (CTLE)

The BUF and CTLE stages equalize the incoming signal. The BUF stage is also used to reduce the impact of common-mode interference. The AFE boost, pole, and bandwidth are controlled using `rxX_eq_ctle_boost`, `rxX_eq_ctle_pole`, and `rxX_rate`, where a higher boost setting means higher equalization, a higher pole setting means a higher pole frequency, and a lower rate setting means a higher bandwidth. The boost control affects both the high-frequency peaking and low-frequency attenuation of the CTLE response. The boost range is approximately 16 dB with a step size of 0.5 dB.

- Voltage Gain Amplifiers (VGA1 and VGA2)

The VGA stages adjust the AFE gain to maintain a constant signal swing at the AFE output. The AFE gain is controlled using `rxX_eq_vga_gain`, where a higher setting means higher gain. The gain range is approximately 14 dB with a step size of 1 dB.

To achieve optimal link performance, the receiver equalization parameters are automatically tuned to their optimal values for protocols that are adapted. For more details about RX AFE equalization adaptation, see Section 3.2.1.7, “RX Equalization and Adaptation”.

3.2.1.2 RX Decision Feedback Equalizer

The post-VGA output is passed to a continuously adaptive Decision Feedback Equalizer (DFE). The output of the DFE is sampled by offset-compensated samplers, deserialized, and sent to the core along with the recovered clock.

The DFE is a half-rate architecture with first tap unrolled. For more details about RX equalization adaptation, see Section 3.2.1.7, “RX Equalization and Adaptation”.

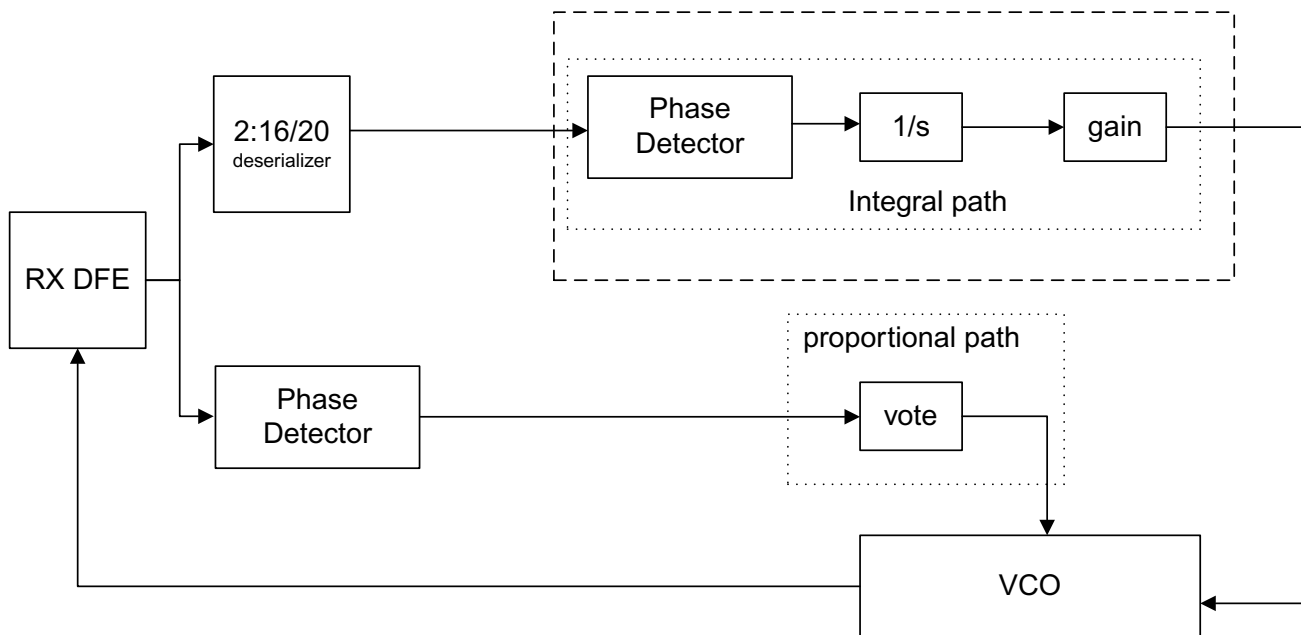
The DFE has eight fixed taps and four floating taps. The floating taps may be individually placed anywhere from tap 9 to tap 20.

The DFE can be disabled (and DFE_BYPASS enabled) through top-level signal or register control:

- `rxX_dfe_bypass`
- `LANEN_DIG_ANA_RX_CTL_OVRD_OUT.RX_ANA_DFE_BYPASS_EN` register bit can be de-asserted to disable DFE and enable DFE_BYPASS
- The DFE and DFE_BYPASS can both be disabled through register, which is used to reset tap2 to tap5, therefore virtually disabling the taps and the DFEs.

3.2.1.3 RX Clock and Data Recovery

The Clock and Data Recovery (CDR) circuit is implemented as a second-order digital loop. The loop includes a proportional path and an integral path. To reduce proportional path latency, the proportional path is implemented separately from the integral path, with much higher speed, as shown in Figure 3-6.

Figure 3-6 RX CDR Diagram

The proportional path and integral path outputs control a dedicated CDR Voltage-Controlled Oscillator (VCO), which generates an RX clock aligned with incoming data. To support Spread-Spectrum Clocking (SSC) and to reduce CDR lock time, proportional path gain and integral path gain are dynamically set during the CDR locking process.

3.2.1.4 RX Deserialization

The deserializer converts serial data from RX and sends the data to the PCS as parallel data. The deserializer qualifies serial input data from the CDR using the high-speed serial recovered clock, and deserializes the data using the low-speed parallel recovered clock. The deserializer forwards the deserialized data to the receiver's PCS channel. The deserializer supports a 16-, or 20-bit interface. The deserializer also optionally supports an 8- or 10-bit interface for lower data-rates as specified in [Table 1-1](#) ("Data Widths Supported").

3.2.1.5 RX Rate Control

The RX high-speed clock is generated by the CDR loop that controls and aligns the high-speed clock frequency and phase with data received by the RX. The CDR-controlled VCO provides a clock rate that can range from 8–16 GHz. In addition, the RX has a flexible data rate control (configured by `rxX_rate`) that allows the RX to reliably receive and process incoming data from 1.25 Gbps to a maximum of 32 Gbps.

For example, by setting `rxX_rate` to `2'b10`, the PHY operates at 2.5 Gbps (the RX clock rate is divided down from 5–1.25 GHz). For detailed PMA configuration settings for all supported protocols, see [Chapter 4, "PHY Usage and Configuration"](#).

3.2.1.6 RX Calibration

The PHY has the ability to calibrate the RX AFE and DFE so as to minimize offset. This is run on start up out of reset and may also be run continuously during mission mode.

3.2.1.6.1 Startup Calibration

The PHY runs startup calibration after the RX is powered up and receiving valid data with the CDR locked. It calibrates the offsets of each stage of the AFE (ATT, CTLE, VGA) in sequence, and then calibrates the offsets of the DFE data, phase, and error slicers.

3.2.1.6.2 Continuous Calibration

The PHY runs continuous calibration in mission-mode. The continuous calibration routine adjusts the offsets of each stage of the AFE (CTLE, VGA, and summer) in sequence. Continuous calibration runs in tandem with continuous adaptation when the latter is supported.

3.2.1.7 RX Equalization and Adaptation

For protocols that require link adaptation (such as PCIe 5.0), the PHY supports adaptation of the receiver equalization parameters, and requests for incrementing or decrementing the far-end transmitter coefficients. The PHY adapts the AFE equalization and gain, the DFE tap coefficients, and the remote TX equalization. It operates in two steps: startup adaptation and continuous adaptation.



Note

Receiver adaptation must NOT be for lower data rates which use a rate of baud/4 or lower.

3.2.1.7.1 Startup Adaptation

The PHY runs startup adaptation after startup calibration is complete, when the RX is receiving valid data and the CDR is locked. It first adapts the AFE, where the ATT level and VGA gain are adapted based on measurements of the average eye opening and the CTLE boost is adapted based on data bit correlation. It then adapts the DFE, where the tap coefficients are adapted based on the same correlation algorithm. It calculates a figure-of-merit based on collected statistics. Finally, it evaluates the TX pre-cursor and post-cursor and makes a decision on whether to adjust remote TX equalization.

3.2.1.7.2 Continuous Adaptation

The PHY runs continuous adaptation in mission-mode for protocols that support adaptation. The continuous adaptation routine adjusts the coefficients of DFE taps and the VGA gain using an algorithm similar to the one used during startup adaptation. Continuous adaptation is run in tandem with continuous calibration.

3.2.2 RX VCO

The Rx VCO frequency is determined by the frequency of the reference clock and the settings of the `rxX_ref_ld_val` and the `rxX_vco_ld_val` through the following equation:

$$\text{Rx_VCOfreq} = \text{rxX_vco_ld_val} / \text{rxX_ref_ld_val} * (\text{lane_ref_clk freq})$$

where $\text{lane_ref_clk freq} = \text{ref_clk freq (at pad or alt pins)} / (1 + \text{ref}\{a,b\}_clk_div2_en)$.

**Note**

The `rxX_vco_ld_val` setting cannot be less than 1332. Other restrictions may apply. If there is a data rate outside the settings as described in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (`dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx` in the doc directory of the PHY release package), contact Synopsys for confirmation of the `rxX_vco_ld_val` and `rxX_ref_ld_val` settings.

3.2.3 Transmitter

The Transmitter functions as follows:

1. The PMA receives parallel data from the relevant lane of the PCS.
2. The data is serialized and sent to the TX driver, which is a high-speed, voltage mode, differential-output driver. The TX driver characteristics, such as amplitude, pre-emphasis, and termination, are user-controllable.
3. The TX driver output is sent to a pair of external pads.

3.2.3.1 TX Serializer

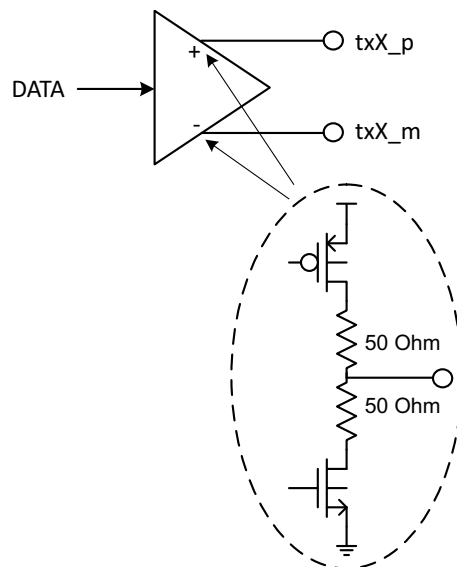
Parallel data from the PCS is latched using a clock that is synchronous to the lane's PCS TX clock, and then serialized by the Parallel-to-Serial (P2S) converter. The serializer supports a 16- or 20-bit interface, or a 8- or 10-bit interface for lower data-rates as defined in [Table 1-1](#) ("[Data Widths Supported](#)").

3.2.3.2 TX Termination

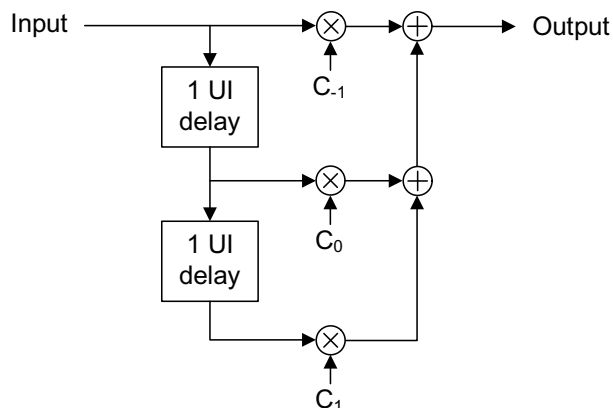
The TX is a voltage mode driver with 50-Ω termination on both `txX_p` and `txX_m` pullup and pulldown, as shown in [Figure 3-7](#). The termination is tuned by the PHY using an off-chip reference resistor.

**Note**

- The PHY supports DC-coupled link into far end floating termination.
- If you require use of DC-coupled link into grounded termination, consult Synopsys.

Figure 3-7 TX Termination**3.2.3.3 TX Equalization**

The TX driver is implemented with a 3-tap Feed-Forward Equalization (FFE), as shown in [Figure 3-8](#).

Figure 3-8 TX Driver Architecture

Where:

- C_0 , C_{-1} , and C_1 are TX driver main tap, pre-cursor tap and post-cursor tap coefficients.
- C_{-1} and C_1 are defined as negative values.

The FFE coefficients C_0 , C_{-1} , C_1 are set by three control signals (txX_eq_main , txX_eq_pre , and txX_eq_post) by the following equations:

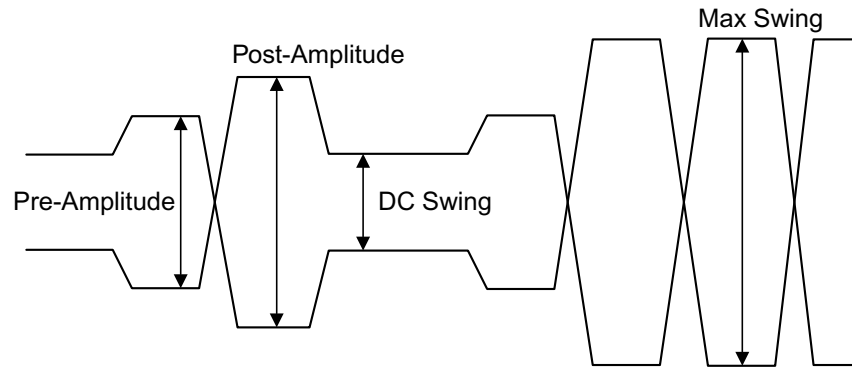
- $C_0 = txX_eq_main$
- $|C_{-1}| = txX_eq_pre/4$
- $|C_1| = txX_eq_post/4$

**Note**

The previous and following equations use the absolute values of C_{-1} and C_1 .

TX Equalization with pre-amplitude, post-amplitude, and DC swing is illustrated in [Figure 3-9](#).

Figure 3-9 TX Equalization



To calculate the differential pre-amplitude, post-amplitude, and DC swing, use the following equations:

- $\text{Pre_amplitude} = \text{Vswing} * (C_0 + |C_{-1}| - |C_1|) / 24$
- $\text{Post_amplitude} = \text{Vswing} * (C_0 - |C_{-1}| + |C_1|) / 24$
- $\text{DC_swing} = \text{Vswing} * (C_0 - |C_{-1}| - |C_1|) / 24$
- $\text{Max_Swing} = \text{Vswing} * (C_0 + |C_{-1}| + |C_1|) / 24$

To maximize the output swing, set it so that $C_0 + |C_{-1}| + |C_1| = 24$, which results in:

$$\text{Max_Swing} = \text{Vswing}$$

To attenuate output swing, set $C_0 + |C_{-1}| + |C_1|$ to be less than 24. For example, to get half swing output, set $C_0 + |C_{-1}| + |C_1| = 12$.

3.2.3.4 TX Rate Control

The TX high-speed clock is generated by the common clock module in the Support block or by the ROPLL in the lane TX. As listed in [Section 3.2.4.3, “MPLL”](#), the Support block provides a clock rate that can range from 4.6875–5.5296 GHz or 16 GHz. The ROPLL in the lane TX can generate clocks from 8-16 GHz. In addition, the TX has a flexible data rate control (configured by controlling `txX_rate`) that allows the TX to operate from 1.25 Gbps to a maximum of 32 Gbps.

The TX high-speed clock is always divided down to be half of the baud rate frequency. For example, if the clock provided by the Support block or TX ROPLL is 16 GHz, by setting `txX_rate` to `3'b010`, the clock is divided down by a factor of 4 to 4 GHz and the TX operates at 8 Gbps. For detailed PMA configuration settings for all supported protocols, refer to [Chapter 4, “PHY Usage and Configuration”](#).

3.2.3.5 TX Ring-Oscillator Phase-Locked Loop

The TX Ring-Oscillator Phase-Locked Loop (ROPLL) generates high frequency clocks, which are used by the lane TX. The TX ROPLL receives a high-frequency reference clock from the MPLL in the Support block. The high reference clock frequency allows a high ROPLL bandwidth to optimize the jitter versus power performance of the ROPLL. You can configure the TX ROPLL independently per lane to provide per-lane frequency independence.

Key features of the TX ROPLL are:

- VCO frequency range of 8-16 GHz
- Integrated reference and feedback dividers for lane frequency generation
- Individual PLLs per lane for per-lane frequency independence
- Support for data-rates from 1.0-32.0 Gbps
- Bypass mode to allow direct clocking from the MPLL in the Support block to the TX (PCIe-only)

The ROPLL VCO frequency is configured as follows, where `ropll_refclk` is the reference clock sent to the ROPLL from the support block:

$$F_{VCO} = \frac{F_{ropll_refclk}}{ropll_refdiv[3:0]} \times ropll_fbdiv[5:0] \times (1 + ropll_fbdiv[6])$$



Note

When `txX_ropll_div_clk_en` is enabled, the frequency of `txX_ropll_div_clk` clock is equal to the ROPLL frequency divided by $(txX_ropll_out_div[7:0] + 4) \times 4$. See [Table 3-1](#) for `txX_ropll_out_div[7:0]` values (in hexadecimal notation) and divider ratio values (in decimal notation). Divider ratio values between 0 to 7 (in decimal notation) are invalid and should not be used.

Table 3-1 ROPLL Divide Output Clock Configuration

<code>txX_ropll_out_div[7:0]</code>	Divider Ratio (in Decimal Notation)			
7'h7F – 7'h7C	524	520	516	512
7'h7B – 7'h78	508	504	500	496
7'h77 – 7'h74	492	488	484	480
7'h73 – 7'h70	476	472	468	464
7'h6F – 7'h6C	460	456	452	448
7'h6B – 7'h68	444	440	436	432
7'h67 – 7'h64	428	424	420	416
7'h63 – 7'h60	412	408	404	400
7'h5F – 7'h5C	396	392	388	384
7'h5B – 7'h58	380	376	372	368
7'h57 – 7'h54	364	360	356	352

txX_ropll_out_div[7:0]	Divider Ratio (in Decimal Notation)			
7'h53 – 7'h50	348	344	340	336
7'h4F – 7'h4C	332	328	324	320
7'h4B – 7'h48	316	312	308	304
7'h47 – 7'h44	300	296	292	288
7'h43 – 7'h40	284	280	276	272
7'h3F – 7'h3C	268	264	260	256
7'h3B – 7'h38	252	248	244	240
7'h37 – 7'h34	236	232	228	224
7'h33 – 7'h30	220	216	212	208
7'h2F – 7'h2C	204	200	196	192
7'h2B – 7'h28	188	184	180	176
7'h27 – 7'h24	172	168	164	160
7'h23 – 7'h20	156	152	148	144
7'h1F – 7'h1C	140	136	132	128
7'h1B – 7'h18	124	120	116	112
7'h17 – 7'h14	108	104	100	96
7'h13 – 7'h10	92	88	83	80
7'h0F – 7'h0C	76	72	68	64
7'h0B – 7'h08	60	56	52	48
7'h07 – 7'h04	x ^a	40	36	32
7'h03 – 7'h00	x ^a	x ^a	x ^a	x ^a

a. Refers to invalid divider setting; do not use.

ROPLL output clock `txX_ropll_word_clk` has a nominal duty cycle of 50%. For `txX_ropll_div_clk`, the systematic duty cycle offset is described in [Table 3-2](#):

Table 3-2 Duty Cycle Offset

Divide Ratio	Minimum Duty Cycle	Maximum Duty Cycle
16	45%	55%
24	Not supported	
32	45%	55%
60	30%	70%
80	35%	65%
96	45%	55%
100	45%	55%
120	40%	60%
160	45%	55%
200	35%	65%

3.2.4 Support Block

The Support block (see [Figure 3-2](#) (“Single-Lane PMA Architecture”)) provides all the common functions for the RX/TX link, TX clock generation, TX/RX termination calibration, biasing voltage, and reference clock buffering.

3.2.4.1 Reference Clock Input

The PHY requires a reference clock from either:

- An external pad input: `ref_pad_clk_p/ref_pad_clk_m`
- Two core-side, on-chip inputs: `ref_alt0_clk_p/m` and `ref_alt1_clk_p/m`

The external or core-side input is selected by `refa/b_clk_sel`. When `refa/b_clk_sel` is set to 1 is asserted, the external reference clock is used; otherwise, the on-chip reference source is used.

For implementation details, see [Chapter 9, “Reference Clock Implementation”](#).

3.2.4.2 Reference Clock Output

The Support block includes a reference clock repeat function. A current-mode logic (CML) differential reference clock is driven on `ref_repeat_clk_{p,m}` when `ref_repeat_clk_en` is asserted. This clock can be used as the reference clock source for other PHYs. For details, refer to [Section 9.1.4, “Repeat Clock Specification”](#).

3.2.4.3 MPLL

The MPLL generates the high-frequency clocks used by the lane TX. There are two MPLLs, MPLL A and MPLL B, which can be either operated individually or combined to support lane bifurcation. The `txX_ropll_refsel` signal allows you to select which MPLL to use.

Key features of the MPLLs are:

- Total output frequency ranges of 1 GHz, 4.6875–5.5296 GHz, 16 GHz
 - With MPLL A enabled the frequency can be 1 GHz or 4.6875–5.5296 GHz
 - With MPLL B enabled, the frequency can be 1 GHz or 16 GHz
- BW requirement compliance for PCIe 5.0 with reference clock of 100 MHz
- Support for lane bifurcation (with two MPLLs enabled)
- Support for Spread Spectrum generation with a range of -5000 ppm to 5000 ppm
- Support for fractional division

3.2.4.3.1 Configuring the MPLL

You need to configure the MPLL so it runs at the correct frequency for your application and protocol. In particular, you need to set the feedback ratio (Feedback N).

To configure the MPLL, perform the following:

1. Enable the correct MPLL (MPLLA or MPLLB), according to the application and the frequency range. For more about enabling the correct MPLL, see Section 3.2.4.3, “MPLL”.
2. Use the configuration settings in [Chapter 4, “PHY Usage and Configuration”](#) to determine the correct setting of `mpll{a,b}_multiplier[11:0]`. The divider has a resolution of 0.25 input clock cycles and thus `mpll{a,b}_multiplier[0]` has a binary weighting of 0.25, `mpll{a,b}_multiplier[1]` has a weighting of 0.5, `mpll{a,b}_multiplier[2]` has a weighting of 1, and so on. There is an additional divide-by-2 or divide-by-4 in the MPLL feedback path configured by the `mpll{a,b}_fb_clk_div4_en` pin.

$$F_{VCO} = \frac{F_{refclk}}{2^{ref_clk_mpll\{a,b\}_div[2:0]}} \times 2^{(1+mpll\{a,b\}_fb_div4_en)} \times \left(\frac{mpll_frac_quot[15:0]}{4} + \frac{mpll_frac_quot[15:0]}{2^{17}} + \frac{mpll_frac_rem[15:0]}{mpll_frac_den[15:0]} \right)$$



Note

For any use of the fractional component in the MPLL VCO frequency reference equation, contact Synopsys Customer Support.

MPLL output clocks `mpll{a,b}_word_clk` and `mpll{a,b}_word_fr_clk` have a nominal duty cycle of 50%.

3.2.4.4 Termination Resistance Tuning

The PHY uses an external precision resistor to calibrate the termination impedances of the high-speed inputs and outputs of the PHY.

A termination calibration is performed any time the PHY is taken out of reset. In addition, you can request the PHY to perform a calibration by asserting `rtune_req`. When the calibration is completed, `rtune_ack` goes low.

The termination calibration is not continuous, so the PHY has the ability to share a single external resistor using a handshake protocol across several PHYs.

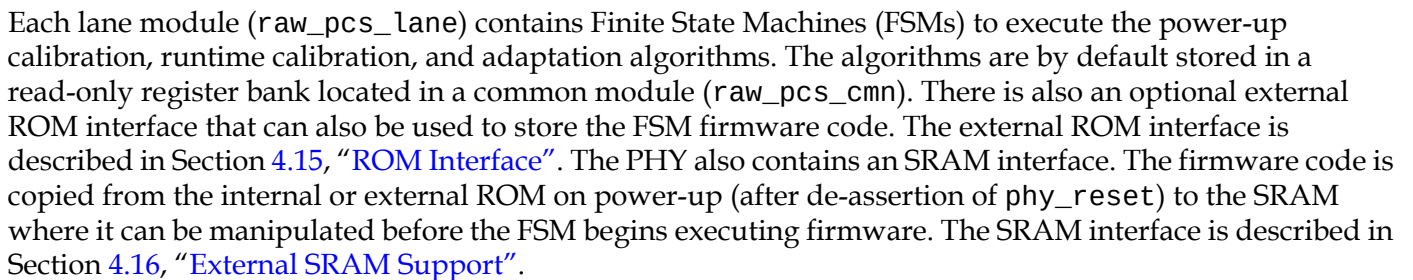
For implementation details, see [Chapter 7, “Physical-Level Implementation”](#).

3.3 Raw PCS

The Raw PCS is a soft macro provided as Verilog RTL. The following features are implemented in the Raw PCS:

- Power-up calibration algorithms for the PMA analog front-end, for example, RX AFE offset cancellation, RX slicer calibration, MPLL calibrations, RX VCO calibrations
- Controls for the adaptation routines in the PMA for modes when adaptation is enabled
- MPLL and ROPLL power-up and power down control based upon `txX_pstate` and `txX_ropll_refsel` and `txX_ropll_bypass` inputs
- JTAG Tap controller for serial register access
- Programmable RX CDR unlock detector
- Register access arbitration between JTAG/CR Parallel access, and the internal power-up and adaptation routines. The selection between JTAG and CR Parallel interfaces is controlled using primary input, `cr_para_sel` (see Section 4.14, “[Control Register Interface](#)”, which means that both of these interfaces cannot operate simultaneously. A round-robin arbitration mechanism exists (except when context restore logic is enabled and active; for details, see Section 4.24, “[Context Restore](#)”) to provide equal priority to different sources requesting access to the shared internal register bus. These sources include multiple internal state machines as well as external register access interfaces (JTAG and CR-Parallel).

Figure 3-10 Raw PCS Block Diagram



The design includes two register access arbiters. The Memory arbiter is used for accessing the register bank whereas the Control Register Arbiter is used to access all other registers in the Raw PCS and the PMA. This mechanism provides efficient use of the registers and state machines within each lane where each lane can independently execute a different portion of the algorithm while reducing the register access bottleneck.

All control registers in the Raw PCS and PMA can be externally accessed using JTAG or CR Parallel Interfaces, with only one interface at any given time.

4

PHY Usage and Configuration

This chapter explains the usage of PHY features and lists the PHY configuration settings for the protocols and data rates that are supported. It includes the following sections:

- “PHY Power-Up”
- “PHY Initialization Sequence”
- “Lane Configuration Update”
- “MPLL Configuration Update”
- “TX Datapath”
- “RX Datapath”
- “RX Signal Detector”
- “RX CDR PPM Detector”
- “Fast Context Switching”
- “TX and RX Power States”
- “TX Equalization”
- “TX and RX Adaptation”
- “Receiver Detection”
- “Control Register Interface”
- “ROM Interface”
- “External SRAM Support”
- “SRAM/ROM Support and Bootloading Options”
- “Firmware”
- “PERST”
- “Aggregation and Bifurcation Support”
- “Power Gating”
- “Disabling a PHY”
- “Configuring the PHY”
- “Context Restore”

4.1 PHY Power-Up

Power up the vp^* and vph power supplies in any order and then proceed with the PHY reset and initialization sequence (vp^* indicates the grouping of all core supplies; vp , $vpdig$, and $vptxN$, where N is the number of lanes).



Note

- The state of the PHY is unknown after power up but before assertion of reset. The PHY may exhibit unwanted behavior such as TX enabled or RX terminations activated, for example.
- While the unknown state is undesirable, the PHY has been designed to prevent damage if it is left in this state.
- Supply ramp should be monotonic with a ramp time of no faster than 10 μ s and a rate no faster than (supply voltage)/10 μ s.
- $vpdig$ may be grounded/disabled while $vddcore$ is enabled.
- $vddcore$ ramping up before $vpdig$ is supported; $vpdig$ ramping down before $vddcore$ is supported.
- $vpdig$ ramping up before $vddcore$ is not supported; $vddcore$ ramping down before $vpdig$ is not supported.
- $vddcore$ and $vpdig$ ramping up/down together is supported.

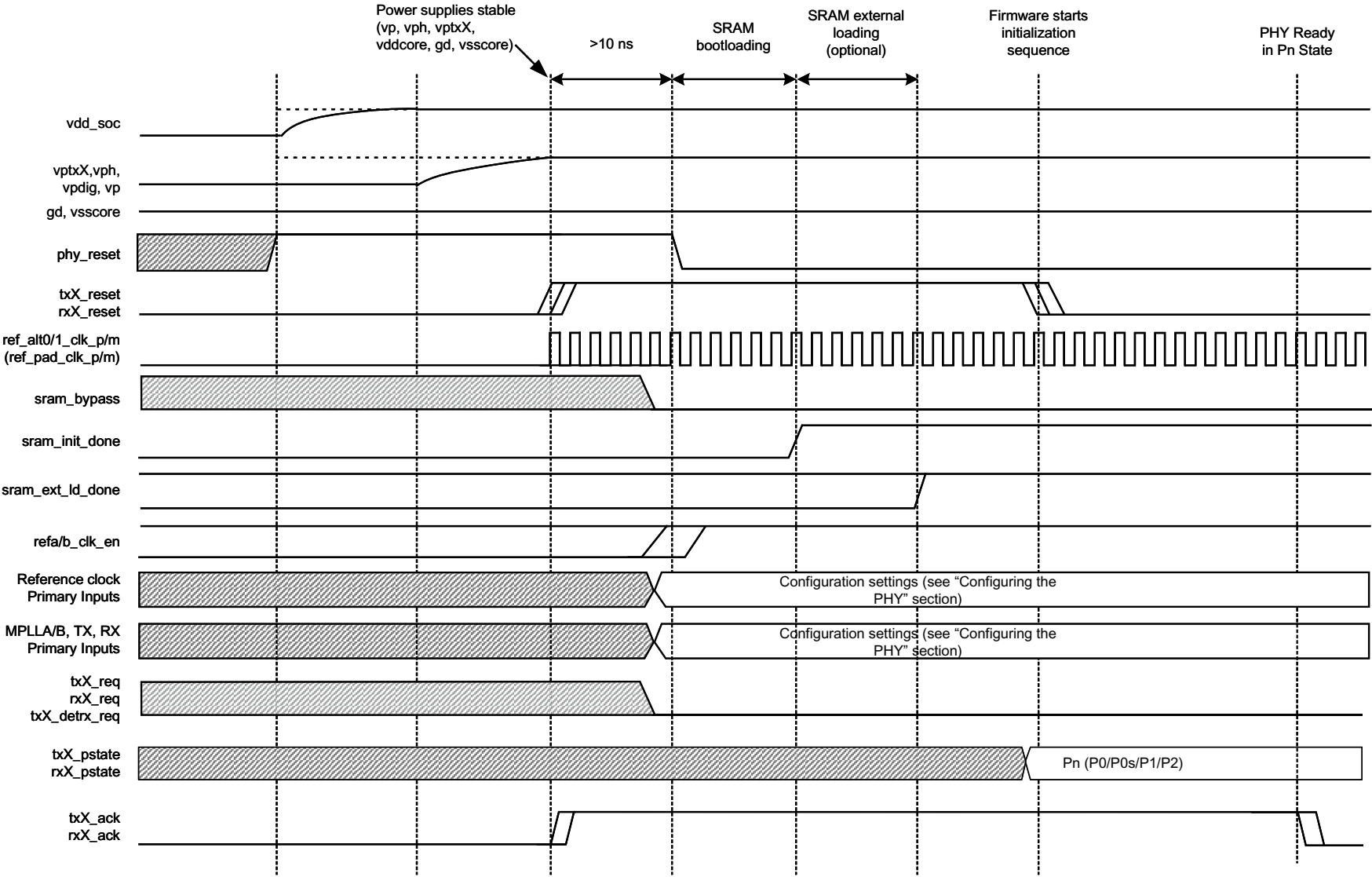
4.2 PHY Initialization Sequence

During PHY reset and initialization sequence, the PHY performs power-up circuit calibrations and enters the configuration mode defined by the primary inputs. The PHY initialization timing diagram when SRAM is enabled is shown in [Figure 4-1](#).

The PHY does not require any clocks to be supplied to enter the reset state (assertion of `phy_reset`). Exit from the reset state is indicated by de-assertion of `txX_ack` and `rxX_ack`, and requires a reference clock to be supplied and `ref_clk_en` input to be asserted. The maximum time from `phy_reset` de-assertion to de-assertion of `txX_ack`/`rxX_ack` can be as high as 5 ms in fast simulation mode and as high as 15 ms in normal simulation mode, because offset calibration is not modeled.

When `phy_reset` is asserted, the TX and RX terminations are disabled and `txX_p/m` are tri-stated for all lanes of the PHY. In addition to global `phy_reset`, `txX_reset`, and `rxX_reset` inputs can be used to reset individual lanes in the PHY. When `txX_reset` is asserted, the `txX_pstate` is similar to P2, the TX terminations are disabled and `txX_p/m` are tri-stated for lane X. When `rxX_reset` is asserted, the `rxX_pstate` is similar to P2 except that the RX terminations are disabled for lane X. Asserting either `txX_reset` or `rxX_reset` during `phy_reset` assertion stalls the RAW_PCS FSM operation before internal registers are updated for lane X after `phy_reset` is de-asserted.

Figure 4-1 PHY Initialization Sequence When SRAM is Enabled



**Note**

- `refa/b_clk_en` must stay asserted, irrespective of `txX_reset/rxX_reset`, if any of the lanes require the reference clock input to be active.
- The `phy_reset` must be asserted before the power supply (VP or VPH) ramps up.

The PHY reset/initialization steps are as follows:

1. Assert reset (`phy_reset = 1`).

The lane-acknowledge outputs are asserted during reset (`txX_ack = rxX_ack = 1`).

2. Set the MPLL and lane (TX/RX) inputs to the desired configuration as defined in Section 4.23, “Configuring the PHY”.

Note that the TX and RX power state control inputs, `txX_pstate[1:0]` and `rxX_pstate[1:0]`, can be set independently.

3. De-assert reset (`phy_reset = 0`). The `phy_reset` needs to be asserted for a minimum of 15 μ s after power-supplies are ramped-up (power-on). For any subsequent `phy_reset` assertions (warm-reset), `phy_reset` needs to be asserted for a minimum of 10 ns.
4. Wait for the lane acknowledge outputs to de-assert (`txX_ack = rxX_ack = 0`). At this point, all the desired inputs set in Step 2 have been applied in the PHY.

**Note**

Initialization will not complete until both enabled PLLs are calibrated and the reference resistor tuning is complete:

- Ensure that the resistor contention bus is connected as shown in Figure 7-8 (“Reference Resistor Contention Bus - Single PHY”) or Figure 7-9 (“Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor”).
- Ensure that both the MPLLs are configured properly or the `mpll{a/b}_init_cal_disable` signal is asserted

5. Assert `txX_clk_rdy` if needed. For usage of the `txX_clk_rdy` input, refer to Section 4.3, “Lane Configuration Update”.

**Note**

If the PHY is powered up for more than 10 μ s, a warm reset requires a minimum of 10 ns.

4.2.1 Resistor Tuning

After `phy_reset` is de-asserted, the PHY performs resistor tuning using the reference resistor. The PHY performs resistor tuning eight times for impedance values ranging from 40–54 Ω in 2- Ω increments. The termination impedance used by the receiver and transmitter are then determined by the value of `rxX_term_ctrl` and `txX_term_ctrl`, respectively. If the termination impedance is to be updated, a `phy_reset` assertion must be applied, and the updated values are provided to the `rxX_term_ctrl` and `txX_term_ctrl` inputs. For a default value of 50 Ohm, set the `rxX_term_ctrl` and `txX_term_ctrl` inputs to 3'd2.

4.3 Lane Configuration Update

Each lane is configured using its TX and RX configuration inputs. [Figure 4-2](#) and [Figure 4-3](#) show the configuration change handshake for the TX and RX signals, respectively. Although the figures show all the inputs that are updated using the txX_req/txX_ack and rxX_req/rxX_ack handshake, it is allowed to change a subset of these inputs. These inputs must stay stable from assertion of txX_req/rxX_req until de-assertion of txX_ack/rxX_ack.

Figure 4-2 Lane TX Configuration Change Handshake

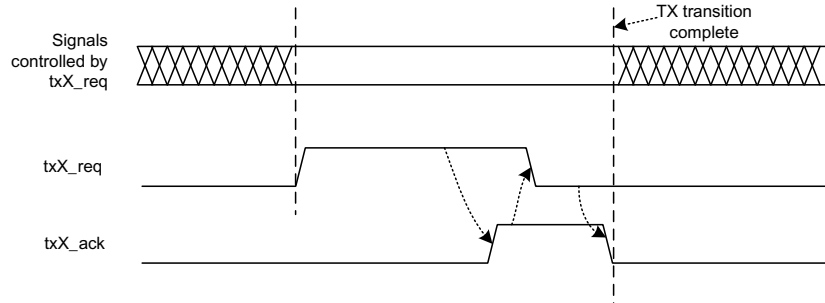
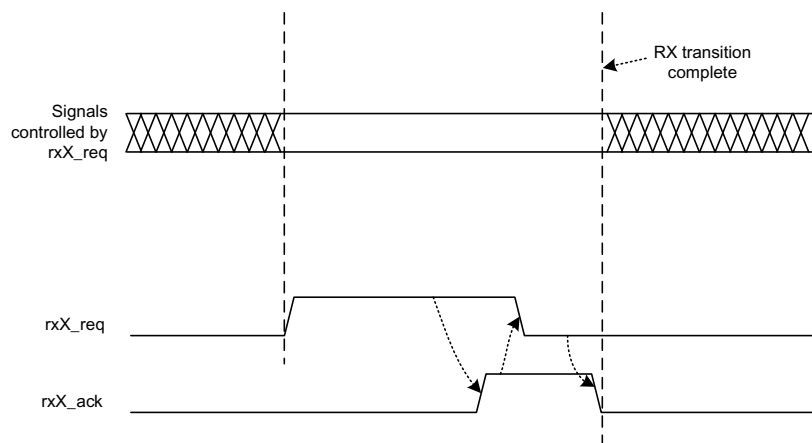


Figure 4-3 Lane RX Configuration Change Handshake



Lanes that are operating as part of one link must comply with the following requirements:

- TX configuration inputs txX_pstate, txXrate, txX_width, txX_ropll_refsel, and txX_ropll_bypass must be set to the same value for all lanes.
- RX configuration inputs rxX_pstate, rxX_rate, and rxX_width must be set to the same value for all lanes.
- A common clock must be supplied to txX_clk inputs for all lanes.
- txX_clk_rdy must be asserted prior to changing the txX_pstate to P0 or P0s and must be asserted synchronously in the same clock cycle for all the lanes in a single link when txX_width=3'b000/001 (8/10-bit). [Figure 4-4](#) ("TX P2-to-P0/P0s Transition When txX_width = 8/10-bit") shows the usage of txX_clk_rdy in 8/10-bit case. [Figure 4-5](#) ("TX P2-to-P0/P0s Transition When txX_width = 16/20/32/40-bit") shows the usage of txX_clk_rdy when txX_width=3'b010/011/100/101 (16/20/32/40-bit), in which case, txX_clk_rdy must be asserted before txX_clk is running and txX_req is asserted.

**Note**

When `txX_width=8/10-bit`, `txX_pstate` transition from P2 (2'b11) to P0/P0s (2'b00/01) is not permitted. As shown in [Figure 4-4](#), `txX_pstate` transitions from P2 to P1 to P0 or P0s. This is because `txX_clk_rdy` input must be asserted synchronously on `txX_clk` prior to the internal transmit clock alignment sequence in the PHY when entering P0/P0s state.

Figure 4-4 TX P2-to-P0/P0s Transition When `txX_width = 8/10-bit`

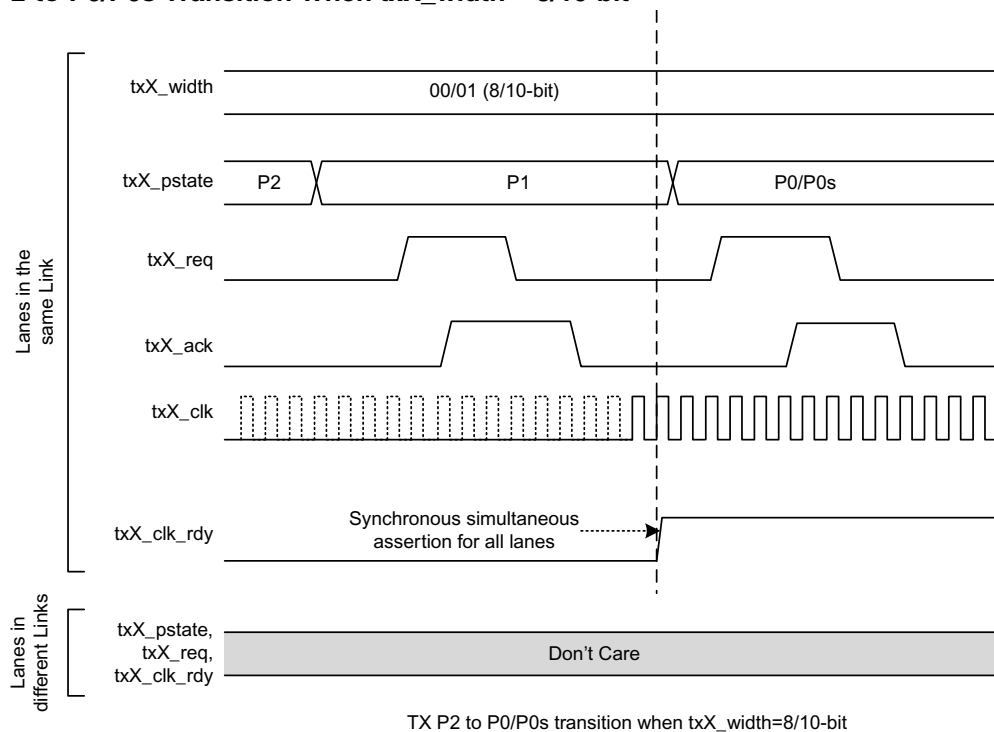
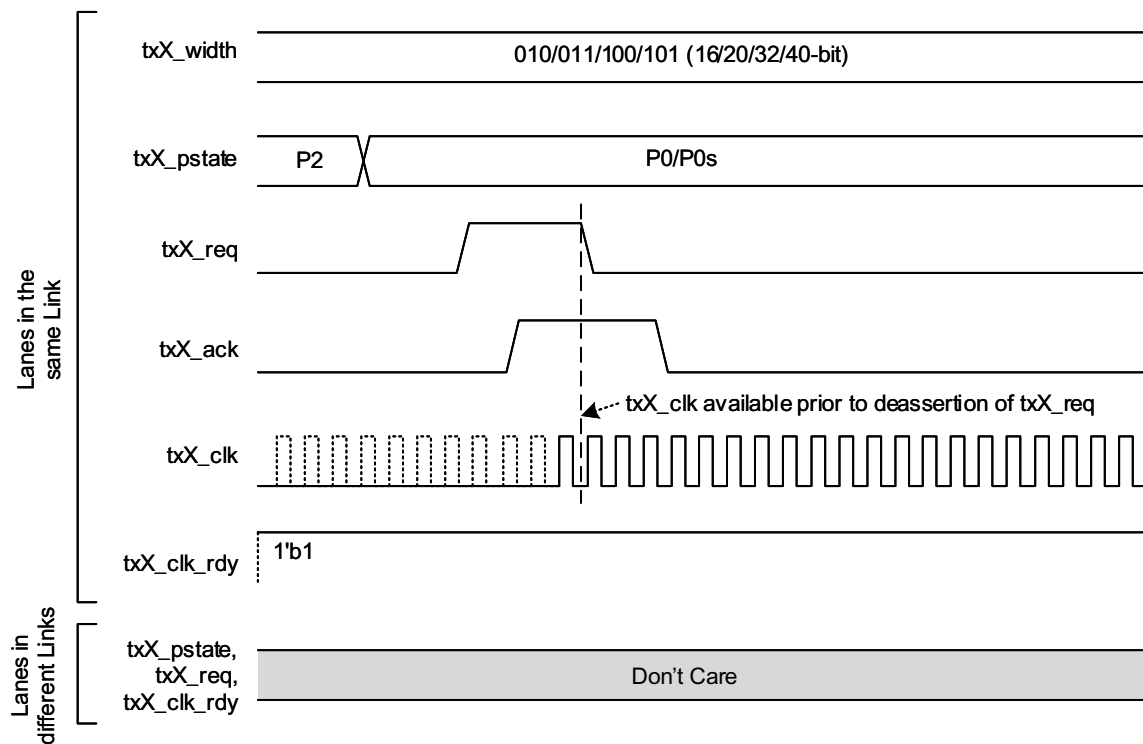


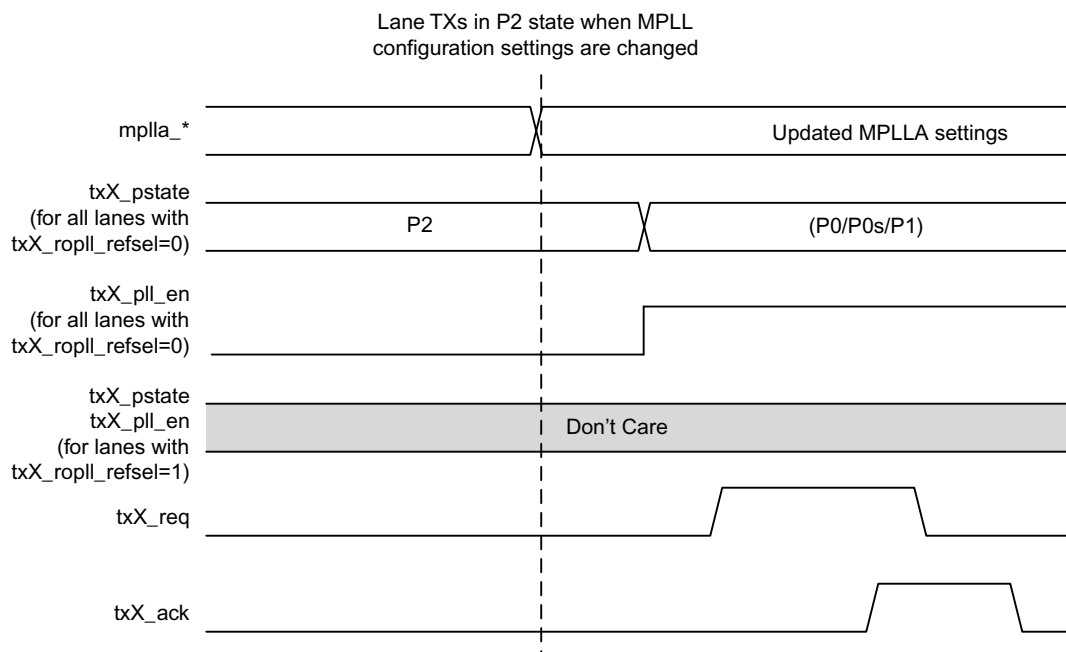
Figure 4-5 TX P2-to-P0/P0s Transition When txX_width = 16/20/32/40-bit

4.4 MPLL Configuration Update

The MPLLA and MPLLB in the PHY are powered up and powered down depending upon the `txX_pstate` and `txX_ropll_refsel` control inputs on each lane. With the default register settings, the MPLLA is powered down when `mplla_force_en` input is 0 and for each lane, `txX_pll_en=0` or `txX_ropll_refsel!=0`. Likewise, MPLLB is powered down when `mpllb_force_en` input is 0 and, for each lane, `txX_pll_en=0` or `txX_ropll_refsel!=1`. Only the used MPLL (based on `txX_ropll_refsel` and `txX_ropll_bypass` settings) is calibrated when `phy_reset` is de-asserted and `mpll{a,b}_init_cal_disable` inputs are de-asserted. Calibration of each MPLL can be skipped if the corresponding `mpll{a,b}_init_cal_disable` input is asserted when `phy_reset` is de-asserted.

The MPLL configuration settings can only be changed when the corresponding MPLL is powered down. As an example, [Figure 4-6](#) shows the requirements for changing the MPLLA configuration settings.

Figure 4-6 MPLLA Configuration Settings Update Followed by a TX Power-State Change



4.5 TX Datapath

The TX datapath is active when the TX lane is in P0 state ($\text{txX_pstate}=2'b00$) and txX_data_en is asserted.

The TX datapath supports serialization of 8/10/16/20/32/40/64-bit data symbol determined by the setting of the input pin txX_width . The valid bits that are transmitted in the 8/10/16/20/32/40/64-bit mode are defined in the signal description of txX_data . The data symbols captured by the PHY on the rising edge of txX_clk when txX_data_en is high are transmitted on $\text{txX_p}/\text{txX_m}$ serial lines. In addition, there are some dribble (invalid) bits transmitted before the first valid word and after the last valid word. The maximum number of dribble bits at the beginning and end of transmission is 4 and 2, respectively.

For lanes belonging to the same link, ensure that the txX_clk inputs are derived from the same source and phase skew balanced to within 400 ps.

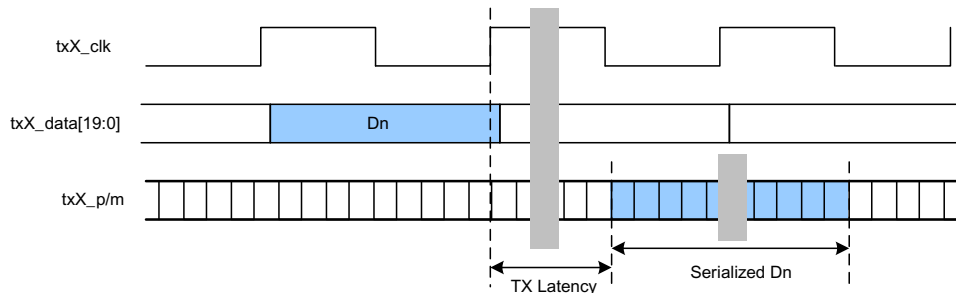
4.5.1 TX Clock Alignment

The PMA contains a circuit that aligns the incoming txX_clk with the internal transmit clock used in the analog portion. This alignment is performed on a 16-bit word when $\text{txX_align_wide_xfer_en} = 1'b0$. When $\text{txX_align_wide_xfer_en} = 1'b1$, this alignment is performed on a 32-bit word to decrease the clock frequency of the data transfer from the txX_clk domain to the internal transmit clock domain. It is recommended that you set the $\text{txX_align_wide_xfer_en}$ bit only when the data rate is > 16 Gbps.

For lanes that are operating as part of one link, there is a clock alignment mechanism in the PHY to align the internal transmit clocks of each lane to the common tx_clk inputs. This alignment results in a maximum TX skew on $\text{txX_p}/\text{txX_m}$ across different lanes in the same link to within 2 unit-intervals (UI) + 500 ps. If the txX_clk clocks are not aligned between lanes, then the serial skew can be up to 1 txX_clk (if 16/20/32/40-b mode) or 2 txX_clk (if 8/10-b mode).

The TX datapath latency for the PHY is measured as per [Figure 4-7](#).

Figure 4-7 Transmit Datapath Latency Measurement In the PHY



[Table 4-1](#) provides the latency values.

Table 4-1 Transmit Datapath Latency

Lane TX I/F width ($\text{txX_width}[2:0]$)	Min Latency (UI)	Max Latency (UI)
16-bit	71	77
20-bit	86	90
32-bit ^a	94	103
40-bit	116	120

Lane TX I/F width (txX_width[2:0])	Min Latency (UI)	Max Latency (UI)
64-bit	94	103

- a. Latency for 32-bit mode can be reduced by 16-UI when voltage and temperature variation of txX_clk clock tree latency is guaranteed to be less than 500 ps instead of 1000 ps mentioned in Section 7.10.2.4, “txX_clk Constraints”. The latency reduction can be achieved by writing a CREG (LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0/TX_ALIGN_R2R=1) during PHY initialization or during txX_reset=1.

**Note**

Maximum frequency of txX_clk is 1.014 GHz. Thus, data rates faster than 16.22 Gbps must use modes 32 bits and greater.

4.5.2 Relationship Between tx_clk, tx_asic_div_clk, and tx_width

This section is only applicable when the UPCS is not used.

The txX_asic_div_clk depends on the tx_width. However, the maximum frequency of txX_asic_div_clk (1 G equal to txX_clk based on edge {1 2 3}) is selected to cover the fastest timing closure scenarios. Detailed information of txX_clk and txX_asic_div_clk is provided as follows:

- Wide_xfer_en = 0
 - 8-, 10-bit interface
 - tx_data[9:0] is used.
 - tx_data[9:0] is sampled by tx_clk to generate tx_data_s01[9:0].
 - tx_asic_div_clk is the div2 of tx_clk.
 - The gasketed data {tx_data_s01[9:0], tx_data[9:0]} is sampled.
 - There are timing arcs for tx_data[9:0] with tx_asic_div_clk and tx_data[9:0] with tx_clk.
 - 16-, 20-bit interface
 - tx_data[19:0] is sampled by tx_asic_div_clk.
 - tx_asic_div_clk is the div1 of tx_clk.
 - There are timing arcs for tx_data[19:0] with tx_asic_div_clk.
 - 32-, 40-bit interface
 - tx_data[39:0] is sampled by tx_asic_div_clk
 - tx_asic_div_clk is the div1 of tx_clk.
 - There are timing arcs for tx_data[39:0] with tx_asic_div_clk
 - 64-bit interface
 - tx_data[63:0] is sampled by tx_asic_div_clk
 - There are timing arcs for tx_data[63:0] with tx_asic_div_clk
- Wide_xfer_en = 1
 - 8-, 10-bit interface
 - tx_data[9:0] is used.

- tx_data[9:0] is sampled by the tx_clk to generate tx_data_s01[9:0], s02 and s03.
- tx_asic_div_clk is the div4 of tx_clk.
- The gasketed data {tx_data_s03[9:0], tx_data_s02[9:0], tx_data_s01[9:0], and tx_data[9:0]} are sampled.
- There are timing arcs for tx_data[9:0] with tx_asic_div_clk and tx_data[9:0] with tx_clk
- 16-, 20-bit interface
 - tx_data[19:0] is sampled by tx_asic_div_clk, and tx_data[19:0] is sampled by tx_clk to generate the tx_data_s01[19:0].
 - tx_asic_div_clk is the div2 of tx_clk.
 - There are timing arcs for tx_data[19:0] with tx_asic_div_clk and tx_clk
- 32-, 40-bit interface
 - tx_data[39:0] is sampled by tx_asic_div_clk.
 - There are timing arcs for tx_data[39:0] with tx_asic_div_clk.
- 64-bit interface
 - tx_data[63:0] is sampled by tx_asic_div_clk
 - There are timing arcs for tx_data[63:0] with tx_asic_div_clk

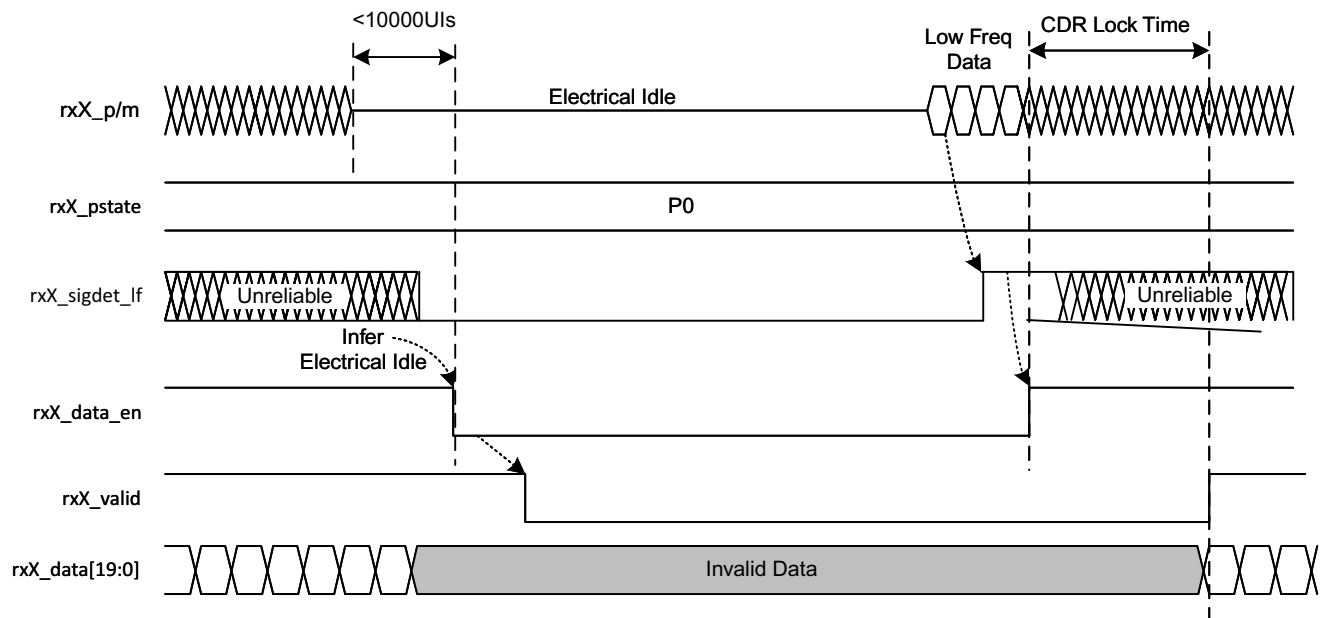
In summary, the .lib timing for tx_data[63:0] is as follows:

- tx_data[63:0] is related to tx_asic_div_clk.
- tx_data[19:0] is related to both tx_asic_div_clk and tx_clk.

4.6 RX Datapath

The RX datapath is active when the RX lane is in P0 state ($rxX_pstate=2'b00$) and rxX_data_en is asserted. You should only assert rxX_data_en when the RX lane is in P0 state and the RX input signal is present (indicated by the assertion of the rxX_sigdet_lf output). If a loss of signal is detected by the de-assertion of rx_sigdet_lf or inferred using higher layer protocol-specific mechanisms, the Clock-and-Data-Recovery (CDR) must be disabled by setting $rxX_data_en = 0$ and re-enabled by setting $rxX_data_en = 1$ after a signal is detected using $rxX_sigdet_lf = 1'b1$. A typical timing diagram of RX datapath is shown in [Figure 4-8](#).

Figure 4-8 RX Datapath and Control Timing



The output rxX_valid is asserted, when a pre-determined bit-locking timer expires after assertion of rxX_data_en . In this case, rxX_valid is de-asserted, when rxX_data_en is de-asserted to 0 or when the CDR frequency drift goes beyond that set by the input $rxX_cdr_ppm_max[4:0]$. The default settings for $rxX_cdr_ppm_max[4:0]$ values provided in this chapter set the drift threshold to 15000 ppm. If the CDR frequency goes beyond this threshold, the rxX_data_en is internally overridden to enable the reference tracking mode until the CDR frequency returns to the ideal frequency. After which the locking to data resumes and rxX_valid re-asserts after the bit-locking timer expires.

When the input rxX_data_en is asserted, the CDR tracking mode is enabled. When $rxX_data_en=0$, the RX VCO frequency is calibrated using the reference clock inputs ($ref_alt0/1_clk_p/m$, $ref_pad_clk_p/m$) to ensure that the VCO frequency stays within the precision required for the CDR to lock to the received incoming data when rxX_data_en is re-asserted.

The rxX_data_en signal must be de-asserted while performing rate changes in P0 state. rxX_data_en must be de-asserted before asserting the rxX_req signal for the rate change and asserted only after the rxX_req/rxX_ack handshaking is completed for the rate change.

If rxX_data_en is enabled during invalid data, the fastest recovery is to change the rxX_pstate from P0 to P1 to P0. This action takes approximately 30 μs and is much faster than negating rxX_data_en for an extended time.

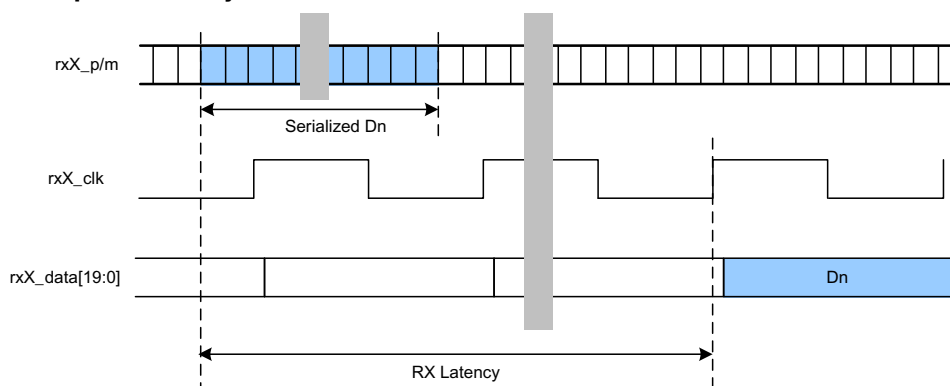
The maximum CDR lock times as measured in [Figure 4-8](#) are provided in [Table 4-2](#) for both Spread-Spectrum-Clocking (SSC) and non-SSC modes.

Table 4-2 CDR Lock Times

CDR Mode (SSC/non-SSC)	Maximum CDR Lock Time (UI)
Non-SSC (rxX_cdr_ssc_en=0)	6,560
SSC (rxX_cdr_ssc_en=1)	36,800

The RX datapath supports de-serialization of 8/10/16/20/32/40/64-bit data symbol determined by the setting of the input pin rxX_width. The valid bits on rxX_data in the 8/10/16/20/32/40/64-bit modes are defined in the signal description of rxX_data.

The RX datapath latency for the PHY is measured as per [Figure 4-9](#). The latency numbers are provided in [Table 4-3](#).

Figure 4-9 Receive Datapath Latency Measurement In the PHY**Table 4-3 RX Datapath Latency**

Lane RX I/F width (rxX_width[2:0])	Min Latency (UI)	Max Latency (UI)
16-bit	56	58
20-bit	70	72
32-bit	72	74
40-bit	90	92
64-bit	136	138

**Note**

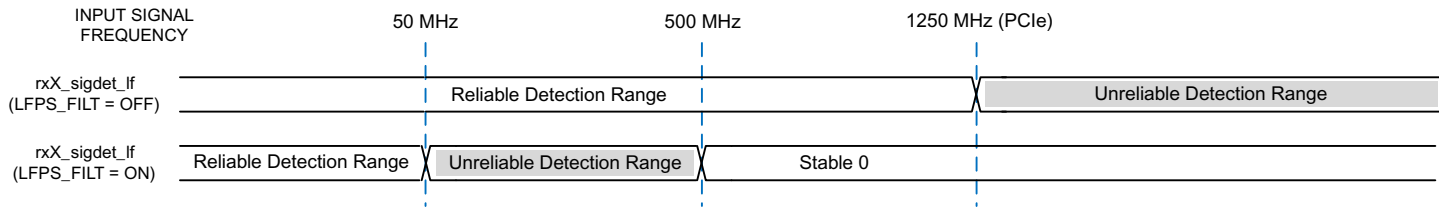
Maximum frequency of rxX_clk is 1.014 GHz. Thus, data rates faster than 16 Gbps must use 32-bit mode.

4.7 RX Signal Detector

The `rxX_sigdet_lf` signal is a low-frequency detect output that denotes the presence of a low-frequency signal up to 1.611 GHz). The 1.611 GHz maximum range for `sigdet_lf` allows for its use in detection of the wake-up signal for 2.5G Ethernet. This circuit is also capable of detecting signal for PCIe1 at 2.5 Gbps (maximum of 1.25 GHz).

The signal detector can also be configured to be used as a true LFPS detector for USB3-like applications when the `rxX_sigdet_lfps_filter_en` input is asserted. This input changes the cut-off frequency of the `rxX_sigdet_lf` signal from 1.25 GHz to 50 MHz. [Figure 4-10](#) shows the behavior of the `rxX_sigdet_lf` signals.

Figure 4-10 Behavior of rxX_sigdet_lf



Note

- A pulse catcher is used in the Synopsys PCS to filter the incoming LF signal to ensure that there is a stable 1 up to 1.611 GHz. On some corners the circuit may transition, but the pulses are not smaller than 0.9 ns.
- For non-filtered LF output, a pulse catcher should be used for PCIe (that is, Gen1 and EIEOS) and Ethernet (that is, ALERT) signal detect applications up to 1.611 GHz
- For LFPS filtered LF output, it is reliable up to 50 MHz. Above this frequency of data it can glitch. Above 500 MHz it does not detect any signal.

The low-frequency signal detection function (`rxX_sigdet_lf`) is enabled when the `rxX_disable` and `rxX_reset` inputs are de-asserted.

The `rxX_sigdet_lf` output is in an undetermined state when the `rxX_disable` input is asserted. When `rxX_disable` is de-asserted, it might take up to 12 μ s for the SigDet to power up, during which time the `rxX_sigdet_lf` output is forced to 0 within the RAW_PCS. Note that these filtering times are based on a 100-MHz input reference clock and can increase by up to 2x for slower reference clocks.

The `rxX_sigdet_lf` output is in an undetermined state when RX terminations are disabled; that is, when `rxX_term_en` = 0.

For cases where the `rxX_sigdet_lf` output is valid, their response times are shown and described in [Figure 4-11](#) and [Table 4-4](#).

Figure 4-11 Response Times for rxX_sigdet_lf Signal Detector Path

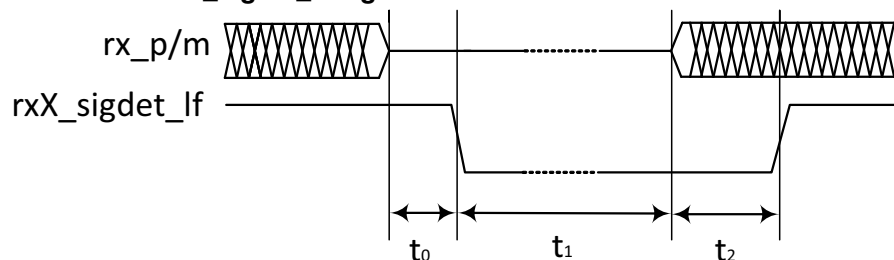


Table 4-4 Response Times for rxX_sigdet_If Signal Detector Path

Response Time	Delay	Unit
t0	< 16 (max)	ns
t0 + t1	> 20 (min)	ns
t2	< 16 (max)	ns
Abs (t2–t0)	< 8	ns

4.8 RX CDR PPM Detector

The PHY includes a CDR PPM detection circuit that contains the following features:

- Programmable PPM offset
- Current status of PPM offset relative to nominal frequency
- Ability to operate simply as a PPM monitor without intervening
- Ability to reset the CDR back to a nominal state when the clock drifts past the programmable PPM offset

The CDR PPM detector contains running counters and compares the incoming rxX_clk to a nominal clock based on the rxX_ref_ld_val and rxX_vco_ld_val inputs. If the difference exceeds the programmable PPM offset, the CDR unlock detector forces the rxX_data_en signal low to the hard-macro PMA and enables the PMA to track back to the nominal frequency.

The RX CDR PPM detector is enabled by default but can be disabled by setting the RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL.RX_CDR_DETECTOR_EN register to 1'b0. If you want to disable the correction mechanisms of the RX CDR PPM detector, you can also enable it in monitoring mode by setting the RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL.RX_CDR_PPM_MONITOR_MODE register to 1'b1.

The rxX_ppm_drift signal is the difference between the current running counter value relative to the programmed rxX_vco_ld_val. The output is a signed value that ranges from -64 (6'b100000) to +63 (6'b011111). This value is valid when rxX_ppm_drift_vld is asserted. The rxX_ppm_drift_vld signal is periodically asserted when the FSM has completed an iteration of comparing the ppm drift.



Note

The rxX_ppm_drift signal must not be used to control an EStore module in the protocol PCS. (Elastic Buffer or EStore supports Nominal Half-Full and Nominal Empty buffer modes and supports common reference clock systems in both of these modes. For more information about EStore, refer to the *Multi-Protocol 32G PCS for the DesignWare Cores 32G PHY Databook*.) This signal provides only a difference from the NOMINAL frequency and does not take into account the actual transmitter.

4.8.1 Configuring the RX CDR PPM Detector

The RX CDR PPM detector can be programmed by changing the rxX_cdr_ppm_max[4:0] input signals. This input specifies the maximum PPM drift to be allowed on the RX clock (relative to the ideal frequency) before the CDR recovery circuit is engaged. This value is available in the PHY configuration section for 15,000 ppm (default setting). However, the value can be updated for each configuration based on the following equation (the example is based on 5600 ppm for PCIe3 @ 100 MHz):

$$FL00R(rxX_vco_ld_val * (ppm_value / 1e6)) - 2 \text{ (for margin)} = (1,360 * 0.015) - 2 = 18$$

4.9 Fast Context Switching

This section describes the Fast Context Switching feature, including initial calibration and rate changes.

4.9.1 Usage Overview

The PHY implements a feature for fast context switching during a rate change, which enables performing calibrations only once for a given PHY TX and RX clock configuration. The calibration results are stored within the always-on (AON) power domain in the RAW_PCS, and subsequent rate changes simply restore the previously calibrated values for that given rate instead of performing a full re-calibration.

Context switching uses the concept of ‘banks’ to store the calibrated values for the MPLL and TX Duty Cycle Correction (DCC) as well as the RX duty cycle correction and IQ correction (IQC). The MPLL calibration results are stored in MPLLA/B calibration banks controlled by the `mpll{a,b}_recal_bank_sel` signal and TX DCC results are stored in the TX cal bank controlled by the `txX_recal_bank_sel` signal. The RX DCC and IQC results are both stored in the RX cal bank controlled by the `rxX_recal_bank_sel` signal. In the case of `mpll{a,b}_recal_bank_sel`, a bank selection is defined by a change in the MPLL clock configuration settings. In the case of `txX_recal_bank_sel`, a bank selection is defined by a change in the TX clock configuration settings. In the case of `rxX_recal_bank_sel`, a bank is defined by the RX VCO clock configuration. [Table 4-5](#) lists the signals that define a bank selection.

Table 4-5 Signals That Define Bank Selection

MPLL Signals	TX Signals	RX Signals
<code>mpll{a,b}_multiplier[11:0]</code>	<code>txX_rate[2:0]</code>	<code>rxX_rate[2:0]</code>
<code>mpll{a,b}_fb_clk_div4_en</code>	<code>txX_ropll_bypass</code>	<code>rxX_ref_id_val[6:0]</code>
	<code>txX_ropll_refsel</code>	<code>rxX_vco_id_val[12:0]</code>
	<code>txX_ropll_refdiv</code>	<code>rxX_dfe_bypass</code>
	<code>txX_ropll_cp_ctl_int[6:0]</code>	<code>rxX_cdr_vco_config[11:0]</code>
	<code>txX_ropll_cp_ctl_prop[6:0]</code>	<code>rxX_dcc_ctrl_range_diff[3:0]</code>
	<code>txX_ropll_rc_filter[2:0]</code>	<code>rxX_dcc_ctrl_range_cm[3:0]</code>
	<code>txX_ropll_v2i_mode[1:0]</code>	<code>rxX_delta_iq[3:0]</code>
	<code>txX_ropll_vco_low_freq[1:0]</code>	
	<code>txX_ropll_fbdiv</code>	
	<code>txX_ropll_postdiv[2:0]</code>	
	<code>txX_dcc_ctrl_range_diff[3:0]</code>	
	<code>txX_dcc_ctrl_range_cm[3:0]</code>	
	<code>mpll{a,b}_tx_clk_div[2:0]</code>	
	MPLL signals	

There are two primary usage cases for fast context switching. The first use case is the initial calibration of a given bank. In this case, the firmware performs all the necessary MPLL, DCC, and IQC calibrations for the

given rate and stores the information within the AON domain in the RAW_PCS so it can be used in subsequent rate changes. The second use case is a subsequent rate change either within or between already calibrated banks. In this case, the relevant information is restored into the PMA, but no calibrations takes place. The following sections describe both use cases in detail.

4.9.2 Initial Calibration

An initial calibration is defined as the first time that the PHY is being configured for a new bank. The initial calibration ensures that all calibrations for the given clock configuration are performed up-front, ensuring that subsequent transitions to this bank do not require re-calibrations. All initial calibrations are performed in PHY state P1 and are triggered when the MPLL is powered up and the corresponding bank has not been previously calibrated.

To perform an initial calibration for a given rate, perform the following procedure.

1. Power the PHY TX and RX into P1 state, and perform four-way request-ack handshaking using txX_req/ack and rxX_req/ack (described in Section 4.3, “Lane Configuration Update”).
 $\text{rxX_pstate} = 2'b10$
 $\text{txX_pstate} = 2'b10$
2. Power down the MPLL, and perform four-way request-ack handshaking using txX_req/ack (described in Section 4.3, “Lane Configuration Update”).
 $\text{txX_pll_en} = 1'b0$
3. Set up PHY inputs to appropriate values based on PHY clock settings for the targeted bank, power up the MPLL, and perform four-way request-ack handshaking using txX_req/ack (for MPLL and TX DCC) and rxX_req/ack (for RX DCC and IQC) (described in Section 4.3, “Lane Configuration Update”).



Note

When changing a bank using the mpla/b_recal_bank_sel signals, you must set the respective MPLL using the txX_ropll_refsel signal for the firmware to properly detect the calibration event. For example, if only mpla_recal_bank_sel is changed, the txX_ropll_refsel signal must be set to 2'b00 to denote that MPLLA is to be powered up during the txX_req/ack handshaking.

For this initial calibration phase, firmware checks the internal calibration done flag and determines that the selected bank has not been calibrated before, then triggers a calibration.

- If the mpll{a,b}_recal_force_en, txX_recal_force_en, or rxX_recal_force_en signals are asserted, the appropriate bank is always re-calibrated.
- If the mpll{a,b}_recal_skip_en, txX_recal_skip_en, or rxX_recal_skip_en signals are asserted, the appropriate bank is not re-calibrated, and any previously stored values are used.

For the following sequences, it is assumed that the bank has not been previously calibrated.

For MPLL/TX DCC calibrations:

- a. MPLLA/B calibration is performed (either one or both depending on which mpll{a,b}_recal_bank_sel buses were updated).
- b. TX DCC calibration is performed at full-rate.
- c. TX DCC calibration is performed at half-rate.

- d. Restore the recently calibrated values for the MPLL and TX DCC based on the txX_ropll_refsel and txX_rate PHY inputs, and release overrides to perform the initial requested TX rate change.

For RX calibrations:

- a. RX DCC calibration is performed at full-rate.
 - b. RX IQC calibration is performed at full-rate.
 - c. RX DCC calibration is performed at half-rate.
 - d. RX IQC calibration is performed at half-rate.
 - e. Restore the recently calibrated codes for RX DCC and IQC based on the rxX_rate and rxX_dfe_by-pass PHY inputs, and release overrides to perform the initial requested RX/TX rate change.
4. Power the PHY TX and RX back to P0 state.

Figure 4-12 Initial Calibration of MPLL and TX DCC

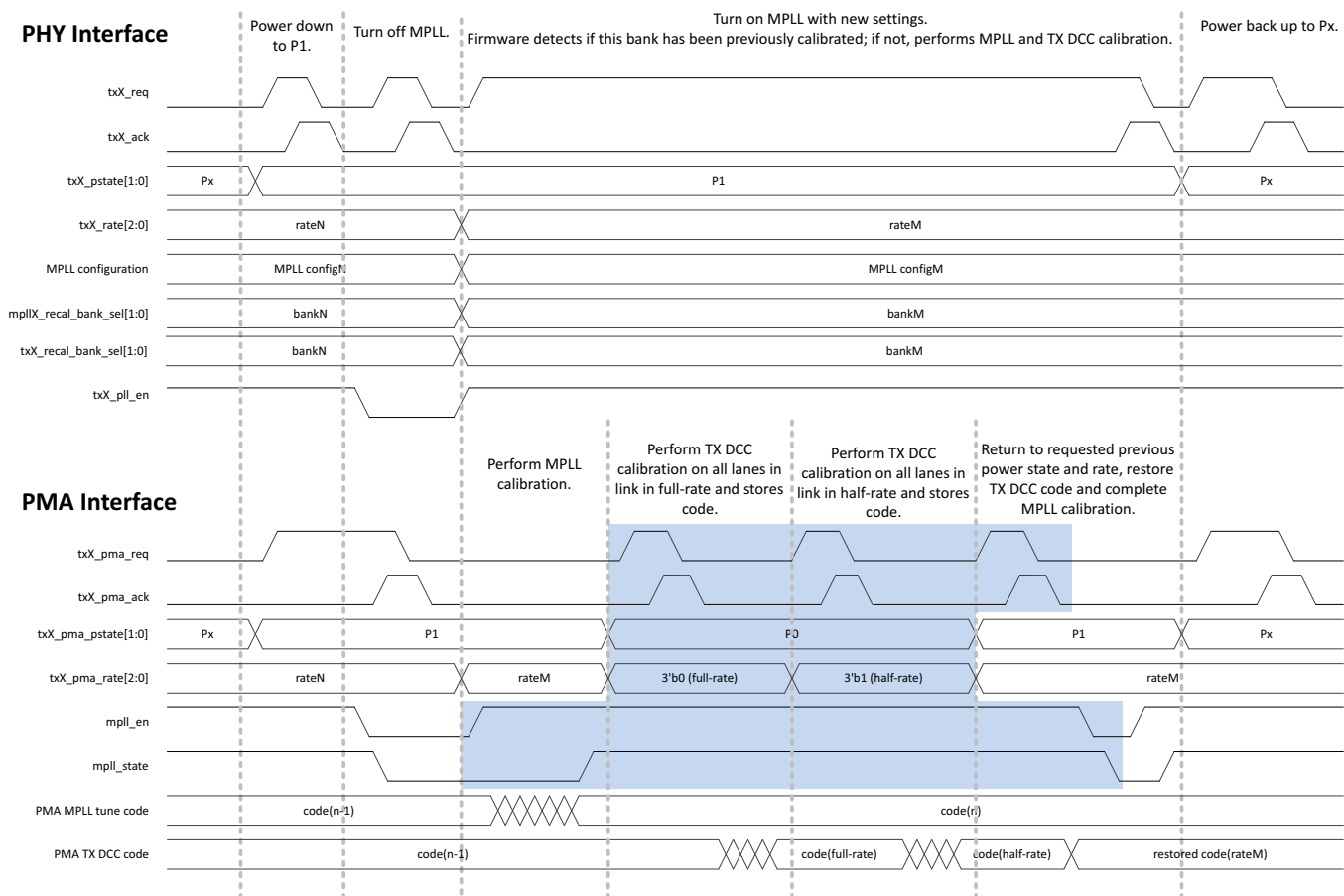
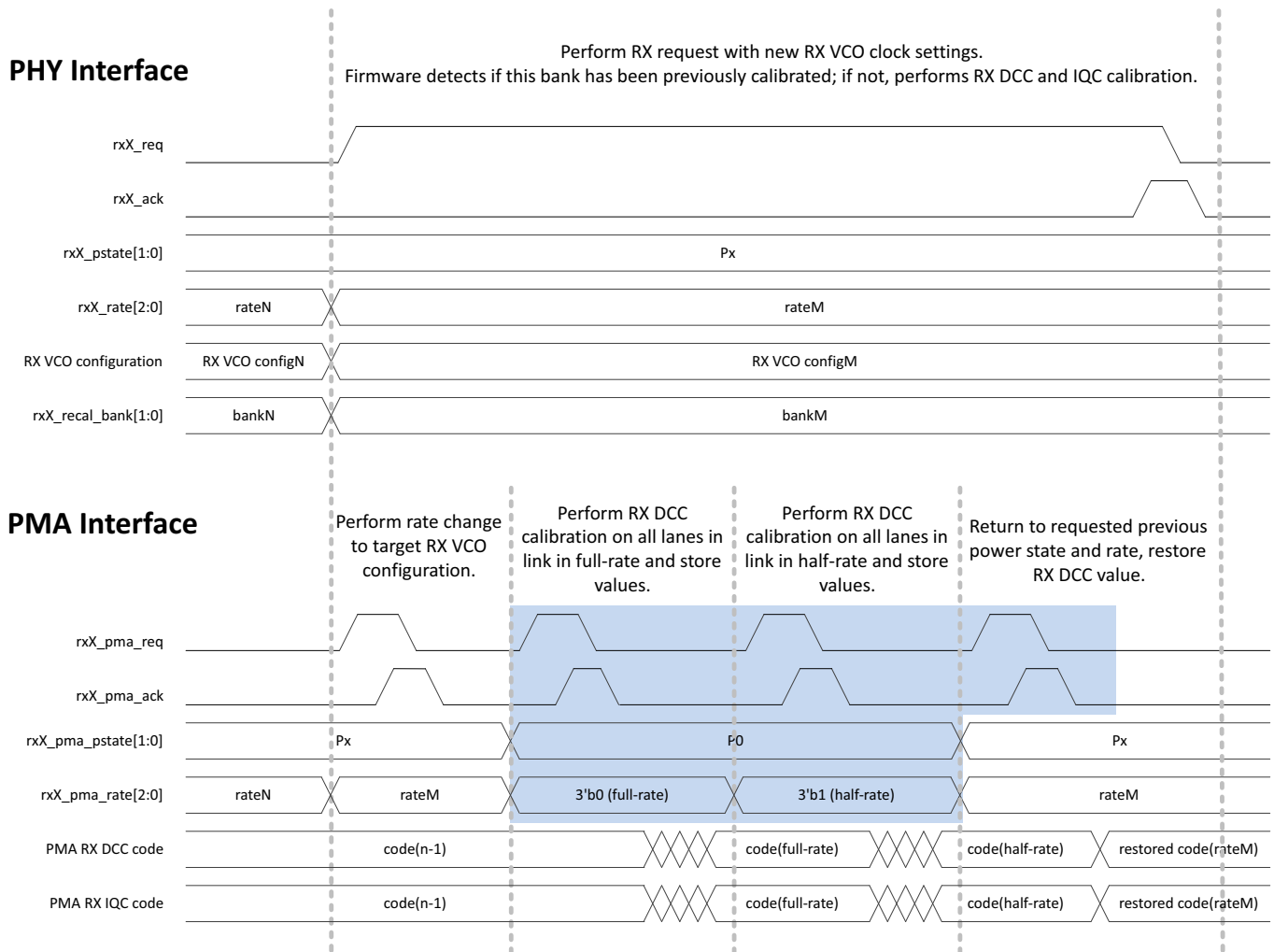


Figure 4-13 Initial Calibration of RX DCC and RX IQC

4.9.3 Subsequent Rate Changes

When performing subsequent rate changes, you have two options depending on whether the destination rate is within the same bank or in another bank.

4.9.3.1 Rate Change to a Different Bank

When performing a rate change to a different bank, a similar procedure to the one described in Section 4.9.2, “Initial Calibration” must be followed. The difference is that because the destination bank has already been calibrated, the firmware does not perform any further calibrations, but only the appropriate settings for the MPLL, TX DCC, RX DCC and RX IQC are restored.

To perform a rate change to a different bank, perform the following procedure.

1. Power the PHY TX and RX into P1 state, and perform four-way request-ack handshaking using txX_req/ack and rxX_req/ack (described in Section 4.3, “Lane Configuration Update”).
 - rxX_pstate = 2'b10

□ txX_pstate = 2'b10

2. Power down the MPLL, and perform four-way request-ack handshaking using txX_req/ack (described in Section 4.3, “Lane Configuration Update”).

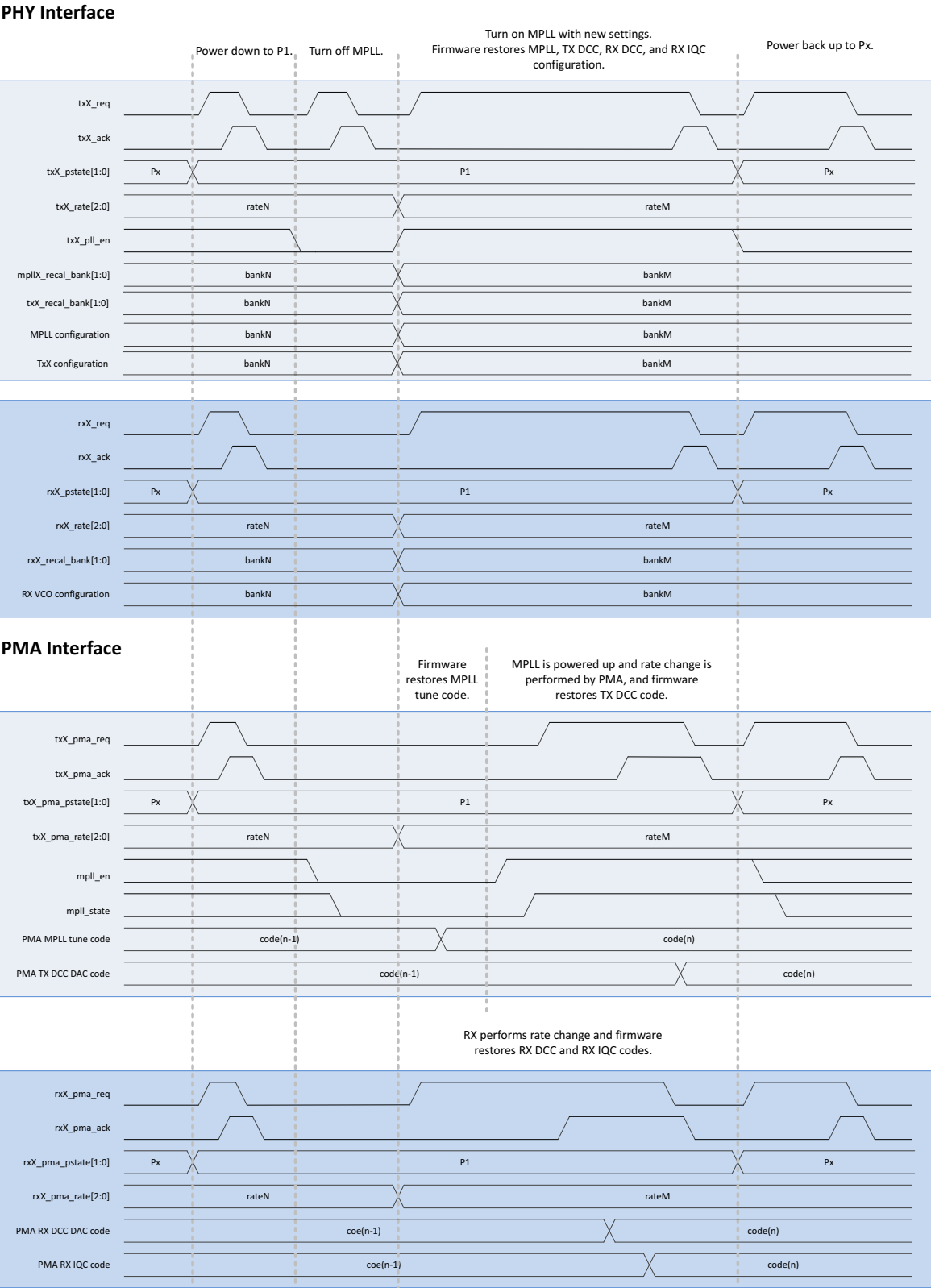
txX_pll_en = 1'b0

3. Set up PHY inputs to appropriate values based on PHY configuration for the destination bank, power up the MPLL, and perform four-way request-ack handshaking using txX_req/ack (for MPLL and TX DCC) and rxX_req/ack (for RX DCC and IQC) (described in Section 4.3, “Lane Configuration Update”).

The firmware determines that the appropriate bank has been previously calibrated and therefore, performs the following tasks.

- a. Restore the calibrated value for the MPLL.
 - b. Perform the requested TX rate change, and restore the previously calibrated values for TX DCC based on the txX_ropll_sel and txX_rate PHY inputs.
 - c. Perform the requested RX rate change, and restore the previously calibrated values for RX DCC and IQC based on the rxX_rate and rxX_dfe_bypass PHY inputs.
4. Power the PHY TX and RX back to P0 state.

Figure 4-14 Rate Change to a Different Bank



4.9.3.2 Rate Change (txX_rate/rxX_rate) Within the Same Bank

When performing a rate change to the same bank, there is no need to power the MPLL down and back up.

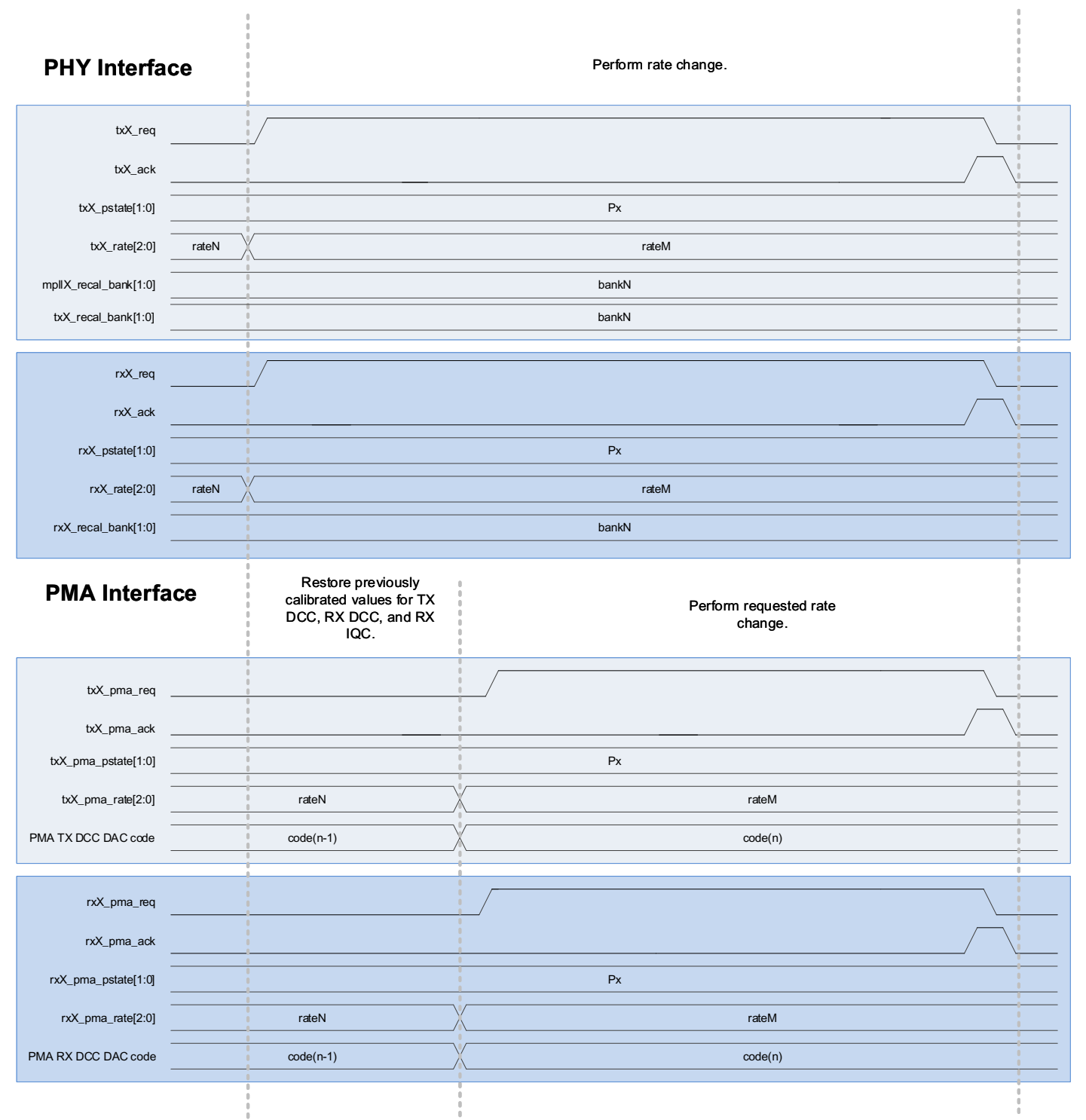
To perform a rate change to the same bank, perform the following procedure.

1. Power the PHY TX and RX into P1 state, and perform four-way request-ack handshaking using txX_req/ack and rxX_req/ack (described in Section 4.3, “Lane Configuration Update”).
 - rxX_pstate = 2'b10
 - txX_pstate = 2'b10
2. Set up PHY inputs to appropriate values based on PHY configuration for destination bank and perform four-way request-ack handshaking using txX_req/ack and rxX_req/ack (described in Section 4.3, “Lane Configuration Update”).
 - txX_rate
 - rxX_rate

The firmware performs the following tasks.

- a. Perform the requested TX rate change, and restore the previously calibrated values for TX DCC based on the txX_rate PHY input.
 - b. Perform the requested RX rate change, and restore the previously calibrated values for RX DCC and IQC based on the rxX_rate PHY input.
3. Power up the PHY TX and RX back to P0 state.

Figure 4-15 Rate Change Within the Same Bank



4.9.4 Force Recalibration

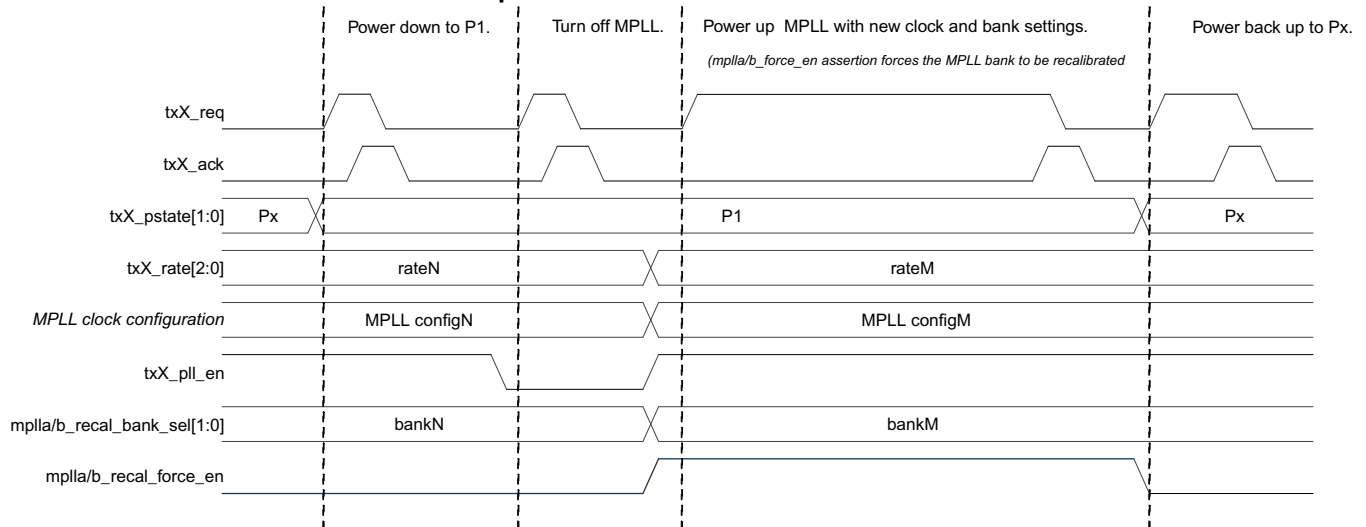
This section describes forcing MPLL and RX recalibration.

4.9.4.1 Force MPLL Recalibration

When using `mplla/b_recad_force_en` to force a recalibration of the MPLL, a rising edge must be detected on the `mplla/b_recad_force_en` signal between subsequent requests. This event implies that `mplla/b_recad_force_en` must first be sampled low with a `txX_req` on a bank that has already been calibrated to ensure that a re-calibration is forced only on the assertion of `mplla/b_recad_force_en` and not on subsequent requests (that is, `pstate` changes) if `mplla/b_recad_force_en` remains asserted.

Only assert `mplla/b_force_recad_en` along with `txX_pll_en` to force an MPLL recalibration, and de-assert `mplla/b_force_recad_en` as soon as the `txX_ack` signal is asserted. The following figure illustrates this sequence.

Figure 4-16 Force MPLL Recalibration Sequence



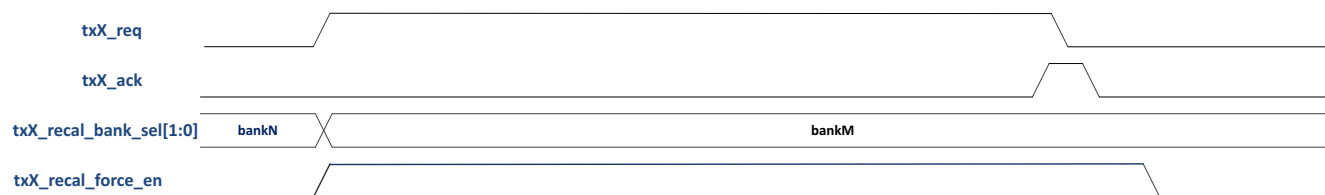
Note

- If multiple lanes are using the same MPLL, reset/req requests for both Tx and Rx at the same time cannot be requested, that is the following are not allowed for any lanes sharing the same MPLL:
 - rxX_reset and txX_reset simultaneously.
 - rxX_reset and txX_req simultaneously.
 - rxX_req and txX_reset simultaneously.
 - rxX_req and txX_req simultaneously.

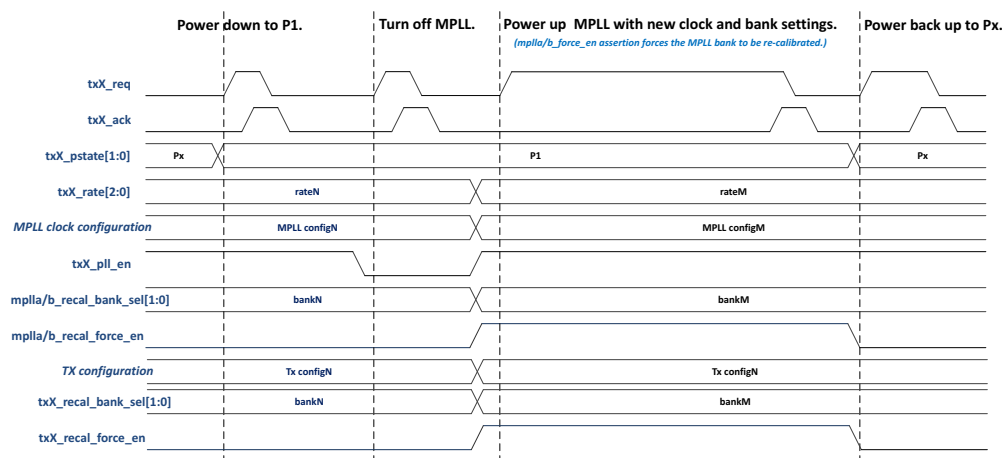
4.9.4.2 Force TX Recalibration

To force a TX recalibration, `txX_recad_force_en` must be asserted and qualified by a TX req to force calibration of the selected TX bank based on the currently selected rate (that is, the stored calibration values are valid only for the selected bank and rate combination only.)

Forced TX calibration causes the link to power down, then up, and then recalibrated.

Figure 4-17 Force TX Recalibration

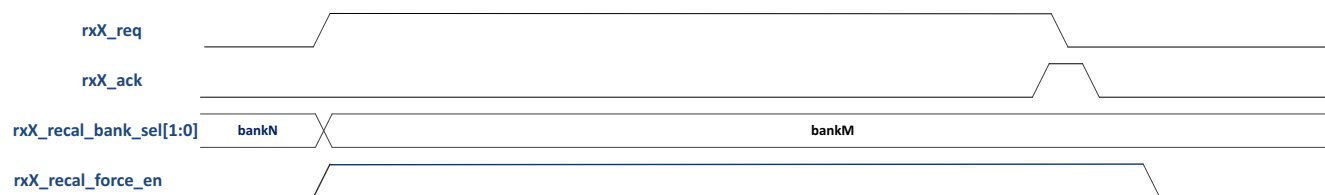
For any lane, if an MPLL is used, then changing the MPLL settings (or recalibrating this MPLL) requires recalibrating the TX.

Figure 4-18 Force TX Recalibration Sequence

4.9.4.3 Force RX Recalibration

To force an RX recalibration, **rxX_recal_force_en** should be asserted and qualified by an RX req to force calibration of the selected RX bank based on the currently selected rate (that is, the stored calibration values are valid only for the selected bank and rate combination only).

Forced RX calibration causes the link to power down, then up, and then recalibrated.

Figure 4-19 Force RX Recalibration

4.10 TX and RX Power States

The PHY supports independent power state controls for each lane and TX and RX within each lane. There are four power states supported for the TX and RX within each lane: P0, P0s, P1, and P2.

4.10.1 TX and RX Power State Description

The operational modes of the lane in each power state are provided in [Table 4-6](#) and [Table 4-7](#).

Table 4-6 TX Power State Descriptions

TX Power State	Description
P0	P0 is the fully operational power state for the TX. In the state: ■ txX_pstate is set to P0. ■ txX_reset, txX_disable inputs must be de-asserted. txX_lpd can be asserted to disable a lane in a link. ■ txX_pll_en must be set to 1. ■ If tx_data_en is asserted, TX driver is enabled and serial data is transmitted. ■ txX_clk input must be valid and of correct frequency prior to de-asserting txX_req input when requesting entry into P0 state (shown in Figure 4-5 (“TX P2-to-P0/P0s Transition When txX_width = 16/20/32/40-bit”). ■ TX common-mode, appropriate MPLL clocks, TX clocks, and TX datapath are enabled. ■ If an MPLL is being used by any active lane (based on txX_ropll_refsel), this MPLL is on in this power state. ■ If the RO-PLL of a lane is used (based on txX_ropll_refsel and txX_ropll_bypass settings), this lane is on in this power state. ■ When the RO-PLL of a lane is powered up, the following clocks are enabled and valid: □ txX_ropll_div16p5_clk, txX_ropll_div33_clk, txX_ropll_div66_clk, when txX_ropll_div16p5_clk_en is asserted □ txX_ropll_div_clk, when txX_ropll_div_clk_en is asserted □ txX_ropll_word_clk, when txX_ropll_word_clk_en is asserted
P0s	Same state as P0 with the following exceptions: ■ txX_pstate is set to P0s ■ tx_data_en must be de-asserted
P1	Same state as P0s with the following exceptions: ■ txX_pstate is set to P1. ■ txX_pll_en/txX_disable may be set to 0/1, respectively. For example, in PCIe L1 state, txX_pstate=P1, txX_pll_en=1, txX_disable=0. In PCIe L1 sub-states, txX_pstate=P1, txX_pll_en=0, txX_disable=1. ■ MPLLs may be powered down if txX_pll_en=0 and no other lane is requesting MPLLs to be powered up, for example, in PCIe L1 sub-states. In this case, MPLL clock outputs are disabled. ■ txX_clk may not be supplied.
P2	Same state as P1, with following exceptions: ■ txX_pstate is set to P2. ■ MPLL{A/B} is powered down if txX_pll_en=0 and no other lane is requesting MPLLs to be powered up. In this case, MPLL clock outputs may be disabled. ■ MPLL{A/B} is powered up if txX_pll_en=1. In this case, MPLL clock outputs are disabled.

Table 4-7 RX Power State Descriptions

RX Power State	Description
P0	All RX circuits are powered up. ■ When rxX_data_enare asserted, the RX data recovery and de-serialization is enabled. ■ rxX_reset and rxX_disable inputs must be de-asserted. rxX_lpd can be asserted for disabling lane while link in P0. rxX_term_en must be asserted. ■ rxX_clk is valid. Note: PCIe P0s state maps to P0 state on the PHY rxX_pstate input.
P0s	Same state as P0 with the following exceptions: ■ RX CDR and datapath are disabled. ■ rxX_reset, rx0_lpd, and rxX_disable inputs may be asserted. ■ rxX_data_en must be de-asserted. ■ rxX_clk is disabled.
P1	Same state as P0s with the following exceptions: ■ RX VCO is powered down. ■ rx0_lpd and rxX_disable inputs may be asserted. For example, in PCIe L1 sub-states, rxX_disable input is asserted.
P2	Same state as P1, with following exceptions: ■ RX analog front end is powered down.

4.10.2 Power State Transition Times for TX and RX

The time taken for the PHY to assert txX_ack and rxX_ack in response to a txX_req and rxX_req corresponding to a txX_pstate and rxX_pstate transition is referred to as the power state transition time. The power state transition times are dependent upon the reference clock frequency, txX_clk/rxX_clk frequency, and ref_range input. In addition, the power state transition times are dependent upon a range of other input parameters. These include, but are not limited to: register settings, reference clock frequency, MPLL clock frequency, RX VCO clock frequency, lane divider settings.

To support faster simulation times, the PHY includes a scaled down mode that can be enabled by a sequence of register writes defined in Section 6.5, “Fast Simulation Mode”. The values assume the TX and RX power states are the same and transitioned at the same time.

For power state transition times for other configurations of the PHY, contact Synopsys customer support.

The protocol specific power state mappings are provided in [Table 4-8](#).

Table 4-8 Protocol Power State Mapping

Protocol	Protocol Power State	PHY Input Settings					
		txX_pstate[1:0]	rxX_pstate[1:0]	txX_pll_en	txX_disable	rxX_disable	refa/b_clk_en
PCIe	P0	P0	P0	1	0	0	1
	P0s	P0s	P0	1	0	0	1
	P1	P1	P1	1	0	0	1
	P1.1	P1	P1	0	0	1	0
	P1.2	P1	P1	0	1	1	0
	P1.CPM	P1	P1	0	0	0	0
	P2	P2	P2	0	0	0	1
	P2.CPM	P2	P2	0	0	0	0
SATA	PHYRDY	P0	P0	1	0	0	1
	Partial	P0	P0	1	0	0	1
	Slumber	P2	P2	0	0	0	1
	DevSleep	P2	P2	0	0	0	1
Ethernet	Active	P0	P0	1	0	0	1
	Refresh	P0	P0	1	0	0	1
	Quiet	P0	P0s	1	0	0	1
	Disabled	P2	P2	0	0	0	1

Table 4-8 shows the settings of PHY inputs after the PHY has entered the corresponding power-state. While transitioning between power states, the PHY inputs require the appropriate sequencing in order to meet the following PHY requirements:

- The txX_disable and rxX_disable signals must not be asserted before the txX_pstate and rxX_pstate transition into the P1 state is completed (indicated by the assertion of txX_ack and rxX_ack).
- The refa_clk_en and refb_clk_en signals must not be de-asserted before txX_pll_en is de-asserted and mppll{a,b}_state output is de-asserted.
- In P0 state, the TX datapath and RX CDR are enabled by asserting txX_data_en and rxX_data_en inputs. These can be enabled only in the active power state (PCIe: P0, SATA: PHYRDY, Ethernet: Active/Refresh) and must be de-asserted in all other power states.

4.11 TX Equalization

TX equalization (EQ) and transmit levels are controlled by the txX_eq_pre, txX_eq_post, and txX_eq_main input signals. These inputs are derived from the coefficients C_{-1} , C_1 , and C_0 (defined in the PCIe Base specification) using the following equations.

TX Equalization Input Mapping to Specification Parameters

$$\text{txX_eq_pre} = 4 \times |C_{-1}|$$

$$\text{txX_eq_post} = 4 \times |C_1|$$

$$\text{txX_eq_main} = C_0 + (\text{txX_eq_pre} \neq 0) + (\text{txX_eq_post} \neq 0)$$

The txX_eq_post, txX_eq_pre and txX_eq_main inputs can be updated in a single txX_clk cycle to a known valid value. Alternately, the inputs can be sequenced to free up resources in the following order:

1. Increment C_1
2. Increment C_{-1}
3. Decrement C_0
4. Decrement C_1
5. Decrement C_{-1}
6. Increment C_0

An invalid value might cause TX serial data corruption in simulation.



Note

The txX_eq_pre and txX_eq_post inputs have two bits of fractional input. The four MSBs are the integer portion of the bus, and the two LSB's are the fractional portion.

4.11.1 TX Equalization Coefficient Mapping

The PHY has a total of 24 legs for TX equalization with $\frac{1}{2}$ - and $\frac{1}{4}$ -leg fractional resolution for PRE and POST legs. For the txX_eq_pre and txX_eq_post buses, bits[5:2] are integer value and bits[1:0] are the fractional value of the coefficient code.

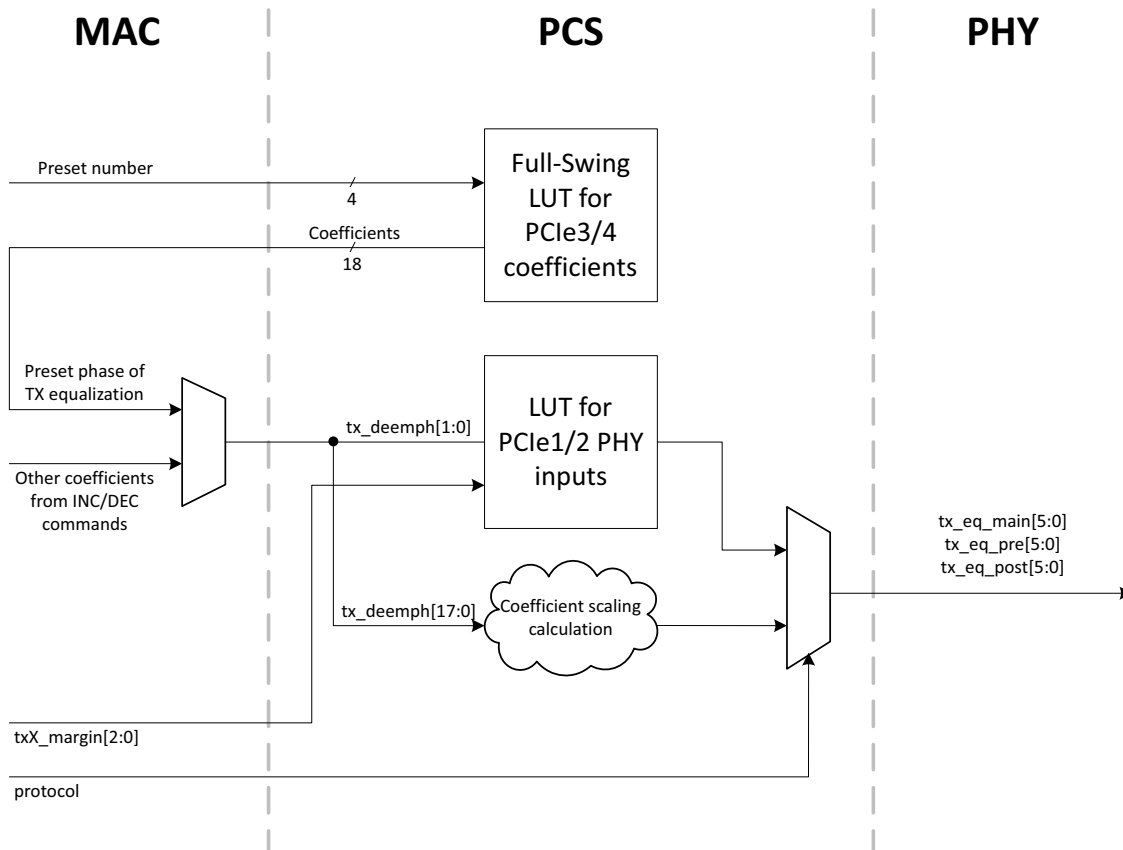
The PHY uses $\frac{1}{2}$ leg fractional resolution for the 11 PCIe preset settings. Because higher-level protocols support only integer values for FS/LF and the three TX coefficients, it is recommended that you add mapping logic in the protocol PCS between the MAC and the PHY. The mapping logic is required when updating PHY TX EQ coefficients based on PIPE TX EQ deemphasis bus changes. A half-leg can be normalized to a weight of 1, and the PHY can be represented as having 48 full legs to the MAC and re-mapping the TX EQ coefficient values.

The input coefficients must be scaled and rounded to ensure that all PCIe coefficient rules can still be met, regardless of the TxMargin setting (1, 0.75, 0.5, 0.25). The protocol PCS must also consider the case of non-preset mode where it receives arbitrary C_{-1} , C_0 , C_{+1} coefficients from the link partner during INC/DEC mode. All values of these coefficients must be mapped to the PHY TX EQ inputs.

The PHY inputs should be generated based on either a lookup table or algebraic equations depending on whether the PHY is operating in PCIe1/2 or PCIe3/4 mode.

Figure 4-20 illustrates this requirement.

Figure 4-20 TX Equalization Coefficient Mapping



When in PCIe Gen1/Gen2 mode, the PHY inputs should be taken from a lookup table that stores the full set of coefficients including all margining values; that is, three PCIe1/2 deemphasis modes (0dB, 3.5dB, 6dB) and four margin settings (1, 0.75, 0.5, 0.25).

When in PCIe Gen3/Gen4/Gen5 mode, the PHY inputs should be calculated from incoming coefficient values from the MAC. The calculation of the txX_eq_main, txX_eq_pre, and txX_eq_post is the same for all modes to ensure full support for both TX margining as well as supporting arbitrary coefficient inputs. The TX EQ mapping equations require two steps and checks if either tx_eq_pre or tx_eq_post requires a fractional value when normalized to 24 full legs. If a fractional value is required, the fractional value is rounded to an integer, resulting in all three PHY inputs tx_eq_pre, tx_eq_main, and tx_eq_post being integer values. This rounding prevents losing resolution when the requested main coefficient results in a fractional value; that is, for odd values. Note that when both tx_eq_pre and tx_eq_post are fractional values, there is no additional rounding necessary.

Stage 1:

$$\text{tx_eq_pre_int} = \text{FLOOR}((C-1 * 2 * \text{margin_factor}) / 2) * 2$$

$$\text{tx_eq_post_int} = \text{FLOOR}((C+1 * 2 * \text{margin_factor}) / 2) * 2$$

Stage 2:

$$\text{tx_eq_pre} = (\text{tx_eq_pre_int}[1] \ \& \ \sim \text{tx_eq_post_int}[1]) ? \text{tx_eq_pre_int} + 2 : \text{tx_eq_pre_int}$$

```

tx_eq_post = (tx_eq_post_int[1] & ~tx_eq_pre_int[1]) ? tx_eq_post_int + 2 :
tx_eq_post_int
tx_eq_main = (24 * margin_factor) - (tx_eq_pre + tx_eq_post) / 4

```

4.11.2 TX EQ Initialization and Presets

The TX equalization input settings and TX margining settings for PCIe are provided in [Table 4-9](#) through [Table 4-12](#).



Note

The following tables outline the ideal coefficients to be applied for each PCIe preset. Due to necessary rounding made in the equations outlined in Section 4.11.1, “TX Equalization Coefficient Mapping”, the values do not exactly match the user codes that result from using the equations.

Table 4-9 PCIe TX Equalization – Codes for Full-Swing Operation (No Attenuation)

Preset	Preshoot	Post Cursor	Attenuation			User Codes (Decimal)		
Number	(dB)	(dB)	(dB)	C ₋₁	C ₊₁	txX_eq_main (C ₀)	txX_eq_pre (4 x C ₋₁)	txX_eq_post (4 x C ₊₁)
P0	0.000	-6.000	0.000	0.000	-0.250	18	0	24
P1	0.000	-3.500	0.000	0.000	-0.167	20	0	16
P2	0.000	-4.400	0.000	0.000	-0.200	19	0	20
P3	0.000	-2.500	0.000	0.000	-0.125	21	0	12
P4	0.000	0.000	0.000	0.000	0.000	24	0	0
P5	1.900	0.000	0.000	-0.093	0.000	22	8	0
P6	2.500	0.000	0.000	-0.125	0.000	21	12	0
P7	3.500	-6.000	0.000	-0.103	-0.196	17	10	18
P8	3.500	-3.500	0.000	-0.125	-0.125	18	12	12
P9	3.500	0.000	0.000	-0.167	0.000	20	16	0
P10	0.000	See note.	0.000	0.000	See note.	16	0	32
Note: For full-swing mode (no attenuation): txX_eq_main + txX_eq_pre / 4 + txX_eq_post / 4 = 24.								

Table 4-10 PCIe TX Equalization - Codes for 3/4 Swing Operation (Attenuation = -2.49 dB)

Preset	Preshoot	Post Cursor	Attenuation			User Codes (Decimal)		
Number	(dB)	(dB)	(dB)	C ₋₁	C ₊₁	txX_eq_main (C ₀)	txX_eq_pre (4 x C ₋₁)	txX_eq_post (4 x C ₊₁)
P0	0.000	-6.000	-2.490	0.000	-0.243	14	0	16
P1	0.000	-3.500	-2.490	0.000	-0.167	15	0	12
P2	0.000	-4.400	-2.490	0.000	-0.189	14	0	16
P3	0.000	-2.500	-2.490	0.000	-0.123	16	0	8
P4	0.000	0.000	-2.490	0.000	0.000	18	0	0
P5	1.900	0.000	-2.490	-0.099	0.000	16	8	0
P6	2.500	0.000	-2.490	-0.123	0.000	16	8	0
P7	3.500	-6.000	-2.490	-0.096	-0.192	13	8	12
P8	3.500	-3.500	-2.490	-0.122	-0.122	13	10	10
P9	3.500	0.000	-2.490	-0.167	0.000	15	12	0
P10	0.000	See note.	-2.490	0.000	See note.	12	0	24
Note: For full-swing mode (no attenuation): txX_eq_main + txX_eq_pre / 4 + txX_eq_post / 4 = 24.								

Table 4-11 PCIe TX Equalization - Codes for Half-Swing Operation (Attenuation = -6.02 dB)

Preset	Preshoot	Post Cursor	Attenuation			User Codes (Decimal)		
Number	(dB)	(dB)	(dB)	C ₋₁	C ₊₁	txX_eq_main (C ₀)	txX_eq_pre (4 x C ₋₁)	txX_eq_post (4 x C ₊₁)
P0	0.000	-6.000	-6.020	0.000	-0.250	9	0	12
P1	0.000	-3.500	-6.020	0.000	-0.167	10	0	8
P2	0.000	-4.400	-6.020	0.000	-0.200	9	0	12
P3	0.000	-2.500	-6.020	0.000	-0.120	10	0	8
P4	0.000	0.000	-6.020	0.000	0.000	12	0	0
P5	1.900	0.000	-6.020	-0.102	0.000	11	4	0
P6	2.500	0.000	-6.020	-0.120	0.000	10	8	0
P7	3.500	-6.000	-6.020	-0.106	-0.213	8	7	9
P8	3.500	-3.500	-6.020	-0.125	-0.125	9	6	6
P9	3.500	0.000	-6.020	-0.167	0.000	10	8	0

Preset	Preshoot	Post Cursor	Attenuation			User Codes (Decimal)		
Number	(dB)	(dB)	(dB)	C ₋₁	C ₊₁	txX_eq_main (C ₀)	txX_eq_pre (4 x C ₋₁)	txX_eq_post (4 x C ₊₁)
P10	0.000	See note.	-6.020	0.000	See note.	8	0	16

Note: For full-swing mode (no attenuation): txX_eq_main + txX_eq_pre / 4 + txX_eq_post / 4 = 24.

Table 4-12 PCIe TX Equalization - Codes for Quarter-Swing Operation (Attenuation = -12.04 dB)

Preset	Preshoot	Post Cursor	Attenuation			User Codes (Decimal)		
Number	(dB)	(dB)	(dB)	C ₋₁	C ₊₁	txX_eq_main (C ₀)	txX_eq_pre (4 x C ₋₁)	txX_eq_post (4 x C ₊₁)
P0	0.000	-6.000	-12.040	0.000	-0.231	5	0	4
P1	0.000	-3.500	-12.040	0.000	-0.167	5	0	4
P2	0.000	-4.400	-12.040	0.000	-0.200	5	0	4
P3	0.000	-2.500	-12.040	0.000	-0.130	5	0	4
P4	0.000	0.000	-12.040	0.000	0.000	6	0	0
P5	1.900	0.000	-12.040	-0.077	0.000	5	4	0
P6	2.500	0.000	-12.040	-0.130	0.000	5	4	0
P7	3.500	-6.000	-12.040	-0.087	-0.217	4	3	5
P8	3.500	-3.500	-12.040	-0.115	-0.115	5	2	2
P9	3.500	0.000	-12.040	-0.167	0.000	5	4	0
P10	0.000	See note.	-12.040	0.000	See note.	4	0	8

Note: For full-swing mode (no attenuation): txX_eq_main + txX_eq_pre / 4 + txX_eq_post / 4 = 24.

The TX equalization input settings and TX margining settings for CCIX20-25G are provided in [Table 4-13](#).

Table 4-13 Codes for CCIX20-25g Full-Swing Operation (No Attenuation)

Preset	Preshoot	Post Cursor	User Codes (DECIMAL)		
Number	(dB)	(dB)	txX_eq_main	txX_eq_pre	txX_eq_post
0	0.000	-6.000	18	0	25
1	0.000	-3.500	20	0	17
2	0.000	-4.400	19	0	21
3	0.000	-2.500	21	0	13

Preset Number	Preshoot (dB)	Post Cursor (dB)	User Codes (DECIMAL) txX_eq_main	txX_eq_pre	txX_eq_post
4	0.000	0.000	23	0	3
5	1.900	0.000	21	9	2
6	2.500	0.000	21	11	1
7	3.500	-6.000	17	9	21
8	3.500	-3.500	18	11	13
9	3.500	0.000	20	15	1
10	0	See note	16	0	32
Note: For full-swing mode (no attenuation): $\text{txX_eq_main} + \text{txX_eq_pre} / 4 + \text{txX_eq_post} / 4 = 24$.					

Table 4-14 provides TX equalization input settings for Long Reach Ethernet (10GBASE-KR, 25GBASE-KR, and 100GBASE-KR4) preset and initialize states.

Table 4-14 TX Equalization Settings for Long Reach Ethernet

State	txX_eq_main[5:0](decimal)	txX_eq_pre[5:0](decimal)	txX_eq_post[5:0](decimal)	Note
PRESET	23	4	0	Natten = 0 ratio_atten = 1
INITIALIZE	16	4	28	Natten = 0 ratio_atten = 1

4.12 TX and RX Adaptation

This section provides details about TX and RX adaptation.

4.12.1 TX Adaptation

This section discusses TX Adaptation

The PHY supports receiver equalization, adaptation of receiver parameters, and commands for incrementing or decrementing the transmitter coefficients of the link partner. For protocols that require DFE and link adaptation (PCIe, 10GBASE-KR, 25GBASE-KR/CR), the receiver adaptation is triggered by asserting the rxX_adapt_req input. Upon completion of the receiver adaptation, the PHY asserts rxX_adapt_ack and uses the following mechanisms to adapt the transmitter coefficients of the link partner:

- **PCIe – Figure of Merit (FOM) Method** – An 8-bit rxX_adapt_fom[7:0] output indicating the quality of the received data eye. A higher value of rxX_adapt_fom [7:0] indicates better link equalization, with 8'd0 indicating worst equalization setting and 8'd255 indicating the best equalization setting. This method is suitable for PCIe.



Note

PCIe provides for an optional direction change method. This is not supported by the PHY.

- **Long Reach Ethernet – Direction Change mechanism** – rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], and rxX_txpost_dir[1:0] outputs providing the increment, decrement, and hold requests for the transmitter coefficients of the link partner. The outputs rxX_adapt_fom[7:0], rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], and rxX_txpost_dir[1:0] outputs are only valid when rxX_adapt_ack is asserted. The maximum time from assertion of rxX_adapt_req to rxX_adapt_ack is dependent on the baud rate and reference clock frequency.

4.12.1.1 PCIe TX EQ Adaptation

As per the PCIe 5.0 specification, the link partner determines the coefficient updates needed for the local TX equalization and encapsulates the pre-cursor and post-cursor coefficients as part of its transmitted TS1 ordered sets. To do so, the link partner uses the local TX's preset-to-coefficient mapping table, as well as the FS and LF values.

These parameters for the PHY TX are as follows:

- **Preset to Coefficient Mapping** (see [Table 4-9](#) through [Table 4-12](#))
- **FS: 24 (decimal)**
- **LF: 8 (decimal)**

Note that these parameters are outputs from the PCIe PCS to the MAC; for more information, refer to the *DesignWare Cores PCS Databook*.

4.12.1.2 Long Reach Ethernet TX EQ Adaptation

As per Long Reach Ethernet specifications (10GBASE-KR, 25GBASE-KR, and 100GBASE-KR4), the link partner determines the increment, decrement, and hold commands for the local TX coefficients and encapsulates these commands for each of the pre-, main-, and post-cursors as part of its transmitted training frames. The algorithm to translate the increment, decrement, and hold commands into the TX EQ coefficient inputs to the PHY is described as follows:

- Number of TX driver legs used for attenuation

$$N_{atten} = 24 - (txX_eq_main[5:0] + txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4)$$

- Full-swing mode (no attenuation)

$$txX_eq_main[5:0] + txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4 = 24$$

The increment, decrement, and hold commands from the link partner are translated into local variables ΔN_{atten} , ΔN_{main} , ΔtxX_eq_pre , and ΔtxX_eq_post as shown in [Table 4-15](#).

Table 4-15 Default TX Coefficient Update Sequence for Long Reach Ethernet

Update Method	EQ Update Request from Link Partner RX			$N_{atten}^k > 0$	TX Configuration			
	C_1	C_0	C_{-1}		Natten [5:0] (decimal)	Nmain [7:0] (decimal)	txX_eq_pre [5:0] (decimal)	txX_eq_post [5:0] (decimal)
1	Increment	Hold	Hold	N/A	+1	0	0	-4
2	Decrement	Hold	Hold	Yes	-1	0	0	+4
3 ^a	Decrement	Hold	Hold	No	0	0	0	0
4	Hold	Increment	Hold	Yes	-1	+4	0	0
5	Hold	Increment	Hold	No	0	0	0	0
6	Hold	Decrement	Hold	N/A	+1	-4	0	0
7	Hold	Hold	Increment	N/A	+1	0	-4	0
8	Hold	Hold	Decrement	Yes	-1	0	+4	0
9 ^a	Hold	Hold	Decrement	No	0	0	0	0

a. These update requests result in a MIN or MAX indication to the link partner as per the status response shown in [Table 4-16](#) ("Status Response to TX Update Requests").

The procedure for updating the TX coefficients must be followed in the following sequence:

1. $N_{atten}[5:0]^{k+1} = N_{atten}[5:0]^k + \Delta N_{atten}$
2. $N_{main}[7:0]^{k+1} = N_{main}[7:0]^k + \Delta N_{main}$
3. $txX_eq_pre[5:0]^{k+1} = txX_eq_pre[5:0]^k + \Delta txX_eq_pre$
4. $txX_eq_post[5:0]^{k+1} = txX_eq_post[5:0]^k + \Delta txX_eq_post$
5. $txX_eq_main[5:0]^{k+1} = 24 - N_{atten}[5:0]^{k+1} - \text{ROUND}(txX_eq_pre[5:2]^{k+1/4} - txX_eq_post[5:2]^{k+1/4})$

Where k is the current coefficient value, k+1 is the next update.

The starting values of N_{main} are equal to $4 \cdot txX_eq_main[5:0]$, where $txX_eq_main[5:0]$ is $txX_eq_main[5:0]$ as defined in Table 4-14 (“TX Equalization Settings for Long Reach Ethernet”).

If simultaneous increment/decrement requests are received, they should be processed in the following order:

1. Increment C_1
2. Increment C_{-1}
3. Decrement C_0
4. Decrement C_1
5. Decrement C_{-1}
6. Increment C_0

To ensure that the PHY meets the transmitter signaling requirements in the Long Reach Ethernet specifications, the following transmitter coefficient constraints must be met before updating the coefficient inputs defined in Table 4-15 (“Default TX Coefficient Update Sequence for Long Reach Ethernet”).

- $0 \leq N_{atten}[5:0] \leq 10$
- $14 \cdot 4 \leq N_{main}[7:0] \leq 24 \cdot 4$
- $0 \leq N_{post}[5:0] \leq 8 \cdot 4$
- $0 \leq N_{pre}[5:0] \leq 3 \cdot 4$
- $N_{main}[7:0] - N_{post}[5:0] - N_{pre}[5:0] \geq 2 \cdot 4$
- $txX_eq_main[5:0]^1 - txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4 \leq 24$



Note

RTL truncates $txX_eq_main[5:0]$ for a sum greater than 24 based on the rounding rule: Round up for main, round down for atten. The result is that both settings for equal or greater than 24 result in the same values to the analog.

Checks should be performed for each coefficient update request to determine if the next $k+1$ set of coefficients is valid. If a coefficient update is within the above limits then the change is accepted, and an UPDATED response can be sent to the link partner. If a coefficient update hits one of the above limits, the change is accepted, and a MIN or MAX response can be sent to the link partner. If a coefficient update tries to go outside the above limits then the update is not performed, and the MIN or MAX previous response is maintained.

Table 4-16 summarizes a system status response after coefficient update requests are received.

Table 4-16 Status Response to TX Update Requests

Coefficient	Status Response to increment/decrement (if rules are violated or limits are hit)		Status Response to Increment and Decrement (if within rules and limits are not hit)
	Increment	Decrement	
C_1	MAX	MIN	UPDATED
C_0	MAX	MIN	UPDATED
C_{-1}	MAX	MIN	UPDATED

4.12.2 RX Adaptation Sequence

Receiver adaptation is supported only when the PHY is in P0 state, electrical idle exit is detected on the RX serial lines, rxX_data_en is asserted, and the Clock and Data Recovery (CDR) circuitry is nominally locked (indicated by the assertion of rxX_valid output). When rxX_data_en is enabled during the equalization phase, for some equalization settings the CDR might drift and not be able to recover. As a result, it is recommended to assert rxX_reset for at least 8 ns between performing RX adaptation requests.

The recommended sequence is as follows.

1. Set rxX_data_en to 1.
The PHY firmware triggers the coarse adaptation algorithm.
2. Wait for rxX_valid to assert.
3. De-assert rxX_data_en.
4. Assert rxX_adapt_in_prog to the PHY.
5. Toggle rxX_reset to the PHY.
6. Wait for de-assertion of rxX_ack from the PHY.
7. Ensure the PHY lane receiver is in P0 state. Transition to P0, if not already in P0 out of reset.
8. Wait for detection of electrical idle exit condition on rxX_sigdet_1f (for low-frequency data)
9. Assert rxX_data_en to the PHY.
10. Wait for assertion of rxX_valid from the PHY.
11. Perform an RX adaptation request and assert rxX_adapt_req.
The PHY performs RX adaptation, then signals the completion by asserting rxX_adapt_ack.
12. De-assert rxX_adapt_req to the PHY.
13. De-assert rxX_adapt_in_prog to the PHY.

Note

- Do not assert rxX_adapt_req unless rxX_dfe_bypass is set to 0.
- Stopping rxX_adapt_req by negating before rxX_adapt_ack is asserted or by asserting rxX_reset can leave the receiver with a less than optimal settings.
- Do not send requests to the PHY using rxX_req and rxX_ack while rxX_adapt_req is asserted.
- The internal RX CDR PPM detector is disabled during adaptation requests. Any external PPM detectors must also be turned off during the adaptation process.
- For suggested presets that must be requested to the remote transmitter during receiver adaptation, see phy/ipxact/*snps_iip.config file in the package and search for the following macros:
 - DEFAULT_GEN3_EQ_PSET_REQ_VEC
 - DEFAULT_GEN4_EQ_PSET_REQ_VEC
 - DEFAULT_GEN5_EQ_PSET_REQ_VEC

4.12.3 PCIe—TX Equalization Link Adaptation Using Figure of Merit

The adaptation of TX equalization can be performed using the Figure of Merit (FOM) calculation in the PHY. Although this mechanism can be used for all protocols, it is recommended for link adaptation when a wide range of TX presets are available for the link partner's TX.

The PCIe MAC iterates through a predefined list of transmitter presets recommended for the PHY, as outlined in [Table 4-17](#).

Table 4-17 Transmitter Presets

PCIe Protocol	Gen3	Gen4	Gen5
Transmitter Presets	P4/5/6/8/9	P4/5/6/8/0	P4/5/6/9

Example of FOM Mechanism

An example of using the FOM mechanism for PCIe 5.0 link adaptation is described below.

The PCIe MAC iterates through the predefined transmitter presets and selects the preset for which the PHY returns the best Figure of Merit (FOM). The terminating condition for PCIe 5.0 link adaptation is for the MAC to go through the presets listed in [Table 4-17](#) (plus 1 preset with the best FOM). The maximum time taken by the PHY for receiver evaluation for each preset is dependent on the baud rate and reference clock frequency. The maximum time in PCIe 5.0 mode for a 100 MHz reference clock input is approximately 1.3 milliseconds. Therefore, the iterations through transmitter presets for the link partner complete deterministically within the timeout limit of 24 milliseconds, as specified in the PCIe base specification.



Note

If `refclk` is slower than 100 MHz, the following is needed to meet the 2 ms equalization time limit as required in the PCIe specification:

- Provide `fw_clk` (100 MHz to 175 MHz) to the PHY for the firmware.
- Use the internal DCO clock (~150 MHz) for the PHY firmware.

A detailed sequence of events for PCIe 5.0 link partner transmitter adaptation is as follows:

1. MAC sets the link partner preset index `N` to 0, best preset index `N_best = 0`, best figure of merit `FOM_best = 0`, `TxPreset = Preset(0)`.
2. The MAC requests the link partner to choose `TxPreset`.
3. The MAC waits for a period of time for the link partner to apply the requested `TxPreset`. It is recommended to wait at least 2 μ s.
4. The PCIe 5.0 PCS asserts `rxX_reset` to the PHY.
5. The PCIe 5.0 PCS waits for de-assertion of `rxX_ack` from the PHY.
6. The PHY lane receiver is transitioned into P0 state.
7. Electrical Idle Exit condition is detected on `rxX_sigdet_lf` (for low-frequency data)
8. The PCIe 5.0 PCS asserts `rxX_data_en` to the PHY.
9. The PCIe 5.0 PCS waits for assertion of `rxX_valid` from the PHY.
10. The PCIe 5.0 PCS requests receiver evaluation and asserts `rxX_adapt_req` to the PHY.

11. The PHY performs receiver evaluation and signals completion of receiver evaluation by asserting rxX_adapt_ack and generating rxX_adapt_fom[7:0].
 12. The PCIe 5.0 PCS de-asserts rxX_adapt_req to the PHY, passes on FOM information to the MAC.
 13. If $FOM > FOM_best$, MAC sets $N_best = N$, $FOM_best = FOM$.
 14. If $N < N_{max}$, set $N = N + 1$, $TxPreset = Preset(N)$, the MAC repeats the procedure from step 2; otherwise, the MAC continues to Step 15.
 15. For $N = N_{max}$, set $N = N_{max} + 1$, $TxPreset = Preset(N_best)$, the MAC repeats the procedure from step 2.
- For $N = N_{max} + 1$, the MAC exits the loop and proceeds to the subsequent Equalization phases mentioned in the PCIe base specification. When $N = N_{max} + 1$ and if there is time, many PCIe MACs use the $Preset(N_best)$ as a coarse remote tune and refines the adjustment by using RX Link Adaptation Using Direction Change (for more information, see Section 4.9.2, “Initial Calibration”).

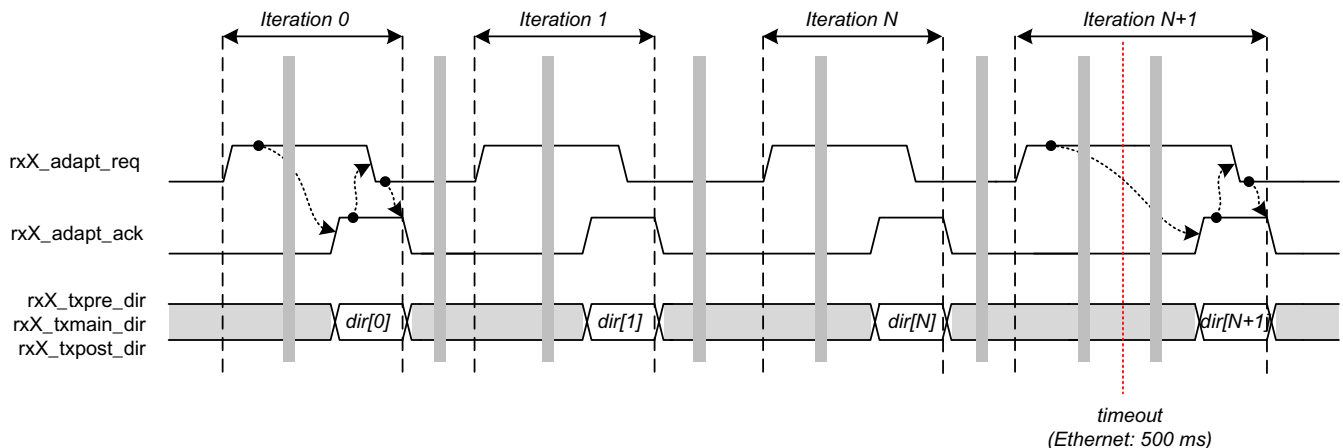
**Note**

During the phase when RX evaluation is in progress or when the link partner is changing its transmitter presets, the recovered data from the PHY can have significant bit errors.

4.12.4 Long Reach Ethernet—TX Equalization Adaptation Using Direction Change

Long reach Ethernet (10GBASE-KR, 25GBASE-KR, and 100GBASE-KR4) adapts the transmitter equalization using the Direction Change mechanism. This mechanism can be used for all protocols supporting link adaptation. The timing diagram for receiver evaluation using the direction change mechanism is shown in Figure 4-21.

Figure 4-21 Link Adaptation Using Direction Change Mechanism



The terminating condition for the link training algorithm should be a programmable time-out (for example, 500 ms for Ethernet).

Direction change outputs (`rxX_txpre_dir[1:0]`, `rxX_txmain_dir[1:0]`, `rxX_txpost_dir[1:0]`) are generated assuming the TX coefficients C_{-1} (pre-shoot) and C_1 (de-emphasis) are encoded as the absolute value of the actual Finite Impulse Response (FIR) coefficient. Incrementing or decrementing either value refers to the magnitude of the actual FIR coefficient. For example, incrementing de-emphasis (`rxX_txpost_dir[1:0]`) means increasing the amount of de-emphasis equalization on serial lines by sending a Decrement command. The main cursor (`rxX_txmain_dir[1:0]`) never indicates an increment or decrement.

**Note**

The PHY never generates an INC and DEC command for the TX main coefficient C_0 ; in other words, the PHY output `rxX_txmain_dir[1:0]` never generates 2'b01/2'b10.

Send the following direction updates for the pre-shoot (`rxX_txpre_dir[1:0]=2'b01`) and de-emphasis (`rxX_txpost_dir[1:0]=2'b01`) from the PHY to the link partner in a two-step sequence as provided in [Table 4-18](#):

Table 4-18 PHY to the Link Command in a Two-Step Sequence

PHY Command	Ethernet Link Command	
<code>rxX_txpost_dir[1:0]=2'b01</code>	Step 1	Decrement C_0
	Step 2	Decrement C_1
<code>rxX_txpre_dir[1:0]=2'b01</code>	Step 1	Decrement C_0
	Step 2	Decrement C_{-1}

This procedure can prevent scenarios where the link partner may not be able to accept equalization update commands and respond with a MIN or MAX status.

**Note**

Implement the two-step sequence on the PHY RX commands in external logic as a programmable feature. By default, enable this two-step sequence.

Example of Direction Change Mechanism

The sequence of events for the Ethernet link partner transmitter adaptation is as follows:

1. PHY is transitioned into P0 state.
2. Electrical Idle Exit condition is detected on `rxX_los`.
3. The BASE-R PCS asserts `rxX_data_en`
4. The BASE-R PCS waits for assertion of `rxX_valid` output from the PHY.
5. The BASE-R PCS sets the link partner preset to the initial state for BASE-R training; in other words, TxPreset = INITIALIZE.
6. The BASE-R PCS resets link training iteration count $i = 0$, and starts the timer.
7. The BASE-R PCS performs link training iteration i , as follows:
 - a. The BASE-R PCS requests receiver evaluation by asserting `rxX_adapt_req`.
 - b. The RAW PCS performs receiver evaluation, and signals completion of receiver evaluation by asserting `rxX_adapt_ack` and generating `rxX_txpre_dir[1:0]`, `rxX_txmain_dir[1:0]`, `rxX_txpost_dir[1:0]`. This is collectively represented by `dir[i]` in [Figure 4-21](#) (“Link Adaptation Using Direction Change Mechanism”).

**Note**

FOM (`rxX_adapt_fom[7:0]`) is also valid when `rxX_adapt_ack` is asserted.

- c. The BASE-R PCS de-asserts rxX_adapt_req to the RAW PCS.
 - d. At this point, the BASE-R PCS can optionally, after waiting for rxX_valid assertion, run a PRBS checker to check the status of the link.
 - e. If the timer has reached (or exceeded) the timeout condition (for example, 300ms) then go to Step 8 to exit the loop; otherwise, continue to the following step.
 - f. For each of pre, main, and post, if the corresponding result in dir[i] is INC or DEC, the BASE-R PCS sends this request to the link partner for it to apply to its TX and go to Step g; otherwise, if all of the coefficient dirs are HOLD (in other words, dir[i] = {HOLD, HOLD, HOLD}), go to Step 8.
 - g. The BASE-R PCS waits for an acknowledgment response from the link partner for each coefficient request it sent (in other words, UPDATED, MAX or MIN).
 - h. Re-evaluate the link partner TX settings by performing a new iteration (for example, set i=i+1) and go to the top of Step 7.
8. The link training algorithm has concluded, and the link is considered "GOOD". If the PRBS checker has been used and it has errors from the last iteration then the link is declared "FAIL".

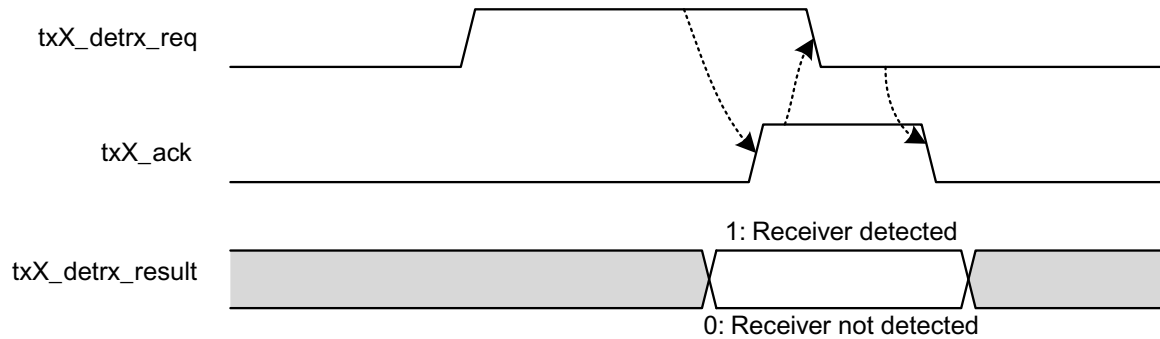
**Note**

The time-out condition should only be checked after the rxX_adapt_ack has been asserted and before sending the resulting dir[i] to the link partner. This is needed to ensure a stable ending to the link training algorithm, with the local receiver and far-end transmitter in known and evaluated states.

4.13 Receiver Detection

Detection of a receiver (with terminations enabled) connected to the TX serial line txX_p/m on lane X in the PHY can be initiated in the TX state (txX_pstate) of P1 or P2 by asserting the input txX_detrx_req. The result of receiver detection is signaled on txX_detrx_result when txX_ack is asserted. The timing diagram of the two-way handshake signals during receiver detection is shown in Figure 4-22.

Figure 4-22 Receiver Detection Handshake



During receiver detection, the common-mode voltage on txX_p/m is distorted and reverted back to nominal when txX_ack is asserted. The receiver detection enable signal txX_detrx_req must stay de-asserted in P0 and P0s states. The time taken by each lane to perform receiver detection (from txX_detrx_req assertion to txX_ack assertion) may vary across lanes to within 1 μ s of other lanes. The maximum time from txX_detrx_req to txX_detrx_result is less than 30 μ s.



Note

Receiver detection function is supported only for AC-coupled links.

4.14 Control Register Interface

The control registers in the PHY can be accessed through a Control Register (CR) Parallel register interface or a serial test interface (JTAG). These interfaces are defined in the following sections.

The PHY register space supports broadcast writes, as follows:

- If you set `cr_para_addr[15:12] = 4'b0101`, PHY broadcasts to all PMA lanes
- If you set `cr_para_addr[15:12] = 4'b0110`, PHY broadcasts to all RAW PCS lanes
- If you set `cr_para_addr[15:12] = 4'b0111`, PHY broadcasts to all RAW PCS AON lanes

The total `cr_para` address space is 64 KB (16-bits) and organized as follows:

- The first 32 KB is CREG.
- The next 16 KB is ROM (unused if ROM is not present).
- The next 16 KB is SRAM.

For 32 KB ROM and SRAM, indirect addressing is used as described as follows.

Table 4-19 CREG Indirect Addressing for 32 KB ROM and SRAM

Register Space Partitions	CREG Address[15:12]
PMA SUP and RAW CMN	0x0
PMA LANE{N}*	0x1
RAW LANE{N}*	0x2
RAW AON LANE{N}*	0x3
PMA LANE Broadcast	0x5
RAW LANE Broadcast	0x6
RAW AON LANE Broadcast	0x7
* where {N} = lane number. It is mapped to CREG Address[11:9].	

ROM (if present) and SRAM are also memory mapped to the CREG address space.

Table 4-20 CREG Address Space for ROM and SRAM

Memory Type	CREG Address[15:0]
ROM (0 to 16 KB)	Write RAWCMN_DIG_LANE_FSM_OP_XTND/ MEM_ADDR_EXT_EN = 0 (default) Then: 0x8000 to 0xBFFF
ROM (16 KB to 32 KB)	Write RAWCMN_DIG_LANE_FSM_OP_XTND/ MEM_ADDR_EXT_EN = 1 Then: 0x8000 to 0xBFFF
SRAM (0 to 16 KB)	Write RAWCMN_DIG_LANE_FSM_OP_XTND/ MEM_ADDR_EXT_EN = 0 (default) Then: 0xC000 to 0xFFFF
SRAM (16 KB to 32 KB)	Write RAWCMN_DIG_LANE_FSM_OP_XTND/ MEM_ADDR_EXT_EN = 1 Then: 0xC000 to 0xFFFF

For details, see the “Register Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*.

For operation during an SRAM update, refer to the Section [Figure 4.16](#) (“External SRAM Support”).

4.14.1 CR Parallel Interface

The CR Parallel interface is a synchronous, 16-bit data/address parallel port provided for on-chip access to control registers inside the PHY. The CR Parallel interface is enabled when input `cr_para_sel` is asserted to 1. While access to these registers is not required for normal PHY operation, this interface is included for users that want to access some of the PHY's diagnostic features during normal operation or to override some of the PHY's control signals.



Note

Wait a minimum of 10 `phy_ref_dig_clk` cycles if `ref_clk_en` = 1, or 10 `cr_para_clk` cycles if `ref_clk_en` = 0.

The CR Parallel interface consists of following PHY interface signals: `cr_para_addr`, `cr_para_clk`, `cr_para_rd_data`, `cr_para_rd_en`, `cr_para_wr_data`, `cr_para_wr_en`, and `cr_para_ack`; and is enabled when input `cr_para_sel` is set to 1.

All control registers can be accessed through either the CR interface or the internal firmware FSM operating on the `cr_clk` domain. A pulse-catching circuit captures the `cr_para_clk` read or write access in this clock domain. As a result, the `cr_para_wr_en` or `cr_para_rd_en` signal must be de-asserted for a minimum of seven `phy_ref_dig_clk` cycles between CR interface requests. Note that the CR interface cannot be accessed during assertion of `phy_reset`. However, any registers that need to be written before powering up the PHY can be done after de-asserting `phy_reset` but before de-asserting the `rxX_reset`/`txX_reset` signal.

A GLCM exists between the external `cr_para_clk` (or `jtag_tck`) and the internal reference clock for accessing registers inside the PHY. This GLCM is controlled by the `ref_clk_en` signal and always uses the internal reference clock when `ref_clk_en` is asserted. As a result, note that the CR interface latency might change when performing CR interface accesses while changing from one clock source to another. When power gating is enabled (`pg_mode_en` = 1'b1), the GLCM feature is disabled, and the `cr_clk` always operates on the internal reference clock.

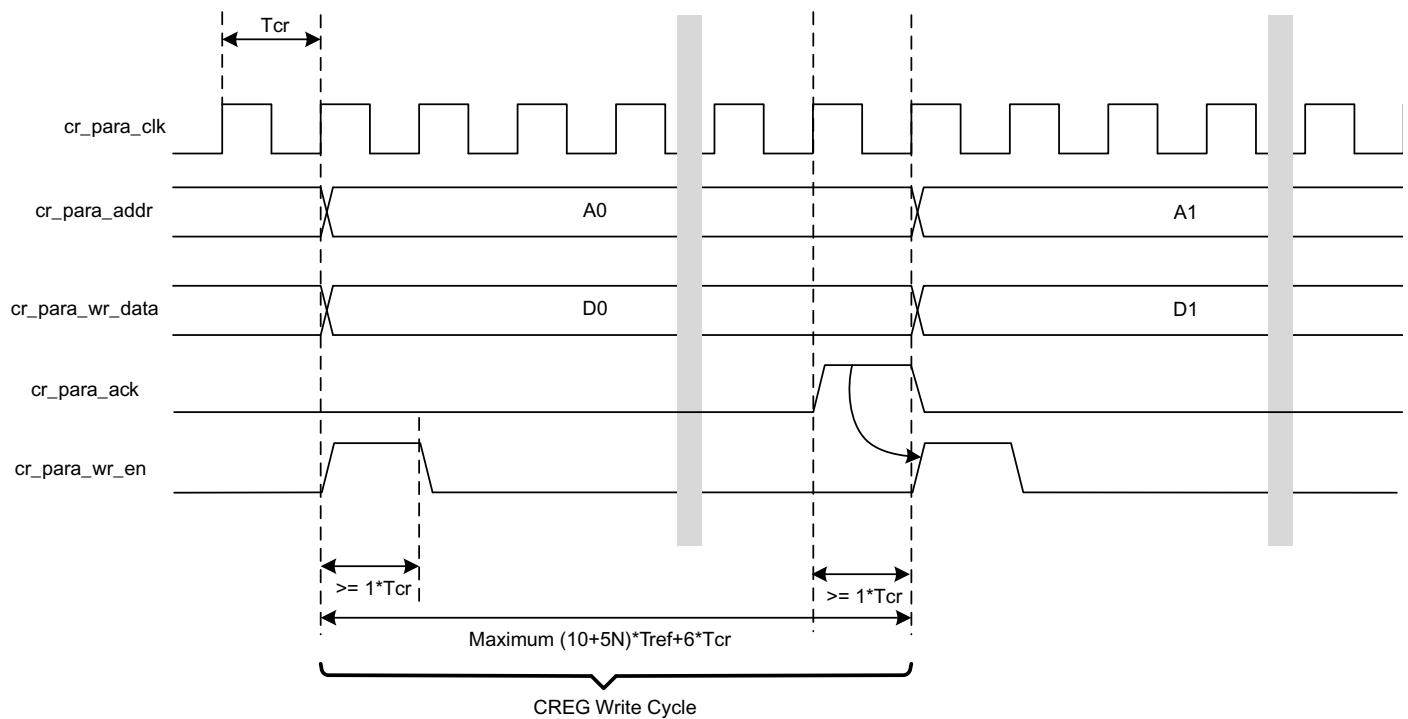
When accessing the CR interface with the reference clock enabled, you must wait at least 2.4 μ s to allow the prescaler to power up, plus at least 50 reference clock cycles after `phy_reset` is de-asserted. Alternatively, you can wait for the `phy_ref_dig_clk` clock to start toggling. This waiting time is required for the clock to propagate through the internal PHY de-glitching circuit and to de-assert internal resets. When accessing the CR interface with the reference clock disabled, you must wait until at least 10 external CR clock cycles have elapsed (`cr_para_clk` or `jtag_tck`) after `phy_reset` has been de-asserted. In this case, the external clock needs to de-assert only internal resets.

The timing diagrams of the CR Parallel register write and read operations are shown in [Figure 4-23](#) and [Figure 4-24](#), respectively. In these diagrams, “N” corresponds to number of lanes in the PHY; for example, for a four-lane PHY, N=4. Parameters `Tref` and `Tcr` are defined as follows:

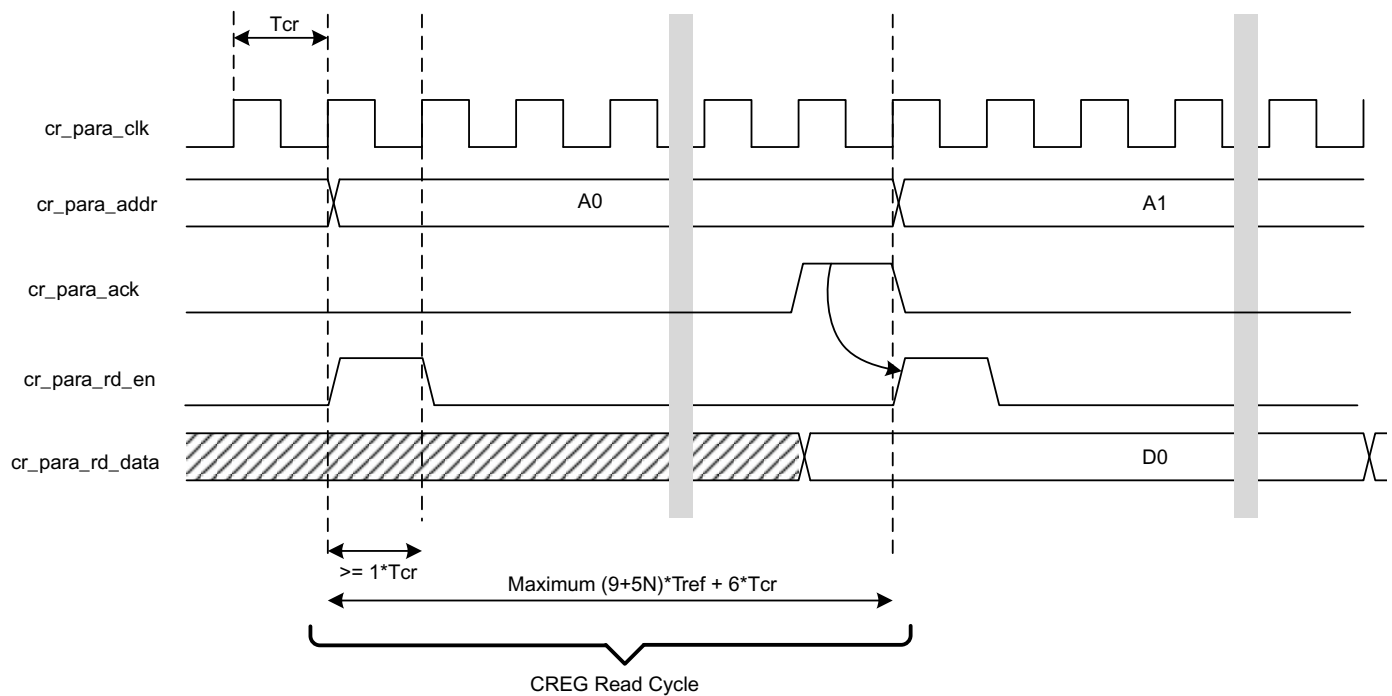
`Tref`: Time Period of `phy_ref_dig_clk` output clock, with a minimum value of 8 ns.

`Tcr`: Time period of the `cr_para_clk`, with a minimum value of 8 ns.

The completion of a write and read cycle is acknowledged by the assertion of the `cr_para_ack` output. The maximum times taken by write and read cycles are provided in [Figure 4-23](#) and [Figure 4-24](#). When context restore is enabled and active, `cr_para_ack` may take an additional time.

Figure 4-23 CR Parallel Register Write Operation

The timing diagram of the CR Parallel register read operation is shown in [Figure 4-24](#).

Figure 4-24 CR Parallel Register Read Operation

**Note**

- Maximum CREG Write/Read Cycle does not apply when the following is true:
 - `refa/b_clk_en` is changed prior or during CREG operation.
 - Context restore is enabled and active (see Section 4.24, “Context Restore”).
- The following registers require two back-to-back read operations to the same register in order to read the register value:
 - `LANEN_DIG_RX_DPLL_FREQ`
 - `LANEN_DIG_RX_LBERT_ERR`
 - `SUP_DIG_MPLL{A,B}_SSC_FRAC_OUT`
 - `SUP_DIG_MPLL{A,B}_SSC_SSC_RAMP`

For data rates slower than 2.5 Gbps where `phy_ref_dig_clk` is faster than `rxX_clk` or `txX_clk`, an extra wait of $6 \times \text{txX_clk/rxX_clk}$ clock cycles is required between back-to-back read operation of the previous list of registers.
- The following register bits need to be toggled from 0 to 1 twice (0-1-0-1-0) for proper operation:
 - `LANEN_DIG_RX_LBERT_CTL[4]` (SYNC bit)

4.14.2 JTAG Interface

The JTAG interface in the PHY provides access to an internal TAP controller which implements the CRSEL instruction used for writing and reading registers in the PHY. The JTAG interface is enabled when input `cr_para_sel` is de-asserted to 0.

There is one JTAG interface per PHY, so for configurations where there are multiple PHYs aggregated (instantiated), there is more than one JTAG interface. To control these with a single TAP controller, these multiple interfaces can be daisy chained externally.

The JTAG maximum clock frequency is 125 MHz (8 ns).

4.14.2.1 IR Codes

Table 4-21 describes the supported JTAG instructions. The LSB of the IR and DR registers is the first bit shifted into `jtag_tdi` input.

Table 4-21 Supported JTAG Instructions

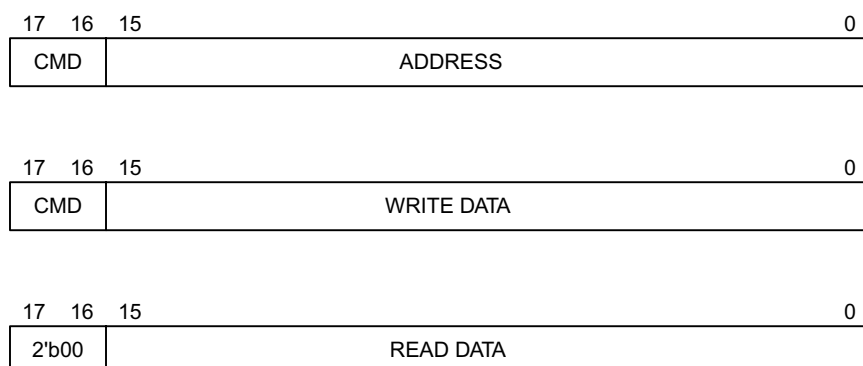
Instruction	IR	DR	Operation
IDCODE	8'h01	32 bits	Shift out ID code. The 32-bit IDCODE can be found in the following registers: { <code>SUP_DIG_IDCODE_HI</code> , <code>SUP_DIG_IDCODE_LO</code> }.
CRSEL	8'h31	18 bits	Control register access
BYPASS	All others	1 bit	<code>jtag_tdi</code> to <code>jtag_tdo</code> bypass (bit is preloaded to 1)

4.14.2.2 CRSEL Instruction

When operating with the CRSEL instruction, internal registers can be accessed through the JTAG interface. When CRSEL instruction is selected, the shifted-in contents of the DR register are loaded into the CRSEL

register during the UPDATE-DR phase of the JTAG state machine. The CRSEL register mapping when CRSEL instruction is selected as shown in [Figure 4-25](#).

Figure 4-25 CRSEL Register Mapping



The CMD field determines what type of access is performed. [Table 4-22](#) shows the encodings which are defined for CMD field of the CRSEL register.

Table 4-22 Command Field Encodings for the CRSEL Register

CMD	Encoding	Description
ADDR	2'b00	Capture ADDRESS field for use in a subsequent WRITE command. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.
WRITE	2'b01	Write the value in the WRITE DATA field to the register selected by the address captured during the most recent ADDR or READ command. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.
DATA	2'b10	During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo. WRITE DATA value shifted in is unused.
READ	2'b11	Perform a read operation to the register selected by the ADDRESS field to make its current value available for subsequent CRSEL accesses. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.

To perform a control register read, shift a READ command into the CRSEL register along with the desired register address in the ADDRESS field. Any subsequent CRSEL access causes the data returned by the READ to be shifted out on jtag_tdo. This allows a limited form of pipelining of READ commands, as the second access may be shifting in the command and address for the subsequent READ while shifting out the data from the previous READ. To ensure reliable read operation, a minimum wait of 100 jtag_tck cycles in the run-test-idle state is needed. This wait is required for the register access to be transferred across clock domains and received by the JTAG controller successfully.

To perform a control register write, the ADDRESS of the register to be written must first be selected by performing either an ADDR or READ command and specifying the register address in the ADDRESS field. A second access using the WRITE command and specifying the data to be written in the WRITE DATA field causes the write to finally be performed to the desired register.

4.15 ROM Interface

The RAW PCS stores the instruction code to execute the calibration and adaptation algorithms. In order to minimize the area of the RAW PCS, this pre-optimized and pre-validated code is by default stored in the form of hard-coded lookup table. To store the RAW PCS instruction code, you have the option to instead use an external ROM. When using an external ROM, the internal lookup table is removed from the design to save area.

The external ROM is enabled by default. To disable it and use only the internal hard-coded lookup table based ROM, define the `DWC_E32MP_X2NS_NO_EXT_ROM_SUPPORT` macro.

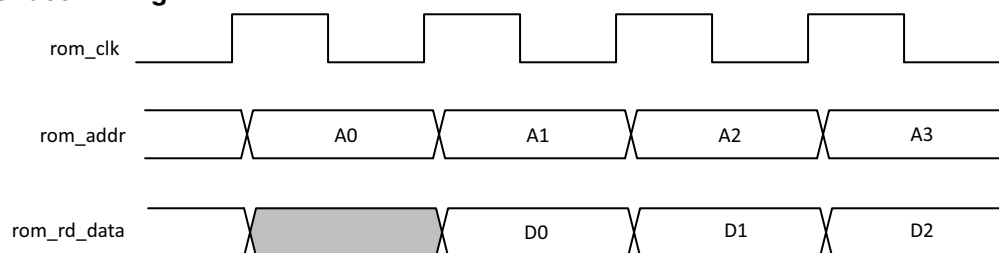


Note

- Do not define this macro if both external ROM and internal LUT are not required (for example, when SoC side loads SRAM). In this case, tie-off PHY input `rom_rd_data` and leave PHY output `rom_clk/addr` unconnected.
Synopsys recommends using an external ROM or no ROM (side loading SRAM by SoC).
- Mandatory ROM size is a 32K single-port ROM consisting of 16-bit locations.

The following figure provides a timing diagram for the ROM interface.

Figure 4-26 ROM Interface Timing



4.16 External SRAM Support

To provide flexibility in updating the firmware code in silicon, an external SRAM interface is provided. During initialization, after de-assertion of `phy_reset`, the boot loader in the RAW PCS loads the code from the internal look-up table to the external SRAM and asserts the output signal, `sram_init_done`. After initialization of the SRAM, you can change the SRAM contents (or access any PHY register) using either the JTAG or the CR Parallel Interface by addressing the SRAM address space in the register map. After external access to the SRAM (or any other PHY register) is complete, input `sram_ext_ld_done` must be set high, allowing the FSMs in the RAW PCS to start executing the code from the SRAM. For an architecture diagram of the SRAM interface, refer to [Figure 3-10](#) (“Raw PCS Block Diagram”).



Note

- Mandatory SRAM size is a 32 K single-port SRAM, consisting of 16-bit locations.
- SRAM address space is 16K from “`RAWMEM_DIG_RAM_CMN0_B0_R0`” through “`RAWMEM_DIG_RAM_CMN31_B15_R31`”
- The SRAM is accessed from 16K address space. The CREG `RAWCMN_DIG_LANE_FSM_OP_XTND.MEM_ADDR_EXT_EN` is needed to select the upper and lower addresses. For more information, refer to the “Register Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*.
- It is mandatory to update the PHY firmware post-tapeout in silicon, therefore, the SoC must be capable of handling this update.

4.16.1 Bypassing SRAM Bootloading

The PHY includes the ability to skip bootloading altogether when it is unnecessary or undesired. The SRAM bootloading can be disabled by asserting the `sram_bootload_bypass` signal. When `sram_bootload_bypass` is asserted, the `sram_init_done` signal is asserted shortly after `phy_reset` and the bootloading phase is skipped. Similar to the bootloading case, after `sram_init_done` is asserted, the PHY waits for the assertion of the `sram_ext_ld_done` signal in the event that any external loading of SRAM is desired.

The following are some use cases when bootloading can be skipped:

- SRAM is sideloaded by the SoC before `phy_reset` de-assertion
- Non-POR `phy_reset` assertion

In both cases, the SRAM already contains the correct firmware program when `phy_reset` is de-asserted. As a result, there is no need to go through the bootloading process and then optionally externally load an updated firmware program.

[Figure 4-27](#) and [Figure 4-28](#) show the timing diagram of SRAM initialization and loading with and without bypass, respectively.

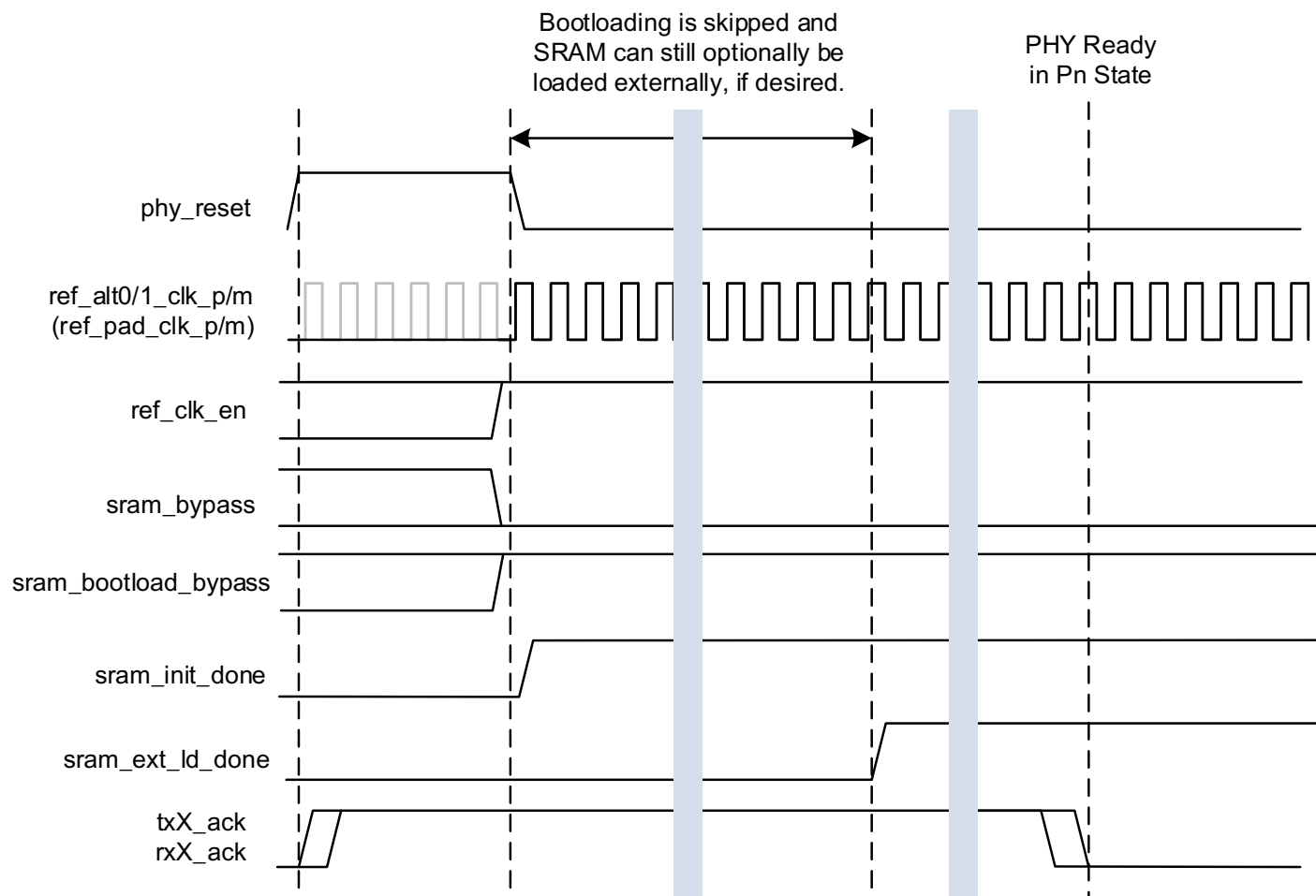
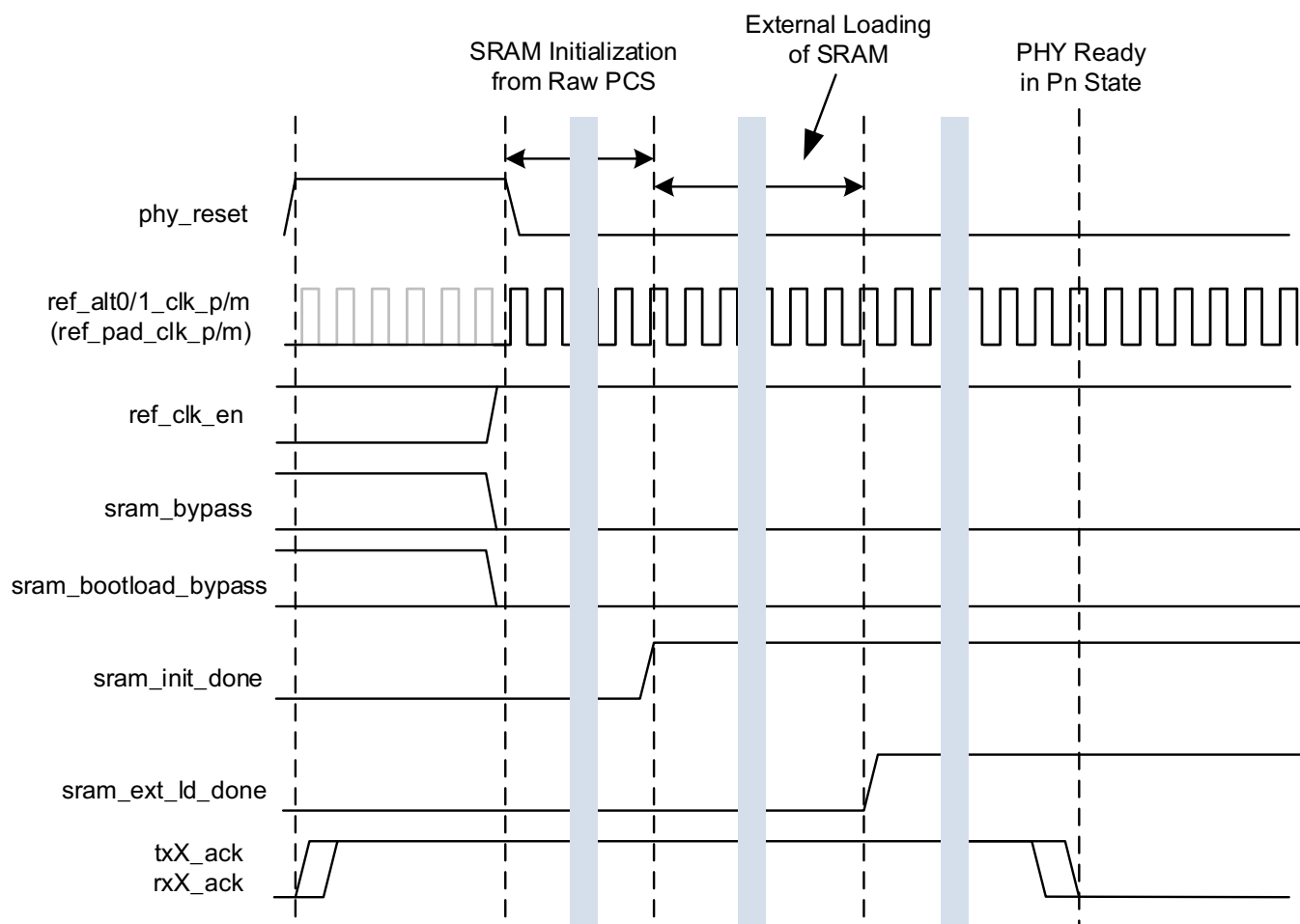
Figure 4-27 SRAM Initialization and Loading Sequence with Bypass

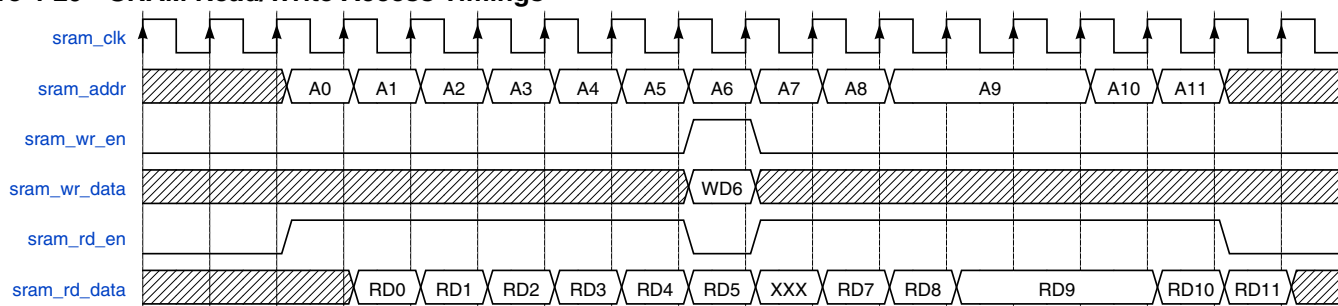
Figure 4-28 SRAM Initialization and Loading Sequence Without Bypass

When `sram_bypass` is asserted, the access to the SRAM is bypassed and the lane FSMs execute the code from the hard coded look up table in the RAW PCS.

**Note**

During the SRAM side loading (before “`sram_ext_ld_done`” is asserted), analog registers write operations are not allowed.

The timing diagram for the SRAM interface is shown in [Figure 4-29](#).

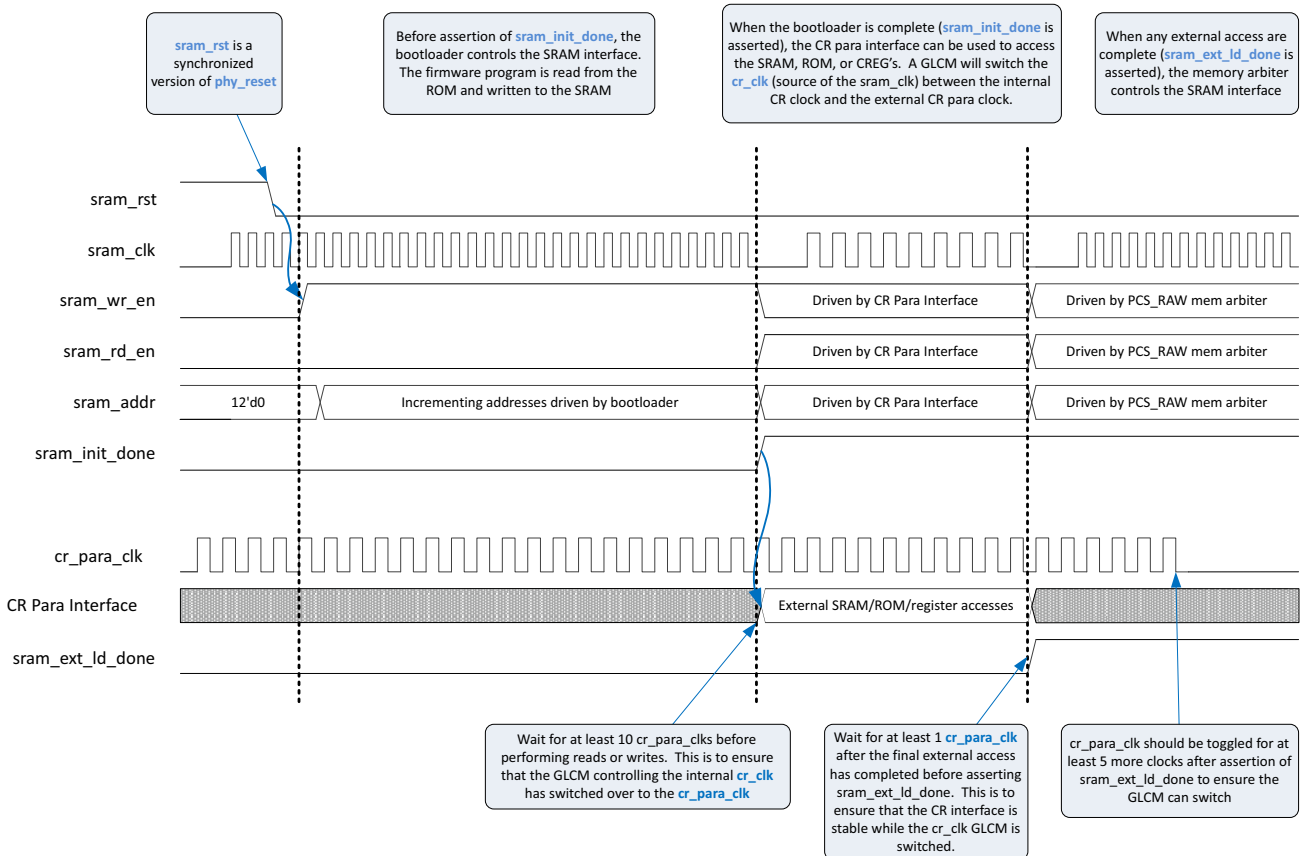
Figure 4-29 SRAM Read/Write Access Timings

**Note**

Maximum clock speed requirement for the SRAM clock is 125 MHz.

The sequence of events on the SRAM interface as part of PHY power up are shown in Figure 4-30, if `sram_bypass` is negated. The SRAM contents after `sram_init_done = 1` is the same as in the file, `phy/include/*_pcs_raw_mem_rst.v`, used during synthesis.

Figure 4-30 Sequence of Events on the SRAM Interface

**Note**

To read ROM through the CR Parallel and JTAG interfaces for debug and diagnostics purposes when the `sram_bypass` input is 0, set the `RAWCMN_DIG_MEM_POWER_SAVING.ROM_POWER_SAVING_EN` register field to 0.

4.17 SRAM/ROM Support and Bootloading Options

Table 4-23 provides all possible options for the bootloading features for SRAM and the internal or external ROM.

Table 4-23 SRAM/ROM Support and Bootloading Options

sram_bootload_bypass	sram_bypass	Boot Method	SRAM or ROM	Firmware
0	0	PHY bootloads internal or external ROM (based on DWC_E32MP_X2NS_NO_EXT_ROM_SUPPORT macro) into SRAM	SRAM	- Internal ROM: pcs_raw_mem_rst.v - External ROM: pcs_raw_ext_rom.bin
0	1	No bootloading by the PHY	Internal or external ROM based on DWC_E32MP_X2NS_NO_EXT_ROM_SUPPORT macro	- Internal ROM: pcs_raw_mem_rst.v - External ROM: pcs_raw_ext_rom.bin
1	0	Customer side loads SRAM using the SRAM image provided by Synopsys	SRAM	Side load SRAM through PHY's cr_para or JTAG interface: pcs_raw_mem_sram_cr_para.fw Side load SRAM directly by customer: - Without ECC: pcs_raw_ext_rom.bin - With ECC: pcs_raw_mem_sram_side_load_ecc.bin
1	1	No bootloading by the PHY	Internal or external ROM based on DWC_E32MP_X2NS_NO_EXT_ROM_SUPPORT macro	- Internal ROM: pcs_raw_mem_rst.v - External ROM: pcs_raw_ext_rom.bin



Note

It is mandatory to update the PHY firmware post-tapeout in silicon, therefore, the SoC must be capable of handling this update.

4.18 Firmware

This section provides information about checking the firmware version, as well as, details on the Firmware Clock Interface.

4.18.1 Checking the Firmware Version

You can use the following registers to obtain the firmware version:

- RAWCMN_DIG_AON_FW_VERSION_0
- RAWCMN_DIG_AON_FW_VERSION_1

For the specific offsets of these PHY registers, refer to the “Register Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*.

A detailed description of these registers is as follows:

- RAWCMN_DIG_AON_FW_VERSION_0

This register stores the firmware version as a 16-bit number in a.b.c. format.

Where:

- a = bits [15:12]
- b = bits [11:4]
- c = bits [3:0]

- RAWCMN_DIG_AON_FW_VERSION_1

This register stores the firmware compile date as a 12-bit number in Day/Month/Year format.

Where:

- Day = bits [11:7]
- Month = bits [6:3]
- Year = bits [2:0] (for example, 2018 = “0”, 2019 = “1”, etc.)

The firmware writes these registers (that have a default value of 0) to the appropriate version based on the firmware file. This is done on initial power-on-reset and during power-gated exit.

Example of Checking Firmware Version

An example of checking the firmware version and compile date is described as follows:

- If the read value of RAWCMN_DIG_AON_FW_VERSION_0 is x2040, then the version is 2.4.0.
- If the read value of RAWCMN_DIG_AON_FW_VERSION_1 is x0829, then the date is May 16, 2019.

If a new firmware is successfully downloaded, it is expected that these values will change to match the newly downloaded firmware after the PHY has executed the power-on reset calibrations.



Note

It is mandatory to update the PHY firmware post-tapeout in silicon, therefore, the SoC must be capable of handling this update.

4.18.2 Firmware Clock Interface

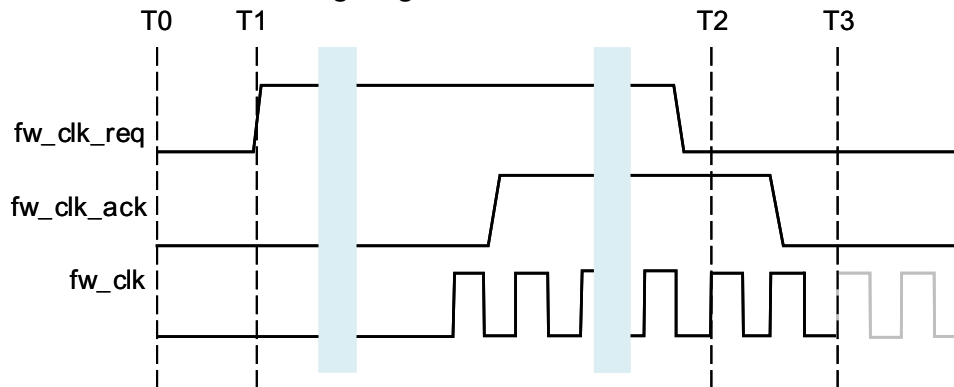
The PCIe PHY can request a `fw_clk` input when the PHY needs to perform a task that impacts the latency in responding to a request at the PHY interface (for example SRAM bootload, POR calibration, power gating exit sequence, rate change, adaptation request, margining, and so on). The following signals at the PHY interface are used to request `fw_clk` input:

- `fw_clk_req`
This output port is driven to a value of 1'b1 when the PHY requires the `fw_clk` to be available. It is asynchronous. The PHY can de-assert this signal irrespective of the value of `fw_clk_ack`.
- `fw_clk_ack`
This input port indicates to the PHY that `fw_clk` is now available and toggling. It is asserted when `fw_clk` is already toggling. This input is de-asserted after `fw_clk_req` is de-asserted. The `fw_clk` remains toggling until `fw_clk_ack` is asserted.
If the feature is not required, tie this input to 1'b0.
- `fw_clk`
This input port is driven from a clock generator to provide a clock frequency from 100 MHz to 175 MHz. The clock is always glitch free when it starts or stops toggling.

Internally, the Glitch Less Free Multiplexer (GLCM) selects between the internal reference clock and `fw_clk` to generate `cr_int_clk`. Another GLCM selects between the `cr_int_clk` and `cr_ext_clk` (`cr_para_clk` or `jtag_clk`) to generate `cr_clk`. Both GLCMs do not switch simultaneously.

Figure 4-31 shows the timing diagram for this interface.

Figure 4-31 Firmware Clock Interface Timing Diagram

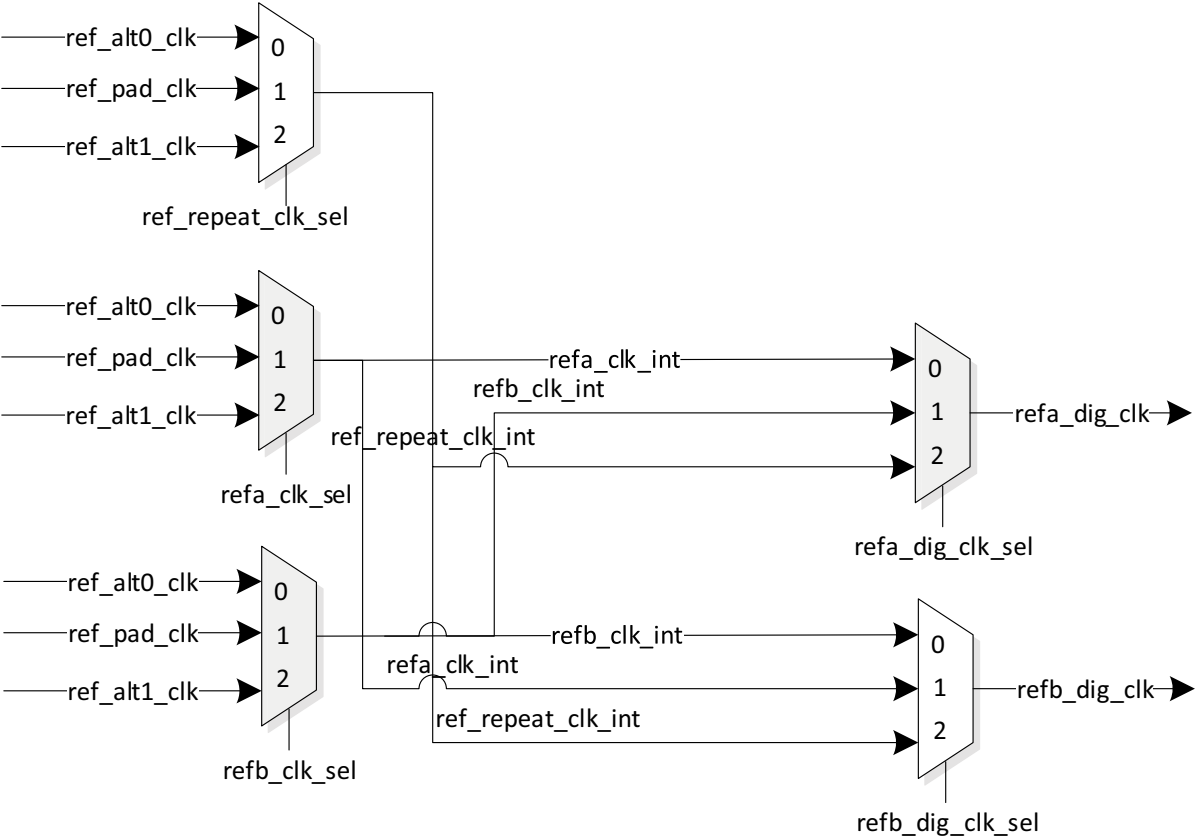


4.19 PERST

PERST can be supported using non-glitchy a ref_repeat_clk multiplexer with ref_dig_clk in the PMA.

If clean, ref_repeat_clk is provided ref_dig_clk source with ref_dig_clk_sel of 2, as shown in [Figure 4-32](#). Only MPLL and DCC calibration need to be restarted after PERST. In this case, the PERST input needs to be tied off (multi_perst_en input set to 0). In POR, the firmware block must reset (tx_reset or rx_reset) until WM. In rate changes, firmware needs to clear the stored previous MPLL and DCC calibration bank, which may be corrupted by a glitchy ref_clk.

Figure 4-32 ref_dig_clk Multiplexer and Select Input



[Table 4-24](#) on page [137](#) summarizes the PERST application cases.

Table 4-24 PERST Application Case

			PHY Input				PHY Action		
Case	ref_dig_clk	mpll_ref_clk	multi_perst_en	PERST	tx_reset/ rx_reset	phy_reset	perst_phy_reset	perst_ref_clk_dis	perst_mpll_recal
#1	Clean	Glitchy	0	N	1	0	N	N	Y

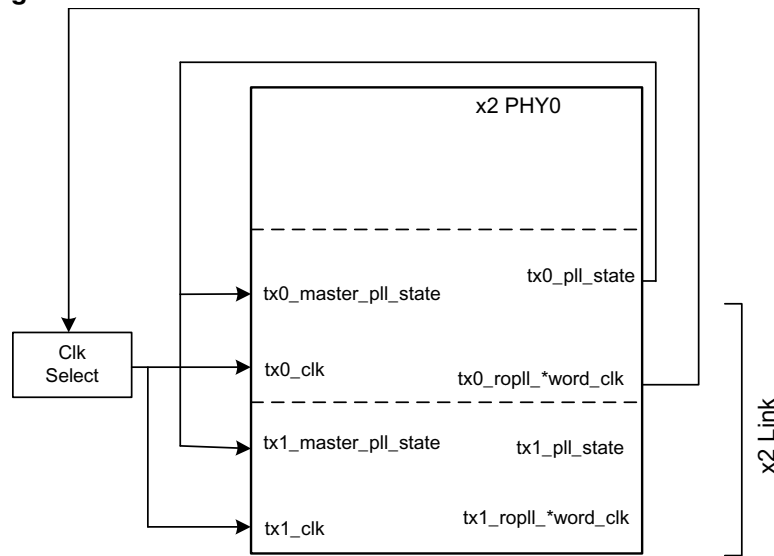
4.20 Aggregation and Bifurcation Support

Combining different instances of the PHY to operate as a common link is referred to as aggregation of PHY. Splitting different lanes in the same PHY across different links is referred to as bifurcation of PHY. The PHYs support aggregation and bifurcation, as well as a combination of both aggregation and bifurcation.

Ensure that the txX_master_pll_state input of lane X is connected to the txX_pll_state output of the master lane in the link from which the txX_clk input is derived from. For lanes belonging to the same link, ensure that the txX_clk inputs are derived from the same source and phase skew balanced to within 400 ps.

Figure 4-33 and Figure 4-34 show connectivity examples of one x2 PHY that is configured to operate as one x2 link and two x1 links, respectively.

Figure 4-33 x2 PHY Configured as x2 Link



In this case, Lane0 is the master lane

Figure 4-34 x2 PHY Configured as Two x1 Links

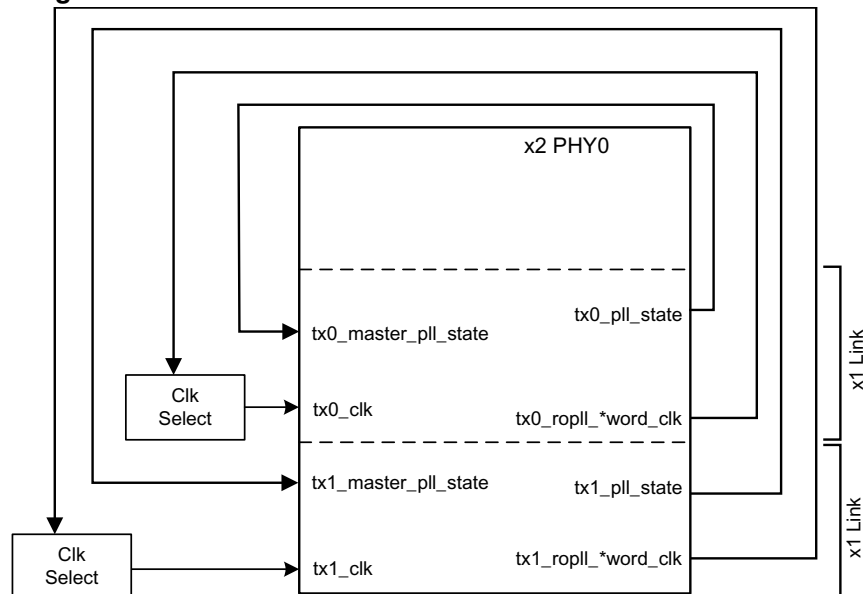


Figure 4-35 shows connectivity example of two x2 PHYs that are configured to operate as one x4 link.

Figure 4-35 Two x2 PHYs Configured as One x4 Link

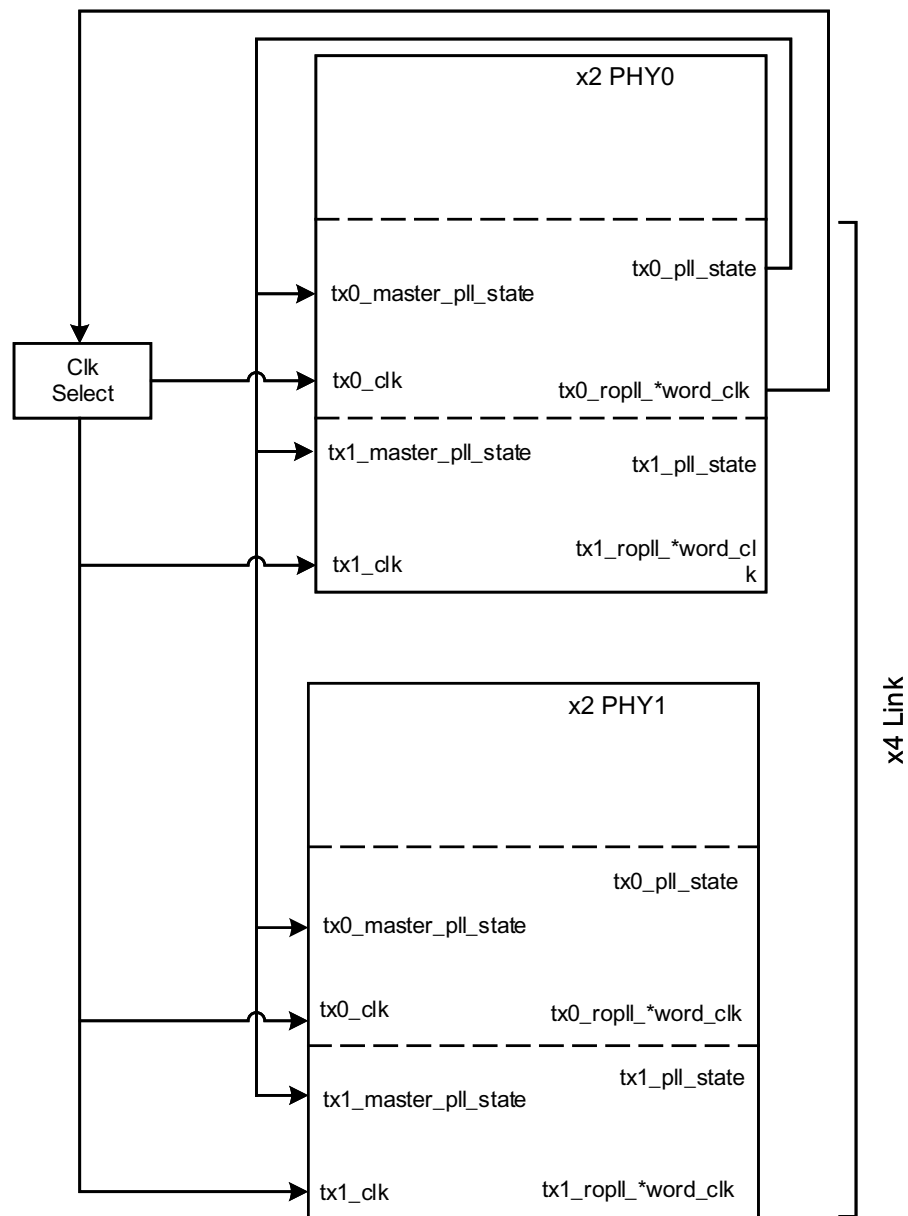
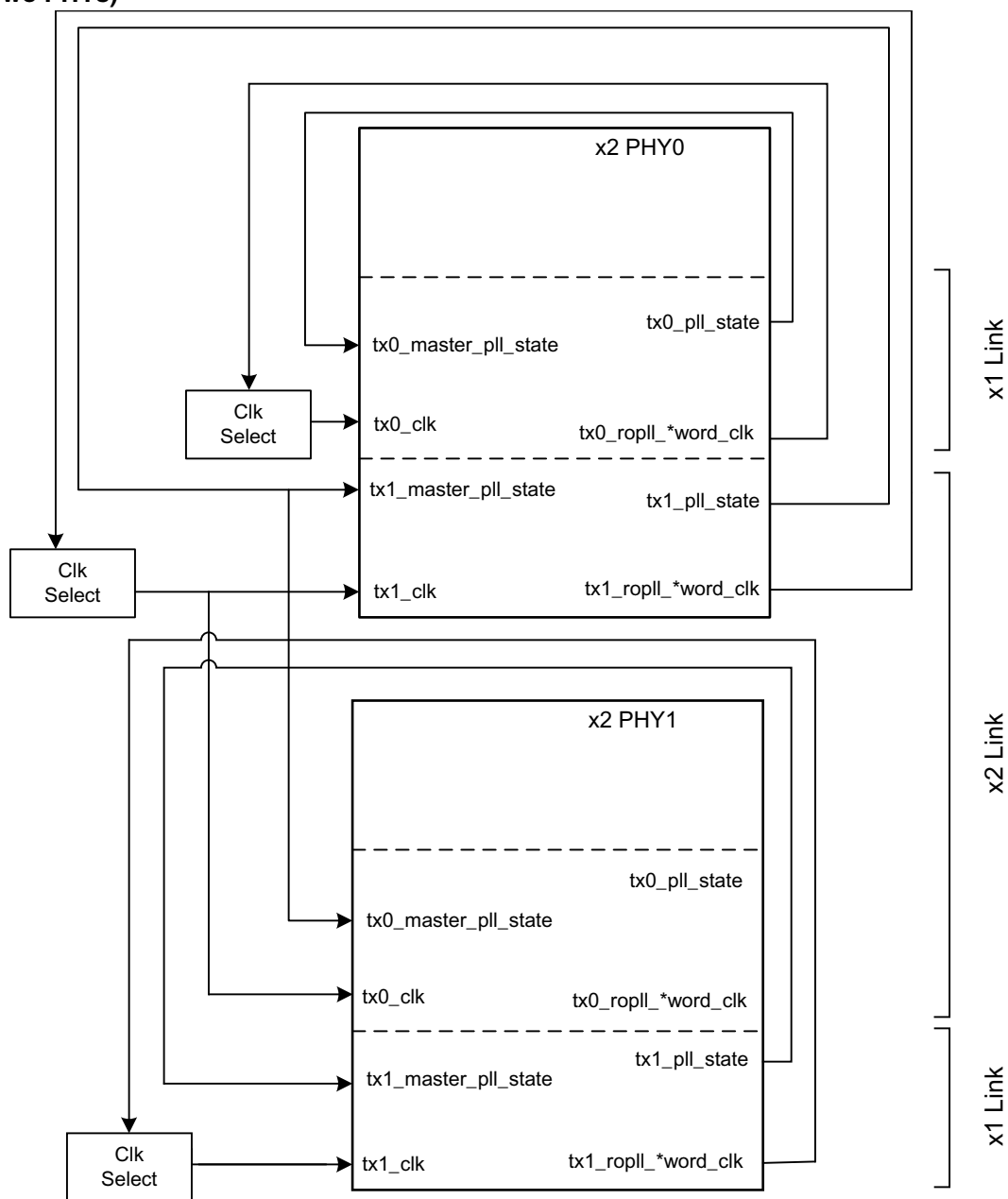


Figure 4-36 shows connectivity example of aggregation and bifurcation where two x2 PHYs are configured to operate as two x1 links and one common x2 link split across the two PHYs.

Figure 4-36 Two x2 PHYs Configured for Aggregation and Bifurcation (Two x1 Links and One x2 Link Split Across Two PHYs)



4.21 Power Gating

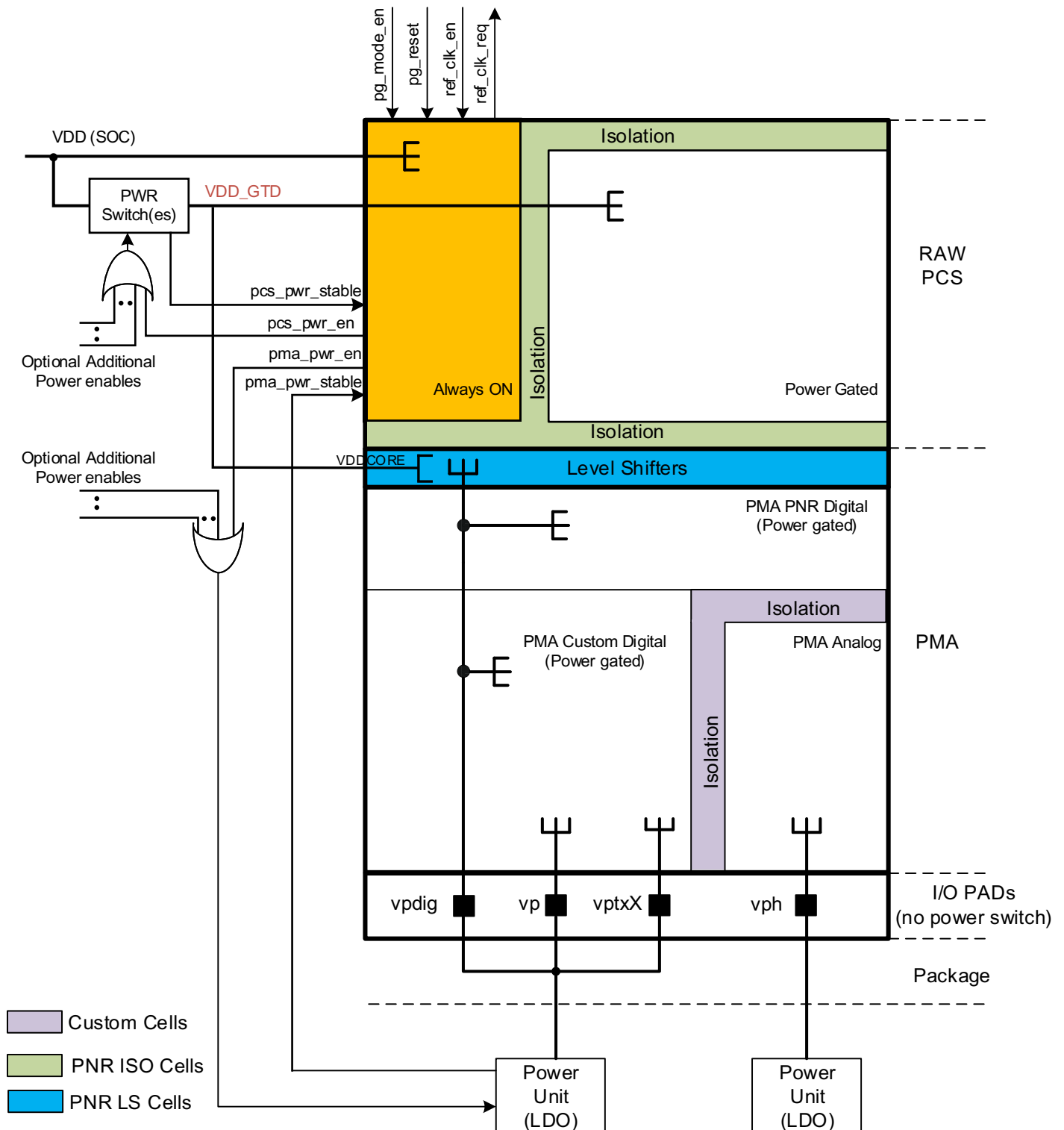
The PHY supports an optional power gating feature through control and status signals for external power switches for the RAW PCS logic and the PMA vp, vptxX, vddcore and vpdig supplies.



Note

- The PHY does not include internal power gating switches. Power gating is only supported using external power switches.
- Signals `ref_repeat_clk_{p,m}`, `refa/b_dig_fr_clk`, and `phy_ref_dig_clk` may be powered down when power gating is enabled. Do not use these signals for logic or PHYs meant to remain operational during power gating.
- A PHY that is repeating the input reference clock should NOT be power gated (that is, `pg_mode = 1'b0`). Power gating the PHY disables the repeating function. This implies that it is only possible to repeat the reference clock across PHYs that are guaranteed to be in a single link.
- The power gating state entry is only allowed from the P1.2 substate if a digitally controlled oscillator (DCO) is used.

[Figure 4-37](#) shows the control signal connectivity for power gating with external power switch for the PMA. The PHY generates power gating enables for the PCS switch (or multiple switches) as well as the PMA switch external to the PHY. The inputs `pcs_pwr_stable` and `pma_pwr_stable` must only be asserted after the supplies ramp up and are at 90% of nominal or higher. The power gating mode is enabled when `pg_mode_en` input is 1. When `pg_mode_en` is 0, the power gating control and status inputs to the PHY (except `ref_clk_en`, `ref_clk_req`, and `txX/rxX_ack`) shown in [Figure 4-37](#) are ignored.

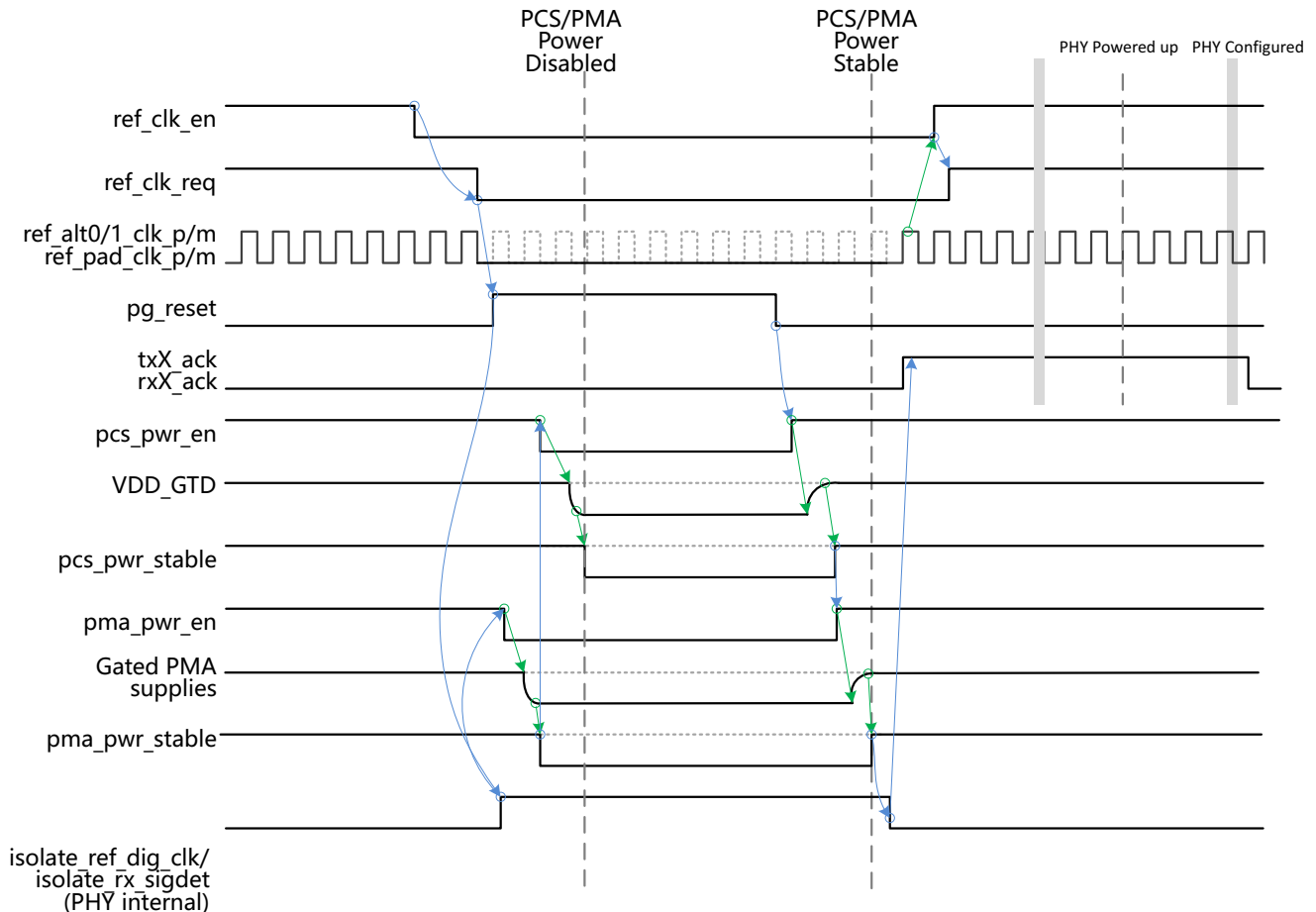
Figure 4-37 Power Gating Connectivity for External PMA Switch

**Note**

- The vp, vptx and vpdig power supplies do not need isolation among themselves as they are tied together either at package or board level.
- The vddcore pin of the PMA must be connected to the gated supply of the SoC (VDD_GTD).

The sequence of powering gating is shown in [Figure 4-38](#).

Figure 4-38 Power-Gating Sequence for the PHY (Internal or External PMA Switch) With pg_mode_en=1



The entry into power gated state starts by the de-assertion of ref_clk_en and ref_clk_req.

The rest of power gating sequence happen when all of the following four conditions are true:

1. pg_reset is asserted
2. mplla_force_en is de-asserted
3. refa_clk_en is de-asserted
4. ref_repeat_clk_en is de-asserted.

Power gating entry sequence is shown as follows:

1. SoC disables the reference clock inputs to the PHY.
2. SoC asserts the `pg_reset` input.
3. PHY performs isolation sequencing in preparation of power gating by retaining states of some calibration registers.
4. PHY de-asserts `pma_pwr_en` and then `pcs_pwr_en`.
5. SoC turns off the power supplies to PMA and RAW PCS. However, it is allowed that these power supplies can stay ON because of any other SOC component requirement.

The exit from power gated state is initiated by de-assertion of `pg_reset`, or assertion of either `mplla<b>_force_en`, `refa<b>_clk_en` or `ref_repeat_clk_en`.

The PHY requests power by assertion of `pcs_pwr_en` and `pma_pwr_en`. It is required that the `ref_clk_en` input be asserted after the PMA and RAW PCS power supplies are stable (`pcs_pwr_stable/pma_pwr_stable=1/1`). The PHY resets the internal power gated logic, disables the isolation on power gated signals, and asserts `txX_ack/rxX_ack` outputs. After the reference clock inputs are supplied, the PHY performs the fast power-up by recovering states of the retained calibration registers, and de-asserts `txX_ack/rxX_ack` outputs. At this point, the PHY is configured into the state defined by its primary configuration inputs.

4.22 Disabling a PHY

To disable a PHY, follow the method outlined in [Table 4-25](#).



Note

The PCIe Controller cannot be re-configured until all configured lanes have asserted PhyStatus when the PHY is Ready after PHY Initialization.

- Make the PCIe Controller default to the minimum lane configuration by default in silicon, then change to the maximum lane width supported after PhyStatus has been asserted on all usable lanes of the configuration.
OR
- Add logic to force phystatus = 1 on all unused lanes.

Table 4-25 Signal Connections If the PHY is Not Used

Signal	All Power Supplies (vp, vph, vptxX, vpdig, vddcore, gd) at Nominal Voltage and IDDQ = 1
PMA inputs from RAW PCS	Not required ^a
PMA outputs to RAW PCS	Not required ^a
rxX_p, rxX_m	Leave floating or tie low
txX_p, txX_m	Leave floating
resref	Leave floating
ref_pad_clk_p, ref_pad_clk_m	Tie low
ref_alt0/1_clk_p, ref_alt0/1_clk_m	Tie low

a. Assuming that VDD_SOC is supplied and at nominal voltage.



Note

- To prevent the PHY circuitry from inadvertently toggling, it is good practice to tie all the inputs to the ground.
- For REXT sharing, the request and ack handshake logic within each PHY must be powered and must allow the handshake signals from other PHYs to propagate through. For more information about resistor sharing, see [Figure 7-9](#) (“Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor”).
- If the clock is being repeated, ensure that you do not power down the PMAs in the chain.
- For more information about IDDQ mode, see [Table 5-1](#) (“PHY Test Mode Selection”).

4.23 Configuring the PHY

For support and lane configuration values, see *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

For the tables in this appendix, make note of the following:

- All values are in decimal
- The following inputs should be stable before assertion of `refa/b_clk_en` and de-assertion of `phy_reset`:
 - `refa/b_range[2:0]`
 - `refa/b_clk_div2_en`
 - `ref_clk_mpll{a,b}_div[2:0]`
 - `rx_vref_ctrl[4:0]`
 - `rxX_term_ctrl[2:0]`
 - `txX_term_ctrl[2:0]`
 - `rxX_term_acdc`
 - `pg_mode_en`
 - `cr_para_sel`
 - `sram_bypass`
- The following inputs should be stable before the de-assertion of `phy_reset` when powering up the PHY. The MPLL-specific inputs can also be changed only when the corresponding MPLL is powered down:
 - `mppll{a,b}_word_clk_div[1:0]`
 - `mppll{a,b}_fb_clk_div4_en`
 - `mppll{a,b}_ssc_en`
 - `mppll{a,b}_ssc_peak[19:0]`
 - `mppll{a,b}_ssc_step_size[20:0]`
 - `mppll{a,b}_ssc_up_spread`
 - `mppll{a,b}_frac_en`
 - `mppll{a,b}_frac_den[15:0]`
 - `mppll{a,b}_frac_quot[15:0]`
 - `mppll{a,b}_frac_rem[15:0]`
 - `mppll{a,b}_div_clk_en`
 - `mppll{a,b}_div_multiplier[7:0]`
 - `mppll{a,b}_multiplier[11:0]`
 - `mppll{a,b}_bw_high[15:0]`
 - `mppll{a,b}_bw_low[15:0]`
 - `mppll{a,b}_ctl_buf_bypass`
 - `rxX_sigdet_lf_threshold[2:0]`

- The following input can be changed only when the TX clock source is disabled (txX_pstate[1:0] = P1/P2, or txX_reset = 1):
txX_disable
- The following inputs can be changed only when the RX clock source is disabled (rxX_pstate[1:0] = P1/P2; or rxX_reset = 1):
 - rxX_cdr_ssc_en
 - rxX_disable

4.24 Context Restore

Context Restore allows the PHY to internally manage the PHY configuration settings. Previously, the PHY configuration settings were managed by the customer and driven through the PHY primary inputs.

Context Restore can be triggered at the start of the PHY initialization or rate change. Context restore is made on link level and when triggered, a state machine in the Raw PCS reads from SRAM or ROM and writes configuration settings in CREGS of all lanes in a link. As a result, the PHY primary inputs used for configuration may be potentially removed (using `DWC_E32MP_X2NS_REMOVE_CNTX_RSTR_INPUTS` macro).



Note

- You must drive any PHY configuration inputs that are not removed (for example, reference clock configurations). For more details, refer to *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (`dwc_32g_phy_tsmc12ffc_x2ns-config_reference_manual.xlsx` in the doc directory of the PHY release package).
- Context Restore is different than Fast Context Switching feature described in Section 4.9, “Fast Context Switching”. Context Restore manages static configuration whereas the Fast Context Switching manages dynamically calibrated settings.

4.24.1 Functional Description

The context restore engine is embedded in the RAW PCS. It services context restore requests from all lanes of the RAW PCS. You can select the mapping between the links and lanes using the `laneX_link_NUM` input. The context requests generated by all links are responded one-by-one using a round-robin arbitration scheme. After receiving the rate change request in a lane, if the context restore is enabled for that lane, the PHY configuration contents stored in SRAM/ROM are read one-by-one from the memory through the memory arbiter and are written to the respective registers through the control register arbiter.

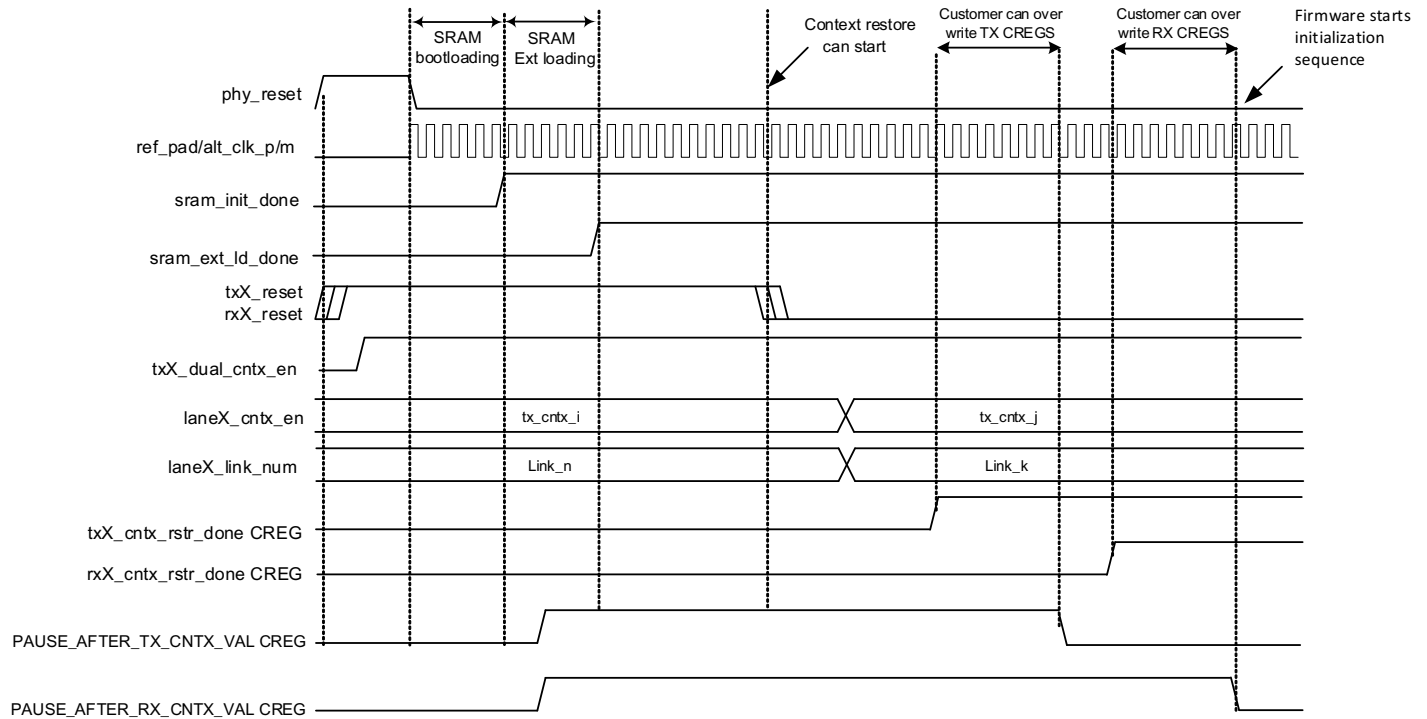
Using context restore, the concept of the selected broadcast CREG write can happen where the CREG of all lanes in a link are restored at the same time. The latency incurred for the context restore of all lanes in a link is approximately 2 μ s (for the reference clock frequency equal to 100 MHz). During this time, external register access through the CR Parallel or JTAG interface is stalled. Any outstanding transaction is serviced after context restore finishes.

4.24.2 Interfaces

For a list of context restore signals, refer to the “Signal Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*. Note that the `laneX_cntx_en` signal is captured when `txX_req` or `rxX_req` are asserted.

4.24.3 Customer Debug Feature

This feature allows you to overwrite control registers (CREG) after the context restore is done and before the firmware starts. This is needed as the top-level inputs are removed to allow you to change the CREG contents, if needed. To exercise this feature, you need to set the `pause_after_tx_res_r` and `pause_after_rx_cntx_res_r` to 1 to hold off the firmware after the context restore is done. When both the `pause_after_tx_cntx_res_r` and `pause_after_rx_cntx_res_r` signals are released, firmware can start. [Figure 4-39](#) shows this concept.

Figure 4-39 Interface Signals for TX Context Restore

In case of any issue, the following signals and registers may be monitored for debugging purposes:

■ **PHY inputs:**

- txX_cntx_sel
- rxX_cntx_sel
- tx_cmnX_cntx_sel
- laneX_link_NUM

For more information about these signals, refer to the “Signal Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*.

■ **Control Registers (CREGs)**

- RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_*/TX_UNIQUE_ID
- RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_*/RX_UNIQUE_ID
- RAWCMN_DIG_CONFIG_MASTER_VERSION (configuration master version)
- RAWCMN_DIG_AON_SRAM_EOF_ADDR
- RAWCMN_DIG_AON_SRAM_BOC_ADDR

For more information about these registers, refer to the “Register Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*.



Note

The RX overwrite values may be overwritten by firmware at the end of POR. Thus, for this “pause” feature to work, you need to issue a second RX reset.

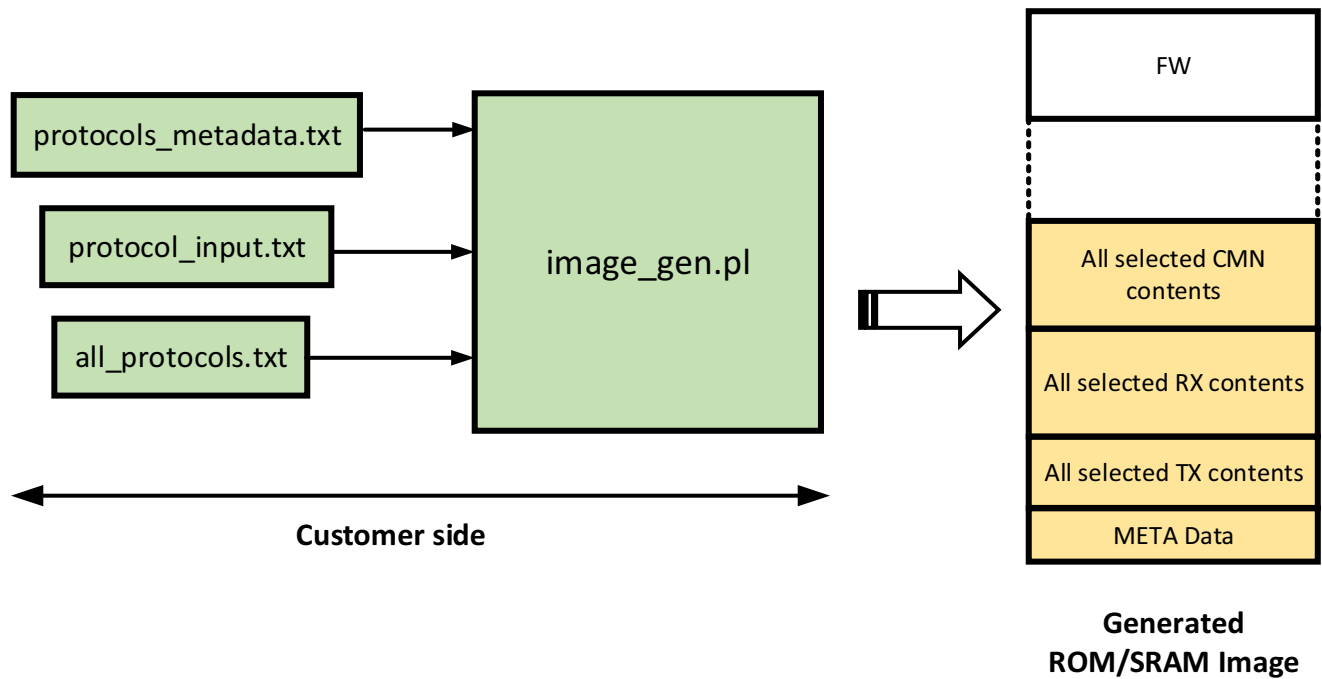
4.24.4 Generating the SRAM and ROM Images

The PHY configuration settings are added to the bottom of the SRAM and ROM, which contain the PHY firmware code. To optimize the SRAM and ROM size, only the specified configuration customer (determined by the protocol, data rate and reference clock frequency) is stored. You can use the image generation script flow (shown in [Figure 4-40](#)) to generate the SRAM and ROM images.

The script flow for generating the firmware image files (along with the description of the input and output files) required for the context restore is shown in [Figure 4-40](#). All files are in the `phy/include` directory of the release package.

Table 4-26 SRAM and ROM Image Generation Script Input and Output Files

File Name	Description
Input Files	
<dwc_prefix>_image_gen.pl	Perl script
protocol_input.txt	A configurable file for defining the protocols and data rates to be included in the output SRAM or ROM image
all_protocols.txt	Binary configuration data of all protocols and data rates supported by the PHY
protocols_metadata.txt	Binary meta data of configurations supported by the PHY
Output Files	
ROM and SRAM images	Includes the context restore content appended to the PHY firmware code
context_sel_ID.txt	Contains the valid range of values for txX_cntx_sel, rxX_cntx_sel, and tx_cmnX_cntx_sel

Figure 4-40 SRAM and ROM Image Generation Script Flow

4.24.4.1 Generating Firmware Image Files – A Sample Flow

To generate the firmware image files, complete the following steps:

1. Change to phy/include directory:

```
% cd phy/include
```
2. Add write permission to the files in this directory:

```
% chmod +w *
```
3. Edit <dwc prefix>_protocol_input.txt (text file) to select the correct protocol, baud-rate and frequency as shown in [Figure 4-41](#).



Note

Only configurations listed in dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package are supported.

Figure 4-41 Selection of Correct Protocol, Baud Rate and Frequency

```

P2      PCIe SRIS/SRNS - MPLLB Cascaded Clocking
#Protocols (Gen1 - Gen5)

#Frequencies
0       25
1       100 Select the frequency
0       125 by changing 0 to 1
0       200

P3      PCIe CC - MPLLA+B Direct Clocking
#Protocols (Gen1 - Gen5)

#Frequencies
0       100
.....
.....

P8      SATA
#Protocols
0       SATA1
0       SATA2
0       SATA3

#Frequencies
0       100

P9      Ethernet
#Protocols
0       1000BASE-KX (10-bit)
0       1000BASE-KX (20-bit)
0       10GBASE-KX4 / XAUI (10-bit)
0       10GBASE-KX4 / XAUI (20-bit)
0       RXAUI
0       2.5GBASE-KX (10-bit)
0       2.5GBASE-KX (20-bit)
0       5GBASE-KR (16-bit)
0       5GBASE-KR (32-bit)
0       10GBASE-KR (16-bit)
0       10GBASE-KR (32-bit)
0       25GBASE@25.78125G
1       AUI@25.78125G Select the protocol and baud rate
1       AUI@26.5625G  by changing 0 to 1

#Frequencies
1       78.125 Select the reference clock frequency
1       125    by changing 0 to 1
0       156.25
0       312.5

```

PCle config is selected if one (or more) frequency is selected

Select the frequency by changing 0 to 1

Protocol unique ID for Synopsys usage

Select the protocol and baud rate by changing 0 to 1

Select the reference clock frequency by changing 0 to 1

4. Run the perl scripts:

```
% <dwc prefix>_image_gen.pl
```
5. Read the context_sel_ID.txt output to determine the values that need to be driven on PHY inputs (txX_cntx_sel, rxX_cntx_sel, and tx_cmnx_cntx_sel), as shown in [Figure 4-42](#).
 - txX__cntx_sel depends on number of unique data rates.
 - The rxX_cntx_sel and tx_cmnx_cntx_sel inputs depend on the number of data rates multiplied by reference clock frequency.

 **Note**

Not all combinations of data rates and reference frequencies are supported. For supported configurations, refer to the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package).

Figure 4-42 PHY Inputs

E32 Config Master Version in P4 is (1000*config_ver in hex): 0x0114 Synopsys internal version for debugging

TX/RX/CMN	tx_cntx_ID	Protocol ID
TX	0	2
TX	1	2
TX	2	2
TX	3	2
TX	4	2
TX	5	9
TX	6	9

PHY input txX.cntx_sel for protocol ID 2 (PCIe Gen1, 2, 3, 4, 5)

PHY input txX.cntx_sel for protocol ID 9 (AUI@25.78125G, AUI@26.5625G)

TX/RX/CMN	rx_cntx_ID	Protocol ID
RX	0	2
RX	1	2
RX	2	2
RX	3	2
RX	4	2
RX	5	9
RX	6	9
RX	7	9

TX/RX/CMN	tx_cm_n_cntx_ID
CMN	0
CMN	0
CMN	0
CMN	0
CMN	0
CMN	1
CMN	2
CMN	3

4.24.4.2 Rerunning Script to Generate Firmware Image Files

If you need to run the script again because you encountered an error in the previous run or you have a new configuration, complete the following steps:

1.

Delete the phy/include directory and copy the entire original phy/include directory from backup.
2.

Repeat steps 1 to 4 listed in “[Generating Firmware Image Files – A Sample Flow](#)”.

 **Note**

Only configurations listed in the dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx in the doc directory of the PHY release package are supported. In case of discrepancies between <dwc_prefix>_protocol_input.txt and *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Configuration Reference Manual* (dwc_32g_phy_tsmc12ffc_x2ns_config_reference_manual.xlsx), use the values in the *Configuration Reference Manual*.

5

Test and Diagnostics

This chapter includes the following sections:

- “Test Modes”
- “Diagnostics”

5.1 Test Modes

The test modes supported in the PHY and the input configuration settings for each mode are provided in [Table 5-1](#).

Table 5-1 PHY Test Mode Selection

PHY Mode of Operation	bs_ce	test_burnin	test_powerdown	test_flyover_en	test_stop_clk_en	scan_mode	scan_shift scan_shift_cg
Boundary Scan (ACJTAG)	1	X	X	X	X	X	X
Burn-In	0	1	0	0	X	0	0
IDDQ	0	0	1	0	X	0	X
Flyover	0	0	0	1	0/1	0/1	0/1
Scan (At-Speed, Stuck-At)	0	0	0	0/1	0	1	0/1
Diagnostics (serial and parallel loopback)	0	0	0	0	0	0	0
Scan Dump	0	0	0	0/1	1	0	0
Normal Operation	0	0	0	0	0	0	0

The following sections describe each test mode in more detail.

5.1.1 Boundary Scan (ACJTAG)

The PHY includes boundary scan registers per lane, which provide direct controllability and observability of the txX_p/m and rxX_p/m lines. The boundary scan instructions supported in the PHY are SAMPLE, PRELOAD, BYPASS, EXTEST_PULSE and EXTEST_TRAIN as defined in IEEE Std 1149.6-2003.

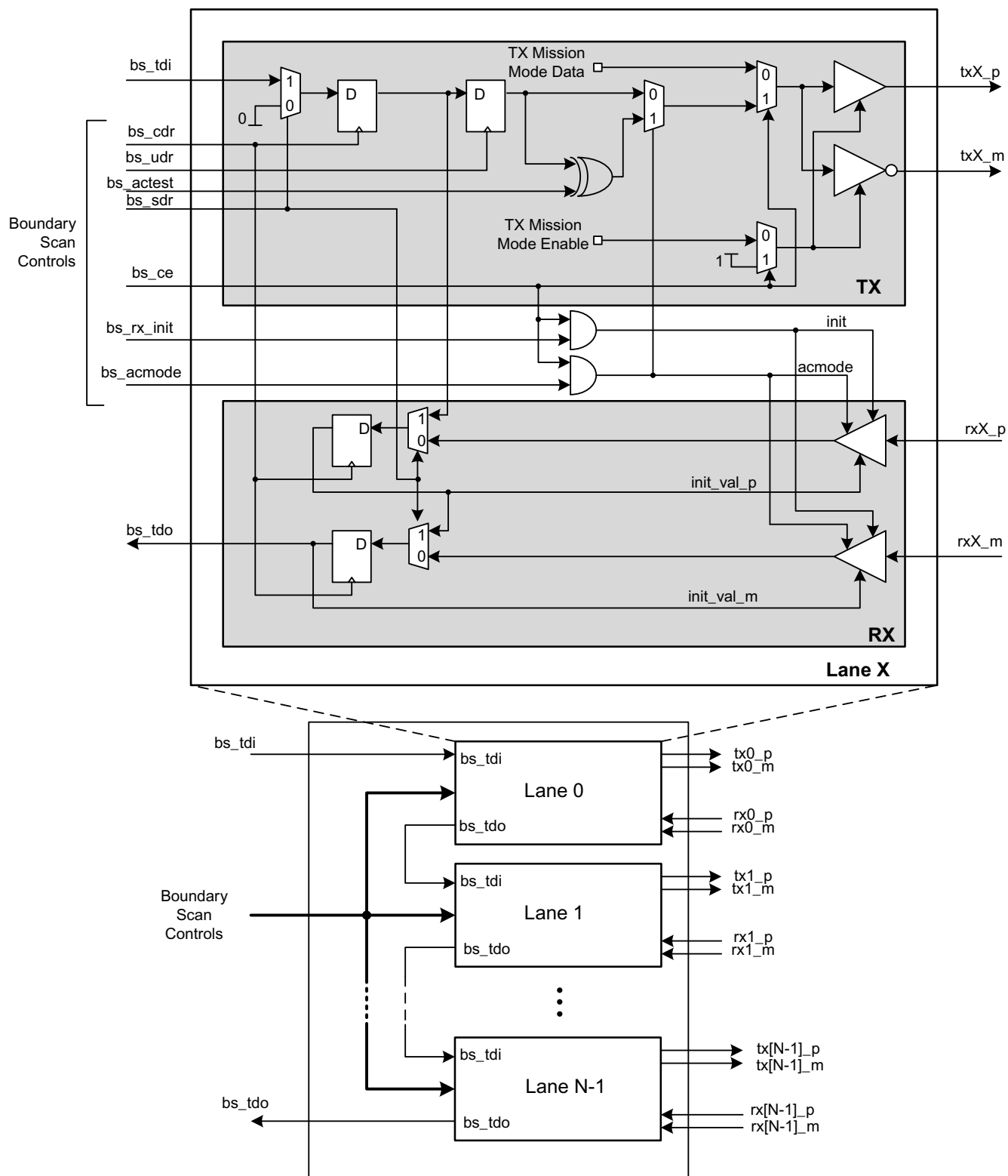
The states of all inputs of the PHY except bs_* are ignored in boundary scan mode.

[Figure 5-1](#) shows the logical structure of the boundary scan implementation in the N-lane PHY. The TX driver control logic is like the Output Driver Example defined in Sec 6.3.2. IEEE Std 1149.6-2003.



Note

There is no single bit register bypass logic within the boundary scan logic of the PHY, but it can be added externally if required.

Figure 5-1 Boundary Scan Structure

5.1.1.1 ACJTAG - AC Operation

The AC JTAG receiver has an adjustable receiver sensitivity that must be pin programmed based on the input signal. For valid settings, see [Table 4-21](#) (“Supported JTAG Instructions”).

Determine the correct settings by doing the following:

1. Identify the lowest differential output from the TX driving the AC JTAG receiver (this is usually on another chip or part of the test setup).
2. [Table 5-2](#) and [Table 5-3](#) determine the row that has high and low values that encompass the transmit level. Values for the `bs_rx_level` and `bs_rx_bigswing` should be set to the appropriate setting. Additionally, `bs_tx_lowswing` should be set to opposite value of `bs_rx_swing` of the table, when applicable.



Note

All the information about the controls and integration can be derived from the bsd file that is delivered in the package.

The low swing mode and big swing mode are intended for half swing (that is, 400-600 mVpp) and full swing (800-1200 mVpp) protocols, respectively.



Note

These tables are only a reference, and the choice of trip level is determined based on your boundary scan implementation.

Table 5-2 ACJTAG - Low Swing Mode Trip Levels

<code>bs_rx_level[4:0]</code>	<code>bs_rx_bigswing</code>	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
5'b00000	0	365	464
5'b00001	0	385	495
5'b00010	0	406	519
5'b00011	0	427	546
5'b00100	0	451	582
5'b00101	0	475	621
5'b00110	0	500	649
5'b00111	0	524	684

Table 5-3 ACJTAG - Big Swing Mode Trip Levels

<code>bs_rx_level[4:0]</code>	<code>bs_rx_bigswing</code>	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
5'b00000	1	683	919.4
5'b00001	1	780	995
5'b00010	1	790	962

bs_rx_level[4:0]	bs_rx_bigswing	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
5'b00011	1	841	1154
5'b00100	1	897	1089
5'b00101	1	953	1321
5'b00110	1	1017	1412
5'b00111	1	1073	1512

5.1.1.2 ACJTAG - DC Operation

For ACJTAG DC mode, use bs_rx_level[2:0]=5b'00111 (decimal 7) for a DC trip level between 85 mV and 115 mV.

ACJTAG DC mode is supported only for common mode of 0 V.

5.1.2 Burn-In

In burn-in mode, all analog blocks are powered up, all clocks are enabled, and the analog front end is put into TX-to-RX Loopback mode. This mode turns on an on-chip oscillator that is used as a reference clock for the MPLLs. This mode is intended for use during High-Temperature Operating Life (HTOL) or Burn-in, and sets up the PHY to toggle as much circuitry as possible (for realistic hot carrier injection degradation - HCI), while unused circuitry is carefully biased in its power-down mode to ensure realistic degradation for Negative Bias Temperature Instability (NBTI).



Note

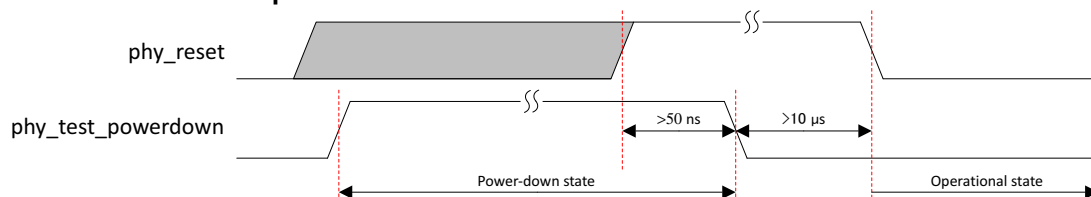
- Prior to enabling the burn-in mode by asserting test_burnin, phy_reset must be asserted and de-asserted to ensure the states of internal registers are reset.
- In burn-in mode: The designed operating temperature and voltage are 125°C and 1.5 times the nominal supply voltage.

5.1.3 IDDQ

In IDDQ test mode, all the analog blocks are powered down to allow leakage measurement. Input settings for enabling the IDDQ mode are provided in [Table 5-1 \("PHY Test Mode Selection"\)](#). States of all other inputs are ignored.

It is recommended that phy_reset be asserted for at least 50 ns before test_powerdown is negated and remain asserted for at least 10 μ s.

Figure 5-2 PHY Power-Down Sequence



5.1.4 Flyover

Settings to enable flyover mode are provided in [Table 5-1](#) (“PHY Test Mode Selection”).

The flyover mode can be used for sending scan data on and off die through the PHY TX and RX I/O pins.

On the TX side, the flyover mode enables you to asynchronously send an on-die signal off-chip from the `txX_flyover_data_p` and `txX_flyover_data_m` ASIC-side inputs to the `txX_p` and `txX_m` outputs. In this mode, a reference clock is not required.

On the RX side, the flyover mode enables you to asynchronously send an off-chip signal on-die through the `rxX_p` and `rxX_m` inputs to the `rxX_flyover_data_p` and `rxX_flyover_data_m` ASIC-side outputs. In this mode, a reference clock is not required as the signal is received asynchronously using the test receiver.

The flyover mode supports up to 200 Mbps single-ended data on both the RX and the TX data paths as specified in [Table 5-4](#). Each of the independent RX inputs on `rxX_p` and `rxX_m` must have a common mode of 0 mV and a signal swing of 400-600 mV. The RX flyover outputs and TX flyover inputs and outputs are all CMOS level signals. For more details, see [Figure 5-3](#) and [Table 5-4](#).



Note

If `scan_mode = 0` during flyover, then you need to drive CR inputs correctly (or driven low).

Figure 5-3 Flyover Mode Signal Parameter Definitions

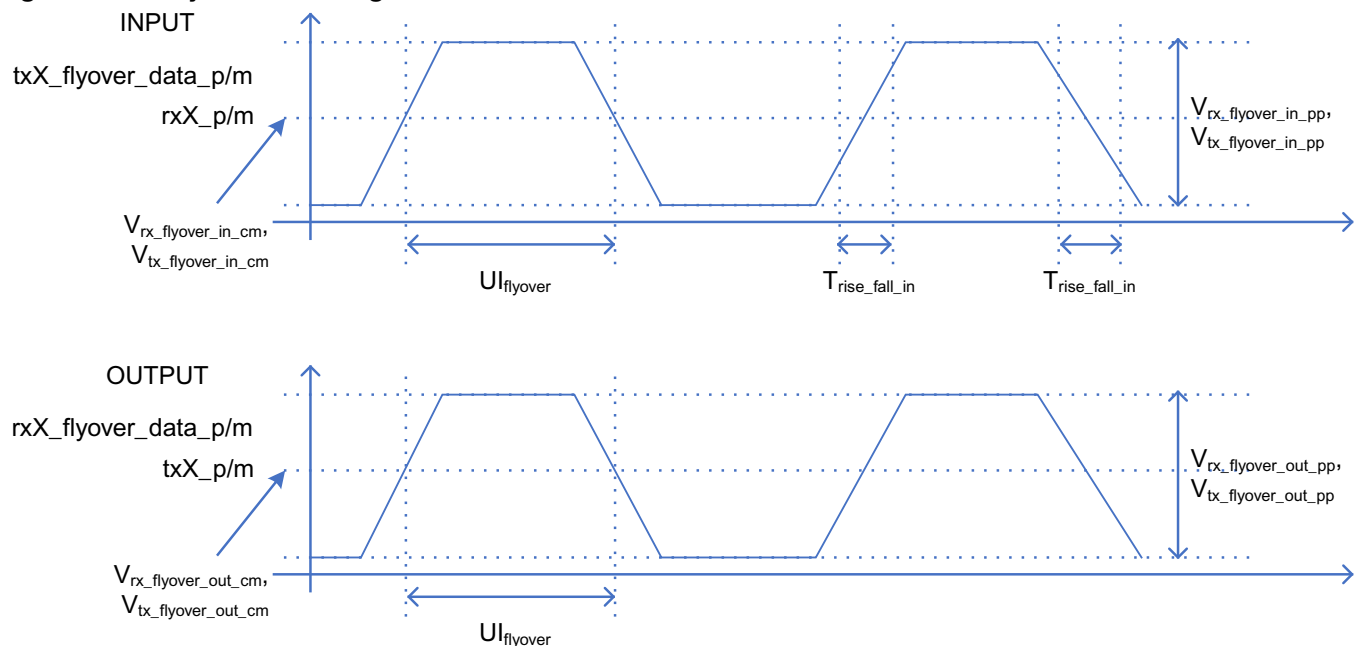


Table 5-4 Flyover Mode Signal Electrical Parameters

Symbol	Parameter description	Specification	Units	Note
Ulflyover	Unit Interval of flyover data	≥ 5	ns	Maximum data rate of 200 Mbps per independent single ended flyover data input/output
Input Specifications				
Trise_fall_in	Rise/Fall Time of TX and RX input signals	$< 0.20 \cdot U_{\text{flyover}}$		Rise/Fall time is measured from 20% to 80% of the full swing input signal.
Vrx_flyover_in_pp	Peak-peak RX flyover input voltage swing	≥ 400 (min) ≤ 600 (max)	mVpp	Single ended input swings between 400mVpp and 600mVpp.
Vrx_flyover_in_cm	Common Mode RX flyover input voltage	0	mV	Common mode of the single ended RX input is 0mV.
Vtx_flyover_in_pp	Peak-peak TX flyover input voltage swing	VPTX	mVpp	Input at CMOS level with VPTX, GD as supplies.
Vtx_flyover_in_cm	Common Mode TX flyover input voltage	$\frac{1}{2} \cdot V_{\text{PTX}}$	mV	Input at CMOS level with VPTX, GD as supplies.
Output Specifications				
Vrx_flyover_out_pp	Peak-peak RX flyover output voltage swing	VP	mVpp	Output at CMOS level with VP, GD as supplies.
Vrx_flyover_out_cm	Common Mode RX flyover output voltage	$\frac{1}{2} \cdot V_{\text{P}}$	mV	Output at CMOS level with VP, GD as supplies.
Vtx_flyover_out_pp	Peak-peak TX flyover output voltage swing	VPTX	mVpp	Output at CMOS level with VPTX, GD as supplies.
Vtx_flyover_out_cm	Common Mode TX flyover output voltage	$\frac{1}{2} \cdot V_{\text{PTX}}$	mV	Output at CMOS level with VPTX, GD as supplies.

5.1.5 Scan (At-Speed, Stuck-At)

Settings to enable scan mode are provided in [Table 5-1 \(“PHY Test Mode Selection”\)](#). To achieve the coverage metrics provided below, all other inputs must be controllable and all outputs must be observable.



Note

PMA .lib ETM modes:

- NORMAL—For Functional mode
- SCAN—For Stuck-at Shift and Capture mode
In this mode, Shift and Capture happens at Scan Clock Frequency.
- SCAN_ASST—For At Speed Scan Testing
In this mode, Shift occurs at Scan Clock Frequency, whereas Capture occurs at Functional mode frequency.

Note the following scan features and requirements in the PHY:

- The PHY provides independent scan clock inputs per internal clock domain.
- Primary clock inputs to the PHY must have scan clocks supplied to them externally in scan mode.
- All clock outputs from the PHY, except `ref_repeat_clk_p/ref_repeat_clk_m`, have a scan clock input muxed on them.
- Scan clocks must have 50% duty cycle.
- Synchronous inputs and outputs are at-speed scannable.
- All scan chains can be shifted in/out at the same time. By default, only one-hot scan clock capture is supported. Customers have the option to externally group the scan clocks, which are logically independent and share the same frequency. The ATPG flow in the deliverables provide an optional static clock grouping of the scan clocks. By default, no clock grouping is enabled for the vectors and scripts that are provided with the product. For more information on enabling clock grouping, see the `./pma/atpg/<metal-stack>/readme_tmax.txt` file.
- All resets (primary input resets or internally generated resets) are over-ridden to primary scan reset input `scan_set_rst` in scan mode.
- Maximum scan shift frequency is 250 MHz.
- Maximum scan chain length is 100 flops.
- All scan chains are positive-edge triggered.
- All scan chains are terminated with lockup latches.
- All analog blocks are powered down in scan mode. To prevent propagation of logic X's in the scan chain, the outputs of all the analog blocks feeding the digital scannable logic are overridden to known states.
- Stuck-At-Fault (SAF) coverage of the PMA is greater than 98%.
- Transition-Delay-Fault (TDF) coverage of the PMA is greater than 85%.
- Coverage of the soft-macro (RAW PCS) is place-and-route implementation dependent.

5.1.6 Scan Dump

The hard-macro PHY provides a scan-dump feature which allows the internal clocks in the design to be stopped and the current state of the PHY to be shifted out and observed. Usage of the scan-dump feature is described as follows:

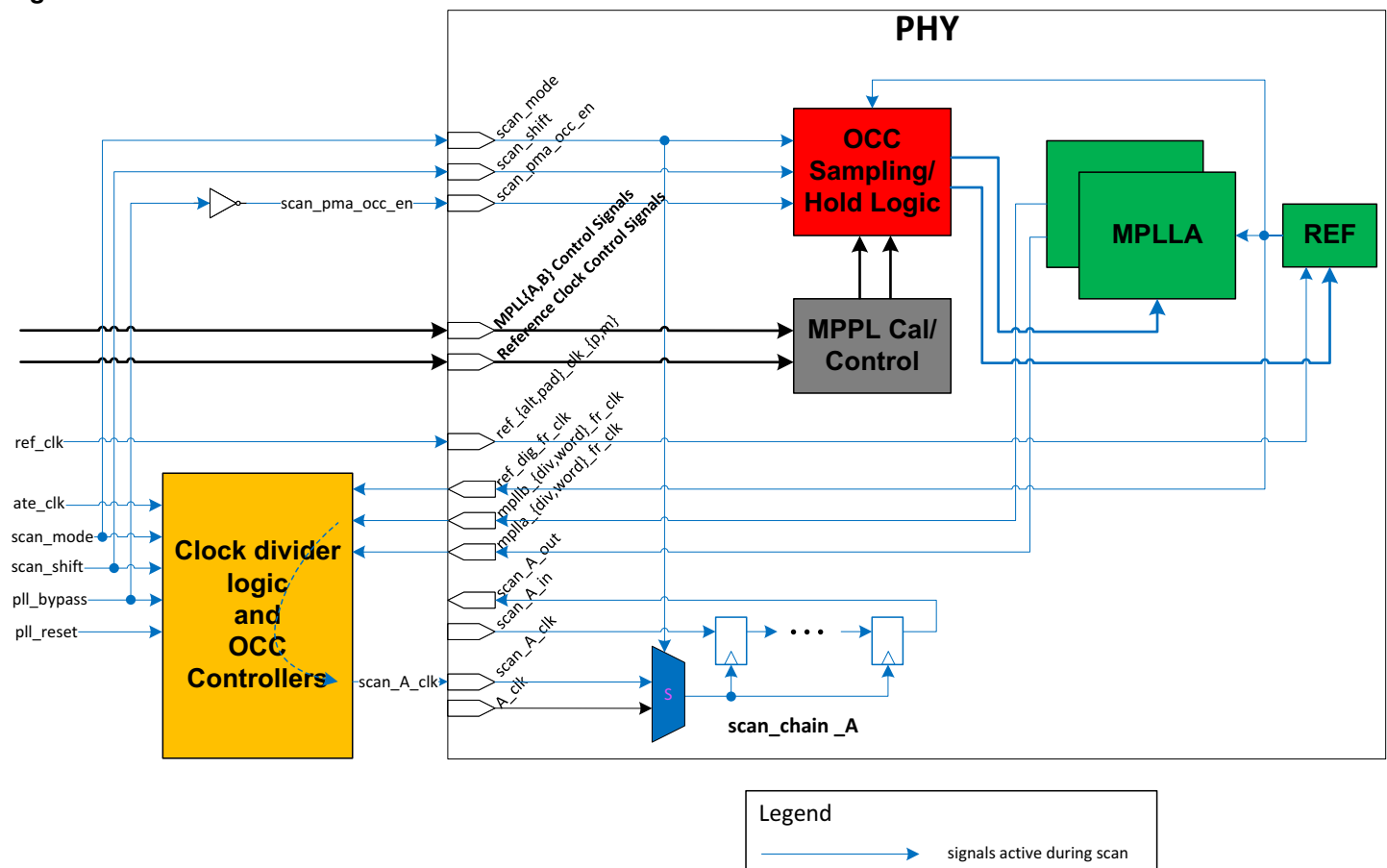
1. Assert `test_stop_clk_en` to stop the clocks.
2. Wait for 250 ns to allow the internal clocks to be gated.
3. Assert `scan_mode` to switch the clock muxes and make clock-gating latches transparent.
4. Assert `scan_shift_cg` and `scan_shift` signals.
5. Toggle scan clocks to shift data out of scan outputs.

The PHY must be reset by pulsing the `phy_reset` signal to exit from the scan-dump mode.

5.1.7 On-Chip Clocking Support

The hard-macro PMA supports On-Chip Clocking (OCC), allowing the internal PLL to be powered up and used as a high-speed clock source for the PHY input scan clocks. However, make sure that the external clock dividers are added outside of the PHY to generate the additional divided frequencies. Also, ensure that you add OCC controllers during DFT insertion to control the scan clocks before they propagate back to the hard-macro PMA – see [Figure 5-4](#) (“OCC Architecture”).

Figure 5-4 OCC Architecture



**Note**

When OCC is enabled, the scan chain associated with the OCC logic is not available.

The sequence to enable OCC for TDF testing is as follows:

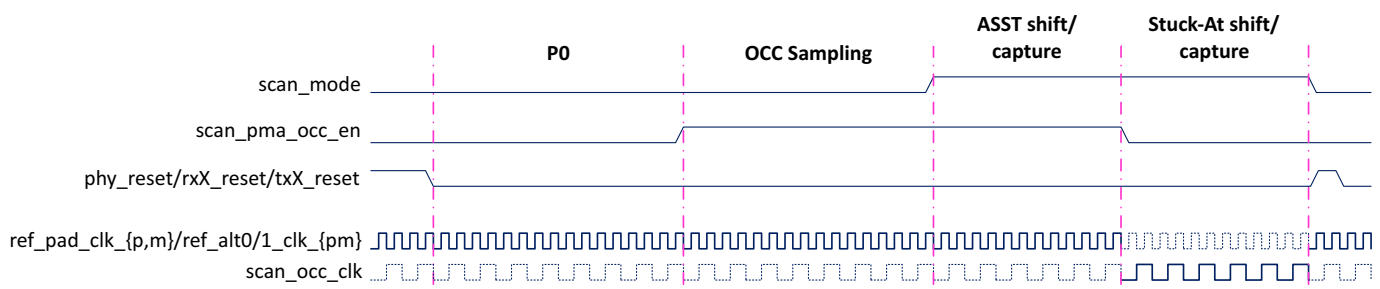
1. Power up PHY to P0 to calibrate and power up the required MPLL using the txX_req/txX_ack handshake.
2. Set scan_pma_occ_en to 1'b1, which latches the MPLLA configuration.
3. Wait for 130 ref_clks + 15μs, which is required to allow internal modulated MPLL signals to stabilize.
4. Set scan_mode to 1'b1.
5. Perform TDF testing by controlling scan_shift and scan_shift_cg.

The sequence to perform SAF testing is as follows:

1. Set scan_pma_occ_en to 1'b0, which disables the OCC latching.
2. Set scan_mode to 1'b1.
3. Perform SAF testing by controlling scan_shift and scan_shift_cg.

A waveform outlining the sequence for OCC with TDF testing, followed by SAF testing is shown in [Figure 5-5](#).

Figure 5-5 OCC_Waves

**Note**

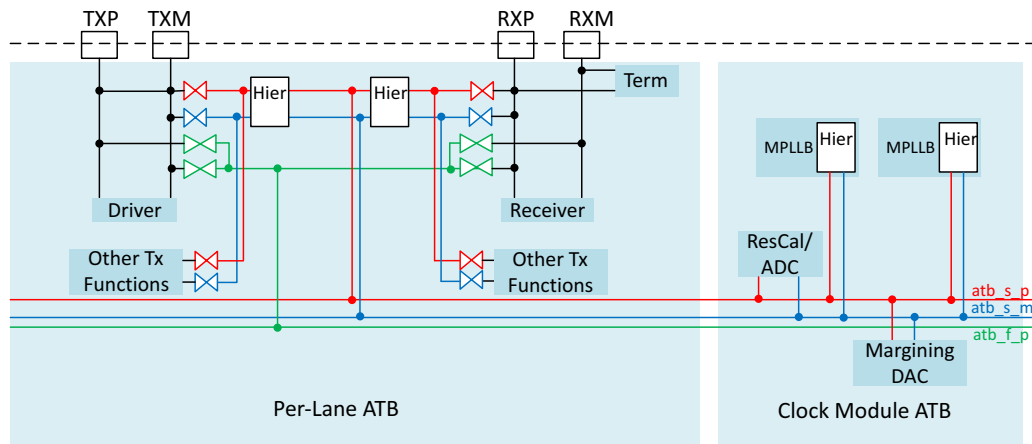
- The example of OCC sequence is implemented in PHY ATE testbench. Refer to the details of ATE Testbench Readme.
- The OCC logic cannot be tested at-speed. For SAF testing, the OCC hold logic is on a separate scan chain so it can be shifted and captured independently. This occurs when the scan_pma_occ_en signal = 1'b0, and scan_mode = 1'b1.

5.1.8 Analog DC Test Capabilities

This section describes the Multi-Protocol 32G PHY PMA's analog DC test capabilities.

5.1.8.1 Analog Test Bus Overview

[Figure 5-6](#) shows a high-level view of the Analog Test Bus (ATB).

Figure 5-6 High-Level View of Analog Test Bus

The ATB comprises three signals routed throughout the Clock module and each lane. Some of these signals can be brought to the internal ADC, while others can be brought to any TX or RX pin. The following table describes the ATB signals.

Table 5-5 Analog Test Bus Signals

Signal	Description	Can be Brought to
atb_s_p	Zero current “sense” connections	ADC, TXP, TXM
atb_s_m	Zero current “sense” connections	ADC, TXP, TXM
atb_f_p	Large current (up to ~1 mA) “force” connection	TXM

Many internal circuits have connections that can be enabled to these buses. Note that these connections are intended to be used in a very-low-frequency (near DC) situation. Series resistance between connections on the bus is typically greater than 30 kOhm, while bus parasitic capacitance can be multiple picoFarads, yielding bandwidths below 2 MHz. When connections are made to nodes carrying high-frequency signals, such as TX or RX serial pins, explicit 10-kOhm resistors are used, so that connecting the ATB bus does not impair the operation of the lane.

Bringing a selected internal signal to either a package pin or the internal ADC is a matter of activating the correct combination of hierarchical switches and leaf-level switches. Control of these switches is distributed across registers in the analog circuitry, as detailed in the following section.

5.1.8.2 ATB Access Through RX

To bring ATB signals out on the RX pins, disable the resistive terminations and the LOS circuit that are otherwise present at these pins.

The TX pins must not be used for this purpose, because the driver contains some impedances and leakage that cannot be completely eliminated.

To disable the RX termination and LOS circuit and connect the ATB to RX pins:

1. Override the RX termination by setting LANEN_DIG_ASIC_RX_OVRD_IN_3.EN to a value of 1.
2. Disable the RX termination by setting LANEN_DIG_ASIC_RX_OVRD_IN_3.TERM_EN to a value of 0.

3. Override the LOS control by setting LANEN_ANA_RX_PWR_CTRL1.OVRD_LOS_EN to 1.
4. Disable the LOS by setting LANEN_ANA_RX_PWR_CTRL1.LOS_EN_REG to 0.
5. Enable connection of atb_s_p to the RXP pin by setting LANEN_ANA_RX_CDR_AFE.MEAS_ATB_RX to a value of 0010.
6. Enable connection of atb_s_m to the RXM pin by setting LANEN_ANA_RX_ATB_MEAS1.MEAS_ATB_RX to a value of 010000.

**Note**

You must stop firmware (FSM_OVRD_EN=1) when measuring RX AFE and DFE internal nodes.

In addition, you must set hierarchical and leaf ATB switches based on the source of voltage or current being output through the RX.

5.1.8.3 ATB Measurement Using 10-Bit ADC

The 10-bit ADC uses circuit components that are already present in the Clock module—the 10-bit DAC required for voltage margining and the offset calibrated comparator present for resistor calibration.

To use the 10-bit ADC:

1. Setup the IP ADC.
 - a. Place the DAC in ADC mode by writing a 0 to SUP_ANA_RTUNE_CTRL.RT_DAC_MODE.
 - b. Connect the ADC to the ATB by writing a 1 to SUP_ANA_RTUNE_CTRL.RT_ATB.
 - c. Sense the ATB sense lines by writing a 0 to SUP_ANA_RTUNE_CTRL.RT_SEL_ATBF.
 - d. Sense on atb_s_p (1) and atb_s_m (0) by writing a 1 or a 0 to SUP_ANA_RTUNE_CTRL.RT_SEL_ATBP.
 - e. Override the power-on signal to force the IP ADC on by writing a 1 to SUP_ANA_RTUNE_CTRL.RT_EN_FRCON.
 - f. Signal RTUNE that you are doing an ADC run by writing a 00 to SUP_DIG_RTUNE_DEBUG.TYPE.
2. Select ATB node in the PHY to sense.
 - a. Enable the block-level switch (if it exists) for the block using CREG.
 - b. Enable the specific ATB switch using CREG.
3. Trigger the ADC run.
 - a. Trigger ADC operation by writing a 1 to SUP_DIG_RTUNE_DEBUG.MAN_TUNE.
 - b. Read the 10-bit result from SUP_DIG_RTUNE_STAT.STAT.
 - c. Reset the ADC engine prior to subsequent use by writing a 0 to SUP_DIG_RTUNE_DEBUG.MAN_TUNE.
 - d. Add 50 μ s delay.
4. Calculate voltage.

$$\text{Voltage} = \text{result} * \text{VPH} / 1,023$$

5.1.8.4 ATB Access Through Top-Level Pins

You can access the ATB through top-level PMA pins.

Due to contention, these pins cannot be wired together. You should isolate the atb* pins from multiple PMA instances from one another with an appropriate pass gate. Doing so ensures that leakage and IR drop from the different instances are isolated from one another.

These top-level pins are not ESD-protected. If they are connected to a bump, use an appropriate I/O cell. Any leakage from the I/O cell can corrupt the ATB voltage.

5.1.8.5 ATB Register Map

[Table 5-6](#) describes the ATB registers.

Table 5-6 ATB Registers Map

Block	Register Name	Master Switch	atb_p_or_m	Description
MPLL	SUP.MPLL_ATB1.MEAS_IV_7	SUP.MPLL_ATB3.ATB_SELECT	p	Measure vreg_left
	SUP.MPLL_ATB1.MEAS_IV_6		p	Measure vreg_s_bg
	SUP.MPLL_ATB1.MEAS_IV_5		p	Measure vreg_vco_bg
	SUP.MPLL_ATB2.MEAS_IV_15		m	Measure reference voltage for VCO magnitude control amp
	SUP.MPLL_ATB2.MEAS_IV_14		p	Measure VCO magnitude control signal vdrive
	SUP.MPLL_ATB2.MEAS_IV_13		p	Measure buffered version of ctrl_fine_int
	SUP.MPLL_ATB2.MEAS_IV_12		p	Measure GD in MPLL
	SUP.MPLL_ATB2.MEAS_IV_11		p	Measure VP / 2 in MPLL
	SUP.MPLL_ATB2.MEAS_IV_10		p	Measure vreg_cp_bg
	SUP.MPLL_ATB2.MEAS_IV_9		p	Measure vreg_outclock_bg
	SUP.MPLL_ATB2.MEAS_IV_8		p	Measure vreg_right_bg
	SUP.MPLL_ATB3.MEAS_IV_22		p	Measure CP vsense
	SUP.MPLL_ATB3.MEAS_IV_21		m	Measure DLL nbias
	SUP.MPLL_VCO_GAIN_CAL.MEAS_VCO_CAL_LVL		p	Measure vco calibration control voltage during calibration; in mission mode set to vp.

Block	Register Name	Master Switch	atb_p_or_m	Description
MPLL	SUP.MPLLB_ATB1.MEAS_IV_7	SUP.MPLLB_ATB3.ATB_SELECT	p	Measure vreg_left_bg.
	SUP.MPLLB_ATB1.MEAS_IV_6		p	Measure vreg_s_bg.
	SUP.MPLLB_ATB1.MEAS_IV_5		p	Measure vreg_vco_bg.
	SUP.MPLLB_ATB2.MEAS_IV_15		m	Measure reference voltage for VCO magnitude control amp.
	SUP.MPLLB_ATB2.MEAS_IV_14		p	Measure VCO magnitude control signal vdrive.
	SUP.MPLLB_ATB2.MEAS_IV_13		p	Measure buffered version of ctrl_fine_int.
	SUP.MPLLB_ATB2.MEAS_IV_12		p	Measure GD in MPLL.
	SUP.MPLLB_ATB2.MEAS_IV_11		p	Measure VP / 2 in MPLL.
	SUP.MPLLB_ATB2.MEAS_IV_10		p	Measure vreg_cp_bg.
	SUP.MPLLB_ATB2.MEAS_IV_9		p	Measure vreg_outclock_bg.
	SUP.MPLLB_ATB2.MEAS_IV_8		p	Measure vreg_right_bg.
	SUP.MPLLB_ATB3.MEAS_IV_22		p	Measure CP vsense.
	SUP.MPLLB.MEAS_IV_28		p	Measure vreg_pfd_bg.
	SUP.MPLLB.MEAS_IV_27		p	Measure vdd_h in regulator.
	SUP.MPLLB.MEAS_IV_26		m	Measure vreg_vdd_drv in regulator.
	SUP.MPLLB.MEAS_IV_25		p	Measure vreg_pmix_bg.
	SUP.ANA.MPLLB.MEAS_IV_24		p	Measure vreg_pmixdiv_bg.
	SUP.MPLLB_VCO_GAIN_CAL.MEAS_VCO_CAL_LVL		p	Measure vco calibration control voltage during calibration; in mission mode set to vp.

Block	Register Name	Master Switch	atb_p_or_m	Description
RX	LANE.RX_ATB_MEAS1.MEAS_ATB_RX_3	LANE.RX_ATB_REGREF.MASTER_ATB_EN_I	p	Measure RX LOS detection threshold voltage.
	LANE.RX_ATB_MEAS1.MEAS_ATB_RX_2		p	Measure RX bias reference voltage.
	LANE.RX_ATB_MEAS1.MEAS_ATB_RX_1		p	Measure RX bias pmos biasing voltage.
	LANE.RX_ATB_MEAS1.MEAS_ATB_RX_0		p	Measure RX bias cascaded pmos.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_7		p	Measure vreg_bypass level; for adapted protocols is 0.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_6		p	Measure RX regulator gd.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_5		p	Measure RX bias pmos biasing voltage.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_3		p	Measure RX bias cascaded pmos.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_2		p	Measure vreg_dfe.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_1		p	Measure RX regulator vph through.
	LANE.RX_ATB_MEAS2.MEAS_ATB_VREG_I_0		p	Measure vreg_clk_vro.
	LANE.RX_CDR_VCO_CTRL.MEAS_VCO_ATB_REG_6		p	Measure VCO vregbg.
	LANE.RX_CDR_VCO_CTRL.MEAS_VCO_ATB_REG_5		p	Measure VCO rx_vref_ext; used in debug only, disregard in MM.

Block	Register Name	Master Switch	atb_p_or_m	Description
RX (Continued ...)	LANEN_ANA_RX_CDR_VCO_CTRL. MEAS_VCO_ATB_REG_4	LANEN_ANA_RX_ATB_REGREF.MASTER _ATB_EN_I	p	Measure VCO rx_vref.
	LANE.RX_CDR_VCO_CTRL.MEAS_V CO_ATB_REG_3		m	Measure VCO Vosc.
	LANE.RX_CDR_VCO_CTRL.MEAS_V CO_ATB_REG_2		m	Measure VCO gd.
	LANE.RX_CDR_VCO_CTRL.MEAS_V CO_ATB_REG_1		p	Measure VCO inref_1.
	LANE.RX_CDR_VCO_CTRL.MEAS_V CO_ATB_REG_0		m	Measure VCO gd.
	LANE.RX_CDR_AFE.MEAS_ATB_SU M_E		p, m	even DFE summer output.
	LANE.RX_CDR_AFE.MEAS_ATB_SU M_O		p, m	odd DFE summer output
	LANE.RX_CDR_AFE.MEAS_ATB_RX _9		p	AFE biasing vbp
	LANE.RX_CDR_AFE.MEAS_ATB_RX _8		m	AFE biasing vbn
	LANE.RX_ATB_REGREF.MEAS_ATB _CAL_MUX_I, bit7		p, m	Measure cal comparator outputs
	LANE.RX_ATB_REGREF.MEAS_ATB _CAL_MUX_I, bit6		p, m	Measure cal_lpf output 1
	LANE.RX_ATB_REGREF.MEAS_ATB _CAL_MUX_I, bit5		p, m	Measure cal_lpf output 2
	LANE.RX_ATB_REGREF.MEAS_ATB _VP_I		p	Measure vp sense

Block	Register Name	Master Switch	atb_p_or_m	Description
RX (Continued)	LANE.RX_DCC_OVRD.MEAS_ATB_CAL_VREF_2	LANE.RX_ATB_REGREF.MASTER_ATB_EN_I	p	Measure vref_sigdet
	LANE.RX_ATB_IQSKEW.MEAS_ATB_CAL_VREF		m	Measure vref_pre
	LANE.RX_ATB_IQSKEW.MEAS_ATB_CAL_VREF		m	Measure vref_lpf
	LANE.RX_MISC_OVRD.MEAS_ATB_RX_12		p, m	Enables sigdet HF sig to atb_s_p_int and HF ref to atb_s_m_int; HF sigdet, not used, disregard the read.
	LANE.RX_MISC_OVRD.MEAS_ATB_RX_11		p, m	Enables sigdet LF sig_lfps to atb_s_p_int and HF ref_lfps to atb_s_m_int; set to ~vp in MM.
	LANE.RX_MISC_OVRD.MEAS_ATB_RX_10		p	Enables sigdet bias test point.
	LANE.RX_ATB_VREG.MEAS_ATB_VREG_8		p	Measure vreg_vco.

Block	Register Name	Master Switch	atb_p_or_m	Description
TX	LANE.TX_ATB1.ATB_GD	LANE.TX_ATB3.ATB_S_ENABLE	p	Measure gd for tx_vreg.
	LANE.TX_ATB1.ATB_VDD		p	Measure vdd_h/3 (charge pump output voltage) for tx_vreg.
	LANE.TX_ATB1.ATB_VPTX		p	Measure vptx.
	LANE.TX_ATB1.ATB_DCC_COMP		p, m	Measure outputs of opamp which amplifies the signal entering DCC comparator; used during calibration, set to 0 for MM.
	LANE.TX_ATB1.ME		p, m	Measure inputs entering AC coupling mux inside tx_cal_top. This would generally be the filtered clock used by the DCC loop; used during calibration, set to 0 for MM.
	LANE.TX_ATB2.ATB_PBIAS		p	Measure TX pbias.
	LANE.TX_ATB2.ATB_NBIAS		p	Measure TX nbias.

Block	Register Name	Master Switch	atb_p_or_m	Description
TX (Continued)	LANE.TX_ATB2.ATB_RXDETREF	LANE.TX_ATB3.ATB_S_ENABLE	p	Measure TX RX detection reference voltage; used during tx rx detect, set to vp in MM.
	LANE.TX_ATB2.ATB_VCM		p	Measure tx vcm.
	LANE.TX_ATB3.MEAS_ATB_VPH_HALF		p	Measure vph/3 for tx_vreg.
	LANE.TX_MISC.MEAS_ATB_BIAS_VPTX		p	Measure TX bias local vptx.
	LANE.TX_OVRD_MEAS.MEAS_ATB_DCC_CAL_VDAC_CM		p	Measure cal.
	LANE.TX_OVRD_MEAS.MEAS_ATB_DCC_CAL_VDAC_DIFF		p, m	Measure voltage on idcc_diff_p/m nodes for diff DAC output voltage.
	LANE.TX_MISC.MEAS_ATB_BIAS_VPTX		p	Measure TX bias local vptx.
	LANE.TX_TERM_CODE_UP.ATB_VREG_TX		p	Measure vreg_tx.
	LANE.TX_TERM_CODE_UP.ATB_VREG_VPTX		p	Measure vreg_vptx/2.
ROPLL	LANE.ROPLL_2.ROPLL_MEAS_ATB_VREG_OUTCLOCKS	LANE.ROPLL_2.ROPLL_ATB_S_ENABLE	p	Measure out-clocks regulator output in ROPLL wrapper.
	LANE.ROPLL_2.ROPLL_MEAS_ATB_VREG_REFCLK		p	Measure ref clk regulator output.

Block	Register Name	Master Switch	atb_p_or_m	Description
SUPPORT	SUP.SWITCH_PWR_MEAS.TEMP_MEAS	SUP.PREA.ATB_SELECT	p, m	Measures bandgap voltages: vbe is sent to atb_s_m and vbg to atb_s_p.
	SUP.PREA.MEAS_VREG, bit7		p	Measure VREG_mplla; used in debug, set to vp in MM.
	SUP.PREA.MEAS_VREG, bit6		p	Measure VREG_DCO; used in HTOL, set to 0 in MM.
	SUP.PREA.MEAS_VREG, bit5		p	Measure VREG_repeat; set to vp if repeater not used in MM.
	SUP.PREA.MEAS_VREG, bit4		p	Measure VREG_mpllb; used in debug, set to vp in MM.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_0		p	Measure slowreg_vref.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_1		p	Measure vph.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_2		p	Measure fastreg_vref (vbg_vref_rx).
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_3		P	Measure vp.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_4		m	Measure gd.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_5		p	Measure vph/2.
	SUP.SWITCH_PWR_MEAS.ATB_SWITCH_6		p	Measure the short between atb_s_p and atb_s_m; used in debug only, disregard in MM.

**Note**

When reading MPLLA/B ATB values, refer to the registers for the active MPLL. Disregard the values for the unused MPLL.

5.2 Diagnostics

The PHY provides the following diagnostic features to enhance testbench characterization and ATE testing.

- Loopback functions (independent per-lane)
- TX-to-RX serial (pre-driver) loopback
- RX-to-TX parallel loopback
- BERT (independent per-lane)
- 7th, 9th, 11th, 15th, 16th, 23rd, and 31st order polynomial pattern generation and recognition
- Generation of simple test patterns
- Error insertion and detection

There are several options to observe the receiver signal within the PHY, outlined in [Table 5-7](#).

Table 5-7 Methods for Receiver Eye Diagnostics

Method	Description	Limitations	BER Depth	Customer Access
Eye Monitor	2D plot of the full eye: observation of equalization and shape of eye; quick measurement	Any rate, non-destructive, accuracy-scope slicers may have timing offset to data slicers	High BER (~1e-5)	Sequence available
PCIe RX Margining	PCIe specification defined RX vertical and horizontal eye margin function for PCIe 4 and 5	Gen4 and Gen5 only, non-destructive, built-in to PCIe controller, accuracy-scope slicers may have timing offset to data slicers	User defined	Built-in to PCIe specification
Four-Point Eye Margin Non-Destructive	Measurement of vertical and horizontal eye margin using scope slicers, preferred for active links	Any rate, non-destructive, accuracy-scope slicers may have timing offset to data slicers	User defined ^a	Sequence available
Four-Point Eye Margin Destructive	Measurement of vertical and horizontal eye margin using data path slicers, for loopback test modes with known patterns.	Any rate, destructive, requires PRBS pattern, for random data requires upper layer ability to detect bit errors	User defined ^a	Sequence available

a. Extrapolation from bathtub curves used to achieve low BER (that is 1e-12).

5.2.1 Loopback Functions

Each lane in the PHY supports a TX-to-RX and an RX-to-TX loopback mode. Only one loopback mode can be enabled at any given time.



Note 64-bit mode is not supported in RX-to-TX loopback mode.

The following conditions must be met prior to enabling the loopback modes:

- Input settings must be set to values provided in [Table 5-1](#) (“PHY Test Mode Selection”)
- phy_reset must be asserted and then de-asserted.
- Reference clock input to the PHY must be supplied.
- MPLLA/B configuration settings must correspond to the required baud rate (for more information, see [Section 4.23](#), “Configuring the PHY”; this can be done by controlling primary inputs or accessing registers.
- Set the signals in “[Loopback Function Settings](#)” accordingly; this can be done by controlling primary inputs or accessing registers.



Note If the BERT pattern generator is used in internal or external cable loopback, use the settings provided in *Appendix A (“ATE Override Control Pin List”) of the ATE Testbench Application Note*. Otherwise, the local Protocol PCS triggers RX Data Recovery and disrupts the RX serial data seen by the BERT pattern checker.

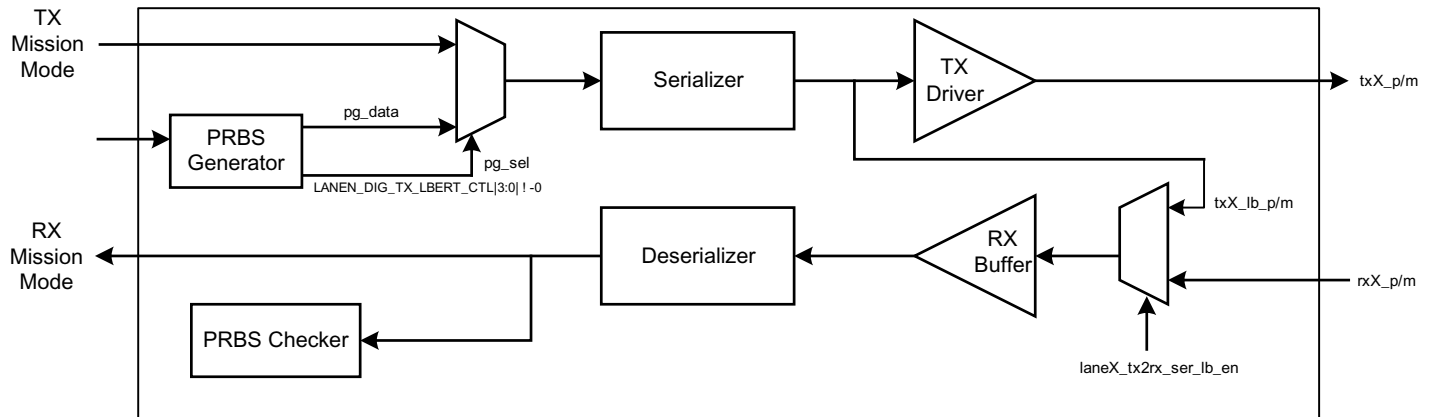
Table 5-8 Loopback Function Settings

Signal	Setting
txX_reset rxX_reset txX_lpd rxX_lpd	De-assert for the corresponding lane X.
txX_pstate rxX_pstate	Set to P0.
txX_rate[2:0] txX_width[1:0] txX_ropll_refsel txX_ropll_bypass rxX_rate[2:0] rxX_width[1:0]	Set to required baud rate (for more information, see Section 4.23 , “Configuring the PHY”).

5.2.1.1 TX-to-RX Serial (Pre-Driver) Loopback

The TX-to-RX loopback path is shown in [Figure 5-7](#).

Figure 5-7 TX-to-RX Serial (Pre-Driver) Loopback



To enable PRBS as a source to the serializer, LANEN_DIG_TX_LBERT_CTL must select the desired pattern and laneX_rx2tx_par_lb_en must be 0. This mode uses either user logic or the built-in PRBS Generator and Checker to generate and check the test data. The serial loopback path to the RX buffer is after the TX pre-driver (before the TX driver). Using this mode, you can verify the operation for all enabled PCS and PMA functional blocks in the transmitter and receiver channel.

The TX-to-RX loopback mode is enabled by asserting the laneX_tx2rx_ser_lb_en inputs, and enabling the TX and RX datapaths by asserting txX_data_en and rxX_data_en inputs (also accessible through override registers LANEN_DIG_ASIC_LANE_OVRD_IN, LANEN_DIG_ASIC_TX_OVRD_IN_0, and LANEN_DIG_ASIC_RX_OVRD_IN_0).



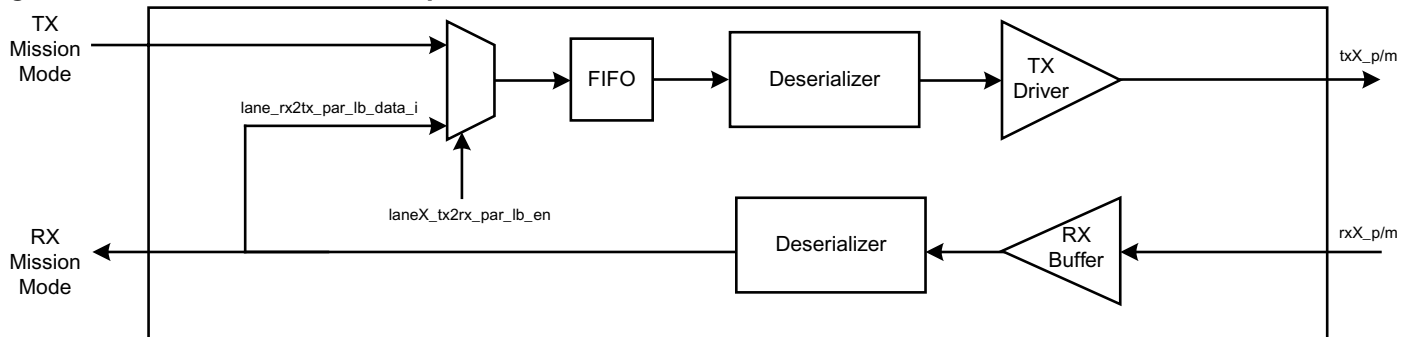
Note

When the TX-to-RX serial loopback is enabled, having any other signal on the RX pins is not supported. For example, if external loopback is still connected, there is a possibility of the two paths, internal and external, interfering with each other.

5.2.1.2 RX-to-TX Parallel Loopback

The RX-to-TX parallel loopback path is shown in [Figure 5-8](#).

Figure 5-8 RX-to-TX Parallel Loopback



The RX-to-TX parallel loopback mode is enabled by asserting the input `laneX_rx2tx_par_lb_en`, and enabling the RX datapath by asserting `txX_data_en` and `rxX_data_en` (also accessible through over-ride registers `LANEN_DIG_ASIC_LANE_OVRD_IN`, `LANEN_DIG_ASIC_TX_OVRD_IN_0`, and `LANEN_DIG_ASIC_RX_OVRD_IN_0`). The `txX_rate` input must also be set to `3'd0` (also accessible through override register `LANEN_DIG_ASIC_TX_OVRD_IN_1`) because the TX uses post-divided clocks from the RX in loopback mode. In this loopback mode, the received recovered data `rxX_p/rxX_m` is transmitted back on the `txX_p/txX_m` outputs. To prevent errors that may get introduced in the RX-to-TX loopback data because of the frequency deviation between the local MPLLs and the external transmitter, the reference clock input to the PHY must be derived from the same source as the reference clock used by the external transmitter feeding data on `rxX_p/rxX_m`.

During RX-to-TX loopback, the TX clock path is sourced from the RX clock path and as a result must be calibrated for TX DCC. Upon entry into RX-to-TX parallel loopback, the firmware re-calibrates the TX DCC for the new clock path. This process takes approximately 100 μ s to complete. The expectation is that loopback data will only be valid after this 100 μ s has elapsed. Upon exiting RX-to-TX parallel loopback, the firmware restores the previous TX DCC calibration code. This process takes approximately 50 μ s to complete. The system should not make any new requests to the PHY until after the 50 μ s has elapsed.

5.2.2 BERT

The Bit Error Rate Tester (BERT) comprises two independently programmed modules: a pattern generator and a pattern matcher/error counter.

5.2.2.1 BERT Pattern Generator

Table 5-9 provides a list of patterns that can be generated with the built-in pattern generator.

Table 5-9 Pattern Generator Mode Control

LANEN_DIG_TX_LBERT_CTL[3:0]	Description
4'b0000	Pattern generator is disabled
4'b0001	LFSR31: $x^{31} + x^{28} + 1$
4'b0010	LFSR23: $x^{23} + x^{18} + 1$
4'b0011	LFSR23: $x^{23} + x^{21} + x^{16} + x^8 + x^5 + x^2 + 1$
4'b0100	LFSR16: $x^{16} + x^5 + x^4 + x^3 + 1$
4'b0101	LFSR15: $x^{15} + x^{14} + 1$
4'b0110	LFSR11: $x^{11} + x^9 + 1$
4'b0111	LFSR9: $x^9 + x^5 + 1$
4'b1000	LFSR7: $x^7 + x^6 + 1$
4'b1001	Fixed word (PAT0)
4'b1010	DC balanced word (PAT0, ~PAT0)

LANEN_DIG_TX_LBERT_CTL[3:0]	Description
4'b1011	Fixed pattern ■ (000, PAT0, 3ff, ~PAT0): 10-bit, 20-bit, and 40-bit TX width ■ (00, PAT0, ff, ~PAT0): 8-bit, 16-bit, 32-bit TX width

The desired pattern PAT0 can be set by writing to the pattern generator control register (LANEN_DIG_TX_LBERT_CTL[14:5] for 10-bit, 20-bit or 40-bit TX width; LANEN_DIG_TX_LBERT_CTL[12:5] for 8-bit, 16-bit or 32-bit TX width). TX width (transmit data width) is selected using the txX_width input or by writing into the control register LANEN_DIG_ASIC_TX_OVRD_IN_0.

**Note**

The PAT0 data pattern has the following restriction:
Patterns of all 0's or all 1's should not be used for “Fixed word” and “Fixed pattern” modes.

A single-bit error can be inserted into the LSB of a transmitted word by writing a 1 on the register LANEN_DIG_TX_LBERT_CTL[4].

5.2.2.2 BERT Pattern Matcher

The pattern matcher can detect and synchronize to errored bytes in multiple types of patterns. Symbols with errors are counted in the error counter. RX width (receive data width) is selected using the rxX_width input or by writing into the control register LANEN_DIG_ASIC_RX_OVRD_IN_0.

Note that the controls for the pattern generator and pattern matcher within a lane are independent.

Table 5-10 describes the LANEN_DIG_RX_LBERT_CTL[3:0] field, which selects the expected pattern and operating mode.

Table 5-10 Pattern Matcher Mode Control

LANEN_DIG_RX_LBERT_CTL[3:0]	Description
4'b0000	Disabled
4'b0001	LFSR31: $x^{31} + x^{28} + 1$
4'b0010	LFSR23: $x^{23} + x^{18} + 1$
4'b0011	LFSR23: $x^{23} + x^{21} + x^{16} + x^8 + x^5 + x^2 + 1$
4'b0100	LFSR16: $x^{16} + x^5 + x^4 + x^3 + 1$
4'b0101	LFSR15: $x^{15} + x^{14} + 1$
4'b0110	LFSR11: $x^{11} + x^9 + 1$
4'b0111	LFSR9: $x^9 + x^5 + 1$
4'b1000	LFSR7: $x^7 + x^6 + 1$
4'b1001	Fixed word (PAT0)

LANEN_DIG_RX_LBERT_CTL[3:0]	Description
4'b1010	DC balanced word (PAT0, ~PAT0)
4'b1011	Fixed pattern ■ (000, PAT0, 3ff, ~PAT0): 10-bit, 20-bit, and 40-bit RX width ■ (00, PAT0, ff, ~PAT0): 8-bit, 16-bit, 32-bit RX width

The pattern matcher generates the expected pattern and synchronizes the generated pattern to the incoming pattern. Synchronization and error counting are initiated by asserting the sync bit (LANEN_DIG_RX_LBERT_CTL[4]). This bit must be toggled twice for reliable operation.

The error counter in the pattern matcher is 22 bits wide. You can read the error count result from the LANEN_DIG_RX_LBERT_ERR register, the content of which is {lbert_err_ovbit14, lbert_err_cnt[14:0]}. The contents of LANEN_DIG_RX_LBERT_ERR are presented according to the following rules:

- When the error count is $< 2^{15}$, the bottom 15 bits are presented in the count field (lbert_err_cnt[14:0]) and the lbert_err_ovbit14 field is de-asserted.
- When the error count is $< 2^{22}$ but $\geq 2^{15}$, the top 15 bits are presented in the count field (lbert_err_cnt[14:0]) and the lbert_err_ovbit14 field is set. This result effectively scales the contents down by a factor of 2^7 .
- When the 22-bit error counter overflows, the count field (lbert_err_cnt[14:0]) is set to 0x7FFF, and the lbert_err_ovbit14 field is asserted.

BERT pattern matcher can be run continuously or can be configured to run for fixed number of bits (which can be used to calculate approximate bit error rate by the user).

Firmware needs to be disabled when using BERT pattern matcher in fixed number of bits mode. A sample sequence to enable this mode:

1. Disable Firmware using FSM_FSM_OVRD_CTL.FSM_OVRD_EN = 1
2. Set LANEN_DIG_TX_LBERT_CTL.MODE = 12 or another mode [TX and RX mode have to be same.]
3. Config registers to enable USE_SAMPLE_CNT mode
 - Set LANEN_DIG_RX_STAT_STAT_CTL0.SC_TIMER_MODE = 1
 - Set LANEN_DIG_RX_STAT_STAT_CTL0.STAT_RXCLK_SEL = 1
 - Set LANEN_DIG_RX_STAT_STAT_CTL1.STAT_CLK_EN = 1
 - Set LANEN_DIG_RX_LBERT_CTL.MODE=12 or another mode
 - Set LANEN_DIG_RX_LBERT_CTL.USE_SAMPLE_CNT = 1
 - Set LANEN_DIG_RX_LBERT_CTL.BER_SEL = 1 [Enable bit Error instead of word errors during pattern matching]
 - Config below registers to load RX START counter with stat_ld_val.
 - stat_ld_val is the number of qwords that user wants pattern matcher to check.
 - Set LANEN_DIG_RX_STAT_LD_VAL_EXT_1.SC1_LD_VAL_EXT_EN = 1 [To extend load value for the sample counter]
 - Set LANEN_DIG_RX_STAT_LD_VAL_1.SC1_LD_VAL = stat_ld_val[14:0]
 - Set LANEN_DIG_RX_STAT_LD_VAL_EXT_1.SC1_LD_VAL_EXT_14_0 = stat_ld_val[29:15]

- Set LANEN_DIG_RX_STAT_LD_VAL_EXT_1.SC1_LD_VAL_EXT_29_15 = stat_ld_val[44:30]
- 4. Set LANEN_DIG_RX_STAT_LD_VAL_1.SC1_START = 1 to start the counter [SC1_START = 1 will start RX STAT counter and RX Pattern matcher. SC1_START is self-clear to 0 after a few clock cycles].
- 5. Polling LANEN_DIG_RX_STAT_SMPL_CNT1.SMPL_CNT1_DONE until a positive edge of SMPL_CNT1_DONE has been observed.
- 6. Read LANEN_DIG_RX_LBERT_ERR and calculate the BER
 - Bit Error = LANEN_DIG_RX_LBERT_ERR.COUNT * 2⁷ [LANEN_DIG_RX_LBERT_ERR.OV14 = 1]
 - Bit Error = LANEN_DIG_RX_LBERT_ERR.COUNT [LANEN_DIG_RX_LBERT_ERR.OV14 = 0]
 - BER = Bit Error / (stat_ld_val * 32 or 40) [32 (when rxX_width[0]=0) or 40 (when rxX_width[0]=1)]
- 7. Go back to step-3 if user needs to continually calculate the BER.

5.2.3 RX Margining

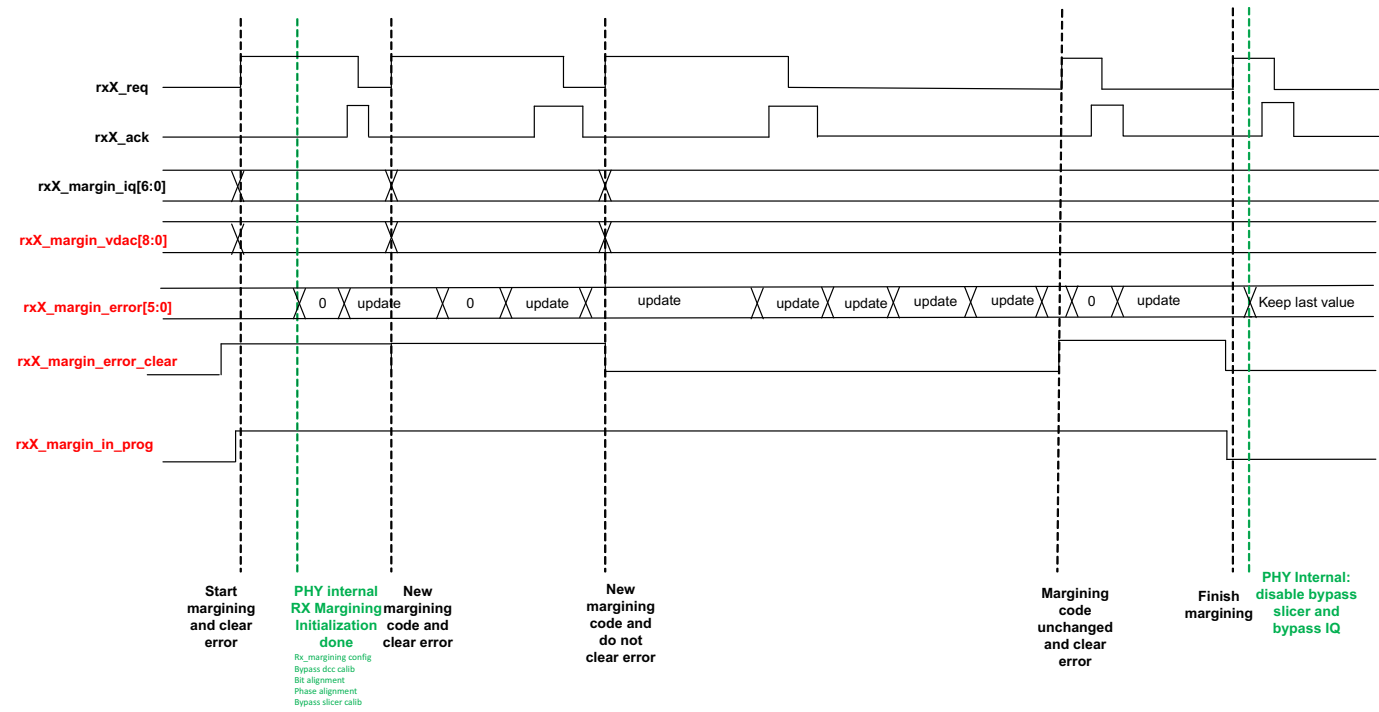
The PHY supports non-destructive horizontal and vertical RX margining when operating at 16 Gbps or 32 Gbps. The RX phase mixer (PMIX) has a nominal step size of 1/35 UI. The PHY supports a maximum horizontal margin offset of +/- 7 codes, which corresponds to a timing offset of +/- 0.2 UI (7 x 1/35). The RX slicer voltage DACs have a nominal step size of 1.85 mV. The PHY supports a maximum vertical margin offset of +/- 32 codes, which corresponds to a voltage offset of approximately +/- 60 mV (32 x 1.85).

The RX margining timing diagram is shown in [Figure 5-9](#).

The following is the general sequence required for this process:

1. To start RX margining, set margin_in_prog=1, error_clear=1 and rx_req=1.
2. The PHY is initialized internally.
3. Firmware clears the margin_error.
4. Firmware updates IQ and VDAC codes.
5. Firmware kicks off sc1_start.
6. Firmware updates margin_error.
7. Firmware informs hardware that margining has started (clears glb_irq).
8. Hardware asserts rx_ack.
9. UPCS de-asserts rx_req, and hardware de-asserts rx_ack.
10. UPCS can read margin_error output.
11. Firmware continues to periodically update error_clear output.

Figure 5-9 RX Margining Timing

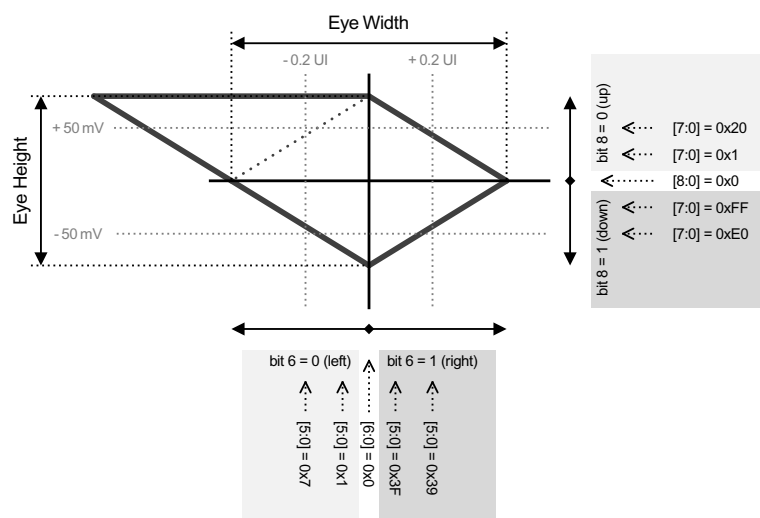


Note

- UPCS can set margin_in_prog=0 and rx_req=1 to request finish margining.
- UPCS can set error_clear=1 or margin_iq/vdac separately to request a dedicated error_clear or margin code change.

RX margining eye is depicted in [Figure 5-10](#).

Figure 5-10 RX Margining Eye



The PHY implementation of the PCIe 5.0 lane margining feature is non-destructive. It uses an error sampler and PMIX that can be aligned to the nominal data sample location and then adjusted independent of the data sampler and PMIX. The eye margins are measured by sweeping the signed `rxX_margin_iq` and `rxX_margin_vdac` inputs from the center position, and detecting mismatches between the data and error paths. Note that the error path is not unrolled and the margining routine observes the 1X (previous bit is 1) receiver eye. This means that only half of the bits received are tested during voltage and timing margining.

To perform RX margining:

1. Configure RX to enable lane margining:
 - a. Start with `rxX_margin_iq` set to 7'd0 and `rxX_margin_vdac` set to 9'd0.
2. Measure left horizontal eye margin:
 - a. Adjust the timing offset left in the eye. Add 1 to `rxX_margin_iq` and perform `rxX_req/rxX_ack` handshaking (described in Section 4.3, “[Lane Configuration Update](#)”).
 - b. Check for mismatches between error and data.
 - c. If no mismatches are found after a specific number of bits have been received (based on target BER), then loop back to step a. Otherwise, if mismatches are found, then the left edge of the eye has been detected.
 - d. Set `rxX_margin_iq` back to 7'd0.
3. Measure right horizontal eye margin:
 - a. Adjust the timing offset right in the eye. Subtract 1 from `rxX_margin_iq` (in two's complement format) and perform `rxX_req/rxX_ack` handshaking (described in Section 4.3, “[Lane Configuration Update](#)”).
 - b. Check for mismatches between error and data.
 - c. If no mismatches are found after a specific number of bits have been received (based on target BER), then loop back to step a. Otherwise, if mismatches are found, then the left edge of the eye has been detected.
 - d. Set `rxX_margin_iq` back to 7'd0.
4. Measure top vertical eye margin:
 - a. Adjust the voltage offset up in the eye. Add 1 to `rxX_margin_vdac` and perform `rxX_req/rxX_ack` handshaking (described in Section 4.3, “[Lane Configuration Update](#)”).
 - b. Check for mismatches between error and data.
 - c. If no mismatches are found after a specific number of bits have been received (based on target BER), then loop back to step a. Otherwise, if mismatches are found, then the left edge of the eye has been detected.
 - d. Set `rxX_margin_vdac` back to 9'd0.
5. Measure bottom vertical eye margin:
 - a. Adjust the voltage offset down in the eye. Subtract 1 from `rxX_margin_vdac` (in two's complement format) and perform `rxX_req/rxX_ack` handshaking (described in Section 4.3, “[Lane Configuration Update](#)”).
 - b. Check for mismatches between error and data.
 - c. If no mismatches are found after a specific number of bits have been received (based on target BER), then loop back to step a. Otherwise, if mismatches are found, then the left edge of the eye has been detected.

- d. Set rxX_margin_vdac back to 9'd0.
6. Configure RX to disable lane margining.

5.2.4 RX Internal Eye Monitor

Provided with the PHY is an internal eye monitor, which can provide data to produce a plot of time and voltage axis using internal DFT hooks built into the AFE. The eye monitor is non-destructive and can be run in parallel with live traffic. The procedure can be used for checking eye shape and the receiver equalization.



Note

- Do not use the scope eye function for screening or wafer/packaged parts sort.
- Scope eye function operates at high BER (1e-4).

5.2.5 Suggested Steps In Debugging Link-Up Issues

To help Synopsys debug your link-up issue, provide the following information:

- Nature of issue (for example, PCIe rate change fall back)
- Frequency of occurrence and reproducibility
- Process/voltage/temperature condition
- Signal integrity analysis (see Section 8.2.2, “[Signal Integrity Analysis](#)”).
- IBIS AMI link simulation data (see Section 8.2.2.2, “[AMI Model](#)”).
- Power integrity analysis (see Section 8.3.1, “[Power Supply Requirements](#)”).
- Debug Registers Dump (see [Table 5-11](#) (“[Key Debug Registers](#)”))

Table 5-11 Key Debug Registers

Register	Description	Range
RAWLANEAONN_DIG_RX_ADPT_ATT	Attenuator at the receiver input	0 to 7
RAWLANEAONN_DIG_RX_ADPT_VGA	Variable Gain Control	0 to 31
RAWLANEAONN_DIG_AFE_RTRIM	AFE resistor trim code	0 to 3
RAWLANEAONN_DIG_RX_ADPT_CTL: POLE	Continuous Time Linear Equalizer transfer function's pole	0 to 3
RAWLANEAONN_DIG_RX_ADPT_CTL: BOOST	Continuous Time Linear Equalizer transfer function's boost	0 to 31
RAWLANEAONN_DIG_RX_ADPT_AFE_RATE	RX Adapted AFE RATE	0 to 8
RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG	RX Adapted AFE CONFIG	0 to 4095
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1	Tap1 of the Decision Feedback Equalizer for Bank 0	–256 to 255
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2	Tap2 of the Decision Feedback Equalizer for Bank 0	–128 to 127
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3	Tap3 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4	Tap4 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5	Tap5 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6	Tap6 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7	Tap7 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8	Tap8 of the Decision Feedback Equalizer for Bank 0	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS	Positions for RX DFE Floating TAPs 0-3 for Bank 0	9 to 20
RAWLANEAONN_DIG_DFE_FLOAT_TAP0	Adapted Value for RX DFE Floating TAP1 and Control for Bank 0	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP1	Adapted Value for RX DFE Floating TAP2 and Control for Bank 0	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP2	Adapted Value for RX DFE Floating TAP3 and Control for Bank 0	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP3	Adapted Value for RX DFE Floating TAP4 and Control for Bank 0	–64 to 63

Register	Description	Range
RAWLANEAONN_DIG_RX_IQ_CAL_BANK_0	RX IQ Calibration Codes Value for Bank 0	0 to 39
RAWLANEAONN_DIG_RX_ADPT_IQ	Phase to data clock separation (nominal 90deg) for Bank 0	0 to 69
RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN	Error slicer reference level for Bank 0	0 to 511
RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD	Error slicer reference level for Bank 0	0 to 511
RAWLANEAONN_DIG_RX_ADPT_ATT_B1	Attenuator at the receiver input for Bank 1	0 to 7
RAWLANEAONN_DIG_RX_ADPT_VGA_B1	Variable Gain Control for Bank 1	0 to 31
RAWLANEAONN_DIG_RX_ADPT_CTLE_B1: POLE	Continuous-Time Linear Equalizer transfer function pole for Bank 1	0 to 3
RAWLANEAONN_DIG_RX_ADPT_CTLE_B1: BOOST	Continuous Time Linear Equalizer transfer function boost for Bank 1	0 to 31
RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B1	RX Adapted AFE RATE for Bank 1	0 to 8
RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B1	RX Adapted AFE CONFIG for Bank 1	0 to 4095
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B1	Tap1 of the Decision Feedback Equalizer for Bank 1	–256 to 255
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B1	Tap2 of the Decision Feedback Equalizer for Bank 1	–128 to 127
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B1	Tap3 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B1	Tap4 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B1	Tap5 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B1	Tap6 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B1	Tap7 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B1	Tap8 of the Decision Feedback Equalizer for Bank 1	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B1	Positions for RX DFE Floating TAPs 0-3 for Bank 1	9 to 20
RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B1	Adapted Value for RX DFE Floating TAP1 and Control for Bank1	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B1	Adapted Value for RX DFE Floating TAP2 and Control for Bank1	–64 to 63

Register	Description	Range
RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B1	Adapted Value for RX DFE Floating TAP3 and Control for Bank1	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B1	Adapted Value for RX DFE Floating TAP4 and Control for Bank1	–64 to 63
RAWLANEAONN_DIG_RX_IQ_CAL_BANK_1	RX IQ Calibration Codes Value for Bank 1	0 to 39
RAWLANEAONN_DIG_RX_ADPT_IQ_B1	Phase to data clock separation (nominal 90deg) for Bank 1	0 to 69
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B2	Tap1 of the of the Decision Feedback Equalizer for Bank 2	–256 to 255
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B2	Tap2 of the of the Decision Feedback Equalizer for Bank 2	–128 to 127
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B2	Tap3 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B2	Tap4 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B2	Tap5 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B2	Tap6 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B2	Tap7 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B2	Tap8 of the of the Decision Feedback Equalizer for Bank 2	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B2	Positions for RX DFE Floating TAPs 0-3 for Bank 2	9 to 20
RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B2	Adapted Value for RX DFE Floating TAP1 and Control for Bank2	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B2	Adapted Value for RX DFE Floating TAP2 and Control for Bank2	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B2	Adapted Value for RX DFE Floating TAP3 and Control for Bank2	–64 to 63
RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B2	Adapted Value for RX DFE Floating TAP4 and Control for Bank2	–64 to 63
RAWLANEAONN_DIG_RX_IQ_CAL_BANK_2	RX IQ Calibration Codes Value for Bank 2	0 to 39
RAWLANEAONN_DIG_RX_ADPT_IQ_B2	Phase to data clock separation (nominal 90deg) for Bank 2	0 to 69
RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN	Error slicer reference level	0 to 511
RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD	Error slicer reference level	0 to 511

Register	Description	Range
RAWLANEAONN_DIG_RX_ADPT_STARTUP_FOM	RX adaptation start-up figure of merit	0 to 255
RAWLANEAONN_DIG_RX_ADPT_MM_FOM	RX adaptation mission mode figure of merit	0 to 255
LANEN_DIG_RX_DPLL_FREQ	RX VCO integral path frequency register value	0 to 16384
LANEN_DIG_ASIC_TX_ASIC_IN_1: TX_MAIN_CURSOR	Value from ASIC for tx_eq_main	4 to 24
LANEN_DIG_ASIC_TX_ASIC_IN_2: TX_POST_CURSOR	Value from ASIC for tx_eq_post	0 to 32
LANEN_DIG_ASIC_TX_ASIC_IN_2: TX_PRE_CURSOR	Value from ASIC for tx_eq_pre	0 to 16
RAWCMN_DIG_AON_RTUNE_TXUP_VAL	Resistor Tune TX Up Value	0 to 1023
RAWCMN_DIG_AON_RTUNE_TXDN_V-AL	Resistor Tune TX down Value	0 to 1023
LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_HIGH_VDAC_OFST LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_HIGH_VDAC_OFST	Offset values for RX DFE Data Even High	0 to 511
LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_LOW_VDAC_OFST LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_LOW_VDAC_OFST	Offset values for RX DFE Data Even Low	0 to 511
LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_HIGH_VDAC_OFST LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_HIGH_VDAC_OFST	Offset values for RX DFE Data Odd High	0 to 511
LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_LOW_VDAC_OFST LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_LOW_VDAC_OFST	Offset values for RX DFE Data Odd Low	0 to 511
LANEN_DIG_RX_ADPTCTL_DFE_ERROR_EVEN_VDAC_OFST	Offset values for RX DFE Error Even vDAC	0 to 511

Register	Description	Range
LANEN_DIG_RX_ADPTCTL_DFE_ERRO R_ODD_VDAC_OFST	Offset values for RX DFE Error Odd vDAC	0 to 511
RAWLANEAONN_DIG_AFE_ATT_IDAC_OFST	Offset value for RX AFE ATT iDAC	0 to 511
RAWLANEAONN_DIG_AFE_BUF_IDAC_OFST	Offset value for RX AFE Buffer iDAC	0 to 511
RAWLANEAONN_DIG_AFE_CTLE_IDA C_OFST	Offset value for RX AFE CTLE iDAC	0 to 511
RAWLANENAONN_DIG_AFE_VGA1_IDA C_OFST	Offset values for RX AFE VGA1 iDAC	0 to 511
RAWLANENAONN_DIG_DFE_PHASE_E VEN_VDAC_OFST	Offset values for RX DFE Phase Even vDAC	0 to 511
RAWLANENAONN_DIG_DFE_PHASE_O DD_VDAC_OFST	Offset values for RX DFE Phase Odd vDAC	0 to 511

5.2.6 SRAM Adaptation Recording

The SRAM adaptation recording feature is the process of reading back the adaptation codes after the PHY receiver is adapted for the incoming data for each rxX_adapt_req and rxX_adapt_ack handshake.

When the SRAM recording feature is enabled, the PHY firmware stores in the SRAM the adaptation codes for each rxX_adapt_req and rxX_adapt_ack handshake. The PHY supports up to a maximum of 16 adaptations. The rxX_adapt_req and rxX_adapt_ack handshakes occur during the link equalization stage on adapted protocols, such as PCIe Gen3, Gen4, and Gen5, and USB3.1 Gen1 and Gen2.

For more details, refer to the *DesignWare Cores 32G SRAM Adaptation Recording Application Note*.

5.2.7 Temperature Sensor

This PHY includes a temperature sensor feature. This feature measures the temperature of the PHY on-die in mission mode and other power modes.



Note

For detailed instructions, refer to the *DesignWare Cores SerDes PHY Temperature Sensor Procedure Application Note*.

6

Verilog Model Description

This chapter includes the following sections:

- [“Overview”](#)
- [“Verilog Function”](#)
- [“Serial Line Modeling”](#)
- [“Verilog Model Files and Simulators”](#)
- [“Fast Simulation Mode”](#)
- [“PCIe Simulation Environment”](#)

6.1 Overview

The product deliverables include Verilog RTL for the Raw PCS and a Verilog behavioral model of the PMA. Contrary to the Verilog RTL for the Raw PCS, which can be synthesized into a gate-level netlist by a synthesis tool, the Verilog behavioral model of the PMA includes a netlist of generic GTECH gates and a non-synthesizable behavioral model of the analog circuits.

In the set of deliverables, the Raw PCS Verilog RTL and the PMA Verilog behavioral model reside at:

- Raw PCS: ./Latest/phy/rtl
- PMA Verilog model: ./Latest/pma/behavior

Together, they form the Verilog model of the PHY (dwc_e32mp_phy_tsmc12ffcll_x2ns.v; N=Number of lanes). The PHY model allows you to run functional verification of your logic along with the PHY.



Note

This PHY model and testbench do not represent an example of exhaustive PHY testing.

6.2 Verilog Function

The Verilog model simulates using a four-state (0, 1, X, Z) simulator. Therefore, several features that require analog voltage and current values transmitted over wires cannot be simulated with this model.

Features that are not accurately modeled in the PHY model are as follows:

- Receive and transmit I/O terminations
- Termination resistor tuning and measurement
- Voltage-level setting for the Receiver Electrical Idle Detector
- AC signal characteristics of the ACJTAG receiver
- Voltage margining
- Analog test buses; the PHY pins atb_f_p, atb_s_m, atb_s_p are inaccurate during simulations
- Receiver detection

The receiver detection status output from the PHY txX_detrx_result is inaccurate during simulations. To simulate the receiver detection feature, you should force the receiver detection status on txX_detrx_result when txX_ack is asserted in response to the assertion of txX_detrx_req, and release the force after txX_ack is de-asserted.

- Transmitter equalization settings (pre-shoot, swing, and post-cursor) on the transmit serial lines txX_p/m
- Receiver equalizer settings (AFE CTLE boost and VGA gain, and DFE taps)

The RX Evaluation outputs rxX_adapt_fom, rxX_txmain_dir, rxX_txpost_dir, and rxX_txpre_dir (described in Section 4.12, “TX and RX Adaptation”) are inaccurate during simulations. To simulate link adaptation, you should force the required settings at these PHY outputs when rxX_adapt_ack is asserted, and release the forces after rxX_adapt_ack is de-asserted.

6.3 Serial Line Modeling

The serial lines (txX_p/m, rxX_p/m) are fundamentally analog signals. Because each wire is limited to the simulator's four states, not all aspects of the analog signals can be reflected. [Table 6-1](#) provides encodings of the analog transmitter model outputs.

Table 6-1 Transmitter Model Serial Encodings

Output	txX_p	txX_m
Differential 1	1	0
Differential 0	0	1
No signal, holding common mode	0	0
No signal, not holding common mode	Z	Z

The analog receiver model interprets only differential 0 or differential 1 encodings as valid signal levels.

6.4 Verilog Model Files and Simulators

The Verilog model is generated for simulations and can be simulated using Synopsys VCS simulators.

Table 6-2 describes each portion of the Verilog model and indicates the product deliverables directory where the portion is stored.

Table 6-2 PHY Model Database Structure

Directory	Description
./Latest/phy/include	Contains macros used in the Raw PCS, PMA and simulation files.
./Latest/pcie/include	Contains macros used in the PCIE PCS and simulation files.
./Latest/phy/rtl	Contains Verilog RTL for the Raw PCS files ■ dwc_e32mp_phy_xn_ns_pcs_raw_xN.v: Top-level RTL file for the xN Raw PCS, where N = 1, 2, 4, 6, or 8
./Latest/pma/behavior	Contains Verilog RTL for the behavioral models for the PMA. ■ dwc_e32mp_phy_xn_ns_gtech.v: Top-level file for the xN PMA instantiating GTECH model of the digital and behavioral model of analog sub-blocks, where N= 1, 2, 4, 6, or 8.

Instructions for the PHY simulations are provided in the TESTBENCH.README file under ./Latest/phy/testbench directory. The simulation database structure for PHY simulations is provided in Table 6-3.

Table 6-3 Simulation Database Structure

Directory	Description
./Latest/phy/sim	VCS regression scripts for GTECH and gate-level simulations of the PHY, which includes the following: ■ run_<simulator>_gtech: Simulation run script for PMA GTECH model and Raw PCS RTL (simulator = vcs) ■ run_<simulator>_gates: Simulation run script for PMA technology specific gates model and Raw PCS RTL (simulator = vcs) ■ regression_gtech.csh: Regressions script with multiple calls to run_<simulator>_gtech for different test cases, where <simulator> is passed as input argument to regression_gtech.csh ■ regression_gates.csh: Regressions script with multiple calls to run_<simulator>_gates for different test cases, where <simulator> is passed as input argument to regression_gates.csh
./Latest/phy/testbench	Sample test-cases for the behavioral and gate-level simulations. ■ TESTBENCH.README: Run instructions for the PHY Testbench.



Note

To simulate the PHY model, define the DWC_E32MP_X2NS_SIM_MODE and DWC_E32MP_X2NS_RANDOM_SEED macros.

6.5 Fast Simulation Mode

Some features, such as power-state transitions, receiver-detect, and RX adaptation, may take long simulation times in normal operating modes. These simulation times can be reduced for simulation purposes by either using fast simulation macros or by writing into the PHY control registers through the Parallel Control Register Interface or through JTAG (for information, see Section 4.14, “Control Register Interface”). In addition, there are certain Verilog force statements that need to be included in the customer testbench to make simulations work.

For more information, see “./phy/testbench/TESTBENCH.README” file.

Within the PHY RTL wrappers ./phy/rtl/<prefix>.v, there exist two modules that are used to force certain values when operating in fast simulation mode. The two modules are described below:

- <phy_prefix>_force_freq_tune.v
Forces the RX VCO calibration code based on the reference clock frequency and the RX VCO configuration inputs
- <phy_prefix>_mpll_coarse_tune.v
Forces the MPLL calibration code based on the reference clock frequency and the MPLL configuration inputs

These modules are automatically loaded when running RTL simulations because they exist in the PHY RTL wrapper. However, if the RAW_PCS has been synthesized, these modules are removed, and as a result must be instantiated and connected as they are in the PHY RTL wrapper.

The fast simulation mode can be enabled by defining the macro “DWC_E32MP_X2NS_SHORT_RESET.” The macro based fast simulation mode is only supported for the RTL model of the Raw PCS. The fast simulation mode is supported for both GTECH and gate-level models of the PMA. In real silicon or with synthesized Raw PCS, the following sequence can be used to modify the internal Raw PCS FSM operation:

1. Set phy_reset and all of rxX_reset and txX_reset to '1'.
2. Wait for 10 ns or a minimum of 10 μ s after the power-supplies are ramped-up (at power-up).
3. Set phy_reset to '0'.
4. Wait a minimum of 10 phy_ref_dig_clk cycles to ensure that the Control Register interface is operational.
5. Write the registers fields as described in Table 6-4 before the assertion of sram_ext_ld_done.
6. Set all of rxX_reset and txX_reset to '0' for lanes, which are to be operational.

Table 6-4 Registers for Fast Simulation Mode

Register	Register Field	Value to be programmed for SHORT_RESET
SUP_DIG_RTUNE_CONFIG2	FAST_RTUNE	1'b1
SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1	FAST_MPLL_LOCK	1'b1
SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1	FAST_MPLL_PWRUP	1'b1
SUP_DIG_BG_CTL_BG_PWRUP_TIME_0	FAST_BG_WAIT	1'b1
SUP_DIG_BG_CTL_REF_PWRUP_TIME_0	FAST_REF_WAIT	1'b1
SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1	FAST_MPLL_LOCK	1'b1
SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1	FAST_MPLL_PWRUP	1'b1
LANEX_DIG_TX_PWRCTL_TX_PWRUP_TIME_0	FAST_TX_CLK_EN_TIME	1'b1
LANEX_DIG_TX_PWRCTL_TX_PWRUP_TIME_0	FAST_TX_REFGEN_EN_TIME	1'b1
LANEX_DIG_TX_PWRCTL_TX_PWRUP_TIME_2	FAST_TX_ROPLL_EN_TIME	1'b1
LANEX_DIG_TX_PWRCTL_TX_PWRUP_TIME_4	FAST_TX_SERIAL_EN_TIME	1'b1
LANEX_DIG_TX_ROPLL_CTL_ROPLL_OVRD	FAST_ROPLL_LOCK	1'b1
LANEX_DIG_TX_ROPLL_CTL_ROPLL_OVRD	FAST_ROPLL_PWRUP	1'b1
LANEX_DIG_RX_PWRCTL_RX_PWRUP_TIME_0	FAST_RX_SIGDET_EN_WAIT	1'b1
LANEX_DIG_RX_PWRCTL_RX_PWRUP_TIME_1	FAST_RX_VREG_EN	1'b1
LANEX_DIG_RX_PWRCTL_RX_PWRUP_TIME_2	FAST_RX_CLK_DCC_EN	1'b1
LANEX_DIG_RX_ADPTCTL_ADPT_CFG_19	FAST_AFE_DFE_SETTLE	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_AFE_ADAPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_AFE_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_DCC_CAL_STARTUP	1'b1

Register	Register Field	Value to be programmed for SHORT_RESET
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_DFE_ADAPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_DFE_CAL_EXT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_DFE_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_FOM_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_IQ_ADAPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_IQ_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_IQ_DELTA_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_PHS_CAL_EXT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_PHS_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_REF_CAL_EXT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_REF_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_TX_INC_DEC_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL	SKIP_RX_VGA_EXT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_AFE_CAL_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_DCC_CAL_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_DFE_ADAPT1_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_DFE_ADAPT25_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_DFE_CAL_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_FOM_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_PHS_CAL_CONT	1'b1

Register	Register Field	Value to be programmed for SHORT_RESET
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_REFLVL_ADAPT_CONT	1'b1
RAWLANEAONX_DIG_RX_CONT_ALGO_CTL	SKIP_RX_VGA_ADAPT_CONT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_ADAPT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_COARSE_ADAPT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_CONT_ADAPT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_CONT_CAL_ADAPT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_PWRUP	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_STARTUP_CAL	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_VCO_CAL	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_RX_VCO_WAIT	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_SUP	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_TX_PWRUP	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_TX_RXDET	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_TX_CMN_MODE	1'b1
RAWLANEAONX_DIG_FAST_FLAGS	FAST_TX_STARTUP_CAL	1'b1
RAWLANEAONX_DIG_SRAM_REC_CTRL	REC_EN	1'b0
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_ADAPT_RELOAD	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_BUF_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_BYPASS_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_DCC_BYPASS_CAL_STARTUP	1'b1

Register	Register Field	Value to be programmed for SHORT_RESET
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_DCC_CAL_RATE	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_DCC_DATA_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_DCC_PHASE_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_ERROR_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_FULL_RATE_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_HALF_RATE_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_IQC_CAL_RATE	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_RX_VGA_SLICER_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_SIGDET_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_TX_DCC_CAL_RATE	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_TX_DCC_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_1	SKIP_VGEN_CAL_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_ATT_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_CTLE_BOOST_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_CTLE_POLE_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_DFE_TAP1_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_DFE_TAP2_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_IQ_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_VGA_COARSE_ADPT_STARTUP	1'b1
RAWLANEAONX_DIG_RX_STARTUP_ALGO_CTL_2	SKIP_RX_MARGINING	1'b1

Register	Register Field	Value to be programmed for SHORT_RESET
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_RATE_TX_DCC_RANGE	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_RX_DFE_FLT_ADAPT_CONT	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_RX_DFE_FLT_ADAPT_STARTUP	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_STARTUP_ATT	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_STARTUP_BUF	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_STARTUP_TX_DCC_RANGE	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_VGA2_CALIB	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_RATE_RX_DCC_RANGE	1'b1
RAWLANEAONX_DIG_SKIP_RX_DFE_FLT_ADAPT	SKIP_STARTUP_RX_DCC_RANGE	1'b1
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE	MPLL_SKIPCAL_COARSE_TUNE	8'h21
LANEX_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2	FREG_TUNE_START_VAL	10'h1d1

**Note**

- Asserting phy_reset resets the registers to default values. To preserve fast simulation mode, the registers enabling the fast simulation mode must be written after every phy_reset de-assertion.
- The DAC behavior is not modeled in the PMA digital model. Therefore, the DAC offset calibration can take a long time, up to 15 ms. For digital simulations, to speed up runtime, it is recommended to define the FAST_RX_STARTUP_CAL macro or write "1" to the FAST_RX_STARTUP_CAL register field.
- For gate-level short_reset simulation, define the macro FAST_RX_VCO_CAL for fast RX VCO calibration.

6.5.1 Interrupt Disable Mode

Even when running in fast simulation mode, the firmware is interrupted any time there is a rate change request to the PHY. The firmware checks each incoming rate change request to determine if any restorations are required (in fast simulation mode, all calibrations are skipped) and performs them accordingly. This process can take up to approximately 10 μ s per request. To disable the interrupts, define the `DWC_E32MP_X2NS_DISABLE_RATE_IRQ` macro in addition to the `DWC_E32MP_X2NS_SHORT_RESET` macro. Defining this macro improves the rate change times in `DWC_E32MP_X2NS_SHORT_RESET` mode by disabling firmware interaction during rate change requests to the PHY.

**Note**

Defining `DWC_E32MP_X2NS_DISABLE_RATE_IRQ` removes the TX and RX DCC calibration and restoration on rate change.

The firmware does not perform any calibrations or restorations during rate changes. The incoming request to change MPLL/TX/RX banks is ignored. Calibration codes for the MPLL are still updated, because they are driven from the `dwc_e32mp_phy_x4_ns_force_mpll_coarse_tune.v` file included in the PHY deliverables. Note that when performing rate changes, the correct MPLL settings must still be applied to the PHY inputs to obtain the correct MPLL clock frequency.

6.5.2 Skip Speed Change Calibration

You can set the following two forces:

```
force `PHY0_TOP.pcs_raw.aon.aon_laneX.rx_cal_done = 1
```

```
force `PHY0_TOP.pcs_raw.aon.aon_laneX.tx_cal_done = 1
```

By setting these, the FSM—which executes the calibration routine the first time a new data rate is achieved—checks and verifies the calibration has already been performed. In this way, the calibration routine is skipped in simulation which shortens the simulation time.

6.6 PCIe Simulation Environment

The deliverables include a PCIe simulation environment including the `dwc_e32mp_phy_xn_ns_pcs_raw` and the Multi-Protocol 32G PHY as the Design Under Test (DUT). For more information on the simulation environment, see the “Simulation” chapter in the *Multi-Protocol 32G PCS for the DesignWare Cores 32G PHY Databook*.

7

Physical-Level Implementation

This chapter includes the following sections:

- “PMA Assembly”
- “PMA Placement”
- “PMA Fill”
- “Operating Conditions”
- “Power Supply Connectivity”
- “ESD”
- “Reference Clock”
- “Reference Resistor”
- “Test Bus Connectivity”
- “Raw PCS Synthesis”

7.1 PMA Assembly

This section provides information about the bump map, pad size, and PMA size.

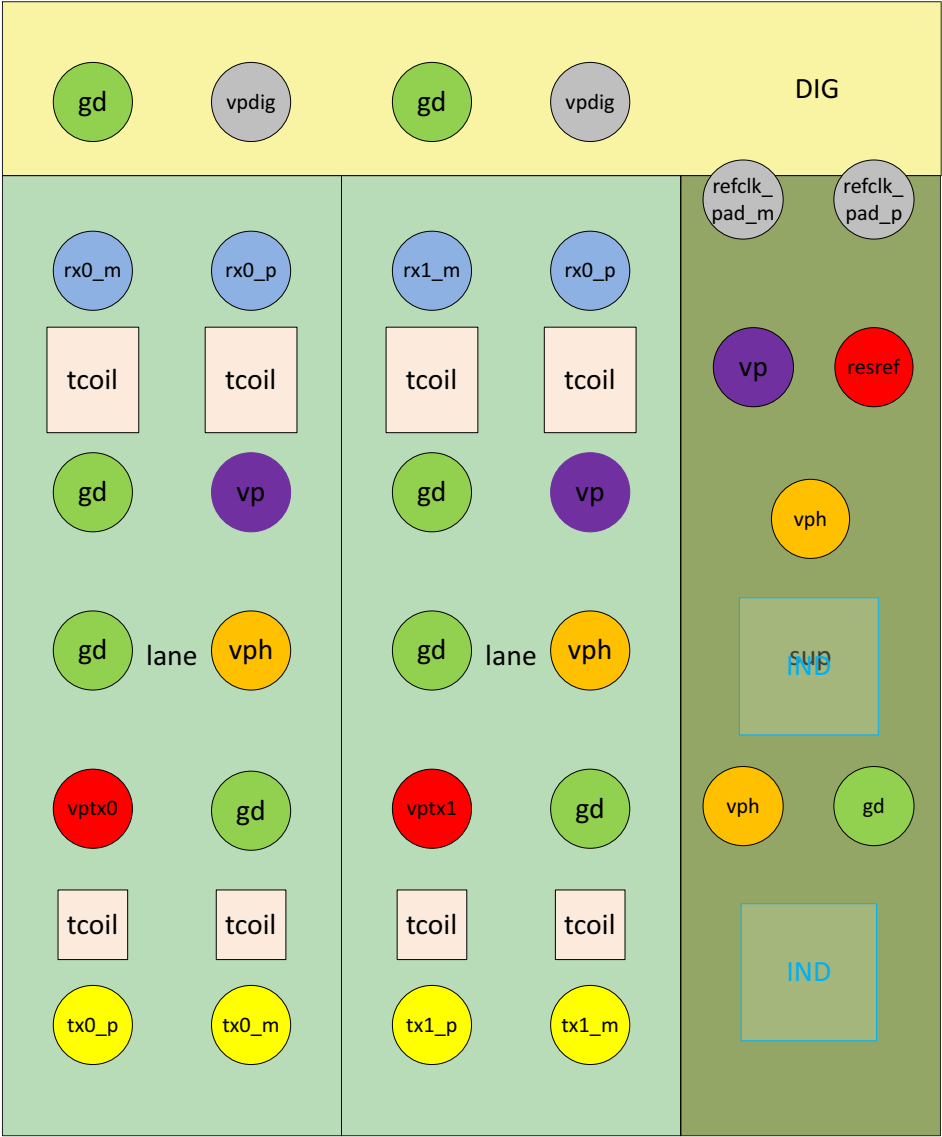
7.1.1 Flip-Chip Pinout

 **Note**

The bump maps shown in this section are for reference purposes only. For dimensions, refer to the .lef in the release package.

An example of the assembled PMA pinout is shown in [Figure 7-1](#).

Figure 7-1 Bump Map Example



7.1.2 Bump Pad RDL and UBM Sizing

PMA Bump pitch, bump type and sizing have been designed based on optimized bump placement for the IP. The bump pad redistribution layer (RDL), passivation opening and under-bump metalization (UBM) sizing are based on foundry required dimensions for the PMA bump pitch.

Table 7-1 Bump Pitch Information

Drawn Bump Pitch (μm)	Bump Type	Drawn UBM (μm)	Drawn RDL (μm)	Technology Shrink Factor
153-163.2	C4 (Controlled Collapse Chip Connection) - Lead Free Flip Chip Build Up	81.6	90.7	0.98



Note

Bump pitches listed in [Table 7-1](#) define the bump type, size, and design rule options which were used for IP verification. This does not mean that all bumps in the IP have the exact same maximum bump pitch or average bump pitch spacing; the bumps on the IP meet the following requirements:

- All bumps are spaced at greater than the minimum pitch spacing.
- All bumps are using the same bump dimensions (associated with the pitch listed in [Table 7-1](#)).
- All bumps meet the design rules related to this bump pitch option.

For PMA bump pin location information for package design, refer to the RDL pin locations in the LEF delivery file.

For different bump pitch, pad type, or spacing customization options, contact Synopsys.



Note

- Ensure that the UBM has the correct size. Foundry and packaging typically require that the UBM shape used for the bump is the same for all bumps within the same die.
- The size of the UBM shape is dictated by the minimum bump pitch in the die.

7.1.3 PMA Size

The size of the PMA is determined by the size of the following:

- PMA hard macro
- Flip-chip pads

The estimated size of the hard macro building blocks for X2 PMA are shown in [Table 7-2](#).

Table 7-2 X2 PMA Size

Block	Drawn X (μm)	Drawn Y (μm)	Drawn Area (mm^2)	Shrink Factor	Actual Area (mm^2)
PMA	1050.48	1255.68	1.319066726	0.9604	1.266831684

7.2 PMA Placement

This section describes things to consider when placing the assembled PMA.

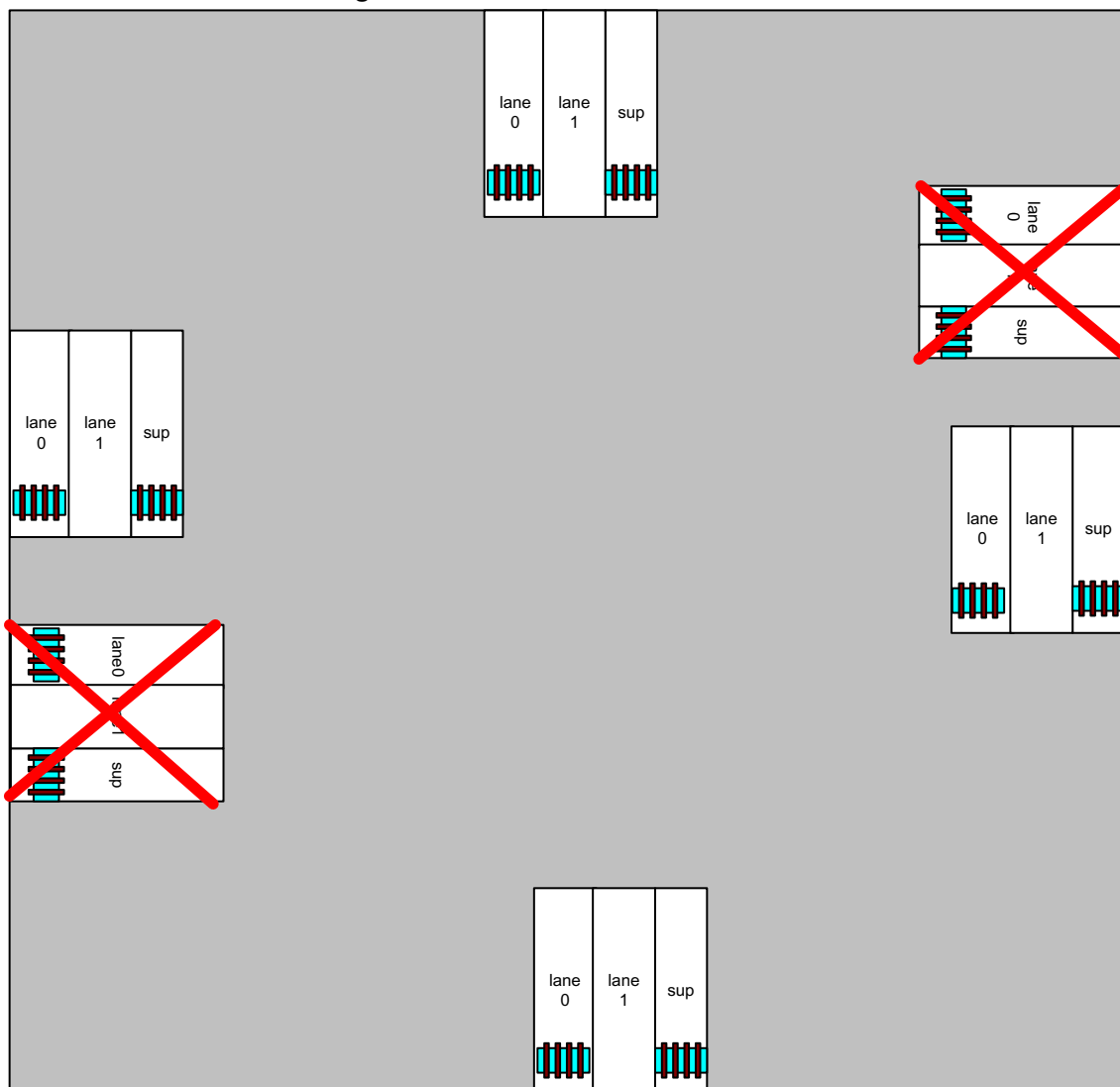
7.2.1 PMA Rotation and Mirroring

The FET POLY gates within the PMA must be vertical according to foundry rules.

The PMA hard macro is designed to be placed with preference for the north or south side of the die, as shown in [Figure 7-2](#).

The preferred placement of the PHY is on the north or south edge of the die. If you place the PHY on the east or west edge, it may complicate the on-die interface routing, as well as, the package routing of the TX and RX pairs. If placing the PMA on the east or west side is required, contact Synopsys customer support.

Figure 7-2 PMA Rotation and Mirroring



The PMA may be rotated by 180°, but not 90° or 270° due to the vertical POLY requirement.

The PMA may be mirrored in either the X or Y direction as this does not change POLY orientation.

7.2.2 PMA Abutment

The following requirements must be met when placing the PMA adjacent to other circuitry:

- Spacing Requirements
 - PMA-to-PMA:
 - When multiple PHYs are being placed adjacent to each other, the minimum spacing requirement must be either 5 μm or the value that is required to meet the minimum bump pitch used (see Section 7.1.2, “Bump Pad RDL and UBM Sizing”). Figure 7-3 shows a PHY abutment example.

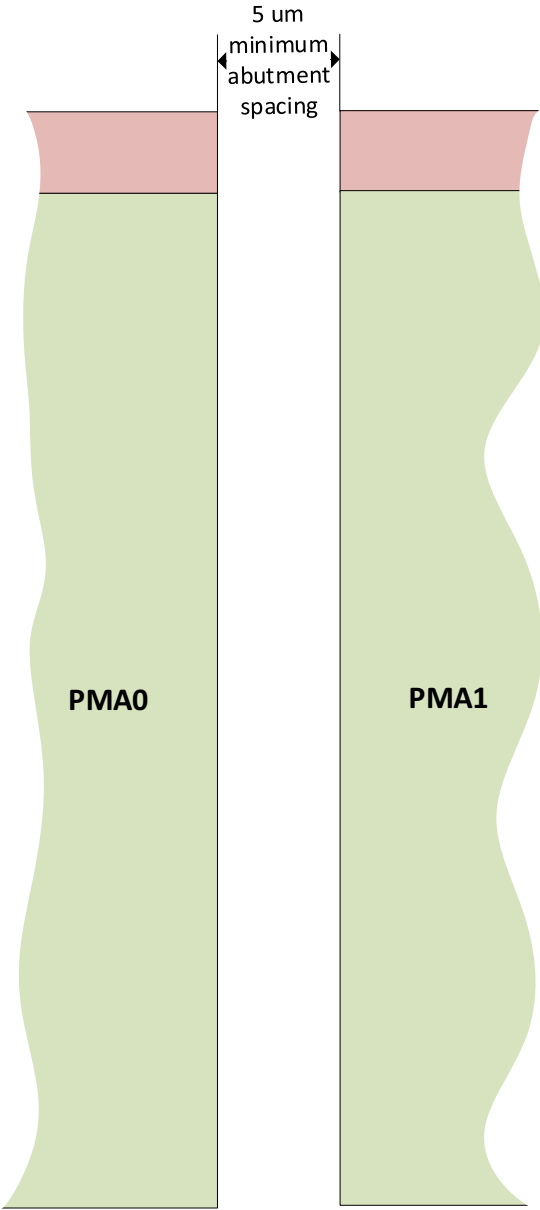


Note

Figure 7-3 (“PMA Abutment”) is for illustration purposes only. Actual bump locations may be different. For accurate details, refer to the LEF view.

- Place a column of GD pads to isolate TX and RX routes of the adjacent PMAs. To accommodate smaller spacing, you can remove the column of GD pads. However, minimum bump pitch requirements must be met.
- Do not place the x1 and x2 PMA such that the clock support block is adjacent to the clock support block of the next PMA (that is, next PMA is instantiated in a flip-X)
- PMA-to-core logic:
 - Spacing for a x1, x2, and x4, and x8 requires a 5- μm “moat” on east and west edge
 - Spacing for x2 and x1 requires a 20- μm moat on the edge of the PMA that has the clock support block exposed to the edge — on the unflipped or rotated PMA, this is the east edge. The other side of the PMA still has the 5- μm requirement.
- Moat requirements:
 - The “moat” cannot contain any core logic circuitry including cells, signal routes, and power meshes.
 - The “moat” cannot contain any substrate taps.

Figure 7-3 PMA Abutment



7.3 PMA Fill

The PMA comes pre-filled. When you run density layer fill at the chip level, no fill shapes are added within the PMA.

7.4 Operating Conditions

This section details the operating conditions of the PMA.

7.4.1 Power Supply Limits

The PMA supports the power supply voltages in [Table 7-3](#).

Table 7-3 Supported Power Supply Limits at the PMA Macro Bumps

Supply	Description	Minimum	Typical	Maximum	Unit
vph	High Voltage I/O supply	1.2 -5%	1.2/1.5/1.8	1.8 +10%	V
vp/vptxX/vpdig	Analog supply	0.9 -5%	0.9	0.9 +5%	V
vddcore	Core supply	0.8 -10%	0.8/0.9	0.9 +5%	V



Note

- Allowed supply variation in [Table 7-3](#) might be lower than the supply variation found in the PMA libs.
- In your design, to use nominal foundry limits for the ASIC core supply, level shifters are inserted in the Synopsys PHY at the RAW PCS-PMA interface.

7.4.2 Temperature

This section provided the operating temperature ranges supported by the PMA and the temperature drift range in mission mode.

7.4.2.1 Operating Temperature

The PMA supports an operating temperature range from -40–125°C.

7.4.2.2 Temperature Drift

The PHY is designed and validated to operate at any junction temperature within the operating temperature range.

During the mission mode (P0 state) operation, the PHY junction temperature can change from an initial link state due to SoC self-heating and a change in ambient conditions.

The PHY supports 125°C temperature drift range in mission mode.

Example:

Start entry into P0 at $T_J = -20^\circ\text{C}$ and change or drift up to $T_J = 105^\circ\text{C}$, which is an overall 125°C drift in temperature.

7.4.3 Electromigration

The electromigration limit of the PMA is 88KPOH@110°C (accounting for localized self heating - SHE).

**Note**

Operating at conditions that exceed the electromigration limits of PMA reduces the lifetime of the PMA. While the PMA can meet functional requirements at higher temperature, this is expected to be a short term condition, such as temporary failure in cooling, and so on.

7.5 Power Supply Connectivity

This section provide details about power and ground isolation, as well as on-chip supply decoupling if indicated after running power integrity simulations.

7.5.1 Power and Ground Isolation

Power supply connectivity must be designed accounting for all noise sources, decap, parasitics, package, and so on. You must perform a power integrity analysis.

To prevent power supply noise from inducing jitter on the transmitted and received waveforms, ensure that the PHY power supplies are isolated from the power supplies used on the rest of the ASIC. For more information about supply connectivity within the package and on the board, see Section 8.3, “[Power Integrity – Supplies Requirements](#)”.

The PHY provides power pins for purpose of additional on-die decap, if needed, as determined by your power integrity analysis of the PHY plus the package plus the board. Due to jitter and isolation requirement these pins must not be used to connect the PHY to other on-die supplies.

The PHY ground, gd, is isolated on the die from the ASIC core ground vsscore_X. Back-to-back diodes between the ASIC ground (connected to vsscore_X pins) and the PHY ground, gd, are provided within the PHY macro as part of the ESD protection strategy (described in Section 7.6.2, “[ESD Protection Strategy](#)”), but no direct metal-metal connection is permitted.

Isolation of all power and ground on the die implies the only connection between the PHY hard macro and the rest of the ASIC are for signal wires. The signal integrity of this interface is maintained using a fully clock-forwarded architecture where both clock and data cross the power domains together, so that subtle differences in supply voltages have no impact on timing.

7.5.2 On-Chip Supply Decoupling

If power integrity simulations indicate that additional PHY power supply decoupling is needed, one option is to include that capacitance on-chip. This may also be advantageous for utilizing space in between PHY instantiations or along the die-edge.

For this purpose, the vp, vph, and gd pins have been exposed on the last two metal layers.



Note

Power integrity of the macro supplies must be considered early in the design process and can impact both on-die implementation (if additional on-die decap is needed) as well as decap design and board and package implementation. For more details on power integrity, including package and board requirements, see Section 8.6, “[Board Requirements—External Components](#)”.

7.6 ESD

This section presents ESD considerations.

7.6.1 ESD Requirements

ESD performance is evaluated by the amount of voltage and current that the protection circuits can absorb during an ESD event while keeping voltages below damaging levels. The protection circuits are designed to meet the protection levels provided in [Table 7-4](#).

Table 7-4 Minimum ESD Protection Levels

Stress Type	Minimum Protection Level	JEDEC Class
Charged Device Model (CDM) [ANSI/ESDA/JEDEC JS-002-2018]	6A peak discharge current	C2a/C1 (500 V/250 V) ^a
Human Body Model (HBM) [ANSI/ESDA/JEDEC JS-001-2017]	2,000 V	2
Latch-up [JEDEC JESD78F]	Signal Pin Test: 100 mA @ 125°C Supply Test: 1.5 x V _{maxSup} @ 125°C	II.A

a. Support for both 250 V or 500 V CDM target level is dependent on maximum discharge current generated in final SOC/package implementation.

7.6.2 ESD Protection Strategy

The ESD protection strategy of the PHY provides the following features:

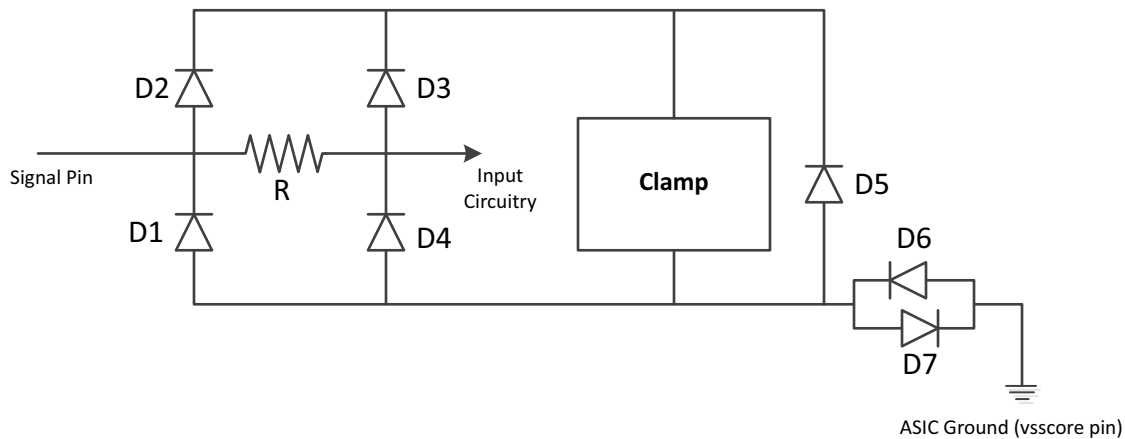
- Large primary diodes between signal pad and PHY power and ground supplies
- A series resistor and secondary diodes between signal pad and PHY power and ground supplies
- An active clamp between the PHY power and ground supplies

The clamp and primary diodes are placed close to each other within the PHY, minimizing the resistance between them and maximizing the effectiveness of the ESD circuits.

- Back-to-back diodes between PHY ground, gd, and interfacing (ASIC) ground, vsscore_X
- Sufficient metal width in all ESD paths

Figure 7-4 shows the ESD strategy to implement in the Multi-Protocol 32G PHY.

Figure 7-4 ESD Protection Strategy

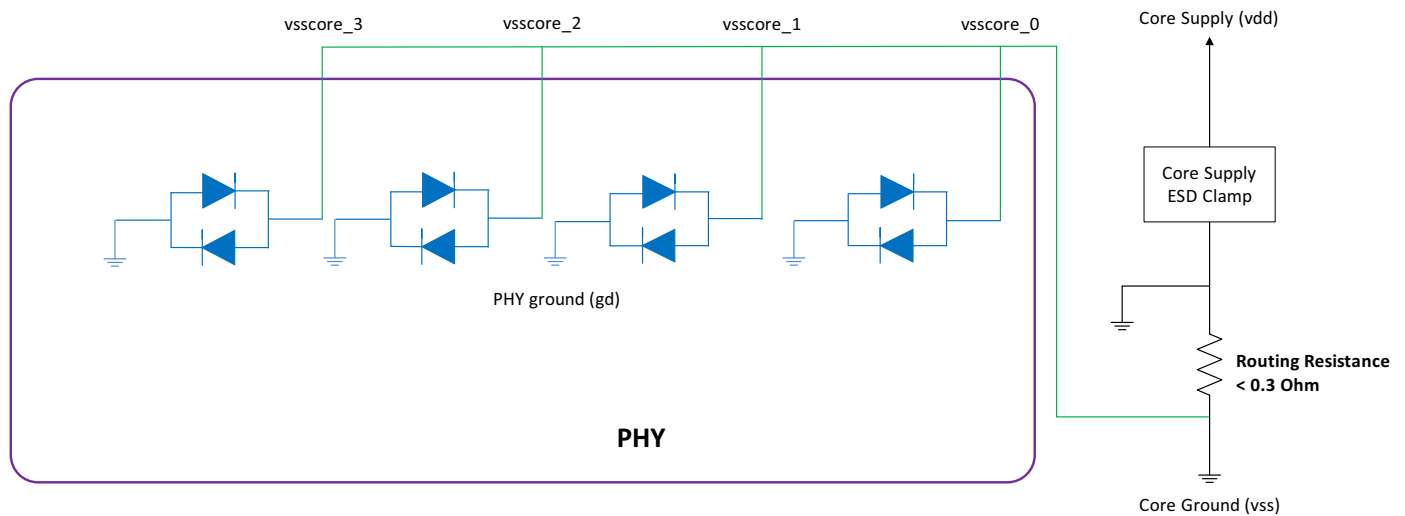


During ESD events, the primary D1 and D2 diodes direct most of the current to the PHY power and ground buses through the clamp. This resistance of this path is minimal, which keeps the signal pad voltage and voltage across the PHY power buses low. The secondary D3 and D4 protection diodes, together with the series resistor R, further reduces the voltage for the input circuitry. The back-to-back D6 and D7 diodes enable the flow of the current between the ground domains, thus preventing high voltage drops across the ground domains. Back-to-back diodes between different ground domains prevent damage between cross-power domain interface circuitry.

7.6.3 ESD Back-to-Back Diode Connection to Core VSS

Although vsscore and gd are to be connected together within the package, for ESD protection of the PHY/ASIC interface a back-to-back diode connection between vsscore and gd is provided within the PHY.

Each of the vsscore pins on the PHY must have a strong connection to the ASIC vsscore supply grid. The vsscore pins provided in the IP must be connected with a solid metal. There must be less than 0.3 Ohms of resistance between the each vsscore pin on the PHY and the closest clamp within the ASIC (see Figure 7-5).

Figure 7-5 Back-to-Back Diodes in PHY**Note**

- The x4 configuration has vsscore_3, vsscore_2, vsscore_1 and vsscore_0; the x2 configuration has vsscore_1 and vsscore_0; the x1 configuration only has vsscore_0.
- Add a vdd-to-vss clamp close to the back-to-back diodes with no more than 0.3 Ω routing resistance from the diodes to the clamp or with a value specified by the foundry Design Rule Manual (DRM).

Additional back-to-back diodes outside of the PHY (or other vss coupling devices) can be used to connect ASIC ground and PHY gd for additional ESD protection if desired.

If non-ground supplies are connected together for ESD protection through back-to-back diodes, special care must be taken to check for, and guard against latch-up, power-up sequencing and leakage issues.

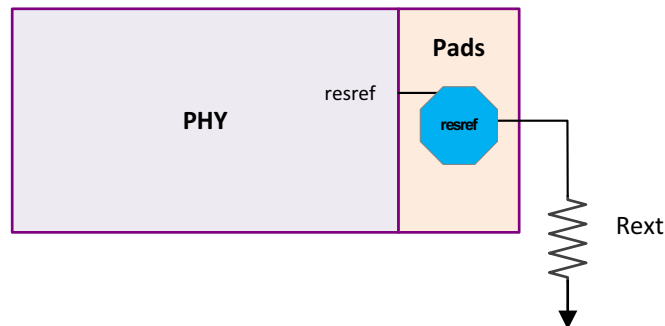
7.7 Reference Clock

For details about reference clock physical implementation, see [Chapter 9, “Reference Clock Implementation”](#).

7.8 Reference Resistor

An external reference resistor is used for calibration of the RX and TX termination. The external resistor is automatically used for calibration when the PHY comes out of reset and when a manual tuning request is made. When a PHY is using the reference resistor, it forces current across the resistor. When a PHY is not using the reference resistor, the current is disabled.

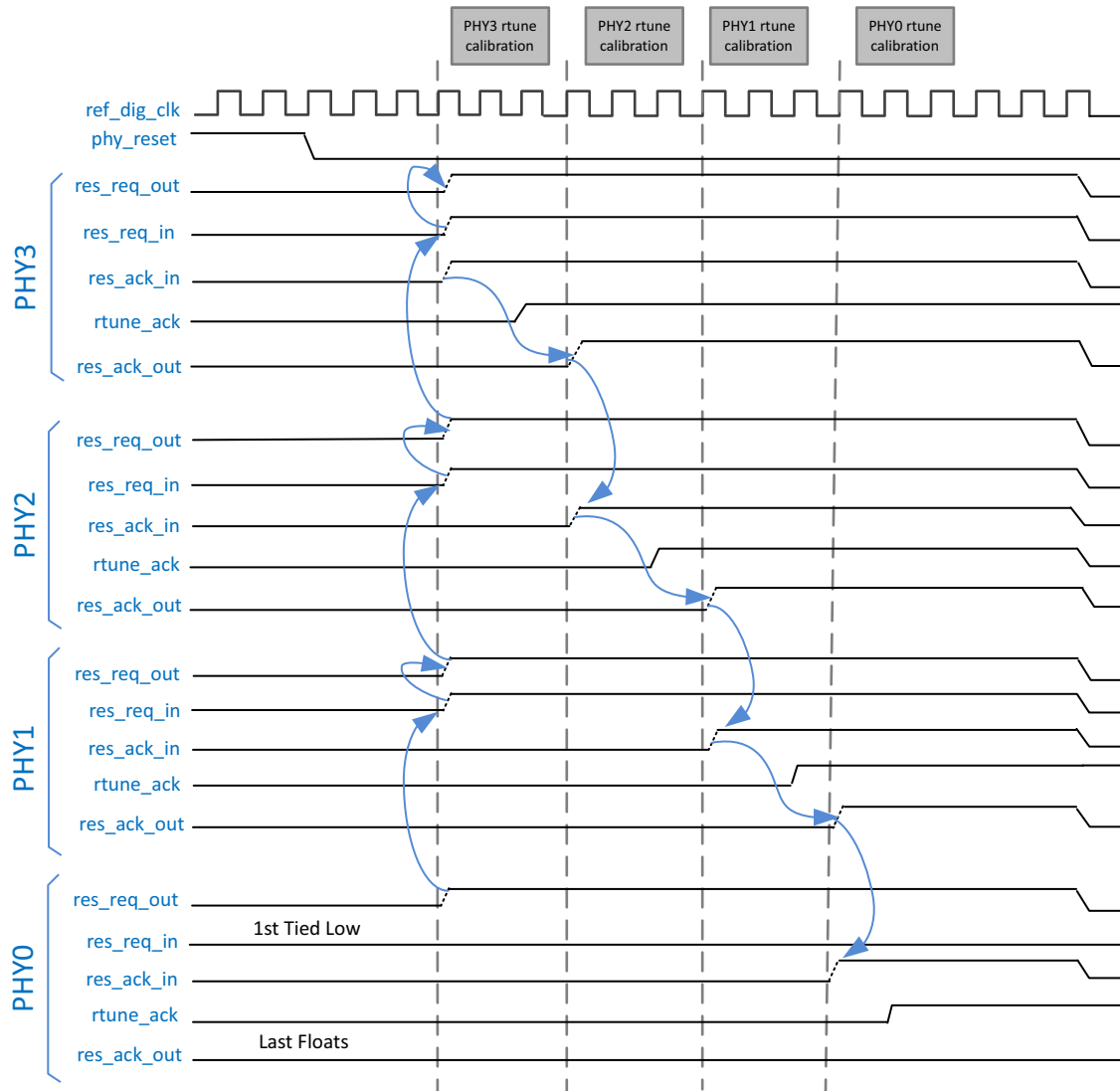
Figure 7-6 Reference Resistor Physical Implementation



7.8.1 Reference Resistor Contention Bus

Each PHY requires access to a reference resistor. This access may be per PHY or it may be shared. A contention bus is used to control PHY access to the external resistor using `res_req_in`, `res_req_out`, `res_ack_in`, and `res_ack_out`.

The reference resistor contention bus diagram is shown in [Figure 7-7](#).

Figure 7-7 Reference Resistor Contention Bus Diagram

For a single-PHY implementation, connect the signals as shown in [Figure 7-8](#).

For sharing a reference resistor among several PHYs, connect the signals as shown in [Figure 7-9](#).

**Note**

- When sharing a reference resistor amongst several PHYs, it may take up to 30 μ s per PHY to sequentially calibrate each PHY.
- Before choosing to share a reference resistor, make sure to be aware of how it is connected in the package and on the board. For information on package and board implementation, see [Chapter 8, "Board- and Package-Level Implementation"](#).

Figure 7-8 Reference Resistor Contention Bus - Single PHY

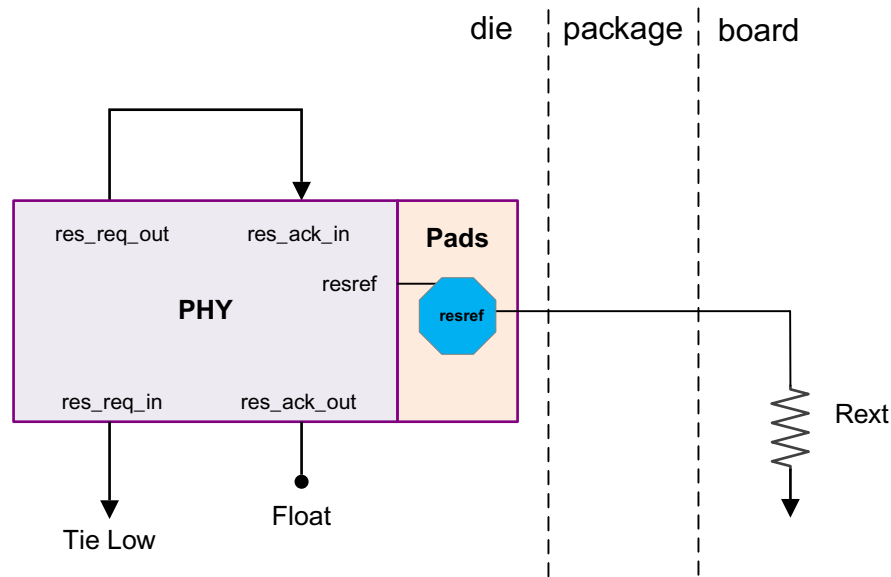
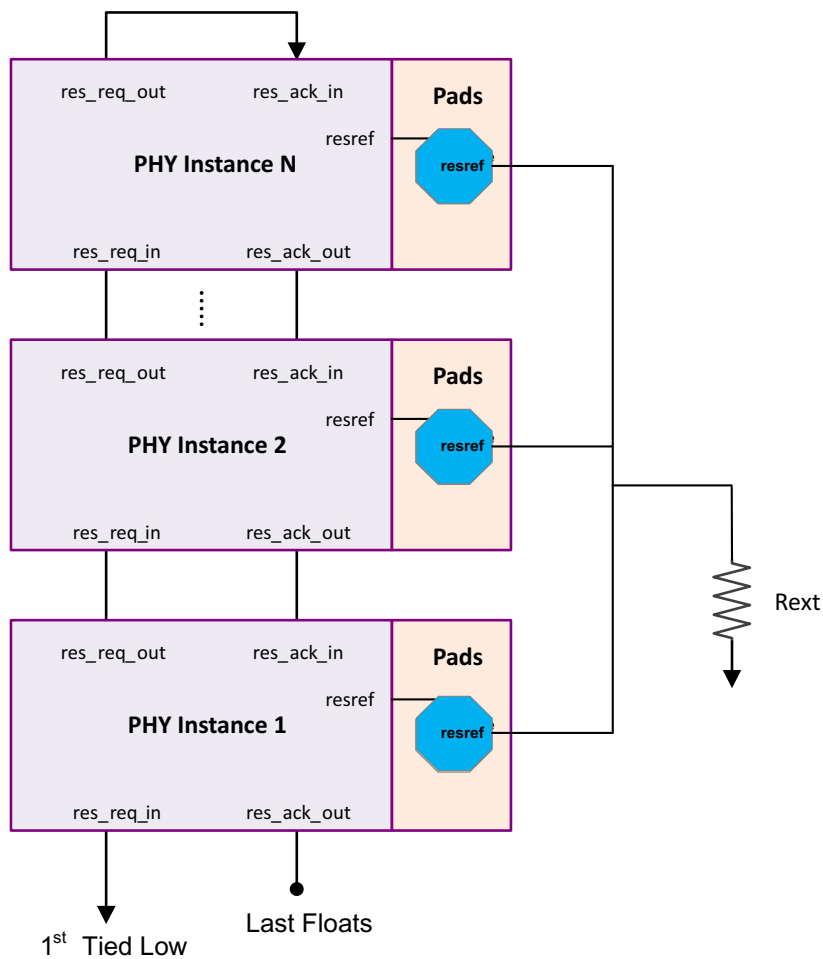


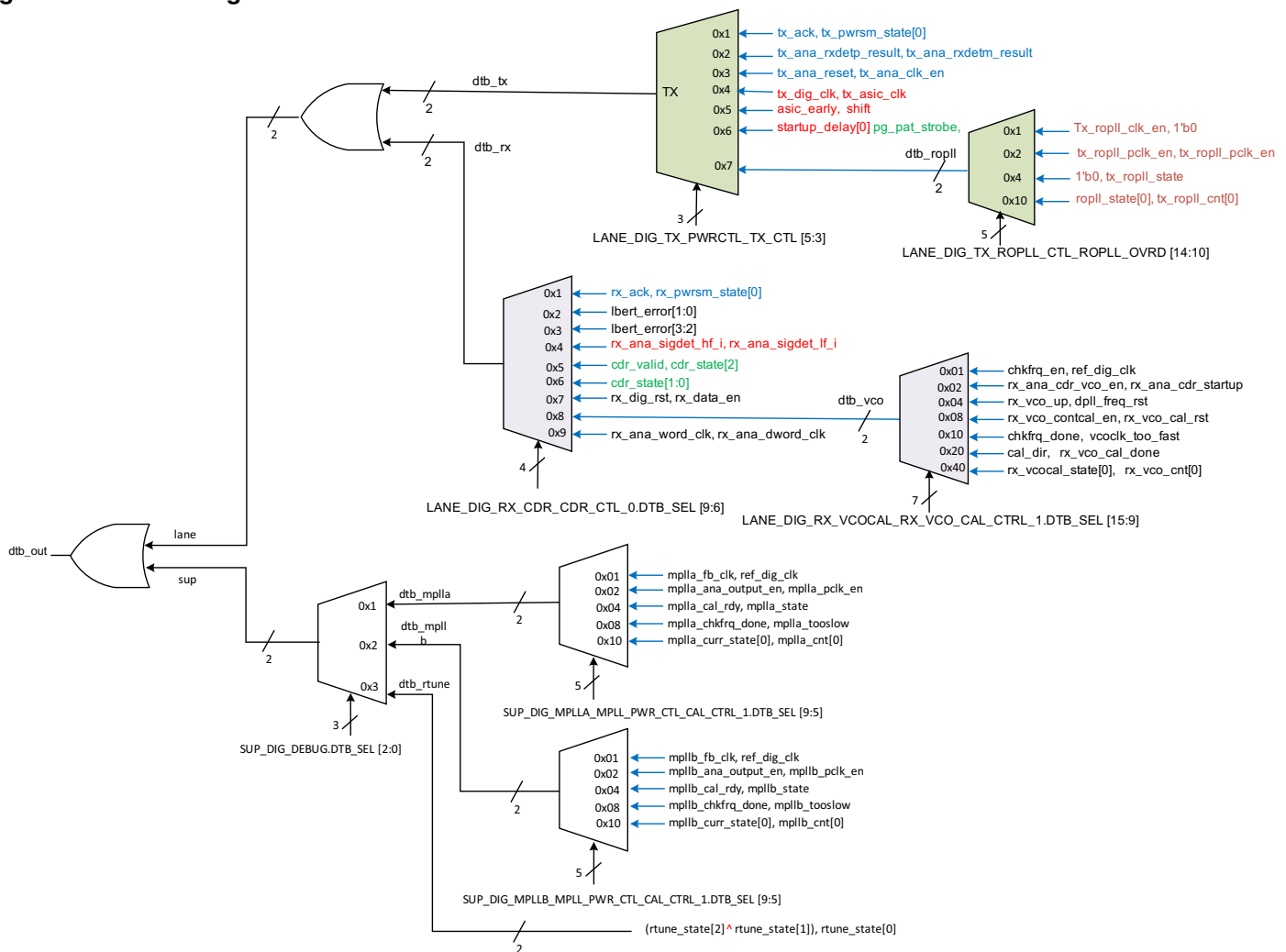
Figure 7-9 Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor

7.9 Test Bus Connectivity

The Analog Test Bus (ATB) is reserved. When integrating the PHY, leave the atb_s_p, atb_s_m, and atb_f_p pins unconnected.

The Digital Test Bus (DTB) is provided as a means of accessing internal digital signals for debug purposes. If this bus is not needed, when integrating the PHY, you can leave the dtb_out[1:0] pins unconnected. The DTB diagram is displayed in Figure 7-10.

Figure 7-10 DTB Diagram



7.10 Raw PCS Synthesis

The Raw PCS is delivered as soft RTL, meaning that it is hardened along with the rest of the RTL of the protocol-PCS controller and ASIC. Synthesizing the Raw PCS requires using a Verilog-2001 compatible synthesis tool.

7.10.1 Synthesis Database

The deliverables include the Design Compiler synthesis scripts in TCL format that can be used to synthesize the Raw PCS logic using the timing model of the PMA. The synthesis database structure is shown in [Table 7-5](#) and the ATPG deliverables are listed in [Table 7-6](#).

Table 7-5 Synthesis Database Structure for PHY

Directory	Description
./Latest/phy/synth	Synthesis directory for PHY ■ run_synth.sh: Shell script to run synthesis using Design Compiler TCL scripts. ■ run_formal.sh: Shell script to run logic equivalence checks between RTL and synthesized netlist of the Raw PCS using Formality. You need to update the variables in the scripts to match the installed technology library location (TECH_DIR, STD_DB). These scripts must be run in the directory ./Latest/phy/synth.
./Latest/phy/synth/scripts	Synthesis and Formal scripts ■ dwc_e32mp_phy_xn_ns_synth.tcl: Design Compiler Synthesis script to synthesize dwc_e32mp_phy_xn_ns_pcs_raw_xN using RTL files and libs for dwc_e32mp_phy_pma_x[1,2]_ns PMA. ■ dwc_e32mp_phy_xn_ns_phy_dft_insert.tcl: DFT synthesis script for Raw PCS and PMA ■ dwc_e32mp_phy_xn_ns_phy_con.tcl: Design Compiler synthesis constraint file for the Raw PCS in TCL format. ■ dwc_e32mp_phy_xn_ns_formal.tcl: Formality script for running logic equivalence check between RTL and the synthesized netlist of Raw PCS. Where N represents the number of lanes.

Table 7-6 ATPG Deliverables for the PHY

Directory	Description
./Latest/pma/atpg/<metal_stack>/ctl	CTL and SPF model
./Latest/pma/atpg/<metal_stack>/models	Verilog scan model of the PMA; only to be used by TetraMax (not to be used for simulations)
./Latest/pma/atpg/<metal_stack>/rpt	TetraMax log file
./Latest/pma/atpg/<metal_stack>/tetramax	run_tmax.sh: Top-level shell script to run TetraMax TCL script (tmax_atpg.tcl) for generating vectors

**Note**

For information on Synthesis flow and database for the PCS, refer to *Multi-Protocol 32G PCS for the DesignWare Cores 32G PHY Databook*.

7.10.2 PHY (RAW PCS and PMA) Constraints

The synthesis constraints are provided in the form of TCL script used by Synopsys Design Compiler. This file (dwc_e32mp_phy_xn_ns_phy_con.tcl) describes the design intent and surrounding constraints for synthesis, clocking, timing, power, test, and environmental and operating conditions. The constraints include the input and output delays of interface pins, clock definitions, and any timing path exceptions in the design, such as false paths or multicycle paths.

In most cases, the Raw PCS is synthesized in conjunction with other logic that resides on the opposite side of the ASIC interface boundary. In this situation, timing constraints can be automatically inferred from the adjoining logic and from the PMA timing model to which the Raw PCS is connected. When synthesizing the Raw PCS as a standalone block, signals on the ASIC interface must be constrained so that sufficient time is budgeted for the adjoining logic to be connected later.

7.10.2.1 Hand-Placed Logic Constraints

There are generic, digital-cell, primitive modules within the Raw PCS that are used to instantiate a specific digital cell that must not be logically optimized during synthesis. These modules are typically used around asynchronous clock domain crossing (CDC) points within the design. It is essential that these modules are processed correctly, not just synthesized as usual with the Raw PCS logic. Failure to do so will likely create glitch-generating and reconvergent logic that can cause problems on the ASIC interface.

The modules are listed in [Table 7-7](#).

Table 7-7 Primitive Modules

Module ^a	Function	Hand-Instantiated Cell Required?
*_gen_sync	3-stage data signal synchronizer	Yes ^b
*_gen_rst_sync	Asynchronous assert, synchronous de-assert reset synchronizer	Yes
*_ani_and2	2 input logic AND gate	Yes
*_ani_and3	3 input logic AND gate	Yes
*_ani_or2	2 input logic OR gate	Yes
*_ani_buf	Logic BUF	Yes
*_ani_clk_gate	Clock-gating cell	Yes
*_ani_clk_and2	2 input clock AND gate	Yes
*_ani_clk_and3	3 input clock AND gate	Yes
*_ani_clk_or2	2 input clock OR gate	Yes
*_ani_clk_mux	2 input clock MUX	Yes

Module ^a	Function	Hand-Instantiated Cell Required?
*_ani_mux	2 input MUX	Yes
*_gen_mux	2 input MUX wrapper with configurable input port width - instantiates ani_mux module	No
*_gen_mux4	4 input MUX wrapper with configurable input port width - instantiates gen_mux module	No
*_gen_and2	2 input logic AND wrapper - instantiates ani_and2 module	No
*_gen_and3	3 input logic AND wrapper - instantiates ani_and3 module	No
*_gen_or2	2 input OR wrapper - instantiates ani_or2 module	No
*_gen_clk_mux	2 input clock MUX wrapper - instantiates ani_clk_mux module	No
*_gen_clk_gate	Clock-gating cell wrapper - instantiates ani_clk_gate module and gen_mux module for scan muxing	No
*_gen_clk_div2	2-division clock logic wrapper	No
*_gen_clk_and2	2 input clock AND wrapper - instantiates ani_clk_and2 module	No
*_gen_clk_and3	3 input clock AND wrapper - instantiates ani_clk_and3 module	No
*_gen_clk_or2	2 input clock OR wrapper - instantiates ani_clk_or2 module	No
*_gen_bus_sync_flag	Bus synchronizer with unstable data flag output - instantiates gen_sync module for each bit	No
*_gen_bus_sync	Bus synchronizer without unstable data flag output - instantiates gen_sync module for each bit	No
*_gen_buf	BUF wrapper - instantiates ani_buf module	No
*_gen_pcatch	Pulse catcher - instantiates a gen_sync module for synchronization and a gen_mux for scan muxing	No
*_pcs_raw_gen_glcm	Functional glitchless clock mux (GLCM) - instantiates the following modules: ■ gen_sync ■ gen_clk_and2 ■ gen_clk_gate ■ gen_clk_or2	No
*_pcs_raw_gen_glcm_psel	Functional glitchless clock mux (GLCM) supporting 2 clock enable ports with different priority - instantiates the following modules: ■ gen_sync ■ gen_clk_and2 ■ gen_clk_and3 ■ gen_clk_gate ■ gen_clk_or2	No

- a. There are two additional cells called `hand_ana_mux` and `hand_ana_buf` in the design. Do not hand instantiate during place and route because they reside within the analog GDS. They exist as analog leaf cells in the behavioral netlist.
- b. The all hand-instantiate 3-stage synchronizer cells must have the same function as the `*_gen_sync` module in RTL, including `RST_VAL` and `RST_LOW` parameters. Raw PCS logic requires certain `*_gen_sync` modules to have a reset value of 1; these synchronizers are identified in RTL as `pcs_raw_gen_sync #(.RST_VAL(1))`.

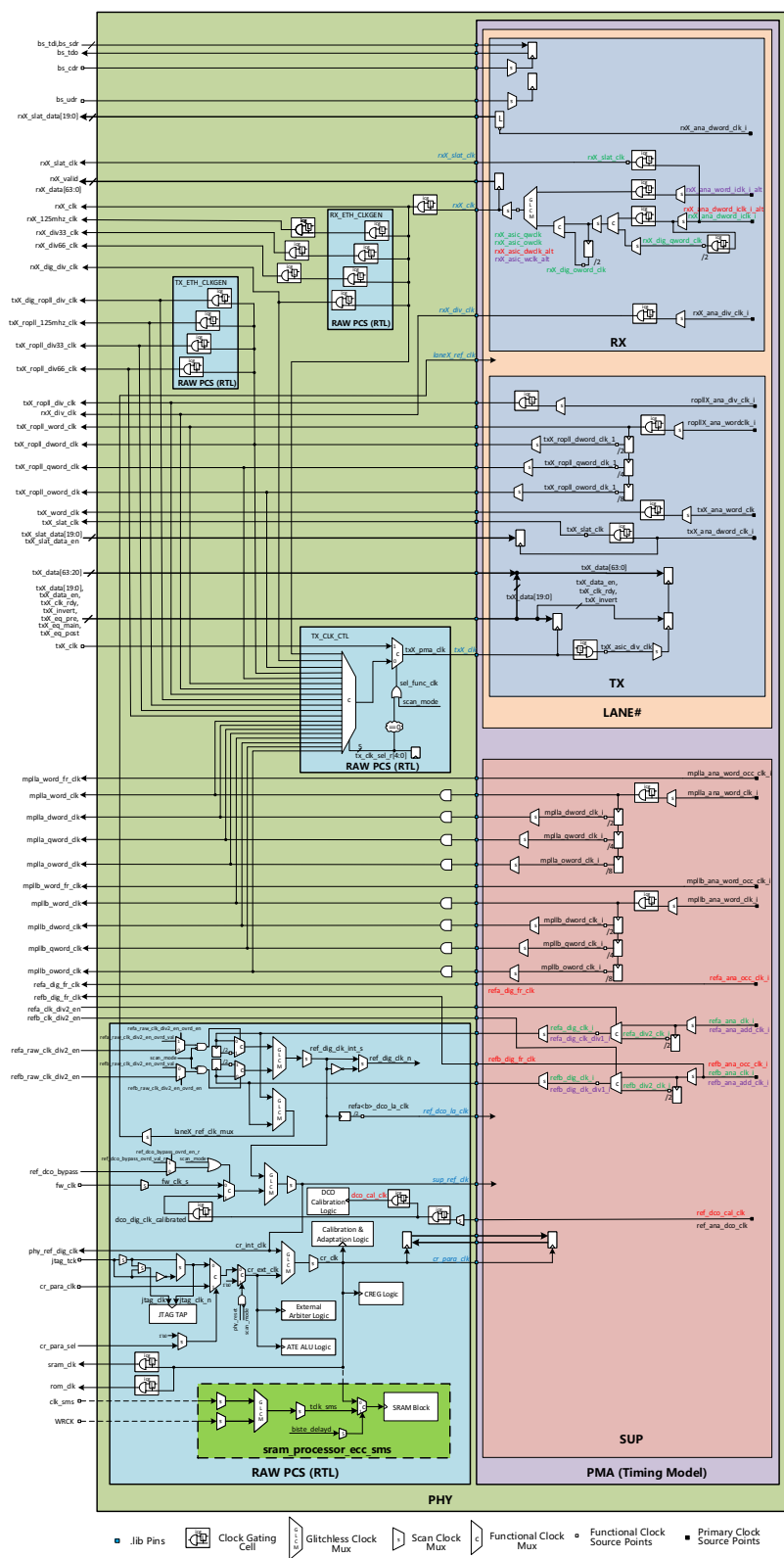
To prevent these modules from being optimized during synthesis, customers can either replace the contents of these modules with an instantiation of the appropriate standard cell from a digital library, or synthesize each module stand-alone with a restricted library, so that only the correct cell is selected by the tool. In both cases, a “`set_size_only`” attribute must be applied to prevent reoptimization of the instance during regular Raw PCS synthesis.

7.10.2.2 Clock Constraints

The `txX_clk` and `jtag_tck` port clocks are used both as functional and scan clocks. For scan design, the external scan multiplexer is needed for `txX_clk` and `jtag_tck`. The `txX_clk` signal cannot be connected to `scan_mpll_dword_clk` because it is a different `SCAN_ASST` clock group.

[Figure 7-11](#) shows the clocking structure for functional clocks, including the legends for clock sources (primary or generated clocks), clock gating cells, and scan clock multiplexers.

Figure 7-11 Functional Clock Definition Diagram for Raw PCS Synthesis^a



a, "sram_processor_ecc_sms" are internal sram for Internal SRAM mode, it does not exist for External SRAM mode.

The clock definitions and associated false paths for the PHY are provided in the constraints file `dwc_e32mp_phy_xn_ns_phy_con.tcl` delivered as part of the synthesis deliverables. The following tables summarize the functional clock and scan clock definitions:

- [Table 7-8 \(“PHY Clocks Definition in RAW PCS^a”\)](#)
- [Table 7-9 \(“PMA .lib Generated Clocks Definition \(It is Automatically Defined When PMA .lib is Linked\)^a”\)](#)
- [Table 7-10 \(“Scan Clocks Definitions at PHY \(PMA plus Raw PCS\) Level^a”\)](#)
- [Table 7-11: \(“Relation between Output Clocks and Scan Clocks”\)](#)
- [Table 7-12 \(“Clock Uncertainty Definitions at PHY \(PMA plus Raw PCS\) level^a”\)](#)

**Note**

The constraints provided are for the highest frequency supported by the PHY (that is 32.44 Gbps). If there is a need to ease the constraints for lower data rates, contact Synopsys for customization.

Table 7-8 PHY Clocks Definition in RAW PCS^a

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
MPLL Clock Definition						
pma/mplla_ana_word_clk_i	pma/mplla_ana_word_clk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA MPLLA clock. (source clock creation)
Reference Clock Definition						
pma/refa_ana_clk_i	pma/refa_ana_clk_i	No	N/A	N/A	2ns {0 1}	PMA REFA clock. (DIV2 source clock creation)
pma/refa_ana_add_clk_i	pma/refa_ana_clk_i	No	N/A	N/A	8ns {0 4}	PMA REFA clock. (DIV1 source clock creation)
pma/refa_ana_occ_clk_i	pma/refa_ana_clk_i	No	N/A	N/A	4ns {0 2}	PMA REFA clock used for OCC. (source clock creation)
pma/refa_dig_clk_div1_i	pma/refa_dig_clk_div1_i/pma/refa_dig_clk_i	Yes	pma/refa_ana_add_clk_i	1	8ns {0 4}	PMA REFA/B digital DIV1 clock, used when source reference clock is less than or equal to 175 MHz.
refa_dig_clk_div2	pcs_raw/aon/aon_cmnr/refa_dig_clk_div2_reg/Q	Yes	pma/refa_dig_clk_i	2	8ns {0 4}	The RAW 2-divided clock for pma/refa_dig_clk_i.
pma/ref_ana_dco_clk	pma/ref_ana_dco_clk	No	N/A	N/A	5.7144ns {0 2.8572}	PMA REF DCO clock used for FW when input reference clock frequency is low.
ref_dco_cal_clk	pma/ref_ana_dco_clk	No	N/A	N/A	2ns {0 1}	DCO calibration clock used for DCO calibration only.
refa_dco_la_clk	pcs_raw/cmnr/dcoctrl/dco_la_clk_pre_reg/Q	Yes	refa_dig_clk_div2	2	16ns {0 8}	DCO latch clock used to sample the DCO tune value.
RX Clock Definition						

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
pma/rxX_ana_dword_iclk_i	pma/rxX_ana_dword_iclk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA RX dword clock inside PMA analog. Used for 32 Gbps mode and below. (source clock creation)
pma/rxX_ana_dword_clk_i	pma/rxX_ana_dword_clk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA RX dword clock inside PMA analog. Used for 32 Gbps mode and below. (source clock creation)
pma/rxX_ana_dword_iclk_i_alt	pma/rxX_ana_dword_iclk_i_alt	No	N/A	N/A	1ns {0 0.5}	PMA RX ALT dword clock inside PMA analog. Only used for 16 Gbps mode and below. (source clock creation)
pma/rxX_ana_word_iclk_i_alt	pma/rxX_ana_word_iclk_i_alt	No	N/A	N/A	1ns {0 0.5}	PMA RX ALT word clock inside PMA analog. Used for 8 Gbps mode and below. (source clock creation)
pma/rxX_ana_div_clk_i	pma/rxX_ana_div_clk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA RX divide clock inside PMA analog. Used for 32 Gbps mode and below. (source clock creation)
rxX_asic_owclk	rxX_clk	Yes	pma/rxX_clk pma/rxX_asic_owclk	1	2ns {0 1}	RX oword clock, used for 32 Gbps mode and below.
rxX_asic_qwclk	rxX_clk	Yes	pma/rxX_clk pma/rxX_asic_qwclk	1	1ns {0 0.25}	RX qword clock, used for 32 Gbps mode and below.
rxX_asic_dwclk_alt	rxX_clk	Yes	pma/rxX_clk pma/rxX_asic_dwclk_alt	1	1ns {0 0.5}	RX dword clock, only used for 16 Gbps mode and below.
rxX_asic_wclk_alt	rxX_clk	Yes	pma/rxX_clk pma/rxX_asic_wclk_alt	1	1ns {0 0.5}	RX word clock, used for 8 Gbps mode and below.

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
rxX_div33_clk_int	pcs_raw/laneX/lane_pcs_xface/rx_div33_clk_gate/hand_clk_gate/U6/X	Yes	pma/rxX_asic_qwclk	edges {1 2 3}	1ns {0 0.25}	RX div33 clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
rxX_div66_clk_int	pcs_raw/laneX/lane_pcs_xface/rx_div66_clk_gate/hand_clk_gate/U6/X	Yes	pma/rxX_asic_qwclk	edges {1 2 3}	1ns {0 0.25}	RX div66 clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
rxX_dig_div_clk_int	pcs_raw/laneX/eth_clkgen_rx_inst/rx_dig_div_clk_gate/hand_clk_gate/U6/X	Yes	pma/rxX_asic_qwclk	edges {1 2 3}	1ns {0 0.25}	RX dig div clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
rxX_125mhz_clk_int	pcs_raw/laneX/lane_pcs_xface/rx_125mhz_clk_gate/hand_clk_gate/U6/X	Yes	pma/rxX_asic_qwclk	edges {1 2 3}	1ns {0 0.25}	RX 125Mhz clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
rxX_slat_clk_int	rxX_slat_clk	Yes	pma/rxX_ana_dword_iclk_i	1	0.5ns {0 0.25}	RX slat clock used for rxX_slat_data.
TX Clock Definition						
pma/txX_ana_word_clk	pma/txX_ana_word_clk	No	N/A	N/A	0.5ns {0 0.25}	PMA TX word clock. (source clock creation)

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
pma/txX_ana_dword_clk_i	pma/txX_ana_dword_clk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA TX dword clock. (source clock creation)
pma/ropllX_ana_div_clk_i	pma/ropllX_ana_div_clk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA TX ROPLL div clock. (source clock creation)
pma/ropllX_ana_wordclk_i	pma/ropllX_ana_wordclk_i	No	N/A	N/A	0.5ns {0 0.25}	PMA TX ROPLL word clock. (source clock creation)
txX_clk	txX_clk	No	N/A	N/A	1ns {0 0.5}	TX clock.
txX_ropll_div33_clk_int	pcs_raw/laneX/eth_clkgen_tx_ropll_inst/tx_ropll_div33_clk_gate/hand_clk_gate/U6/X	Yes	pma/txX_ropll_dword_clk_1	edges {1 2 3}	1ns {0 0.5}	TX div33 clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
txX_ropll_div66_clk_int	pcs_raw/laneX/eth_clkgen_tx_ropll_inst/tx_ropll_div66_clk_gate/hand_clk_gate/U6/X	Yes	pma/txX_ropll_dword_clk_1	edges {1 2 3}	1ns {0 0.5}	TX div66 clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
txX_dig_ropll_div_clk_int	pcs_raw/laneX/eth_clkgen_tx_ropll_inst/tx_dig_ropll_div_clk_gate/hand_clk_gate/U6/X	Yes	pma/txX_ropll_dword_clk_1	edges {1 2 3}	1ns {0 0.5}	TX dig ropll div clock. It is the pulse gated clock. To simplify the constraint, here the division is 1. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
txX_ropll_125mhz_clk_int	pcs_raw/laneX/eth_clkgen_tx_ropll_inst/tx_ropll_125mhz_clk_gate/hand_clk_gate/U6/X	Yes	pma/txX_ropll_dword_clk_1	edges {1 2 5}	2ns {0 0.5}	TX 125Mhz clock. It is the pulse gated clock. To simplify the constraint, here the division is 2. For the detailed clock definition, refer to the constraints. Exact clock edges definition based on RAW RTL design is preferred if necessary.
txX_slat_clk_int	txX_slat_clk	Yes	pma/txX_ana_dword_clk_i	1	0.5ns {0 0.25}	TX slat clock used for txX_slat_data.
Others						
bs_cdr	bs_cdr	No	N/A	N/A	10ns {0 5}	The boundary scan clock for capture DR, directly driving PMA.
bs_udr	bs_udr	No	N/A	N/A	10ns {0 5}	The boundary scan clock for update DR, directly driving PMA.
cr_para_clk ^b	cr_para_clk	No	N/A	N/A	8ns {0 4}	CR parallel interface clock.
jtag_clkb	jtag_tck	No	N/A	N/A	8ns {0 4}	JTAG interface clock.
fw_clk	fw_clk	No	N/A	N/A	5.7144ns {0 2.8572}	Fast FW clock to lower the FW's latency.

a. The clock periods are based on PCIe5 32 Gbps data rate. For JESD 32.44 Gbps, the period of "pma/mplla_ana_word_clk_i" for example is 0.4932ns instead of 0.5ns, for all of the detail clock periods of JESD 32.44 Gbps, refer to the PHY synthesis clock report in which the synthesis setting option of DATA_RATE is set to 32440. JESD 32.44 Gbps support is available as a customization, contact Synopsys if such customization is required.

b. The maximum period can be up to 5.7144ns if supporting 700-MHz reference clock. 700-MHz reference clock can be supported with a harden macro customization.

Table 7-9 PMA .lib Generated Clocks Definition (It is Automatically Defined When PMA .lib is Linked)^a

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
PMA MPLL Clock						
pma/mplla_word_fr_clk	pma/mplla_word_fr_clk	Yes	pma/mplla_ana_word_clk_i	1	0.5ns {0 0.25}	PMA MPLLA/B word free-running clock.
pma/mplla_dword_clk_i	pma/mplla_dword_clk_i	Yes	pma/mplla_ana_word_clk_i	2	1ns {0 0.5}	PMA MPLLA/B dword clock.
pma/mplla_qword_clk_i	pma/mplla_qword_clk_i	Yes	pma/mplla_ana_word_clk_i	4	2ns {0 1}	PMA MPLLA/B qword clock.
pma/mplla_oword_clk_i	pma/mplla_oword_clk_i	Yes	pma/mplla_ana_word_clk_i	8	4ns {0 2}	PMA MPLLA/B oword clock.
PMA Reference Clock						
pma/refa_div2_clk_i	pma/refa_div2_clk_i	Yes	pma/refa_ana_clk_i	2	4ns {0 2}	PMA REFA/B DIV2 clock.
pma/refa_dig_clk_div1_i	pma/refa_dig_clk_div1_i	Yes	pma/refa_ana_clk_i	1	2ns {0 2}	PMA REFA/B digital DIV1 clock, used when source reference clock is less than or equal to 100Mhz.
pma/refa_dig_clk_i	pma/refa_dig_clk_i	Yes	pma/refa_div2_clk_i	1	4ns {0 2}	PMA REFA/B digital DIV2 clock, used when source reference clock is greater than 100Mhz.
PMA TX Clock						
pma/txX_asic_div_clk	pma/txX_asic_div_clk	Yes	pma/txX_clk txX_clk	edges {1 2 3}	1ns {0 0.5}	PMA TX ASIC div clock.
pma/txX_ropll_dword_clk_1	pma/txX_ropll_dword_clk_1	Yes	pma/ropllX_ana_wordclk_i	2	1ns {0 0.5}	TX ROPLL dword clock.
pma/txX_ropll_qword_clk_1	pma/txX_ropll_qword_clk_1	Yes	pma/ropllX_ana_wordclk_i	4	2ns {0 1}	TX ROPLL qword clock.

Clock Name ^a	Source	Generated Clock	Master Source and Master Clock	Division	Period	Description
pma/txX_ropll_oword_clk_1	pma/txX_ropll_oword_clk_1	Yes	pma/ropllX_ana_wor dclk_i	8	4ns {0 2}	TX ROPLL oword clock.
pma/txX_slat_clk	pma/txX_slat_clk	Yes	pma/txX_ana_dword _clk_i	1	0.5ns {0 0.25}	TX slat clock for short latency mode
PMA RX Clocks						
pma/rxX_dig_qword_clk	pma/rxX_dig_qword_clk	Yes	pma/rxX_ana_dword _iclk_i	edges {1 2 5}	1ns {0 0.25}	RX qword clock, used for 32Gbps mode and below.
pma/rxX_dig_oword_clk	pma/rxX_dig_oword_clk	Yes	pma/rxX_dig_qword_ clk	2	2ns {0 1}	RX oword clock, used for 32Gbps mode and below.
pma/rxX_asic_owclk	pma/rxX_asic_owclk	Yes	pma/rxX_dig_oword_ clk	1	2ns {0 1}	RX oword clock, used for 32Gbps mode and below.
pma/rxX_asic_qwclk	pma/rxX_asic_qwclk	Yes	pma/rxX_dig_qword_ clk	1	1ns {0 0.25}	RX qword clock, used for 32Gbps mode and below.
pma/rxX_asic_dwclk_alt	pma/rxX_asic_dwclk_alt	Yes	pma/rxX_ana_dword _iclk_i_alt	1	1ns {0 0.5}	RX dword clock, only used for 16Gbps mode and below.
pma/rxX_asic_wclk_alt	pma/rxX_asic_wclk_alt	Yes	pma/rxX_ana_word_i clk_i_alt	1	1ns {0 0.5}	RX word clock, used for 8Gbps mode and below.
pma/rxX_slat_clk	pma/rxX_slat_clk	Yes	pma/rxX_ana_dword _iclk_i	1	0.5ns {0 0.25}	RX slat clock for short latency mode

a. The clock periods are based on PCIe5 32Gbps data rate. For JESD 32.44Gbps, the period of "pma/mplla_word_fr_clk" for example is 0.4932ns instead of 0.5ns, for all of the detail clock periods of JESD 32.44 Gbps, refer to the PHY synthesis clock report in which the synthesis setting option of DATA_RATE is set to 32440. JESD 32.44Gbps support is available as a customization, contact Synopsys if such customization is required.

The PHY supports two scan clock modes for stuck-at and at-speed (ASST) scan.

- `scan_clk_sel=1`

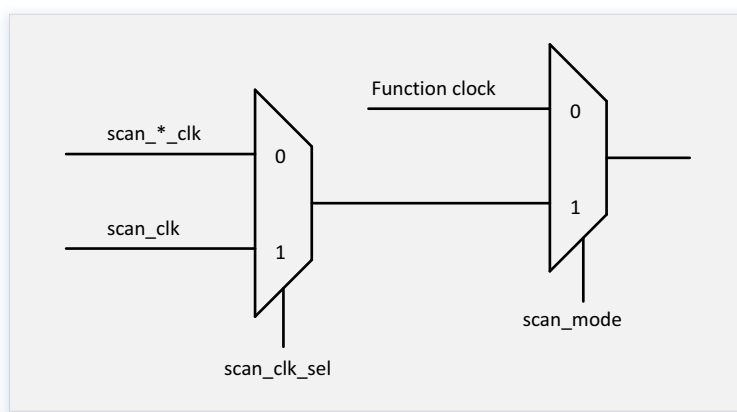
The scan clock MUX selects the `scan_clk` input clock. This clock is used by entire PHY. Only stuck-at (shift and capture) is supported.

- `scan_clk_sel=0`

The scan clock MUX selects the at-speed scan input clocks. Both stuck-at and at-speed (shift and capture) are supported. The scan clock inputs are required to be one-hot.

The scan clock multiplexer is not without glitches, therefore, `scan_clk_sel` must not change dynamically when the clocks are active.

Figure 7-12 Scan Clock Multiplexer Structure



Scan clocks definitions at the PHY (PMA plus Raw PCS) level are indicated in [Table 7-10](#).

Table 7-10 Scan Clocks Definitions at PHY (PMA plus Raw PCS) Level^a

Clock Name ^a	Creation Port	Max Frequency MHz	Description
<code>scan_sa_clk</code>	<code>scan_clk</code>	125	Stuck-at scan capture clock.
<code>scan_shift_clk</code>	<code>scan_clk</code>	250	Stuck-at and ASST scan shift clock.
<code>scan_cr_clk</code>	<code>scan_cr_clk</code>	175	ASST scan clock for <code>cr_clk</code> . The frequency is dependent on the maximum functional clock frequency of <code>cr_clk</code> sources.
<code>scan_jtag_clk</code>	<code>jtag_tck</code>	125	ASST scan clock for <code>jtag_tck</code> . <code>jtag_tck</code> is used for both scan and functional modes, so external scan clock mux may be needed.

Clock Name ^a	Creation Port	Max Frequency MHz	Description
scan_mpll_word_clk	scan_mpll_word_clk	2000*	ASST scan clock for the scan chain with MPLL word clock frequency.
scan_mpll_dword_clk	scan_mpll_dword_clk	1000*	ASST scan clock for the scan chain with MPLL dword clock frequency.
scan_ref_clk	scan_ref_clk	250	ASST scan clock for the scan chain with reference clock frequency.
scan_tx_clk	txX_clk	1000*	ASST scan clock for TX scan chains. For ASST scan test, the txX_clk can be connected together with one scan_tx_clk. txX_clk are used for both scan and functional modes, so external scan clock mux may be needed.

a. The frequencies of scan_mpll_word_clk, scan_mpll_dword_clk, and scan_tx_clk are based on PCIe5 32 Gbps data rate. For PCIe Gen4 and PCIe Gen3 applications, the frequencies of scan_mpll_word_clk should be 2000 MHz/2 and 2000 MHz/4. For PCIe Gen4 (16-bit mode) and PCIe Gen3 (16-bit mode) applications, the frequencies of scan_mpll_dword_clk and scan_tx_clk should be 1000 MHz and 1000 MHz/2, respectively. For other PCIe Gen4 and PCIe Gen3 applications, the frequencies of scan_mpll_dword_clk and scan_tx_clk should be 1000 MHz/2 and 1000 MHz/4, respectively. For JESD 32.44 Gbps, maximum frequencies are 1014 MHz and 2028 MHz instead of 1000 MHz and 2000 MHz, respectively. JESD 32.44 Gbps support is available as a customization. If such customization is required, contact Synopsys. The frequency of scan_cr_clk is based on fw_clk or ref_dco_clk frequency which is maximum 175 MHz.

The relation between output clocks and scan clocks are indicated in [Table 7-11](#).

Table 7-11 Relation between Output Clocks and Scan Clocks

Output Clock	Corresponding Scan Clock Input
mplla_word_clk	scan_mplla_word_clk
mplla_dword_clk	scan_mplla_dword_clk ^a
mplla_oword_clk	scan_mplla_dword_clk ^a
mplla_qword_clk	scan_mplla_dword_clk ^a
phy_ref_dig_clk	scan_ref_clk
rxX_125mhz_clk	scan_mpll_dword_clk
rxX_clk	scan_mpll_dword_clk
rxX_div33_clk	scan_mpll_dword_clk
rxX_div66_clk	scan_mpll_dword_clk

Output Clock	Corresponding Scan Clock Input
rxX_div_clk	scan_mpll_word_clk
rxX_dig_div_clk	scan_mpll_dword_clk
txX_ropll_125mhz_clk	scan_mpll_dword_clk
txX_ropll_div33_clk	scan_mpll_dword_clk
txX_ropll_div66_clk	scan_mpll_dword_clk
txX_ropll_div_clk	scan_mpll_word_clk
txX_ropll_dword_clk	scan_mpll_dword_clk
txX_ropll_oword_clk	scan_mpll_dword_clk
txX_ropll_qword_clk	scan_mpll_dword_clk
txX_ropll_word_clk	scan_mpll_word_clk
txX_word_clk	scan_mpll_word_clk
txX_dig_ropll_div_clk	scan_mpll_dword_clk
txX_slat_clk	N/A
rxX_slat_clk	N/A

a. For PCIe Gen4 (MPLLA+B Direct clocking mode) and Gen3(MPLLA+B Direct clocking mode) applications, if these clocks are used, group mplla/b_word_clk and mplla/b_dword_clk into higher frequency scan clock domain in your design. For PCIe Gen3 (MPLLA+B Direct clocking mode) applications, if these clocks are used group mplla/b_qword_clk into higher frequency scan clock domain in your design.

Table 7-12 Clock Uncertainty Definitions at PHY (PMA plus Raw PCS) level^a

Clock Name	Clock Uncertainty ^a (ns)		Duty Cycle Uncertainty ^b	
	Setup uncSU	Hold uncHD	Duty Cycle Unc(ns) uncDC	Duty Cycle Function Clocks
pma/mplla_ana_word_clk_i	0.04	0.05	0.05	40% ~ 60%
pma/refa_ana_clk_i	0.1	0.05	0.3	35% ~ 65%
pma/refa_ana_add_clk_i	0.1	0.05	0.3	35% ~ 65%
pma/refa_ana_occ_clk_i	0.1	0.05	0.3	35% ~ 65%
pma/refa_dig_clk_div1_i	0.1	0.05	0.3	35% ~ 65%
refa_dig_clk_div2	0.1	0.05	0	50%
pma/ref_ana_dco_clk	0.1	0.05	0.228576	46% ~ 54%
ref_dco_cal_clk	0.1	0.05	0.08	46% ~ 54%
refa_dco_la_clk	0.1	0.05	0	50%

Clock Name	Clock Uncertainty ^a (ns)		Duty Cycle Uncertainty ^b	
	Setup uncSU	Hold uncHD	Duty Cycle Unc(ns) uncDC	Duty Cycle Function Clocks
pma/rxX_ana_dword_iclk_i	0.06	0.05	0.015	47% ~ 53%
pma/rxX_ana_dword_clk_i	0.06	0.05	0.015	47% ~ 53%
pma/rxX_ana_dword_iclk_i_alt	0.1	0.05	0.03	47% ~ 53%
pma/rxX_ana_word_iclk_i_alt	0.1	0.05	0.15	35% ~ 65%
pma/rxX_ana_div_clk_i	0.06	0.05	0.1	30% ~ 70%
rxX_asic_owclk	0.1	0.05	0	50%
rxX_asic_qwclk	0.1	0.05	0.015	47% ~ 53%
rxX_asic_dwclk_alt	0.1	0.05	0.03	47% ~ 53%
rxX_asic_wclk_alt	0.1	0.05	0.15	35% ~ 65%
rxX_div33_clk_int	0.1	0.05	0.015	47% ~ 53%
rxX_div66_clk_int	0.1	0.05	0.015	47% ~ 53%
rxX_dig_div_clk_int	0.1	0.05	0.015	47% ~ 53%
rxX_125mhz_clk_int	0.1	0.05	0.015	47% ~ 53%
rxX_slat_clk_int	0.06	0.05	0.015	47% ~ 53%
pma/txX_ana_word_clk	0.04	0.05	0.075	35% ~ 65%
pma/txX_ana_dword_clk_i	0.04	0.05	0.025	45% ~ 55%
pma/ropllX_ana_div_clk_i	0.04	0.05	0.1	30% ~ 70%
pma/ropllX_ana_wordclk_i	0.04	0.05	0.025	45% ~ 55%
txX_clk	0.06	0.05	0.1	40% ~ 60%
txX_ropll_div33_clk_int	0.1	0.05	0	50%
txX_ropll_div66_clk_int	0.1	0.05	0	50%
txX_dig_ropll_div_clk_int	0.1	0.05	0	50%
txX_ropll_125mhz_clk_int	0.1	0.05	0	50%
txX_slat_clk_int	0.04	0.05	0.025	45% ~ 55%
bs_cdr ^c	0.1	0.05	0	50%
bs_uds ^c	0.1	0.05	0	50%
cr_para_clk ^c	0.1	0.05	0	50%

Clock Name	Clock Uncertainty ^a (ns)		Duty Cycle Uncertainty ^b	
	Setup uncSU	Hold uncHD	Duty Cycle Unc(ns) uncDC	Duty Cycle Function Clocks
jtag_clk ^c	0.1	0.05	0	50%
fw_clk ^c	0.1	0.05	0	50%
Scan Clocks				
scan_sa_clk ^c	0.06	0.05	0	50%
scan_shift_clk ^c	0.06	0.05	0	50%
scan_cr_clk ^c	0.06	0.05	0	50%
scan_jtag_clk ^c	0.06	0.05	0	50%
scan_mpll_word_clk ^c	0.06	0.05	0	50%
scan_mpll_dword_clk ^c	0.06	0.05	0	50%
scan_ref_clk ^c	0.06	0.05	0	50%
scan_tx_clk ^c	0.06	0.05	0	50%

a. Total Uncertainty:

```
set sm [expr {$uncSU + $uncDC}]
```

```
set hm [expr {$uncHD + $uncDC}]
```

```
set_clock_uncertainty -setup $sm -rise_from $ck -fall_to $ck
```

```
set_clock_uncertainty -setup $sm -fall_from $ck -rise_to $ck
```

```
set_clock_uncertainty -hold $hm -rise_from $ck -fall_to $ck
```

```
set_clock_uncertainty -hold $hm -fall_from $ck -rise_to $ck
```

b. The duty cycle uncertainty setting is based on one of the source clocks which have the maximum frequency.

c. These clocks are customer input clocks. You can determine the clock uncertainty and duty cycle uncertainty.

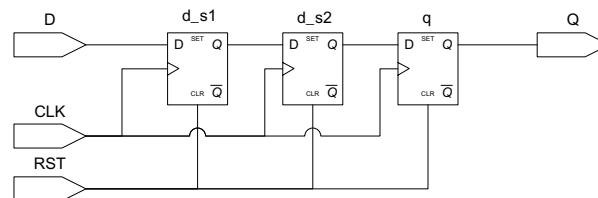
7.10.2.3 Synchronizers

Soft IP (Raw PCS) instantiates synchronizer modules used for clock domain crossing protection. There are two types of synchronizer modules used in the design.

7.10.2.3.1 Data Synchronizers

For single-bit signals that must be passed from one clock domain to another, an instance of the module(s) `dwc_e32mp_phy_xn_pcs_raw_gen` is used to synchronize the source signal into the destination clock domain (where N is a placeholder for the number of lanes). The structure of a single-bit data synchronizer is shown in [Figure 7-13](#).

Figure 7-13 `dwc_e32mp_phy_xn_pcs_raw_gen_sync` Module



The second and third registers in the `dwc_e32mp_phy_xn_pcs_raw_gen_sync_sync` module are `d_s2` and `q`. To ensure stable synchronization, constrain the maximum delay to these cells, from `d_s1/CK` to `d_s2/D` and from `d_s2/CK` to `q/D` to 20% of the sampling clock frequency.

For ease of timing constraint definition, this delay can be implemented as 200 ps (20% of 1000 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instances. You can ignore the clock-tree delay if the paths are constrained explicitly from `d_s1/CK` to `d_s2/D` and from `d_s2/CK` to `q/D`.

For multiple-bit signals that must be passed from one clock domain to another, an instance of the module(s) `dwc_e32mp_phy_xn_pcs_raw_gen_sync_bus_sync` or `dwc_e32mp_phy_xn_pcs_raw_gen_sync_bus_sync_flag` are used to synchronize the source bus signal into the destination clock domain.

The modules `dwc_e32mp_phy_xn_pcs_raw_gen_sync_bus_sync` and `dwc_e32mp_phy_xn_pcs_raw_gen_sync_bus_sync_flag` instantiated single-bit synchronizer module `dwc_e32mp_phy_xn_pcs_raw_gen_sync_sync` to synchronize each bit of bus signal, and the module `dwc_e32mp_phy_xn_pcs_raw_gen_sync_bus_sync_flag` has unstable data flag output.

The allowed maximum skew for multiple-bit synchronizer inputs is 80% of the destination clock period.

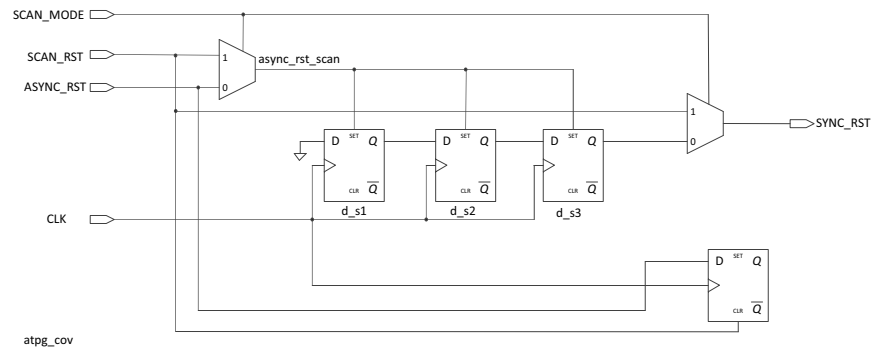
Metastability behavior is modeled in the synchronizers and is activated by defining the macro `DWC_E32MP_X2NS_SIM_MODE` during simulations.

7.10.2.3.2 Reset Synchronizers

For synchronizing the asynchronous reset signal, an instance of the modules, "dwc_e32mp_phy_xn_pcs_raw_gen_sync_rst_sync," is used. As shown in [Figure 7-14](#), any time an external or internal reset condition is passed from one clock domain to another domain, the reset condition must pass through a `dwc_e32mp_phy_xn_pcs_raw_gen_sync_rst_sync` module to do the following:

- Asynchronously capture the reset signal

- Stretch the reset pulse to be active for at least one full clock cycle in the receiving domain
- Synchronously de-assert the reset signal with respect to the receiving domain

Figure 7-14 dwc_e32mp_phy_xn_pcs_raw_gen_rst_sync Module

Because assertion of a reset causes all flip-flops in the synchronizer module to capture the pulse, the de-assertion edge of the asynchronous reset input has no effect on the final flip-flop that drives the reset into the receiving domain. Therefore, all asynchronous paths through the d_s3 register of this module can be considered false.

The second and third registers in the dwc_e32mp_phy_xn_pcs_raw_gen_sync_rst_sync module are d_s2 and d_s3. To ensure stable synchronization, constrain the maximum delay to these cells, from d_s1/CK to d_s2/D and from d_s2/CK to d_s3/D to 20% of the sampling clock frequency.

For ease of timing constraint definition, this delay can be implemented as 200 ps (20% of 1000 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instance.

Metastability behavior is modeled in the synchronizers and is activated by defining the macro DWC_E32MP_X2NS_SIM_MODE during simulations.

7.10.2.3 Synchronizer Constraints

Care must be taken when defining constraints related to data synchronizer cells. It is recommended that you use only clock groups or false paths across asynchronous clock domains instead of globally adding false path constraints to the inputs of data synchronizer cells. Doing so ensures that valid intra-clock domain paths are still timed as synchronous paths. For example, the internal FSM synchronizes all read data, but in the case where the read data is coming from the internal cr_clk, the data is expected to be synchronous. Only timing paths from different asynchronous clock domains must be false paths through the data synchronizer cells.

7.10.2.4 txX_clk Constraints

TX parallel data to the PHY is transferred from the external ASIC txX_clk domain to the internal txX_ana_dword_clk domain. These two clocks are generated from the same MPLL source clock, but asynchronous in phase. As a result, the clocks are aligned within the PMA by the TX Clock Alignment module to be able to accurately transfer data from one domain to the other. The TX Clock Alignment function is triggered whenever the PHY enters P0/P0s state from a low-power state (P1/P2). The ASIC's txX_clk is aligned to the internal txX_ana_dword_clk to provide $\frac{1}{2}$ cycle of setup and hold margin between the clocks.

However, it is possible for the incoming ASIC txX_clk to drift relative to the internal txX_ana_dword_clk domain as a result of voltage and temperature variation. Because TX clock alignment is performed only

upon entry to P0/P0s state, the variation on the txX_clk must be bound to ensure that this drift does not cause txX_clk to misalign relative to the txX_ana_dword_clk domain, resulting in the PHY's inability to capture incoming data. Because the ASIC txX_clk is derived from the output PMA MPLL clocks, the entire path from the PMA lib outputs, through the external PCS logic and back to the PMA, the txX_clk input must be analyzed to determine the worst-case clock tree latency variation across all voltage/temperature (VT) corners.

This external variation must be less than 100 ps when PMA internal TX FIFO is disabled, and less than (0.5 QWORD clock + 100 ps) for 32-bit mode or (0.5 DWORD Clock + 100 ps) for 16-bit mode which translates to ~600 ps when FIFO is enabled.



Note The TX FIFO is enabled by default.

7.10.2.5 rxX_clk Constraints

To support 8, 16, and 32-bit interfaces in synthesis, there are additional generated clocks with the .lib model of the PMA. The PHY RX interface can operate at up to 1 GHz for 8/16/32-bit mode. To maintain a common .lib model for all modes, there are internal pins for primary clock definitions for 8/16-bit operation and for 32-bit operation in the same .lib model:

8/16-bit operation:

- pma/rxX_ana_word_iclk_i_alt (for 8-bit)
- pma/rxX_ana_dword_iclk_i_alt (for 16-bit)

32-bit operation:

- pma/rxX_ana_dword_iclk_i

The result is three functional output clocks at the rxX_clk output pin of the PMA .lib model:

8/16-bit operation:

- pma/rxX_asic_wclk_alt (for 8-bit)
- pma/rxX_asic_dwclk_alt (for 16-bit)

32-bit operation:

- pma/rxX_asic_qwclk

Perform functional timing analysis for all three clocks coming out of the rxX_clk output because these clocks have different paths inside the PMA and as a result, have slightly different delays relative to the rxX_clk output pin. If only a single mode is being used (such as always operating in 32-bit mode), then it is possible to consider timing analysis results from a single clock only (for example, for a 32-bit interface, only the rxX_asic_qwclk is necessary), and ignore the other clocks.

7.10.2.6 Reference Clock Constraints

No input-to-output arc exists for the refa_ana_clk_i in the PMA .lib file. This is an internal node within the PMA .lib file that is used to generate the refa_dig_clk and refa_dig_clk_div1 output of the PMA. The reference clock must be defined on this internal refa_ana_clk_i pin of the PMA instead of on ref_*_clk_p/m pins because these pins are not used within the PMA .lib file.

Both of two generated clocks are available to support different reference clock input.

The PMA `refa<b>_dig_clk` is selected from either DIV1 `refa<b>_dig_clk_div1_i` or DIV2 `refa<b>_dig_clk_i` path based on the setting of `refa<b>_clk_div2_en` signal.

- When `refa<b>_clk_div2_en` is 0, the DIV1 path `refa<b>_dig_clk_div1_i` is selected. The propagation from `refa<b>_dig_clk_i` to `refa<b>_dig_clk` needs to be stopped.
- When `refa<b>_clk_div2_en` is 1, the DIV2 path `refa<b>_dig_clk_i` is selected. The propagation from `refa<b>_dig_clk_div1_i` to `refa<b>_dig_clk` needs to be stopped.

For the PMA `refa<b>_dig_clk` selection and propagation stopping, refer to PHY constraint `<phy_prefix>_phy_con.tcl`.

7.10.2.7 PMA .lib ETM Timing Modes

The PHY contains a multi-mode .lib file for the PMA module that provides timing information for various modes of operation. The following table outlines the modes.

Table 7-13 PMA .lib ETM Timing Modes

Timing Mode	ETM Mode in .lib	Additional Constraints
Functional	NORMAL	N/A
Scan Shift	SCAN	False path all non scan_*_in/out ports
Stuck-At Scan Capture	SCAN	False path all scan_*_in/out ports
At-Speed Scan Capture	SCAN_ASST	False path all scan_*_in/out ports



Note

The hard-macro PMA .lib file contains generated clock definitions for various functional clocks. If timing is performed using .lib ETM timing modes for SCAN/SCAN_ASST modes without defining the functional clocks, it may be necessary to remove these generated clocks (using `remove_clock` constraints) to prevent clock propagation errors.

7.10.2.7.1 Unused Signals

Unused inputs to the raw PCS must be tied to ground potential.

Unused outputs from the raw PCS must be left floating.

8

Board- and Package-Level Implementation

This chapter includes the following sections:

- “Package Type”
- “Signal Integrity – RX and TX Differential Signal Guidelines”
- “Power Integrity – Supplies Requirements”
- “External Reference Resistor”
- “Reference Clock”
- “Board Requirements – External Components”

8.1 Package Type

The PMA is designed to support flip-chip packaging technology. For PMA bump-map diagrams, see Section 7.1.1, “Flip-Chip Pinout”.

8.2 Signal Integrity – RX and TX Differential Signal Guidelines

You must design and simulate each of the RX and TX differential routes to ensure that signal integrity requirements are met.



Note

For more information, see the requirements listed in the “SERDES Differential Channel Guidelines” chapter of the *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*. Also, see the “Package and PCB checklist” in Appendix A of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*.

8.2.1 Package Guidelines – RX and TX Maximum Allowed Package Loss

For Ethernet protocols, the TX eye requirements must be met at the package pins. Both TX output amplitude and equalization are affected by package loss. These considerations can limit the maximum amount of package loss for certain protocols. The TX amplitude and equalization should be adjusted to account for package loss.

For PCIe, the transmit launch amplitude should be adjusted to account for package loss.

- For 2.5 Gbps and 5 Gbps, the PCIe 5.0 specification states that VTX-DIFF-PP should be measured as the average transition bit amplitude at the 0.5 UI location using across 500 repetitions of the compliance pattern.
- For 8 Gbps, 16 Gbps, and 32 Gbps VTX-DIFF-PP should be measured with the TX P4 equalization preset as the average amplitude of bits 57-62 of the 64-ones/64-zeros segment of the averaged across 500 repetitions of the compliance pattern.

For 10GBASE-KR, adjust both the amplitude and the de-emphasis level to account for package loss. Increase the de-emphasis level to account for the amount of package loss at Nyquist rate.

8.2.2 Signal Integrity Analysis

To enable signal integrity (SI) analysis of the RX and TX routes, s-parameter models, as well as, an AMI model is provided with the PHY deliverables.

8.2.2.1 S-Parameter Model

To assist in package design, an S-parameter model of the RX input and TX output are provided. These models should be simulated along with your extracted package model to ensure that the protocol specific requirements for return loss and other signal integrity requirements are being met.

8.2.2.2 AMI Model

To assist link modeling, an IBIS Algorithmic Modeling Interface (IBIS-AMI) model of the PHY transmitter and receiver is provided along with a representative channel model.

The provided channel model is simply an example, and you should measure the system board and connectors you are using to get a more accurate representation of the link performance.

8.3 Power Integrity – Supplies Requirements

You must design and simulate the Power Distribution Network (PDN) for each of the PHY supplies to ensure that power integrity requirements are met.



Note

For more information, see the requirements listed in the “Supply Guidelines” chapter of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*.

Also, see the “Package and PCB checklist” in Appendix A of *DesignWare Cores SERDES PCB and Packaging Design Guide Application Note*.

8.3.1 Power Supply Requirements

You must perform a power integrity (PI) analysis to ensure that the supply noise requirements are met at the die bumps. There are separate excursion limits for DC noise and lower frequency (LF) noise, as well as AC high frequency (HF) noise provided in [Table 8-1](#). For frequency specific details, see sections “[Supply Requirements at Die Bumps for vp, vptxX, and vpdig](#)” and “[Supply Requirements at Die Bumps for vph](#)”.

Figure 8-1 Excursion Limits for DC and AC Noise

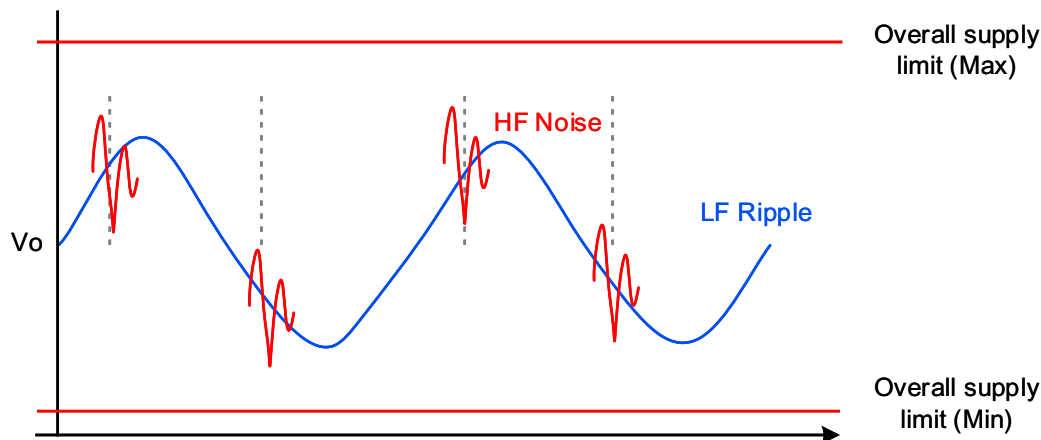


Table 8-1 Power Supply Requirements at Die Bumps

Pin	DC and Low Frequency Noise (Maximum excursion for all components)	AC High Frequency Noise (Maximum excursion for all components)	Overall Supply Limit Including both AC and DC Variation
vph	1.2/1.5/1.8 V -5%/+10%	< 3% Vpp of nominal DC level	1.2/1.5/1.8 V -6.5%/+11.5%
vp/vptxX/vpdig	0.9 V -5%/+5%	< 5% Vpp of nominal DC level	0.9 V -7.5%/+7.5%

Pin	DC and Low Frequency Noise (Maximum excursion for all components)	AC High Frequency Noise (Maximum excursion for all components)	Overall Supply Limit Including both AC and DC Variation
vddcore	0.8/0.9 V -10%/+5%	AC noise requirements are determined by dynamic voltage drops in the digital block that do not cause violation of soft logic digital timing. You should run analysis to confirm.	

**Note**

- You can request a script from Synopsys that allows you to check your supply noise against the DC and AC limits.
 - You can request mode-change waveforms for this product that allow you to check supply noise during power-state change events.
 - This product has a lane staggering feature that can be used to reduce di/dt excursions by staggering the enable and disable of each lane RX and TX.
- For more information about the script, mode-change waveforms, and the lane staggering feature, contact Synopsys Customer Support.

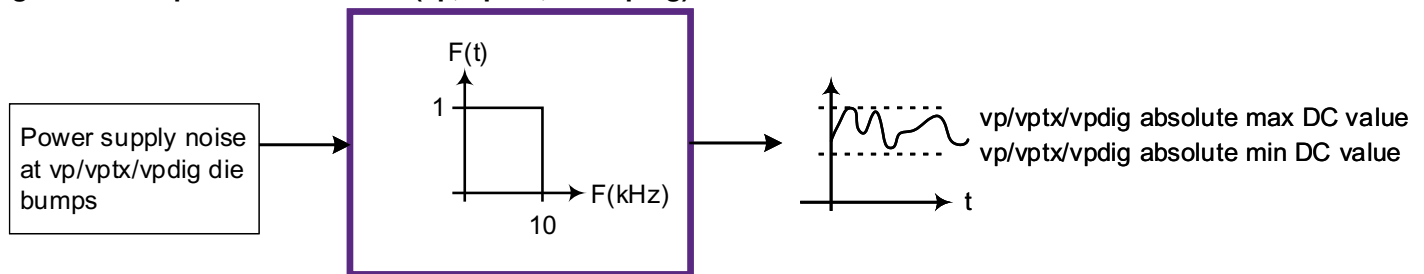
8.3.1.1 Supply Requirements at Die Bumps for vp, vptxX, and vpdig

DC Limits for vp, vptxX, and vpdig

You can check the DC level by applying a low pass filter with a cutoff frequency at 10 kHz on measured or simulated noise ripple.

As illustrated in [Figure 8-2](#), the total amount of tolerated noise in this band depends on the nominal DC level.

Figure 8-2 Specified DC Limits (vp, vptxX, and vpdig)



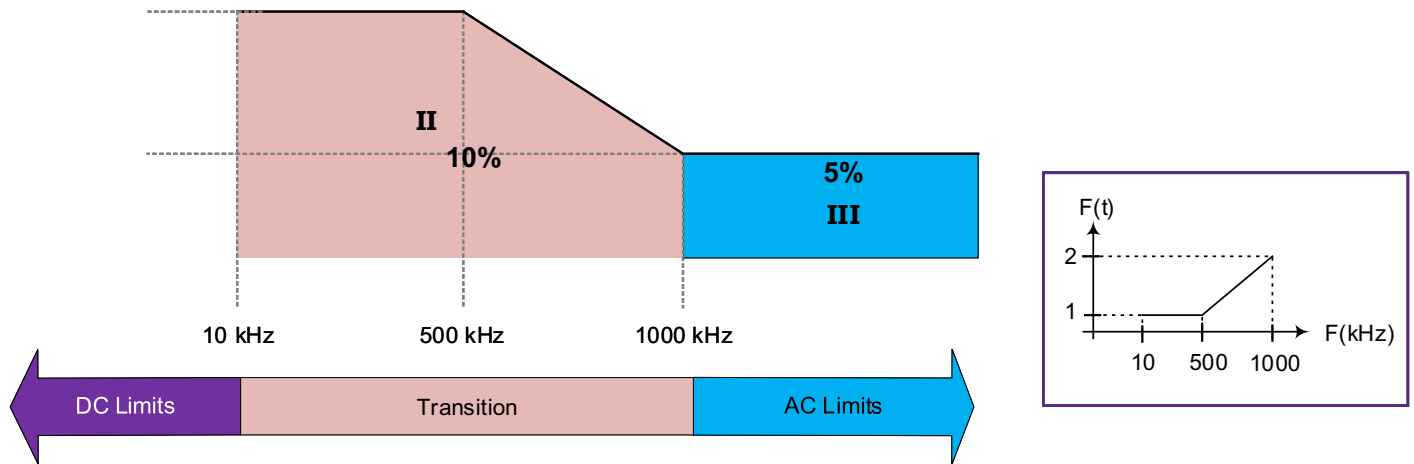
Transition Zone Limits for vp, vptxX, and vpdig

[Figure 8-3](#) illustrates the transition zone limits for vp, vptxX, and vpdig. Within the transition zone band, consider the following guidelines:

- 10 kHz to 1000 kHz Vp, Vptx, and Vpdig peak-to-peak voltage noise must be less than 10% of the “nominal vp DC level” but limited to the overall supply limit.
- 500 kHz to 1000 kHz noise ripple can increase 20dB/DEC towards 500 kHz from the AC limits zone III (see [Table 8-3](#)) but limited to the overall supply limit, including the AC and DC variations defined in [Table 8-1](#).

You can check the transition zone noise by applying a band pass filter as shown in [Figure 8-3](#).

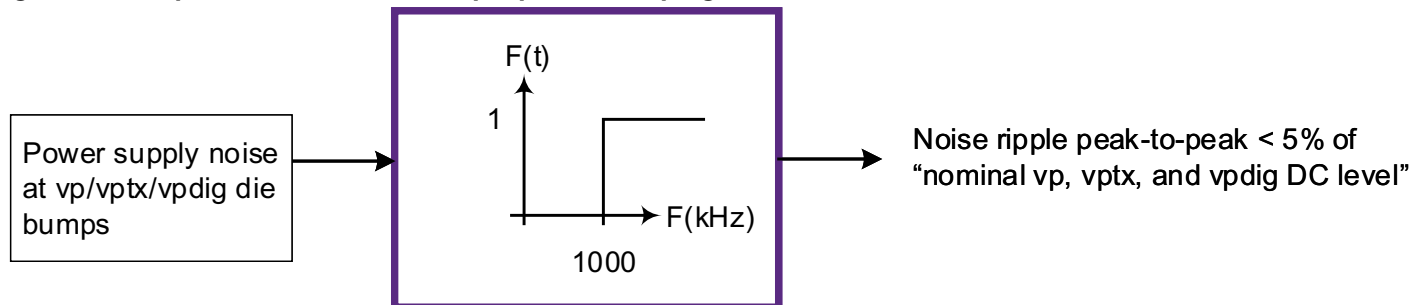
Figure 8-3 Transition Zone Limits for vp, vptxX, and vpdig



AC Limits for vp, vptxX, and vpdig

The maximum allowed peak-to-peak voltage is relative to the “nominal vp, vptx, and vpdig DC level”. You can check AC limits by applying a high pass filter with 1 MHz corner frequency as shown in [Figure 8-4](#).

Figure 8-4 Specified AC Limits for vp, vptxX, and vpdig



8.3.1.2 Supply Requirements at Die Bumps for vph

DC limits for vph

You can check the DC level by applying a low pass filter with a cutoff frequency at 10 kHz on measured or simulated noise ripple.

The total amount of tolerated noise in this band depends on the nominal DC level, as illustrated in [Figure 8-5](#).

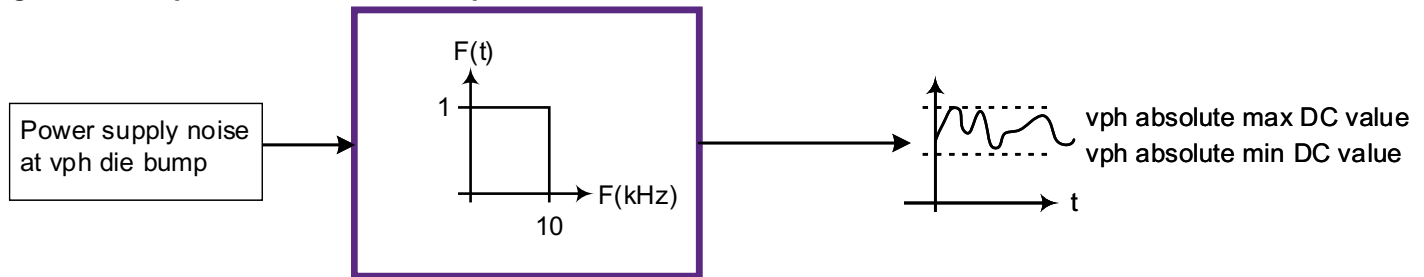
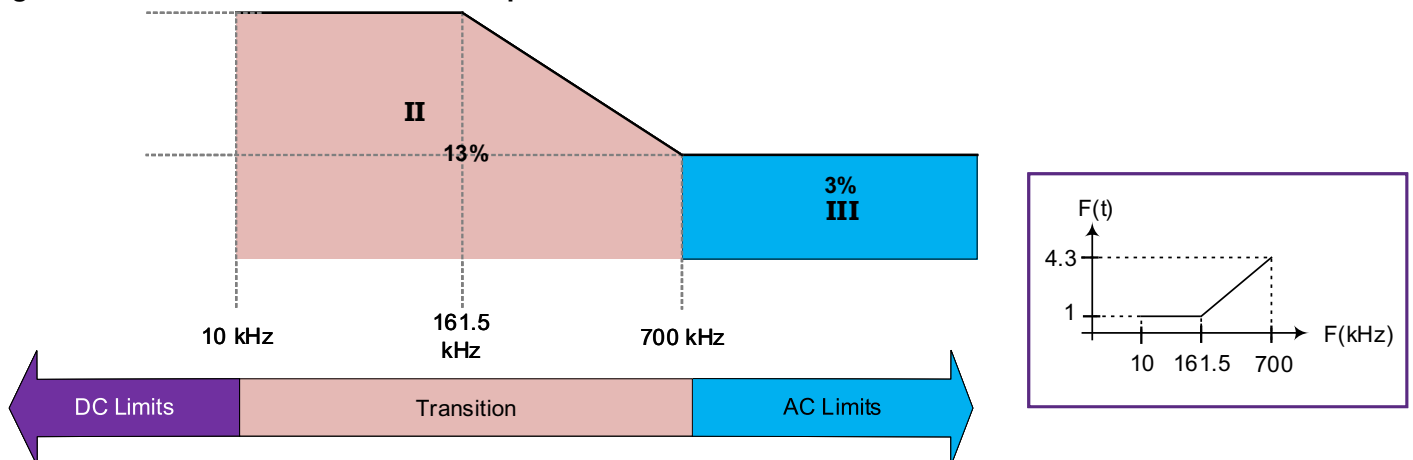
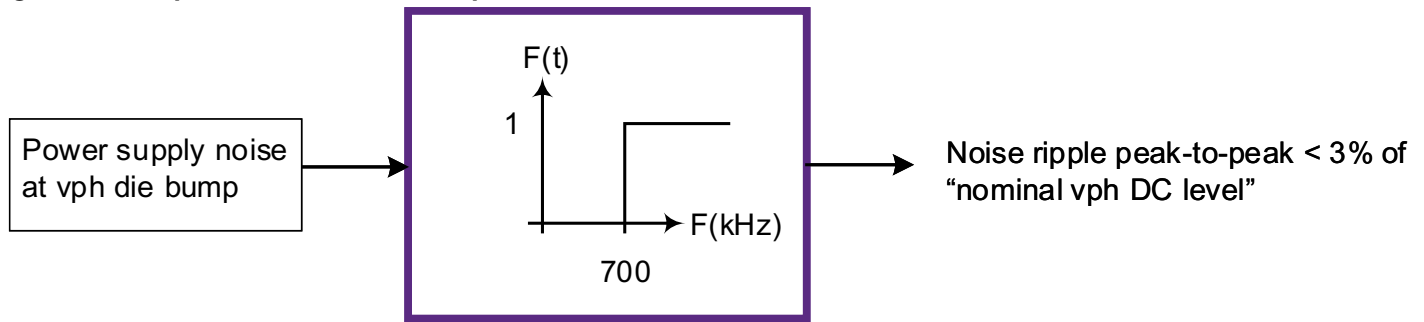
Figure 8-5 Specified DC Limits for vph**Transition Zone Limits for vph**

Figure 8-6 illustrates the transition zone limits for vph. Within the transition zone band, consider the following guidelines:

- 10 kHz to 700 kHz vph peak-to-peak voltage noise is less than 13% of “nominal vph DC level” but limited to overall supply limit.
- 161.5 kHz to 700 kHz noise ripple can increase 20dB/DEC towards 161.5 kHz from the AC limits zone III (see Table 8-6) but limited to the overall supply limit, including the AC and DC variations defined in Table 8-1.

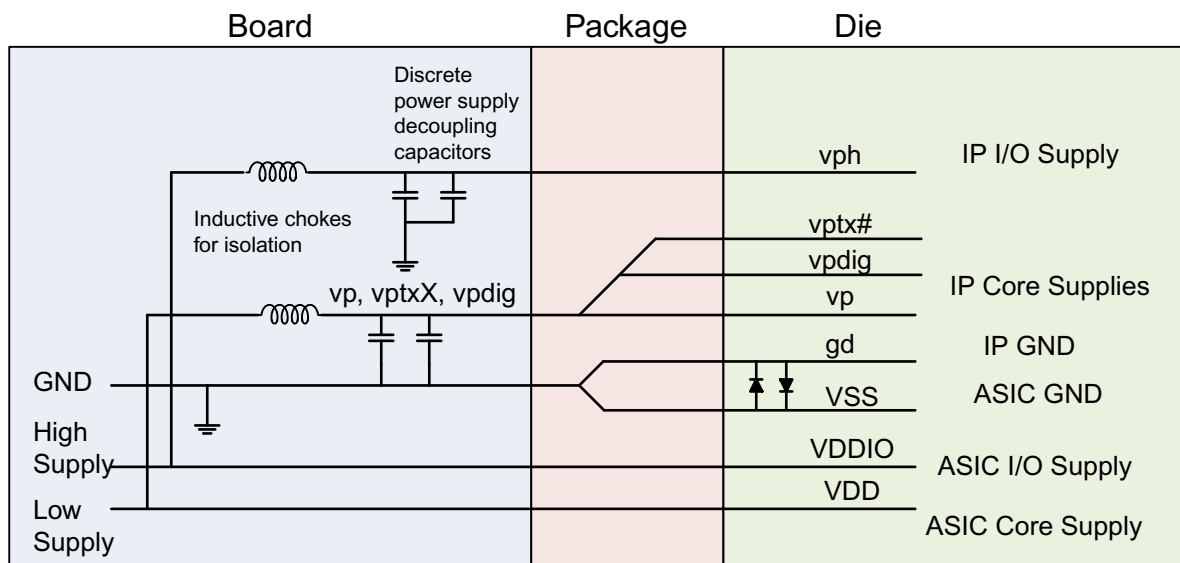
Figure 8-6 Transition Zone Limits for vph**AC Limits for vph**

The maximum peak-to-peak voltage is relative to the nominal vph DC level. You can check AC limits by applying a high pass filter with 700 kHz corner frequency as shown in Figure 8-7.

Figure 8-7 Specified AC Limits for vph

8.3.2 Package and Board Guidelines—Power Supply Sharing

A diagram of the power supplies is shown in [Figure 8-8](#).

Figure 8-8 Power Supply Connections

Note

There exists timing constraints between IP Core Supplies and ASIC Core Supply. For more details, refer to Section 4.1, "PHY Power-Up".

To ensure good jitter performance, the PHY is designed to completely isolate power and ground supplies from the rest of the ASIC on-die. Per the power supply connections diagram, you need to isolate the PHY power supplies from other ASIC supplies by using inductive chokes (such as, ferrite bead).

To ensure good ESD performance, the PHY requires a common ground in the package with the rest of the ASIC.

The following supplies may be connected/shorted together within the package:

- PHY core supplies (vptx0 through vptxX, vpdig, vp, where X represents the number of lanes)
- PHY ground (gd) and ASIC ground

To reduce noise coupling from the ASIC, vsscore may be connected to PHY gd outside the package on the board.

When multiple PHYs are instantiated, their vp and vph may be connected together within the package according to the following requirements:

- If you are using multiple reference clocks with different sources, particularly if they are SSC and each independently spread, then recommendation is not to share the VPH supply in order to mitigate the risk of PLL pulling. Supplies may be shared if the PHYs are aggregated. If the PHYs are bifurcated or independently controlled, then the recommendation is to keep the supplies separated to reduce effects from startup/power-down transients.
- Per-PHY supply parasitics and bypass capacitance are maintained.
- Power integrity simulation has been performed to confirm AC supply noise requirements are observed for all PHYs.

8.3.3 Recommended Maximum Off-Silicon Die Power Supply Inductance

The AC noise power supply is a tied, first-order consideration of the off-silicon die power supply inductance from the die bumps outwards, comprised mainly of the following two parts:

- Package: loop inductance from silicon die bumps to BGA balls
- PCB: loop inductance from BGA power and ground balls to PCB power and ground planes, considering local PCB decoupling capacitor. The main contributors usually are the inductance from BGA balls to power and ground planes; number, distribution and length of vias; and decoupling capacitors location, footprint, and parts choice.

Recommended maximum off-silicon die supply inductance for each of vp and vph is given in [Table 8-2](#). A power integrity simulation must be performed to ensure supply voltage noise requirements and specifications are met with the implemented system power distribution network.

Table 8-2 Supply Inductance Targets

Parameter	Target	Comments
Off-Silicon Die inductance Vp	<500 pH	Total loop inductance looking out into PCB from silicon die bumps
Off-Silicon Die inductance Vph	<500 pH	Total loop inductance looking out into PCB from silicon die bumps



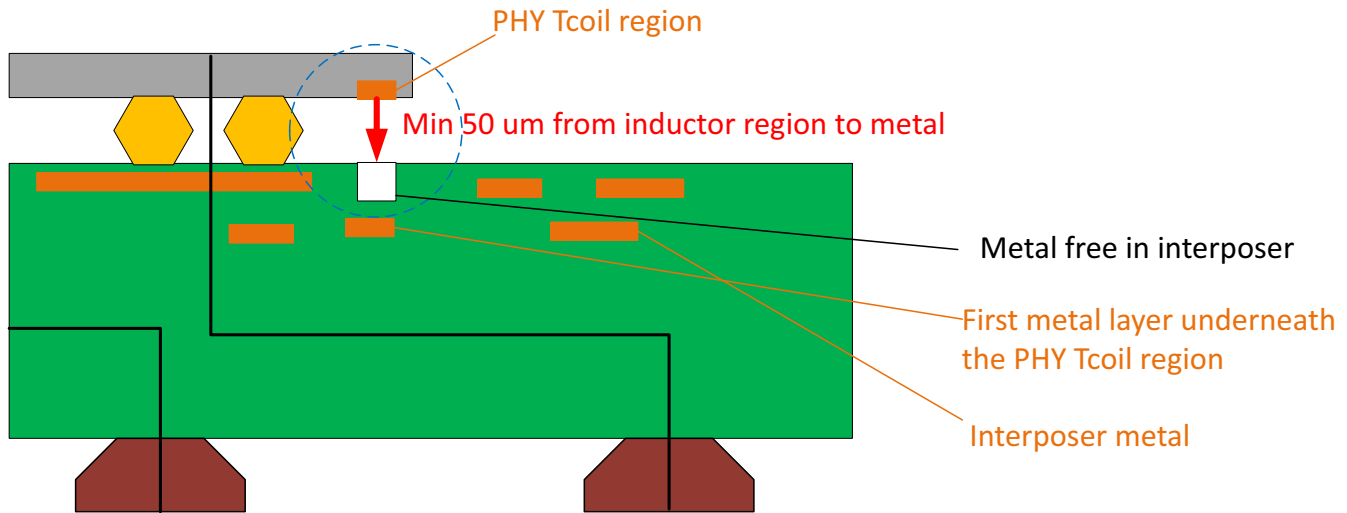
Attention

[Table 8-2](#) is to be used as a guideline for package and PCB power supply design. This is not a specification.

8.3.4 Package Guidelines—Avoidance of Substrate Routing Above Inductors

There are on-die inductors implemented in the PHY: Tcoil inductors in transmit and receive paths, and in MPLL. In the silicon interposer or package layout, keep a minimum of 50 μm clearance between the on-chip inductors and the metal plane underneath, as shown in [Figure 8-9](#).

Figure 8-9 Silicon Interposer Layout



Synopsys requires that you avoid any routing of signals, for example, TX or Rx differential lines or reference clock lines, or any other toggling signals immediately beneath the MPLL on-chip inductors due to noise coupling concerns. Synopsys recommends to have a ground metal reference copper or an I/O Supply reference copper populated under MPLL inductor at 50 μm or more vertical spacing on package substrate or interposer. This helps to shield inductor from noise coupling and at the same time ensure on chip inductance not affected. For Tcoil inductors in transmit and receive path, ensure a minimum 50 μm vertical spacing between Tcoil on chip and metal shape on package or interposer to avoid impact on Tcoil inductance. [Figure 8-10](#) shows an example of the location of the inductors and tcoils within the lanes. For the coordinates of the keep-out areas illustrated in [Figure 7-1](#) (“Bump Map Example”), contact Synopsys.

Figure 8-10 Example of On-Die Inductors

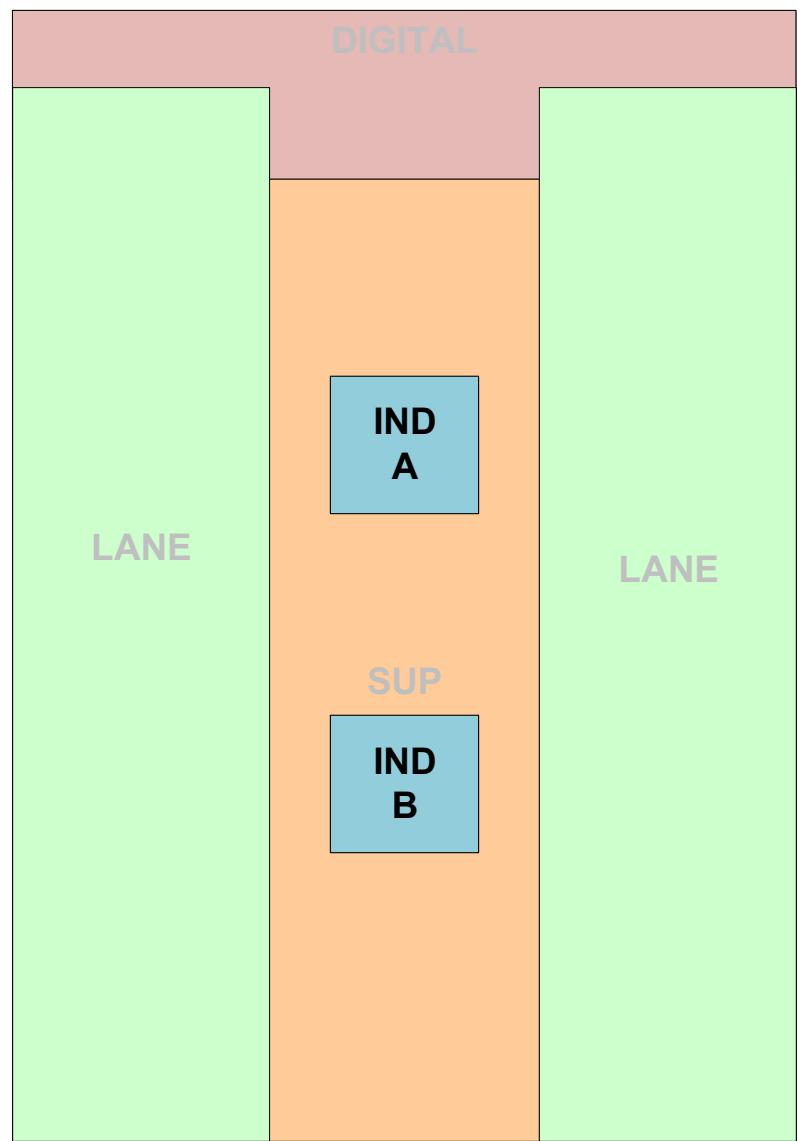


Table 8-3 provides the coordinates for the inductors.

Table 8-3 Inductor Coordinates

Inductor	Position	Coordinate
A	Upper-right corner	(935,570)
	Lower-left corner	(804,432)
B	Upper-right corner	(935,194)
	Lower-left corner	(804,64)

8.4 External Reference Resistor

An external reference resistor is used for calibration of the RX and TX termination. The external resistor is automatically used for calibration when the PMA comes out of reset and when a manual tuning request is made. When a PMA is using the reference resistor, it forces current across the resistor. When a PMA is not using the reference resistor, the current is disabled. The requirements for the reference resistor are covered in [Table 8-4](#).

For details on reference resistor implementation for a single PMA, see Section [8.4.2, “Reference Resistor Implementation for a Single PMA”](#).

For details on reference resistor sharing between multiple PMAs, see Section [8.4.3, “Sharing a Reference Resistor Between Multiple PMAs”](#).



Note

For additional details on physical implementation and contention bus connectivity see Section [7.8, “Reference Resistor”](#).

8.4.1 Package and Board Requirements—Reference Resistor

Reference resistor requirements are outlined in [Table 8-4](#).

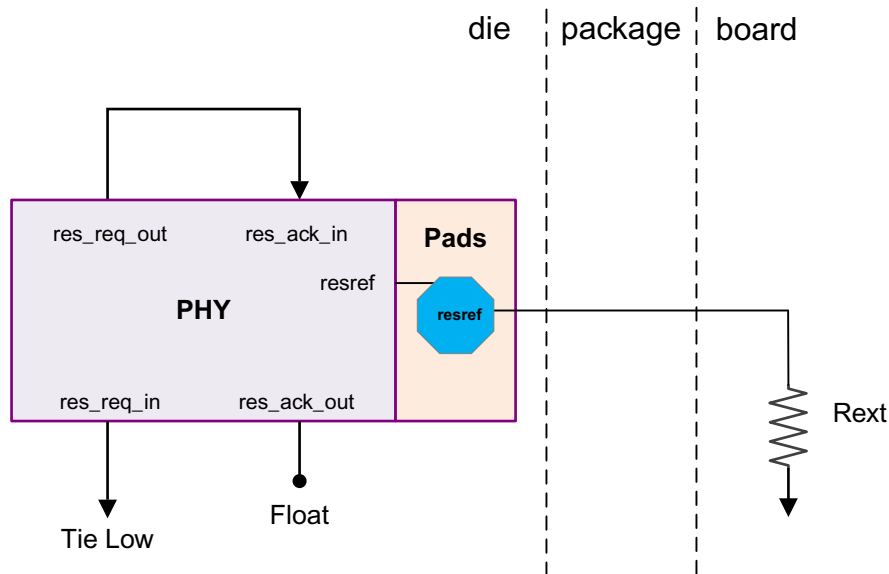
Table 8-4 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 Ohm precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2 mA	None
External Reference Resistor parasitic capacitance C_{load}	<20 pF	■ The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. ■ Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA can support up to 18.5 pF (20 pF - 1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 14 pF (20 pF - 4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 Ohm	Take care to minimize the resistance of the connection through the package to the external reference resistor as this affects the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board's external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals (for example, clocks) should be spaced away from resref or should be shielded with ground, if possible.

8.4.2 Reference Resistor Implementation for a Single PMA

Package and board connectivity for a single PMA implementation is shown in [Figure 8-11](#).

Figure 8-11 External Reference Resistor Connectivity for a Single PMA



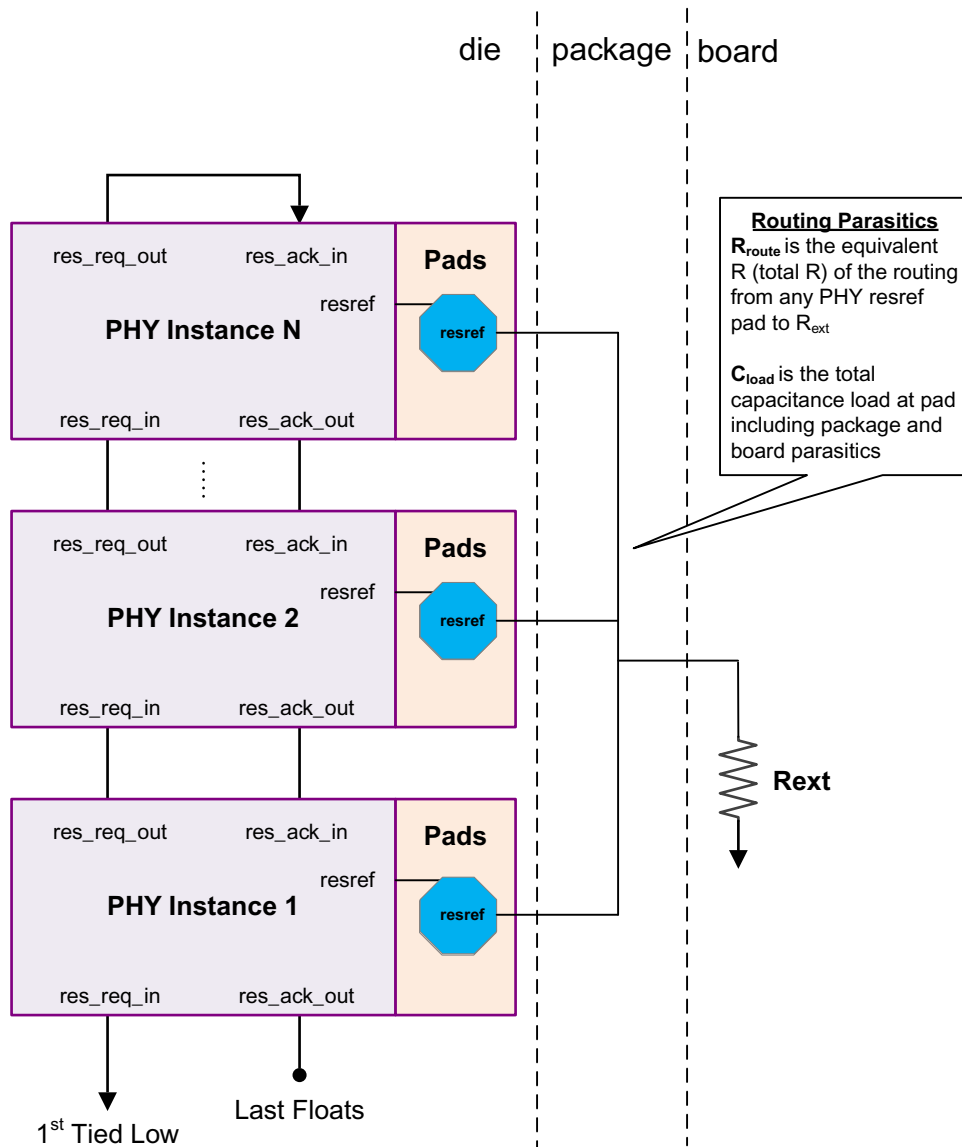
8.4.3 Sharing a Reference Resistor Between Multiple PMAs

The external reference resistor may be shared among a maximum of four instances of the Multi-Protocol 32G PHY x2 for TSMC 12FFC PMA by connecting the `resref` signals together in the package substrate. Sharing an external reference resistor by tying the `resref` bus together in the package is shown in [Figure 8-12](#).



Note

- If multiple PHYs are sharing a single reference resistor and one or more of the PHYs are being put in a disabled state, see the specific requirements provided in [Section 4.22](#), “Disabling a PHY”
- If multiple PHYs are sharing a single reference resistor, do not use the VP supply power-gating on any of the PMAs because it disables the contention bus
- Reference resistors may be shared with other instantiations of the same PMA, but should not be shared with different PMAs
- Take care to meet the parasitic resistance and capacitance requirements noted in [Section 8.4.1](#), “Package and Board Requirements—Reference Resistor”

Figure 8-12 External Resistor Sharing

8.5 Reference Clock

For details about reference clock board and package requirements, see [Chapter 9, “Reference Clock Implementation”](#).

8.6 Board Requirements—External Components

The PHY requires the following external components:

- AC-coupling capacitors as specified by the standard; for reference, see [Table 8-5](#).
- Supply bypass capacitors selected to minimize supply noise
- Precision 200 Ω ($\pm 1\%$) resistor used for biasing and termination impedance calibration; the precision resistor requirement is shown in [Table 8-6](#).

Table 8-5 AC-Coupling Capacitors per Protocol

Protocol	AC Coupling Capacitor	Units
PCIe 1.1 and PCIe 2.1	200	nF
PCIe 3.1, PCIe 4.0, and PCIe 5.0	176-265	nF
CCIX	176-265	nF
1000BASE-KX	4.7	nF
10GBASE-KX4	4.7	nF
XAUI	4.7	nF
RXAUI	4.7	nF
10GBASE-KR	100	nF
40GBASE-KR4	100	nF
40GBASE-CR4 100GBASE-CR10	100	nF
XLAUI C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
XLAUI C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
XLPP/CPPI (nPPI)	Specification: In actual implementations, there may be various methods for AC-coupling.	
CAUI-10 C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
25GBASE-CR	100	nF
25GBASE-CR-S	100	nF
25GBASE-KR	100	nF

Protocol	AC Coupling Capacitor	Units
25GBASE-KR-S	100	nF
100GBASE-CR4	100	nF
100GBASE-KR4	100	nF
CAUI-4 C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
CAUI-4 C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
25GAUI C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
25GAUI C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
50GAUI-2 C2C 100GAUI-4 C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
50GAUI-2 C2M 100GAUI-4 C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
200GAUI-8 C2C 400GAUI-16 C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
200GAUI-8 C2M 400GAUI-16 C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
LAUI-2 C2C	Specification: In actual implementations, there may be various methods for AC-coupling.	
LAUI-2 C2M	Specification: In actual implementations, there may be various methods for AC-coupling.	
SGMII	Specification does not provide for a specific value.	
QSGMII	Specification does not provide for a specific value.	
2.5G-SXGMII	100	
5G-SXGMII, 5G-DXGMII	100	
10G-SXGMII, 10G-DXGMII, 10G-QXGMII	100	
JESD204C	Specification does not provide for a specific value.	
CPRI	Specification does not provide for a specific value.	

Table 8-6 External Component Requirements

Component	Value
resref	200 Ohm ($\pm 1\%$)

9

Reference Clock Implementation

The PHY is designed to handle a wide range of input clock frequencies to support a wide range of applications.

This chapter includes the following sections:

- [“Reference Clock Requirements”](#)
- [“Reference Clock Implementation”](#)

9.1 Reference Clock Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

9.1.1 Reference Clock Requirements PCIe (100 MHz)

The 100-MHz PCIe reference clock jitter and swing requirements are listed in [Table 9-1](#).



Note

With respect to PCIe specifications, the PMA uses an Independent Refclk (IR) architecture, also known as a data clocked receiver architecture.

Table 9-1 PCIe Reference Clock Requirements

Application	Reference Clock Jitter and Swing Requirements
PCIe 1.1 - 2.5 Gbps	Refer to the PCI Express 3.0 Card Electromechanical specification.
PCIe 2.1 - 5.0 Gbps	Refer to the PCI Express 3.1 base specification.
PCIe 3.1 - 8.0 Gbps	Refer to the PCI Express 3.1 base specification.
PCIe 4.0 - 16.0 Gbps	Refer to the PCI Express 4.0 base specification.
PCIe 5.0	PCI Express Base Specification, Revision 5.0 Version 1.0, May 2019, PCI-SIG

9.1.2 Reference Clock Amplitude Requirements

[Table 9-2](#) provides the reference clock amplitude requirements.

Table 9-2 Reference Clock Amplitude Requirements

Symbol	Parameter	Min	Max	Units	Conditions
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC
VDREF_CLK_ALT	Differential input swing for ref_alt0/1_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_DC and VDSEREF_CLK_AC
VDSEREF_CLK_DC	Input Voltage Level	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_DC or both VDREF_CLK_ALT and VDSEREF_CLK_DC
VDSEREF_CLK_AC ^a	Absolute voltage level at pin including overshoot	-0.3	1.15	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_AC or both VDREF_CLK_ALT and VDSEREF_CLK_AC

a. Allowed overshoot time: 10% of the reference clock period.

9.1.3 Generic Reference Clock Requirement for Protocols

Differential reference clock requirements for protocols other than PCIe are provided in [Table 9-3](#) for data rates below and above 8 Gbaud.

Table 9-3 Reference Clock Input Requirements

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	200	MHz	Not continuous; per protocol as noted in the databook
FREF_OFFSET	Reference clock frequency offset	-100	100	ppm	
RMSJREF_CLK <=8 Gbps	Reference clock random Jitter		1.6	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			1.2		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK <=8 Gbps	Reference clock deterministic jitter		2.8 5.6	ps, pp	0.75-10 MHz (offset) ^a 0.2-50 MHz (offset) ^a
RMSJREF_CLK <=16 Gbps, > 8 Gbps	Reference clock random jitter		1.0	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			0.8		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK <=16 Gbps, > 8 Gbps	Reference clock deterministic jitter		1.7 3.4	ps, pp	0.75 to 10 MHz (offset) ^a 0.2 to 50 MHz (offset) ^a
RMSJREF_CLK > 16 Gbps	Reference clock random jitter		0.45	ps _{rms}	Integrated RJ from 12 kHz to 20 MHz
			0.3		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK > 16 Gbps	Reference clock deterministic jitter		0.4 0.85	ps, pp	0.75 to 10 MHz (offset) ^a 0.2 to 50 MHz (offset) ^a
DCREF_CLK	Duty cycle	40	60	%	
VCMREF_CLK	Common mode input level	0	VP	V	Differential inputs
VDREF_CLK	Differential input swing for ref_pad_clk_{p,m}	300		mVpp	Differential inputs must meet VDREF_CLK, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDREF_CLK_ALT	Differential input swing for ref_alt0/1_clk_{p,m}	200		mVpp	Differential inputs must meet VDREF_CLK_ALT, VDSEREF_CLK_DC and VDSEREF_CLK_AC.
VDSEREF_CLK_DC	Input Voltage Level	-0.15	0.85	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_DC or both VDREF_CLK_ALT and VDSEREF_CLK_DC

Symbol	Parameter	Min	Max	Units	Conditions
VDSEREF_CLK_AC ^b	Absolute voltage level at pin including overshoot	-0.3	1.15	V	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK_AC or both VDREF_CLK_ALT and VDSEREF_CLK_AC
SKEWDIF_CLK	Differential clock input skew		100	ps	
SWREF_CLK	Differential clock input slew rate	0.6		V/ns	20–80%

a. DJ noise limits are for noise within an offset from the divided reference clock and harmonics of the divided reference clock.

b. Allowed overshoot time: 10% of the FREF_CLK period.

**Note**

For PCIe and CCIX, follow the specification requirements.

9.1.4 Repeat Clock Specification

The PHY is able to repeat the input reference clocks through up to six PHY macros. A reference clock repeater is provided to output the reference clock through ref_repeat_clk_p and ref_repeat_clk_m. The specifications for the repeater are provided in [Table 9-4](#).

**Note**

A PHY that is repeating the input reference clock should NOT be power gated (that is, pg_mode = 1'b0). Power gating the PHY disables the repeating function.

Table 9-4 Repeater Clock Output Specification

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	200	MHz	
DCREF_CLK	Duty cycle	40	60	%	No degradation is allowed
VCMREF_CLK	Common mode output	VP-0.2	VP-0.1	V	Differential CML output
VDREF_CLK	Differential output swing	400	800	mVpp	Recommended constant swing design, which allows for 400 mVpp swing all the time
SKEWDIF_CLK	Differential clock output skew		100	ps	No extra skew is allowed; instead, the PHY corrects the incoming clock skew (if needed)
CL_CLK	Single-ended output loading		2.5	pF	Specifies the maximum loading that the PHY can drive

Symbol	Parameter	Min	Max	Units	Conditions
SWREPT_CLK	Differential clock output slew rate at far end	0.6		V/ns	Measured over 20% to 80% of signal transition
LVL_OFF_CLK	Differential signal level when disabled		GD		Disabled clock repeater is pulled to 0 V
REPT_CLK	Maximum repeat times allowed		5		Repeat in daisy-chain connections through five PMAs to the final sixth PMA

9.2 Reference Clock Implementation

There are two types of reference clock implementations:

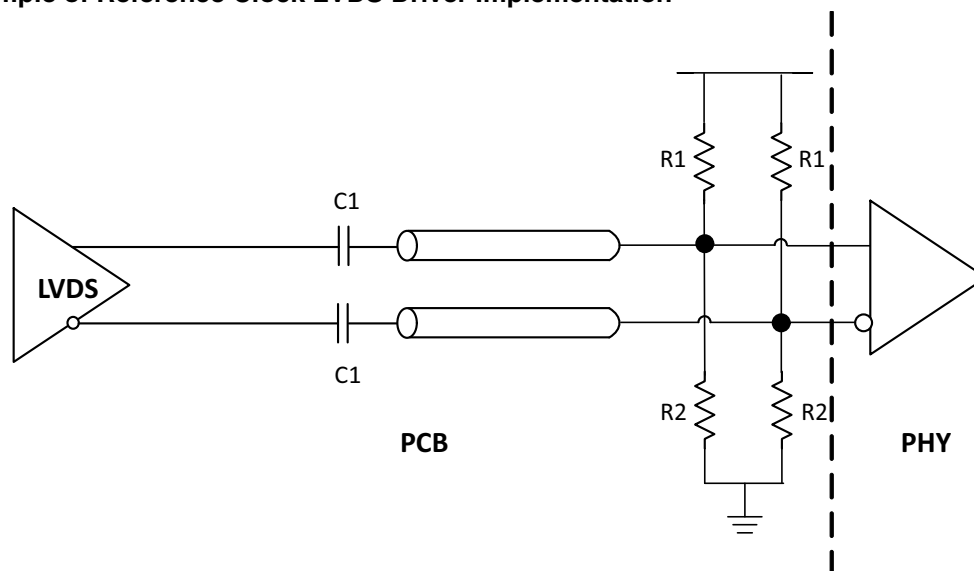
- External off-chip reference clocks using `ref_pad_clk_p` and `ref_pad_clk_m`
- On-chip reference clocks using `ref_alt0_clk_p/m` and `ref_alt1_clk_p/m`

Both reference clock implementations support either a differential clock or a CMOS single-ended clock. The reference clock input is an unterminated, AC coupled input. If the reference clock is a single-ended CMOS clock, one of the clock inputs (such as `ref_pad_clk_p`) must be a full-swing CMOS clock, and the other input (such as `ref_pad_clk_m`) should be tied to either high or low. If the reference clock is differential, the Host Clock Signal Level (HCSL) is the preferred signaling specification.

It may be also Common Mode Logic (CML), Low Voltage Differential Signaling (LVDS) or any differential specification that meets the requirements outlined in [Table 9-3](#) ("Reference Clock Input Requirements").

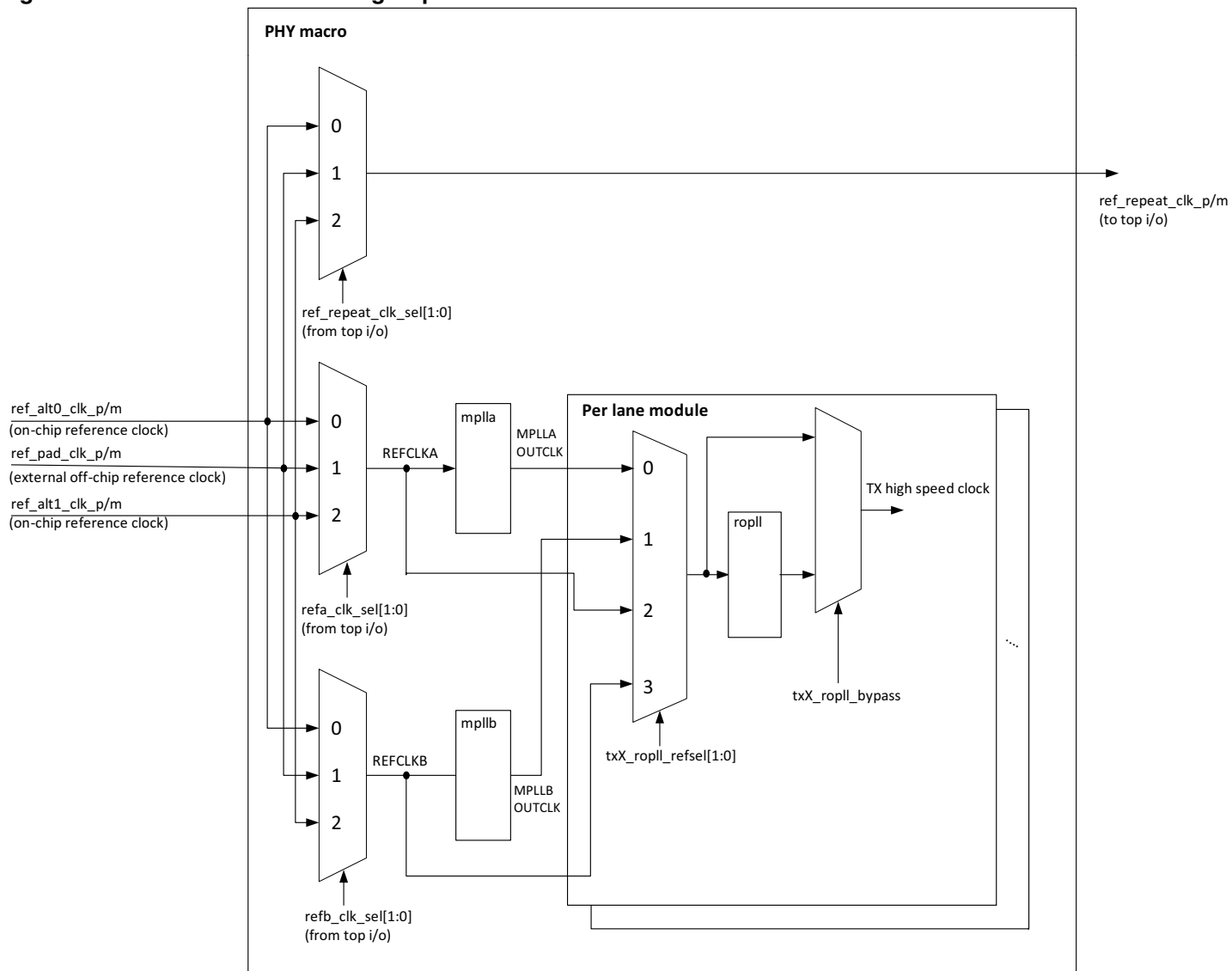
If the signaling used is other than HCSL, special care must be taken to meet the common mode level specification. An example showing level shift of LVDS signal to the common mode range supported by the `refclk` input is shown in [Figure 9-1](#). The `R1`, `R2`, `C1` should be chosen to achieve the desired common mode termination and bandwidth.

Figure 9-1 Example of Reference Clock LVDS Driver Implementation



Any reference clock inputs that are not being used should be tied to ground.

The selected reference clock(s), either on-chip or external, can be used in generating the MPLL clock for either direct clocking mode or cascaded clocking mode. For PCIe Gen1 and Gen2, the reference clock can also be used as the input reference for the lane ROPLL. In cascaded clocking mode the ROPLL uses a high-speed reference clock supplied from the MPLL output. These connections are shown in [Figure 9-2](#).

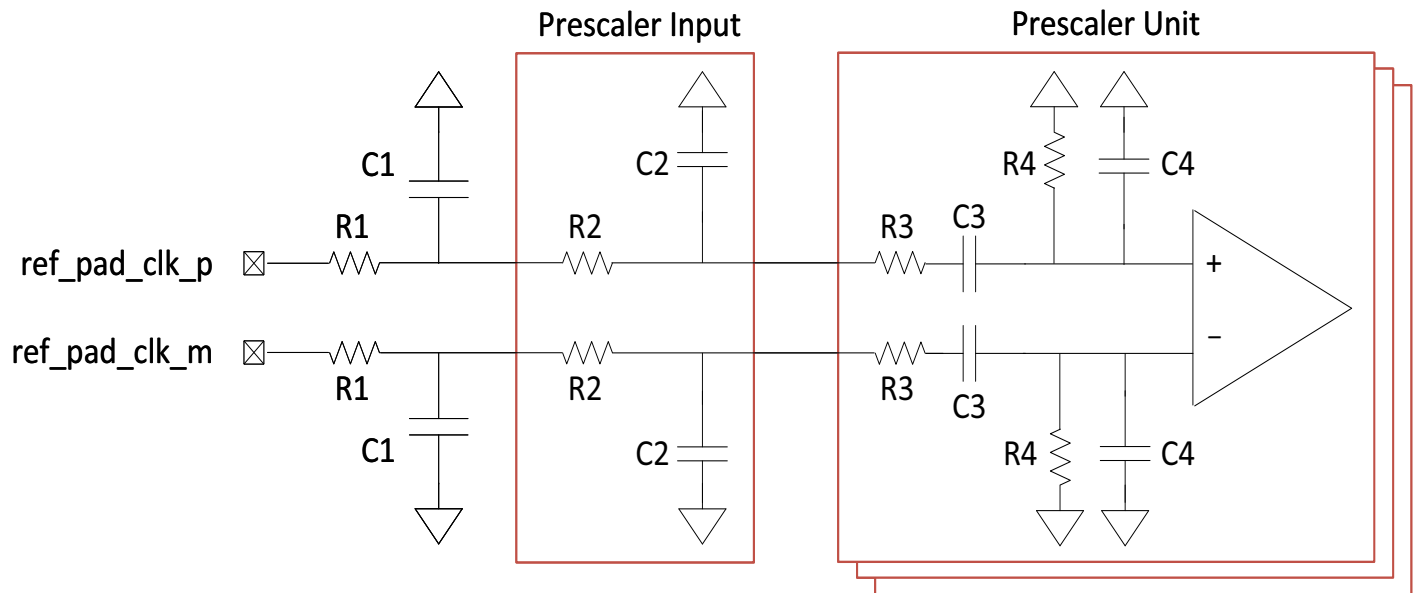
Figure 9-2 Reference Clock Usage Options for Direct and Cascaded PLL Modes

9.2.1 External Off-Chip Clock Sources

When routing reference clocks in from off-chip, a signal integrity (SI) analysis should be performed using a model of the clock input, package, board traces, and clock source. This analysis should confirm that the reference clock input specification is being met across the external on-chip reference clock input (ref_pad_clk_p/ref_pad_clk_m), as outlined in Section 9.1, “Reference Clock Requirements”.

A model of the reference clock input is provided in Figure 9-3 with parameters provided in Table 9-5.

Figure 9-3 Model of Reference Clock Pad Input Loading



Note

Loading of the prescaler input depends on the application-specific number of the prescaler units.

Table 9-5 Reference Clock External Pad Input Parasitics

Location	Parasitic Device	Value	Unit	Notes
Within PHY: bump-to-prescaler input	R1	≤ 20	Ω	
Within PHY: bump-to-prescaler input	C1	≤ 1.2	pF	
Within prescaler: second ESD Prot resistor	R2	250	Ω	Required for 6A peak discharge current ESD rating
Within prescaler: second ESD Prot, clock routes, and input cap of multiplexer	C2	180	fF	
Within prescaler unit: 3-1 multiplexer turn-on resistor	R3	150	Ω	
Within prescaler unit: AC cap	C3	880	fF	

Location	Parasitic Device	Value	Unit	Notes
Within prescaler unit: DC Bias resistor	R4	460k	Ω	
Within prescaler unit: input cap of prescaler input RX	C4	≤ 120	fF	

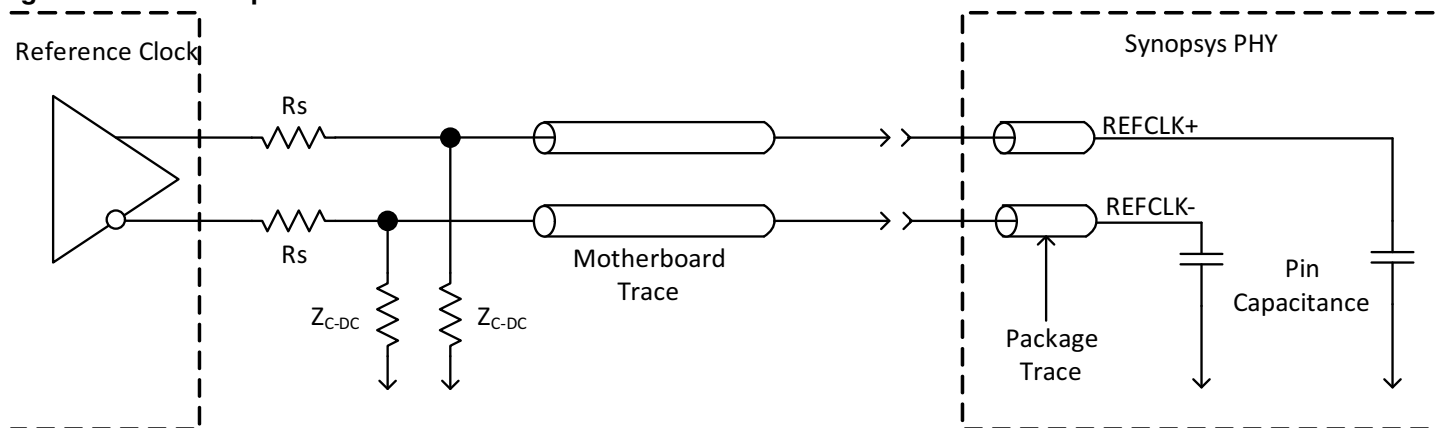
**Note**

To improve the Common-Mode Rejection Ratio (CMRR), differential traces should be as well matched as possible.

To enable connecting reference clocks to multiple PHYs at the same time, the reference clock inputs are not terminated. If termination is used, place it on your board as close to the chip as possible. For PCIe, see the PCIe CEM specification for details on how to implement the on-board termination.

If you use off-chip reference clock (that is, `ref_pad_clk_p` or `ref_pad_clk_m`), you must terminate the off-chip clock source to suppress reflection. Also, make sure that the clock swing at the Synopsys PHY input is within reference clock specification. An example of this implementation is shown in [Figure 9-4](#).

Figure 9-4 Off-chip Clock Source Termination



9.2.2 On-Chip Clock Sources

Many applications require an on-chip reference clock. The challenge is getting the reference clock to the PHY without adding supply-noise-induced jitter to the clock.

The following general guidelines should be followed when routing an on-chip reference clock:

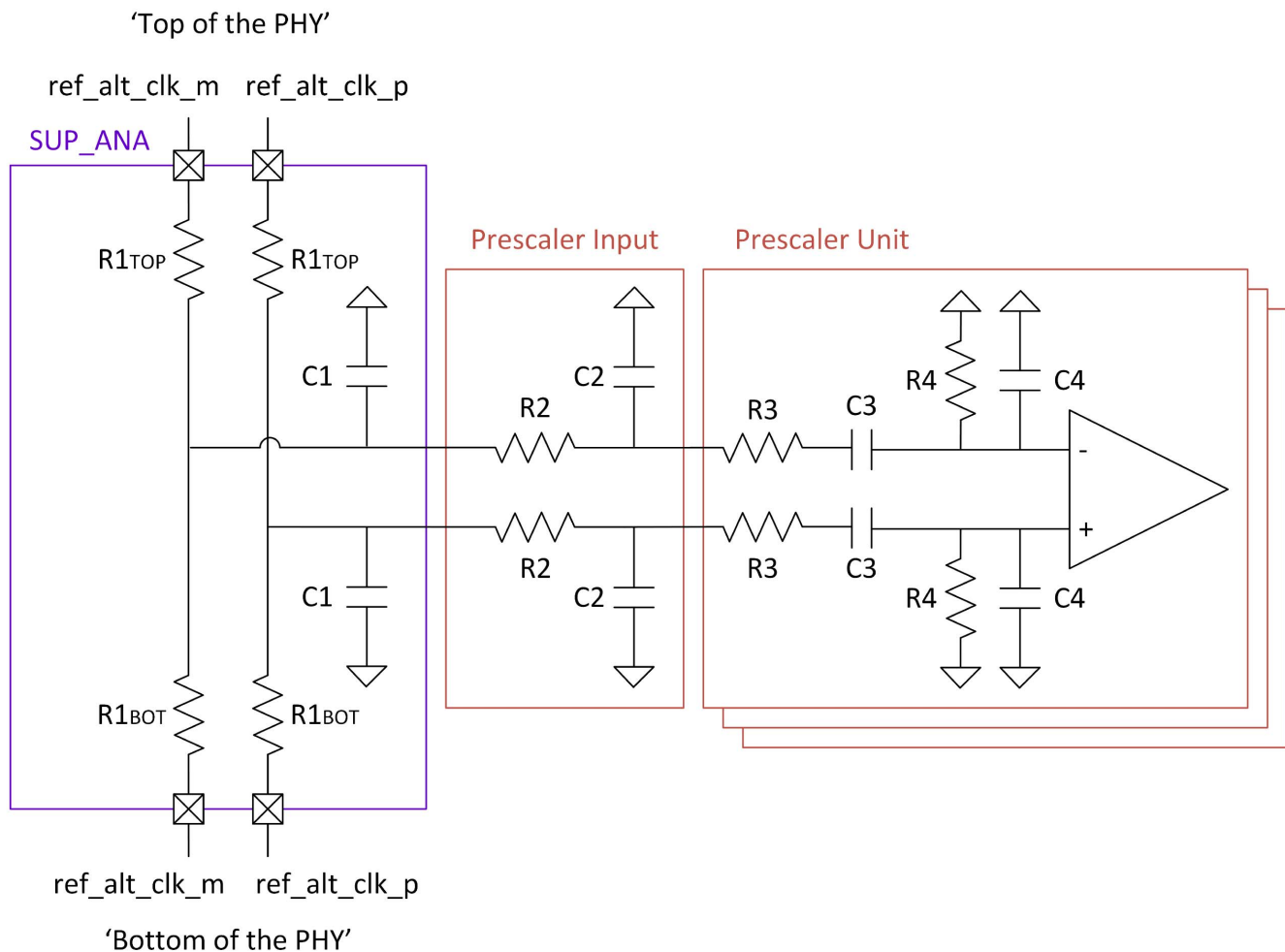
- Route manually (do not use PNR tool router).
- Design differential routing to optimize RC.
- Consider higher-level metal to get best RC factor (lower-level metals tend to be too resistive).
- Avoid noisy signals directly above and below the routes.
- If you do route signals near the routes, do not have them parallel – use perpendicular routes only, so that noise is common mode.
- Reserve keep-out space as necessary around diff routes to minimize coupling and parasitic loading.

- Density layer fill for routes on routing layer (none or targeted fills to Min.Density per DRC) in both sections.
- Simulate to ensure proper signal integrity.

A model of the reference clock input is provided in [Figure 9-5](#) with parameters provided in [Table 9-6](#).

When routing reference clocks on-chip, you should consider the parasitic loading of the clock distribution lines and the PHY clock input. This analysis should confirm that the reference clock input specification is being met across the on-chip reference clock input (ref_alt0_clk_p/m and ref_alt1_clk_p/m) as outlined in [Section 9.1](#), “Reference Clock Requirements”.

Figure 9-5 Model of Reference Clock ASIC Side ALT Input Loading



Note

Loading of the prescaler input depends on the application-specific number of the prescaler units.

Table 9-6 Reference Clock ASIC-Side ALT Input Parasitics

Location	Parasitic Device	Value	Unit	Notes
Clock routes before prescaler from “Top of the PHY”	R1 _{TOP}	25	Ω	
Clock routes before prescaler from “Bottom of the PHY”	R1 _{BOT}	90	Ω	
Clock routes before prescaler	C1	200	fF	
Within prescaler: clock routes	R2	280	Ω	
Within prescaler: clock routes and input cap multiplexer	C2	80	fF	
Within prescaler: 3-1 multiplexer turn-on resistor	R3	150	Ω	
Within prescaler: AC cap	C3	880	fF	
Within prescaler: DC bias resistor	R4	460k	Ω	
Within prescaler: input cap of prescaler input RX	C4	≤ 120	fF	

**Note**

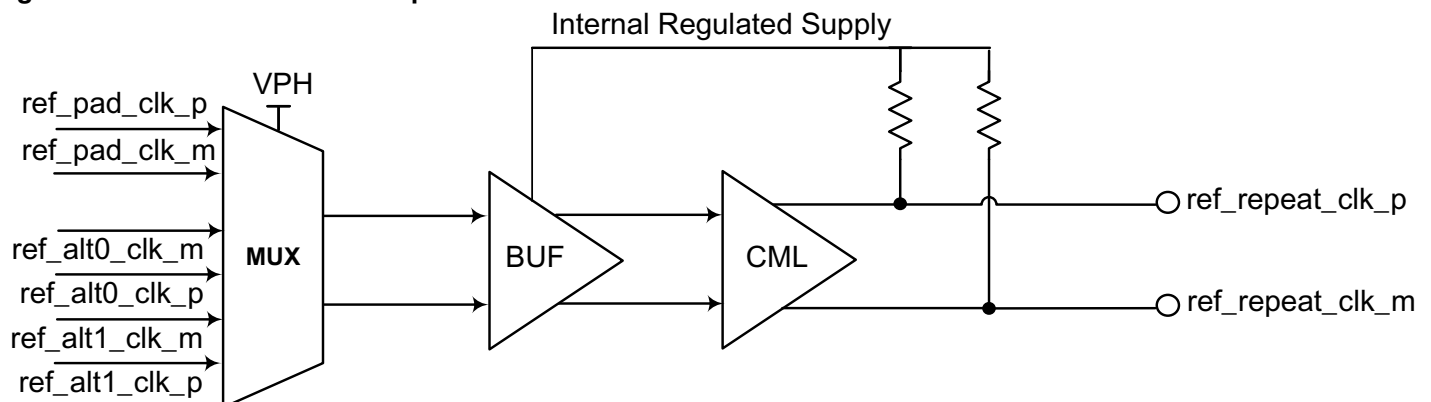
To improve the Common-Mode Rejection Ratio (CMRR), differential traces should be as well matched as possible.

9.2.3 Reference Clock Repeater

There is a reference clock repeater output (ref_pad_clk_m/ref_pad_clk_p) that allows the PHY to buffer the reference clock and send it on to another PHY. The driver is an internally regulated supply referenced CML output clock driver, as shown in [Figure 9-6](#). The repeater is designed to meet the specifications given in [Section 9.1.4](#), “Repeat Clock Specification”.

**Note**

Ref_clk repeat buffer and single-ended buffers are not in an always-on power domain. If power-gating is enabled, these outputs are powered down. Do not use these signals for logic or PHYs intended to remain operational during power-gating.

Figure 9-6 Reference Clock Repeater

**Note**

To improve common-mode rejection ratio (CMRR), differential traces should be as well matched as possible.

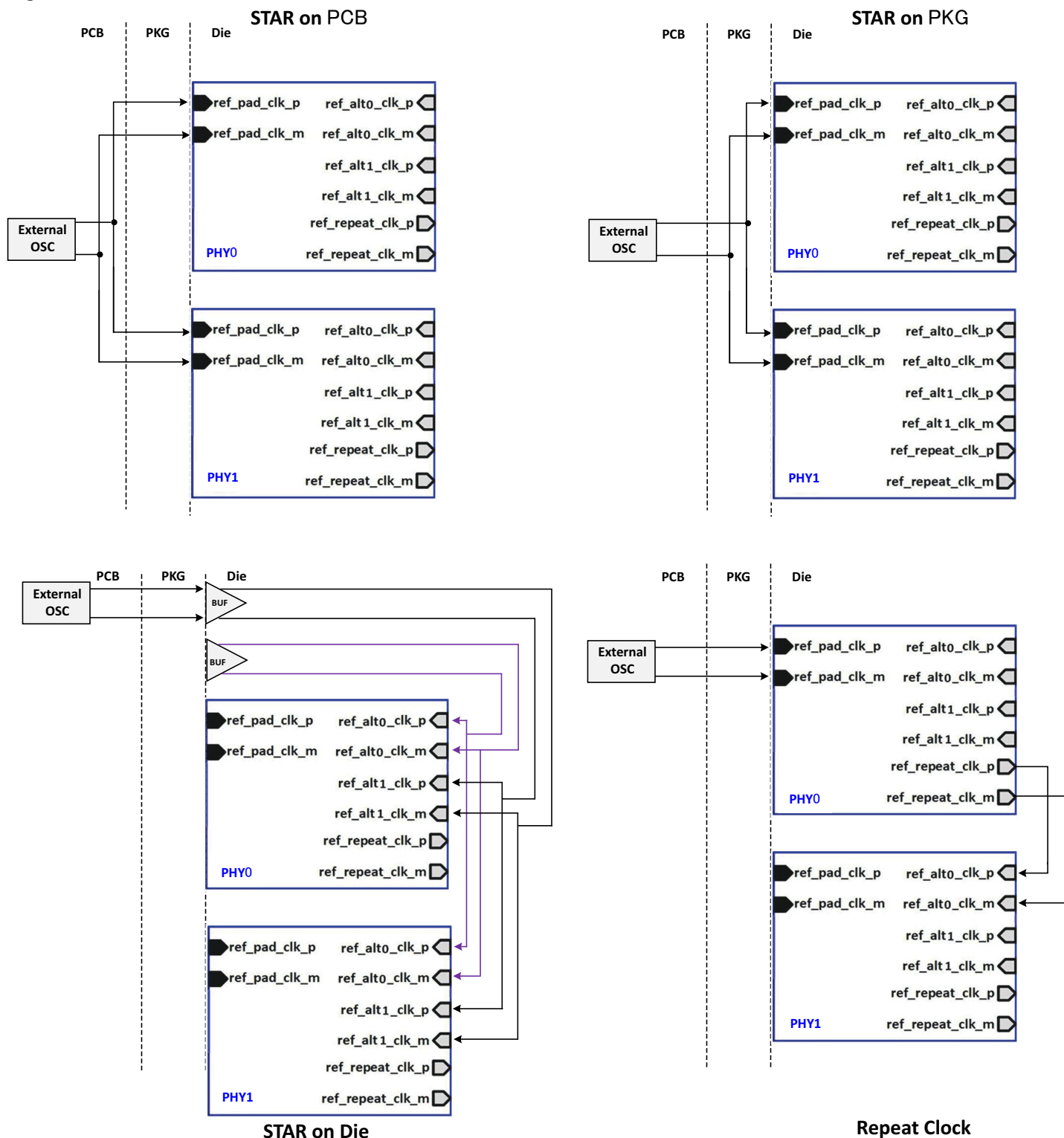
9.2.4 Reference Clock Distribution

To minimize the amount of jitter on the reference clock, Synopsys recommends STAR configuration in the clock distribution. [Figure 9-7](#) provides examples of the clock distribution on board, in the package and on the die. [Table 9-7](#) provides impact of each of the reference clock configurations.

Table 9-7 Clock Distribution Impact

No.	Item	Clock Distribution Example				Remark
		STAR on PCB	STAR on Package	STAR on Die	Repeat Clock	
1	REFCLK input	PAD	PAD	Core	PAD/Core	
2	Jitter	-	-	Increase	Increase	Jitter increases if you use active devices
3	ATE Testbench	As is	As is	As is	Customize	Two PHYs setting becomes different if you use repeat clock. Then, you need to customize your ATE testbench.
4	REFCLK SI analysis	ref_pad_clk only	ref_pad_clk only	ref_alt0/1_clk only	ref_pad_clk ref_alt0/1_clk	
5	Additional buffer	-	-	Necessary	-	Additional REFCLK buffer is necessary for STAR on Die

Synopsys recommends to use STAR on PCB or PKG. It results in better reference clock quality (less jitter) and less implementation efforts. Depending on your design, you can use STAR on Die or repeat clock.

Figure 9-7 Reference Clock Distribution



- Connect **repeat** clock to either **alt0** or **alt1** (not both). Unused **alt** signal connect to gnd.
- Set *ref_repeat_clk_sel[1:0]* according to your selection (see table “Reference Clock Control Signals” in the “Signal Descriptions” chapter in the *DesignWare Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC Reference Manual*).
- When off-chip reference clocks (such as, pad clocks) are not used, make note of the following:
 - Tie the unused clocks to ground.
 - OR
 - There is no issue if clocks are toggling on the reference clock pads.

10

Electrical Specifications

This chapter includes the following sections:

- [“Power Dissipation Overview”](#)
- [“Power Dissipation for PCIe”](#)
- [“Power Dissipation for CCIX”](#)
- [“Power Dissipation for Ethernet”](#)

10.1 Power Dissipation Overview

The PHY supports the power supply voltages in [Table 10-1](#).

Table 10-1 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Minimum	Typical	Maximum	Unit
vph	High voltage IO supply	1.2 -5%	1.2/1.5/1.8	1.8 +10%	V
vp/vptx_X/vpdig	Analog supply	0.9 -5%	0.9	0.9 +5%	V
vddcore	Core supply	0.8 -10%	0.8/0.9	0.9 +5%	V

The following sections provide simulated power consumption values for the PHY.

- The typical (Typ) values are for TT process corner, 25°C junction temperature, and typical supplies for specified data rate.
- The worst-case (Wc) values are for FF process corner, 125°C junction temperature, and maximum voltage supplies for specified data rate.



Note

The power numbers in the following tables

- are based on a direct PLL clocking configuration, and
- do not include raw PCS power consumption.

10.2 Power Dissipation for PCIe

Table 10-2 PCIe x2 Direct Power Dissipation

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P0	32Gbps Typ	180.300	167.800	177.900	40.101	0.289	566.391	563.093	616.463	669.833
	32Gbps Wc	250.000	213.600	210.600	154.832	4.351	833.383	866.522	936.020	1005.518
	16Gbps Typ	90.970	95.230	115.400	22.861	0.197	324.657	326.792	361.412	396.032
	16Gbps Wc	142.900	122.600	141.700	137.904	4.430	549.534	572.447	619.208	665.969
	8Gbps Typ	62.060	57.250	97.150	15.541	0.095	232.097	238.022	267.167	296.312
	8Gbps Wc	108.600	77.050	121.800	121.372	3.257	432.079	453.990	494.184	534.378
	5Gbps Typ	42.830	37.130	75.590	10.761	0.092	166.403	172.430	195.107	217.784
	5Gbps Wc	84.760	56.080	97.710	117.345	3.286	359.181	376.067	408.311	440.556
	2.5Gbps Typ	36.620	25.020	70.560	10.253	0.099	142.552	149.455	170.623	191.791
	2.5Gbps Wc	77.250	41.820	90.780	115.602	3.343	328.794	344.753	374.710	404.668

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P0s	32Gbps Typ	149.600	93.050	184.800	34.522	0.256	462.227	471.419	526.859	582.299
	32Gbps Wc	222.100	127.400	218.800	147.143	4.323	719.766	762.229	834.433	906.637
	16Gbps Typ	80.740	52.890	123.100	20.164	0.185	277.079	286.282	323.212	360.142
	16Gbps Wc	132.100	75.420	151.000	133.834	4.375	496.729	526.034	575.864	625.694
	8Gbps Typ	56.960	31.780	104.800	13.886	0.081	207.506	218.188	249.628	281.068
	8Gbps Wc	102.700	50.350	131.200	119.639	3.267	407.157	433.963	477.259	520.555
	5Gbps Typ	39.850	18.130	83.140	9.910	0.085	151.115	160.937	185.879	210.821
	5Gbps Wc	81.370	37.530	106.900	116.217	3.290	345.307	366.402	401.679	436.956
	2.5Gbps Typ	35.120	11.480	78.290	9.852	0.096	134.838	144.831	168.318	191.805
	2.5Gbps Wc	75.470	29.630	100.500	114.917	3.342	323.859	343.734	376.899	410.064

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P1	32Gbps Typ	4.150	10.280	32.940	5.248	0.058	52.676	57.285	67.167	77.049
	32Gbps Wc	27.990	26.520	37.510	112.049	4.131	208.200	210.816	223.194	235.572
	16Gbps Typ	4.110	10.240	32.940	5.495	0.071	52.856	57.445	67.327	77.209
	16Gbps Wc	28.270	26.150	37.500	116.489	4.312	212.721	215.084	227.459	239.834
	8Gbps Typ	3.856	4.295	32.940	6.872	0.063	48.026	53.099	62.981	72.863
	8Gbps Wc	28.040	20.400	37.520	111.140	3.369	200.469	203.513	215.894	228.276
	5Gbps Typ	3.292	4.296	17.150	5.350	0.059	30.146	32.271	37.416	42.561
	5Gbps Wc	27.120	20.390	22.190	111.306	3.363	184.369	182.550	189.873	197.196
	2.5Gbps Typ	3.293	4.295	17.150	7.383	0.079	32.200	34.117	39.262	44.407
	2.5Gbps Wc	27.130	20.390	22.190	112.925	3.451	186.086	184.173	191.495	198.818
P2	Typ	2.687	0.623	3.806	3.795	0.035	10.945	10.989	12.131	13.273
	Wc	26.420	16.210	4.886	109.764	4.078	161.358	154.316	155.928	157.540
P1.cpm	Typ	2.525	2.525	2.525	2.415	0.189	10.180	9.900	10.658	11.415
	Wc	30.290	30.290	30.290	4.695	4.541	100.107	105.959	115.955	125.951
P1.1	Typ	0.899	1.299	0.509	3.803	0.189	6.699	6.162	6.315	6.468
	Wc	28.250	15.720	0.688	108.216	4.541	157.415	149.015	149.242	149.469
P1.2	Typ	0.879	0.289	0.507	3.603	0.177	5.454	5.043	5.195	5.347
	Wc	28.260	14.420	0.688	108.629	4.703	156.699	148.338	148.565	148.792

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
Power Down	Typ	0.465	0.289	0.000	1.344	0.054	2.152	1.931	1.931	1.931
	Wc	26.060	14.020	0.043	108.333	4.635	153.090	144.686	144.700	144.714

Table 10-3 PCIe x2 Cascaded Power Dissipation

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P0	32Gbps Typ	179.300	167.900	226.200	41.534	0.290	615.223	621.532	689.392	757.252
	32Gbps Wc	248.800	213.900	269.700	154.856	4.336	891.592	943.692	1032.693	1121.694
	16Gbps Typ	89.870	93.720	166.700	21.461	0.142	371.893	384.700	434.710	484.720
	16Gbps Wc	141.900	125.400	204.100	135.196	4.239	610.835	653.777	721.130	788.483
P0s	32Gbps Typ	148.600	93.190	233.500	35.035	0.216	510.542	529.516	599.566	669.616
	32Gbps Wc	220.900	128.000	277.800	148.358	4.238	779.295	840.609	932.283	1023.957
	16Gbps Typ	79.630	51.250	174.700	18.271	0.121	323.972	343.973	396.383	448.793
	16Gbps Wc	131.200	78.710	209.800	131.219	4.201	555.130	603.273	672.507	741.741
P1	32Gbps Typ	9.212	7.823	68.190	6.466	0.069	91.760	103.034	123.491	143.948
	32Gbps Wc	26.810	26.510	91.930	112.104	4.114	261.468	281.561	311.898	342.235
	16Gbps Typ	4.557	5.530	45.910	4.089	0.032	60.119	67.877	81.650	95.423
	16Gbps Wc	26.780	25.400	72.740	113.940	4.221	243.081	256.989	280.993	304.997

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P2	Typ	2.675	0.623	2.559	4.616	0.043	10.516	10.228	10.996	11.763
	Wc	26.390	16.220	3.307	109.564	4.063	159.543	152.009	153.100	154.191
P1.cpm	Typ	2.525	1.300	1.129	3.802	0.186	8.943	8.369	8.707	9.046
	Wc	30.290	15.720	1.510	108.205	4.502	160.227	151.981	152.479	152.977
P1.1	Typ	0.899	1.299	0.509	3.803	0.188	6.697	6.161	6.314	6.467
	Wc	28.250	15.720	0.688	108.216	4.633	157.507	149.102	149.329	149.555
P1.2	Typ	0.879	0.289	0.507	3.603	0.174	5.452	5.041	5.193	5.345
	Wc	28.260	14.420	0.688	108.629	4.662	156.658	148.300	148.527	148.754
Power Down	Typ	0.465	0.289	0.000	1.345	0.046	2.145	1.926	1.926	1.926
	Wc	26.060	14.020	0.043	106.933	3.708	150.764	142.488	142.502	142.516

10.3 Power Dissipation for CCIX

Table 10-4 CCIX x2 Cascaded Power Dissipation

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
P0	25 Gbps Typ	146.800	135.600	183.200	34.408	0.249	500.257	505.166	560.126	615.086
	25 Gbps Wc	214.500	171.100	223.900	145.967	4.226	759.693	801.873	875.760	949.647
	20 Gbps Typ	126.400	111.000	154.000	28.328	0.114	419.842	424.046	470.246	516.446
	20 Gbps Wc	189.900	142.100	198.700	137.986	4.212	672.898	710.401	775.972	841.543
P0s	25 Gbps Typ	123.500	74.900	191.200	29.162	0.219	418.981	434.421	491.781	549.141
	25 Gbps Wc	187.500	101.700	232.900	140.193	4.105	666.397	717.083	793.940	870.797
	20 Gbps Typ	107.700	60.460	167.500	24.855	0.093	360.607	374.787	425.037	475.287
	20 Gbps Wc	168.700	84.960	207.800	133.406	4.183	599.049	644.026	712.600	781.174
P1	25 Gbps Typ	9.912	8.735	70.700	6.909	0.068	96.324	107.895	129.105	150.315
	25 Gbps Wc	27.160	24.650	86.910	112.848	4.159	255.727	274.253	302.934	331.614
	20 Gbps Typ	5.193	7.447	66.550	6.472	0.064	85.726	97.112	117.077	137.042
	20 Gbps Wc	28.980	23.340	81.800	111.304	4.138	249.562	266.511	293.505	320.499
P2	Typ	2.667	0.614	2.554	4.744	0.045	10.624	10.324	11.090	11.856
	Wc	26.370	15.830	3.309	109.742	4.073	159.324	151.802	152.894	153.986
Power Down	Typ	0.465	0.289	0.0002	1.322	0.052	2.129	1.911	1.911	1.911
	Wc	26.060	14.020	0.043	109.727	4.792	154.641	146.151	146.165	146.179

10.4 Power Dissipation for Ethernet

Table 10-5 Ethernet x2 Power Dissipation

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
Active	28 Gbps Typ	153.200	151.000	197.700	38.015	0.281	540.196	545.458	604.768	664.078
	28 Gbps Wc	226.800	189.200	242.000	153.133	4.389	815.521	861.418	941.278	1021.138
	26 Gbps Typ	127.600	141.900	187.100	37.756	0.295	494.651	501.286	557.416	613.546
	26 Gbps Wc	193.100	182.300	228.700	151.680	4.496	760.276	804.223	879.694	955.165
	10.3 Gbps Typ	71.170	64.270	138.500	16.181	0.091	290.212	302.731	344.281	385.831
	10.3 Gbps Wc	119.900	87.440	174.600	127.690	3.380	513.011	550.270	607.888	665.506
	5.15 Gbps Typ	53.060	39.610	126.400	12.945	0.109	232.124	246.821	284.741	322.661
	5.15 Gbps Wc	98.620	57.810	160.800	118.938	3.315	439.483	475.612	528.676	581.740
	3.125 Gbps Typ	38.060	31.050	136.000	9.285	0.081	214.475	233.820	274.620	315.420
	3.125 Gbps Wc	80.270	46.240	168.400	120.374	3.528	418.812	458.927	514.499	570.071

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
Refresh	28 Gbps Typ	129.000	84.540	197.600	32.780	0.258	444.178	459.015	518.295	577.575
	28 Gbps Wc	195.800	112.600	241.200	146.606	4.448	700.654	752.568	832.164	911.760
	26 Gbps Typ	110.800	78.510	187.200	32.597	0.259	409.366	424.563	480.723	536.883
	26 Gbps Wc	172.100	109.400	227.800	143.871	4.448	657.619	706.875	782.049	857.223
	10.3 Gbps Typ	64.420	34.280	138.500	14.316	0.060	251.575	267.962	309.512	351.062
	10.3 Gbps Wc	112.200	55.160	174.400	124.989	3.384	470.133	509.676	567.228	624.780
	5.15 Gbps Typ	49.590	20.680	126.500	12.375	0.110	209.254	226.268	264.218	302.168
	5.15 Gbps Wc	94.850	39.030	160.800	117.923	3.343	415.946	453.369	506.433	559.497
	3.125 Gbps Typ	36.040	16.540	136.100	8.786	0.076	197.542	218.611	259.441	300.271
	3.125 Gbps Wc	77.880	32.620	168.500	119.810	3.545	402.355	443.413	499.018	554.623

Mode		Current from vp (mA)	Current from vptx (mA)	Current from vph (mA)	Current from vpdig (mA)	Current from vddcore (mA)	Total current (mA)	Total Power Consumption (mW) vph= 1.2/1.32	Total Power Consumption (mW) vph= 1.5/1.65	Total Power Consumption (mW) vph= 1.8/1.98
Quiet	28 Gbps Typ	119.200	84.540	197.800	12.012	0.093	413.645	431.611	490.951	550.291
	28 Gbps Wc	183.800	112.600	241.300	120.583	4.325	662.608	716.652	796.281	875.910
	26 Gbps Typ	101.700	78.510	187.400	13.127	0.099	380.836	398.963	455.183	511.403
	26 Gbps Wc	160.700	109.400	227.800	120.850	4.407	623.157	674.308	749.482	824.656
	10.3 Gbps Typ	60.830	34.280	138.600	6.823	0.047	240.581	258.098	299.678	341.258
	10.3 Gbps Wc	107.800	55.160	174.400	116.447	3.552	457.360	497.605	555.157	612.709
	5.15 Gbps Typ	47.820	20.680	126.600	8.369	0.080	203.549	221.166	259.146	297.126
	5.15 Gbps Wc	92.630	39.030	160.900	113.708	3.451	409.719	447.522	500.619	553.716
	3.125 Gbps Typ	35.080	16.540	136.200	6.679	0.057	194.557	215.955	256.815	297.675
	3.125 Gbps Wc	76.690	32.620	168.500	117.750	3.633	399.193	440.425	496.030	551.635
Disabled	Typ	2.586	0.549	2.561	4.730	0.051	10.477	10.192	10.960	11.729
	Wc	26.230	15.720	3.318	111.638	4.225	161.132	153.514	154.609	155.704
Power Down	Typ	0.465	0.289	0.0002	1.322	0.052	2.128	1.910	1.910	1.910
	Wc	26.060	14.020	0.043	108.248	4.625	152.996	144.597	144.611	144.625

A

Support and Lane Configuration Settings

For Support configuration values for configuring the PHY for your application, refer to the <phy installation directory>/doc/dwc_32g_phy_<process>_config_reference_manual.xlsx (SUPPORT_CONFIG_DATA_BOOK tab).

For Lane configuration values for configuring the PHY for your application, refer to the <phy installation directory>/doc/dwc_32g_phy_config_<process>_reference_manual.xlsx (LANE_CONFIG_DATA_BOOK tab).

B

Macro ifdef Settings

This appendix contains the configuration and compile macro ifdef settings for your PHY, provided in the following tables:

- [Figure B-1, “Configuration and Compile Macro ifdef Settings”](#)
- [Table B-1, “Design Mode Macros”](#)



Note

- Instances of <PREFIX> in the tables are placeholders for DWC_E32MP_X2NS.
- Do not use Scaledown0 simulation mode as an SoC sign-off criteria. For SoC-level sign-off criteria, run a few Non Scale Down (also called normal reset or long reset) simulations without the <PREFIX>_SHORT_RESET macro defined. For normal reset simulations, set FAST_RX_STARTUP_CAL bit to '1' and all other FAST bits to '0'.
- Limitations of the <PREFIX>_ANA_BEHAV macro are as follows:
 - <PREFIX>_ANA_BEHAV uses a fast analog behavioral mathematical model. Hence, do not use <PREFIX>_ANA_BEHAV for ATE, Gate-Level, and Boundary Scan simulations.
 - Do not use <PREFIX>_ANA_BEHAV for SATA simulations because of a mathematical precision issue for certain clock frequencies, such as 600 MHz, which has period of 1.6666.... ns.
 - When <PREFIX>_ANA_BEHAV is used and when rxX_p and rxX_m are in an idle state, the PMA model does not drive any random data, hence, 'X' is observed on the RxData and RxDataValid signals. This is a known behavior.
 - When <PREFIX>_ANA_BEHAV is not used and when rxX_p and rxX_m are in an idle state, the PMA model drives random data on rxX_p and rxX_m. In this case, use Scale-Down1 mode.

Figure B-1 Configuration and Compile Macro ifdef Settings

Legend														
✓	Indicates that the macro must be set to achieve this Scaledown Mode.													
✓	Indicates that the macro is automatically set in the Scaledown Mode by a macro that must be set in this mode.													
Simulation Scaledown Mode			ScaleDown0		Scaledown1		Scaledown2		Scaledown3		Non-ScaleDown			
Analog Model			Behavioral		Netlisted		Netlisted		Netlisted		Netlisted			
TX/RX Power-up timing			Fast		Fast		Silicon		Silicon		Silicon			
TX/RX Interrupts			Disabled		Disabled		Enabled		Enabled		Enabled			
Adaptation			Fast		Fast		Fast		Silicon		Silicon			
MPLL Power-up timing			Fast		Fast		Fast		Fast		Silicon			
Calibrations			Skipped		Skipped		Skipped		Skipped		Enabled			
Superset	Macro ^a	Description	PMA GTECH	PMA GATE	PMA GTECH	PMA GATE	PMA GTECH	PMA GATE	PMA GTECH	PMA GATE	PMA GTECH	PMA GATE		
	<PREFIX>_SIM_MODE	Enables synchronizer metastability modeling and analog model functionality. Mandatory for running simulations.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
	<PREFIX>_RANDOM	Randomizes behavioral modeling, such as clock tree delays. Mandatory for running simulations.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
	<PREFIX>_RANDOM_SEED	Randomizes behavioral modeling, such as clock-tree delays. Mandatory for running simulations.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
	<PREFIX>_SIM_GATES	Required for PMA gate-level simulations.		✓		✓		✓		✓		✓		
	<PREFIX>_SIM_GTECH	Required for PMA GTECH simulations.	✓		✓		✓		✓		✓			
	<PREFIX>_ANA_BEHAV	Use fast analog behavioral model.	✓	✓										
	<PREFIX>_SHORT_RESET	Run in FAST Simulation Mode.	✓	✓	✓	✓								
	<PREFIX>_DISABLE_RXTX_REQ_IRQ	Disable RX/TX request interrupts.	✓	✓	✓	✓								
	<PREFIX>_FAST_RX_ADAPT	FAST RX Start-up Adaptation	✓	✓	✓	✓	✓	✓						
	<PREFIX>_FAST_RX_STARTUP_CAL	FAST RX Start-up Calibration	✓	✓	✓	✓	✓	✓	✓	✓				
	<PREFIX>_FAST_TX_STARTUP_CAL	FAST TX Start-up Calibration					✓	✓	✓	✓				
SUP	<PREFIX>_FAST_SUP	Skip MPLL Calibration and Fast Rtune, bandgap, and prescaler powerup.	✓	✓	✓	✓	✓	✓	✓	✓				
	<PREFIX>_FAST_BG_WAIT	Fast bandgap powerup	✓	✓	✓	✓	✓	✓	✓	✓				
HX	<PREFIX>_FAST_RX_PWRUP	Fast sigdet time, vreg_en, and clk_dcc_en	✓	✓	✓	✓								
	<PREFIX>_FAST_RX_VCO_CAL	Fast cdr_startup, vco_cntr_pd, clk_en and vco_update.	✓	✓	✓	✓								
	<PREFIX>_FAST_RX_VCO_WAIT	Skip VCO Calibrations.	✓	✓	✓	✓								
	<PREFIX>_FAST_RX_CONT_ADAPT	Fast Continuous Adaptation.	✓	✓	✓	✓								
	<PREFIX>_FAST_RX_CONT_CAL_ADAPT	Fast Continuous calibration/adaptation.	✓	✓	✓	✓								
TX	<PREFIX>_FAST_TX_PWRUP	Fast TX Powerup	✓	✓	✓	✓								
	<PREFIX>_FAST_TX_VCM_HOLD	Fast common-mode charge-up time	✓	✓	✓	✓								
	<PREFIX>_FAST_TX_RXDET	Fast Receiver Detect time	✓	✓	✓	✓								
Additional Macros	<PREFIX>_FW_FAST_SIM	Reduces calibration and adaptation firmware counts.												
	<PREFIX>_FW_OVRD	Half firmware												
	<PREFIX>_DISABLE_SRAM_REC	Disable SRAM adaptation parameter recording.												
	<PREFIX>_INT_SRAM_SUPPORT	Enables instantiation of SRAM inside PHY (Raw PCS). Applies to projects where SRAM is instantiated inside the PHY as a customization. Contact Synopsys, if this customization is required.												
	<PREFIX>_REMOVE_CNTX_RSTN_INPUTS	Removes PHY configuration input ports for reduced pinout. See Section "Context Restore".												
	<PREFIX>_SYNTH_MODE	Used to synthesize the digital design. Not to be used in simulations.												
	<PREFIX>_TOP_PG_PINS	Defines the power-pins at the top-level instance.												
	<PREFIX>_PG_PINS	Defines the power-pins at the hard-macro PMA level.												
	<PREFIX>_PHY_HARDENING	Add additional ports for projects where Raw PCS is hardened by Synopsys as a customization. Contact Synopsys if this customization is required.												

a. For more information on macros, see "/.phy/testbench/TESTBENCH.README" file.

Table B-1 Design Mode Macros

ifdef Macro Name	Description	Reference
<PREFIX>_SIM_MODE	Sets the model to operate in simulation mode.	Section 6.4, “Verilog Model Files and Simulators” on page 194
<PREFIX>_RANDOM_SEED	Sets the seed for randomization.	
<PREFIX>_NO_EXT_ROM_SUPPORT	Disables external ROM and allows you to use only the internal hard-coded lookup table based on ROM.	Section 4.15, “ROM Interface” on page 128
<PREFIX>_SHORT_RESET	Enables the fast simulation mode.	Section 6.5, “Fast Simulation Mode” on page 195
<PREFIX>_DISABLE_RATE_IRQ	Disables the interrupts; removes the rate change times in <PREFIX>_SHORT_RESET mode by disabling firmware interaction during rate-change requests to the PHY; removes TX/RX DCC calibrations/restorations on rate changes.	Section 6.5.1, “Interrupt Disable Mode” on page 201

C

Installation Quick Reference

This appendix contains the installation quick reference for your PHY.

PHY Installation: Quick Reference		
Installation Process	Key Commands	Key URLs
1. Prerequisites - A valid Synopsys Solvnet ID - A valid Project ID - A valid license running on your license server - Ensure that system requirements are met. 2. Downloading the PHY a. Navigate to the product component page---from the mydesignware notification e-mail or search for IP on https://www.designware.com. b. On the product component page, download the image files: .run file and .pm files (if applicable) on to the same directory. 3. Installing the PHY a. On the UNIX server, navigate to the directory where you have downloaded the files. b. Make sure that the image files (.run and .pm) have execute permissions: <code>% chmod 555 <filename></code> c. Execute the .run file to unpack the image. <code>% ./<filename>.run --dir <install_directory></code> d. Enter y when you are prompted to install to the specified directory. e. Enter y to confirm that the installation directory is a project-specific directory. f. Enter the project ID to install the PHY.	Setting Up Your Environment <pre>% unsetenv SNPSLMD_LICENSE_FILE % unsetenv LM_LICENSE_FILE % setenv SNPSLMD_LICENSE_FILE port@host % setenv LM_LICENSE_FILE port@host port: TCP port specified in the license file host: Hostname of the license server</pre> Verifying the License Setup <pre>% echo \$SNPSLMD_LICENSE_FILE% echo \$LM_LICENSE_FILE% lmsstat lmsstat -a -c port@host port: TCP port specified in the license file host: Hostname of the license server</pre> Installing the PHY <pre>% ./<filename>.run --dir <install_directory></pre> Running Simulation PHY: <pre>% <install_folder>/synopsys/<product_name>/<version>/phy/sim/./run_cust_tb_regression.sh -sim_type gtech -sim_tool vcs -grid 0</pre> UPCS: <pre><install_folder>/synopsys/<product_name>/<version>/upcs/sim/./run_cust_tb_regression_upcs.sh -sim_tool vcs -grid 0</pre>	Setting Up Your Environment Solvnet Registration: https://solvnet.synopsys.com/ProcessRegistration Download Image: https://www.designware.com or https://www.mydesignware.com Licensing Information: https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html Supported Platforms: Visit the Synopsys support page at: https://www.synopsys.com/support.html Resources Readme files, Databooks, Application notes (if any), and other documents (if any) that are packaged with the product Log files that are generated when you install the product or run simulation Locations: Readme files, Databooks, Application notes: <install_folder>/synopsys/<product_name>/<version>/doc/ Product Information: https://www.synopsys.com/dw/ipdir.php?ds=dwc_multi_protocol_32g_phy

Customer Support

Website: <https://www.synopsys.com/support/global-support-centers.html>
Enter a Call: <http://solvnet.synopsys.com/EnterACall>
Email: support_center@synopsys.com
Telephone: North America: Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, M-F

Ensure that you have the following details:

- Product name and Sub Product name
- Version of the product
- Issue Severity
- Problem type
- Problem
- Licensed Product
- Customer Tracking Num
- Any files: debug files, waveforms